

Mathematical Methods for Physics 2

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Goal of these notes is to give a succinct introduction both to the mathematical theory of operators on Hilbert space and to that of distributions. As much as linear algebra and calculus play a pivotal role in understanding classical mechanics and general physics, so these topics are paramount for understanding all quantum theories (quantum mechanics, quantum field theory, quantum statistical mechanics) both at a conceptual and at a computational level.

I am not so sure that there will ever be a reader of these notes, barring of course the author himself. Yet, if I am proven to be incorrect, I am sure that, after a couple of pages, such a reader will be complaining and cursing the author since he is going through yet another long series of definitions, theorems and proofs. I can hear him/her saying “I am here to study physics, I do not care about all this math”. My answer is “Sorry pal, life is not fair. You have to endure it, because, all this math, all these useless theorems are the hearth of what makes the experiments

reliable. The consistency of a physical theory is what differs us from an alchemist (I am ready to make an exception for Edward Elric)”.
Anyway, enough blathering! Let’s get started, shall we?

Acknowledgments

No, we shall not! Because fair is fair! After so many years, it is due time that I acknowledge all those who have helped me improving these notes by spotting a seemingly unending series of typos, a word which across the multiverse is always used in place of mistake whenever the ego of the author should not be shattered, be it for one reason or the other. I am especially grateful in rigorous random order to Valter Moretti and to the many phone calls during which we discussed the theory of operators, to Nicolò Drago, especially for the many discussions on the theory of distributions and on deficiency indices. I and Nicolò are grateful to Mario Carante for drawing our attention to the Laplace transform, forcing us not only to study it but to admit that it is indeed useful... sometimes. I am very much indebted to Rubens Longhi who was the first student who ever read these notes, years after taking the exam and finding the first mist... typos, especially on the Fourier transform. I am afraid that I have corrected them only now and not out of my good will. Many thanks also to Alberto Bonicelli and Paolo Rinaldi each of them for enduring two years of exercise classes using my notes, a deed worth one or two songs. I am indebted with Giovanni Bracchi who helped me finding new proofs, shorter ones and more elegant ones especially since he really read and learned what is written in [2]. At last but only in chronological order, I am very grateful to Beatrice Costeri, who is probably the first one who has really read from the first to the last line, ok maybe appendix excluded, these notes. She had the patience of going through the proofs and of pointing out unclear explanation and several typos, finding them even after many generations of students went through these notes and mysteriously did not notice them.

1 Preliminary concepts – a summary

In this section we recall the basic definitions, structures and properties of Hilbert spaces. We shall not dwell into many details, leaving to the course of Mathematical Methods for Physics 1 the duty of a more advanced analysis.

1.1 Banach Spaces

Our starting point is $\mathbb{R}^n$, which plays for example the role of configuration space in classical mechanics. Besides being a vector/linear space, it can be endowed with additional important structures which allow to assign a notion of length to each of its elements and of relative angle to each pair of its elements. This is nothing but the well-known Pythagoras theorem, but there is more to it than it meets the eyes. As a matter of fact, while, for finite dimensional real vector spaces, these structures can always be constructed mimicking the example of $\mathbb{R}^n$, it is

more complicated to mimic them as soon as we allow for infinite dimensions. First of all we must translate the notion of assigning a length to a vector into that of a precise mathematical structure. Bear in mind that, in these notes, the vector spaces are built over the complex field unless it is explicitly stated that we consider real vector spaces.

Definition 1. Let V be any real or complex vector space. A **norm** is a map $\|\cdot\| : V \rightarrow \mathbb{R}$ such that

1. $\|v\| \geq 0$, for any $v \in V$,
2. $\|\lambda v\| = |\lambda| \|v\|$, for all $v \in V$ and for all $\lambda \in \mathbb{R}/\mathbb{C}$,
3. $\|v + w\| \leq \|v\| + \|w\|$ (triangle inequality) for all $v, w \in V$,
4. $\|v\| = 0$ if and only if $v = 0$.

The pair $(V, \|\cdot\|)$ is called a **normed space**. If condition 4. is not met, but the others are, we call $\|\cdot\|$ a **seminorm**.

One can construct easily several examples of various interest in practical applications:

- Let $V = \mathbb{C}^n$ and let $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}$ be such that

$$\|v\| = \left(\sum_{i=1}^n |v_i|^2 \right)^{\frac{1}{2}}, \quad \forall v \in V \quad (1)$$

where $v = \sum_{i=1}^n v_i e_i$, e_i , $i = 1, \dots, n$ being the standard basis of $\mathbb{C}^n$ seen as a vector space. It is straightforward to verify that this is a norm and we leave it as an exercise to the reader.

- Let $V = C_0^\infty(\mathbb{R}; \mathbb{C}) \equiv C_0^\infty(\mathbb{R})$ be the space of complex, valued, smooth and compactly supported functions on $\mathbb{R}$. It is a normed space if endowed with $\|\cdot\| : C_0^\infty(\mathbb{R}) \rightarrow \mathbb{R}$

$$\|f\|_p = \left(\int_{\mathbb{R}} dx |f(x)|^p \right)^{\frac{1}{p}}, \quad \forall f \in C_0^\infty(\mathbb{R}), \quad (2)$$

where $p \in \mathbb{N}$. Notice that condition 1. is met since the integrand is strictly positive, while condition 2. is the one responsible for the factor $\frac{1}{p}$. The third requirement is a by-product of the inequality $|a + b|^p < |a|^p + |b|^p$ valid for all complex numbers a, b and for all strictly positive integers p . To conclude condition 4. descends from noticing that $\|f\|_p = 0$ implies that integral of $|f(x)|^p$ vanishes, a condition which is tantamount to $|f(x)| = 0$ since the integrand is everywhere non-negative.

At this stage it would be natural to ask ourselves whether Definition 1 is really satisfactory and necessary. The answer is no for a reason which is at the same technical and practical. On the one hand, if we consider a normed space $(V, \|\cdot\|)$, we would like to make sense of a notion of convergence of sequences of elements $v_n \in V$, $n \in \mathbb{N}$. In other words, can we define $\lim_{n \rightarrow \infty} v_n$ and does it identify an element of V ? On the other hand, while the notion of length of a vector is important per se, the main reason for defining it stems from the necessity of making precise at a mathematical level the heuristic concept of “small with respect to” and “close to”. These ideas, often used in physical models, can be formalized only if a quantifier such as the norm is introduced. Yet, one has to bear in mind that, when saying that two objects are close to each other, both objects need to be part of the same space on which the norm is defined, else the whole discussion would be pointless. Assuming only Definition 1 this does not happen necessarily. We need a few more definitions. Bear with me.

Definition 2. Let $(V, \|\cdot\|)$ be a normed space. For any $v \in V$, we call **open ball** of centre v and radius $r > 0$, the set

$$B_r(v) \doteq \{v' \in V \mid \|v' - v\| < r\}.$$

A set $W \subseteq V$ is called **bounded** if there exists $B_r(v)$, an open ball of finite radius such that $W \subseteq B_r(v)$.

Exercise 1. Prove that the following two statements descend from Definition 2:

1. Let $v, v' \in V$ and let $r, r' \in (0, \infty)$. Then

$$B_r(v) \cap B_{r'}(v') = \emptyset, \quad \text{if } 0 < r + r' < \|v - v'\|.$$

2. Let $v, v' \in V$ and let $r, r' \in (0, \infty)$. Then

$$B_r(v) \subset B_{r'}(v'), \quad \text{if } v \in B_{r'}(v'), \quad \text{and} \quad 0 < r + \|v - v'\| < r'.$$

In addition to the notion of open ball, we can construct the following additional structure on a normed space:

Definition 3. Let $(V, \|\cdot\|)$ be a normed space. Then we say that $W \subset V$ is **open** if it is either empty or the union of a, possibly infinite but countable, number of open balls. The collection of all open sets identifies the **norm topology**¹ of $(V, \|\cdot\|)$.

In order to understand why we insisted on these concepts, a reader should recall the definition of continuous map learned during the course of calculus. It was based on the so-called (ϵ, δ) -prescription. This was nothing but a statement on how close were two elements in the domain and in the range of the map under consideration. Such structure did make use of the norm structure of $\mathbb{R}$ and it can be easily generalized to generic normed spaces as follows:

¹For a mathematically inclined reader (indeed, up to now, we only stated the basics), we observe that the norm topology makes V a topological space which enjoys, moreover, the Hausdorff property.

Definition 4. Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|')$ be two normed spaces. A map $f : V \rightarrow W$ is **continuous** at $v \in V$, if $\forall \epsilon > 0$, there exists $\delta > 0$ such that

$$\|f(v') - f(v)\|' < \epsilon, \quad \text{whenever} \quad \|v' - v\| < \delta.$$

If this property holds for all points in V , we say that f is **continuous**² and we write for simplicity $\lim_{v \rightarrow v'} f(v) = f(v')$.

In between the continuous maps between normed spaces, there is a special class which plays a key role in all physical models:

Definition 5. Let $(V, \|\cdot\|)$ and $(W, \|\cdot\|')$ be two normed spaces. We say that a linear map $U : V \rightarrow W$ is **isometric** or equivalently an **isometry** if

$$\|U(v)\|' = \|v\| \quad \forall v \in V.$$

If U is, moreover, both injective and surjective, we call it an **isomorphism of normed spaces** and we say that V and W are isomorphic under U .

Exercise 2. Prove that every isometry as per Definition 5 is both continuous and injective.

At the same time, we can use the norm to make sense of a notion of convergence of sequences of elements, much in the same spirit of what is done on $\mathbb{R}$:

Definition 6. Let $(V, \|\cdot\|)$ be a normed space and let $\{v_n\}_{n \in \mathbb{N}}$ be a sequence of elements in V . We say that v_n **converges** to $v \in V$, i.e., $\lim_{n \rightarrow \infty} v_n = v$ if, for all $\epsilon > 0$, there exists $N_\epsilon \in \mathbb{N}$ such that for all $n > N_\epsilon$

$$\|v_n - v\| < \epsilon \quad \text{or equivalently} \quad \lim_{n \rightarrow \infty} \|v_n - v\| = 0.$$

Having under control the notion of converging sequences allows us to mimic the theory on $\mathbb{R}$ noticing that every converging sequence $\{v_n\}_{n \in \mathbb{N}}$ in a normed space $(V, \|\cdot\|)$ enjoys the so-called *Cauchy property*: For every $\epsilon > 0$, there exists $N_\epsilon \in \mathbb{N}$ such that, for all $n, m > N_\epsilon$, it holds

$$\|v_n - v_m\| < \epsilon.$$

The collection $\{v_n\}_{n \in \mathbb{N}}$ is called a **Cauchy sequence**. The key advantage of considering this property is that on $\mathbb{R}$ one can prove easily that a sequence of numbers is convergent if and only if it is a Cauchy sequence. This is a desirable property since it guarantees that a collection of elements in a normed space, which are close to each other in a controlled way, is the approximation of a certain element of the space itself. Unfortunately such property is not necessarily valid in normed spaces and we have to add it to our requirements leading thus to

²Notice that the above notion of continuity can be read also as the statement that the pre-image of an open ball is itself an open ball. In other words the definition of a continuous map requires only a notion of open sets and, thus, of topology. The existence of a norm is not at all necessary.

Definition 7. We call **Banach space** a normed space which is **complete**, that is all its Cauchy sequences are convergent to an element of the space itself.

Exercise 3. Prove that $\mathbb{R}^n$ endowed with (1) is a Banach space, while $C_0^\infty(\mathbb{R})$ endowed with (2) is not, no matter the choice of p .

This last exercise raises naturally the question whether, given a space which is normed but not Banach, it is possible to complete it by adding a sufficient number of elements to make every Cauchy sequence convergent. The answer is positive and we omit the proof which is very similar to the one used to completing $\mathbb{Q}$ to $\mathbb{R}$ – see [5, Th. 2.15]:

Theorem 8. Let $(V, \|\cdot\|)$ be a normed space. Then

1. There exists a Banach space $\mathbb{B}(V)$, called **completion** of V such that V is isometric to a dense subspace³ of $\mathbb{B}(V)$ under a linear injective map $J : V \rightarrow \mathbb{B}(V)$,
2. if $(J_1, \mathbb{B}_1, \|\cdot\|_1)$ is a triple such that $(\mathbb{B}_1, \|\cdot\|_1)$ is a Banach space while $J_1 : V \rightarrow \mathbb{B}_1$ is a linear isometry whose image is dense in $\mathbb{B}_1$, then there exists a unique isomorphism of normed spaces $U : \mathbb{B}(V) \rightarrow \mathbb{B}_1$ such that $J_1 = U \circ J$.

1.2 Hilbert Spaces

While the notion of norm generalizes the idea of assigning a length to each element of a vector space and it leads to the notion of Banach space, we are still falling short of one key structure, namely we need to generalize the notion of angle between two vectors. In the following all vector spaces that we shall consider are assumed to be complex.

Definition 9. Let V be a complex vector space. We call **Hermitian inner product** (or *Hermitian scalar product*) a map $S : V \times V \rightarrow \mathbb{C}$ such that

1. $S(v, v) \geq 0$ for all $v \in V$,
2. $S(v, \alpha v' + \beta w) = \alpha S(v, v') + \beta S(v, w)$ for all $v, v', w \in V$ and for all $\alpha, \beta \in \mathbb{C}$,
3. $S(v, w) = \overline{S(w, v)}$ for all $v, w \in V$,
4. $S(v, v) = 0$ if and only if $v = 0$

Notice that the map S is often indicated simply as $(\cdot, \cdot)$, although we try to avoid this nomenclature at this stage. The following theorem characterizes a few notable properties of inner products, which are of paramount relevance to answer rather natural questions such as whether an inner product also identifies a norm.

³The notion of a dense set is developed in the Appendix, see Definition 215.

Theorem 10. *Let (V, S) be a vector space endowed with an Hermitian inner product. The Cauchy-Schwarz inequality holds:*

$$|S(v, w)|^2 \leq S(v, v)S(w, w), \quad \forall v, w \in V,$$

where the equality applies if and only if v and w are linearly dependent.

Proof. Let $v, w \in V$ be different from 0 and let $\lambda \in \mathbb{C}$. In view of 1. and 3. in Definition 9, it holds

$$0 \leq S(v + \lambda w, v + \lambda w) = S(v, v) + 2\Re(\lambda S(v, w)) + |\lambda|^2 S(w, w),$$

where in the equality we used also that S is bilinear. Choose $\lambda = -\frac{\overline{S(v, w)}}{S(w, w)}$ and the above inequality becomes

$$0 \leq S(v, v) - \frac{|S(v, w)|^2}{S(w, w)},$$

which entails the sought inequality. If either v or w vanish, the Cauchy-Schwarz inequality becomes trivial. Suppose now that v is proportional to w , say $w = \lambda v$, $\lambda \in \mathbb{C}$. Then the equality holds since

$$|S(v, w)|^2 = |\lambda|^2 S(v, v)^2 = S(v, v)S(\lambda v, \lambda v) = S(v, v)S(w, w).$$

Conversely, assume that there exists $w, v \in V$, both non vanishing, such that $|S(v, w)|^2 = S(v, v)S(w, w)$. Then, if we introduce $v' = \frac{v}{\sqrt{S(v, v)}}$ and $w' = \frac{w}{\sqrt{S(w, w)}}$, it holds that $|S(u', v')|^2 = 1$, hence $|S(u', v')| = 1$ being the absolute value of a complex number a non negative number. In other words, there exists $a \in \mathbb{R}$ such that $S(u', v') = e^{ia}$. We look for $\lambda, \mu \in \mathbb{C}$ such that $\lambda u' + \mu v' = 0$. By condition 4. in Definition 9, this is tantamount to claiming that

$$0 = S(\lambda u' + \mu v', \lambda u' + \mu v') = |\lambda|^2 + |\mu|^2 + 2\Re(\bar{\lambda}\mu e^{ia}),$$

where we used that $S(u', u') = S(v', v') = 1$ per construction. It suffices to choose $\lambda = e^{ib}$, $\mu = e^{ic}$ with $-b + c + a = \pi$ and we have found the sought constants for which v' and u' and, consequently also u and v are proportional. $\square$

Exercise 4. Prove the Cauchy-Schwarz inequality for a vector space V endowed with an Hermitian semi-inner product, namely a map S which enjoys all properties of Definition 9 barring 4. (Suggestion: Follow the proof of Theorem 10, but notice that $S(w, w)$ can vanish even if $w \neq 0$.)

Thanks to Theorem 10, we are ready to spell out the following remark

Remark 1. *Every vector space with an Hermitian inner product (V, S) is also a normed space since one can define $\|\cdot\| : V \rightarrow \mathbb{R}$ as*

$$\|v\| = \sqrt{S(v, v)}, \quad \forall v \in V. \quad (3)$$

Comparing with Definition 1, we see that (3) meets automatically the requirements 1., 2. and 4. The only one which is not obvious is the third. Yet, if $v, w \in V$, it holds that

$$\begin{aligned}\|v + w\|^2 &= S(v + w, v + w) = S(v, v) + S(v, w) + S(w, v) + S(w, w) = \\ &= S(v, v) + S(w, w) + 2\Re(S(v, w)).\end{aligned}$$

Furthermore, in view of the Cauchy-Schwarz inequality $\Re(S(v, w)) \leq |S(v, w)| \leq \|v\| \|w\|$ which entails the triangle inequality $\|v + w\|^2 \leq \|v\|^2 + \|w\|^2 + 2\|v\| \|w\| = (\|v\| + \|w\|)^2$.

In addition, since every vector space with an Hermitian inner product is also a normed space, we can apply Definition 2 and 3 to endow it with norm topology.

Remark 2. A second notable feature of an Hermitian inner product S on a vector space V is the possibility of defining a notion of orthogonality between two vectors $u, v \in V$, which we indicate by $u \perp v$ and which is characterized by the equation $S(u, v) = 0$. More generally if $W \subseteq V$ is a vector subspace of V , we introduce the **orthogonal space** to W as

$$W^\perp \doteq \{v \in V \mid S(w, v) = 0, \forall w \in W\}.$$

Observe that, if we consider a third vector subspace $W' \subseteq W \subseteq V$, then $W^\perp \subseteq W'^\perp$.

Exercise 5. Let (V, S) be a vector space endowed with an Hermitian inner product. Show that

1. the **parallelogram rule** holds: For all $v, w \in V$

$$\|v + w\|^2 + \|v - w\|^2 = 2(\|v\|^2 + \|w\|^2), \quad (4)$$

where $\|v\|^2 = S(v, v)$ for all $v \in V$.

2. the **polarisation identity** holds: For all $v, w \in V$

$$S(v, w) = \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2 - i\|v + iw\|^2 + i\|v - iw\|^2). \quad (5)$$

Since all vector spaces with an Hermitian inner product are also normed spaces, the notion of a continuous map between two such spaces is taken verbatim from Definition 4. Nonetheless the notion of isometry and of isomorphism need a few rearrangements:

Definition 11. Let (V, S) and (V', S') be two vector spaces endowed with an Hermitian inner product. A linear map $U : V \rightarrow V'$ is called an **isometry** if

$$S'(U(v), U(w)) = S(v, w), \quad \forall v, w \in V.$$

If U is moreover injective and surjective we call it an **isomorphism** and we say that (V, S) and (V', S') are isomorphic as inner product spaces under the action of U .

Exercise 6. Show that every isometry between two vector spaces endowed with an Hermitian inner product is necessarily injective.

Notice that a vector space endowed with an Hermitian inner product suffers of the same problem of a normed space, namely we can introduce a notion of convergence of sequences following Definition 6, but we have no guarantee a priori that all Cauchy sequences are converging to an element of the vector space. This leads to the following definition:

Definition 12. We call Hilbert space $\mathcal{H}$ a complex vector space equipped with an Hermitian inner product S which is a Banach space with respect to the norm induced by S .

Notice that an isomorphism between Hilbert spaces is also called a **unitary operator**, a nomenclature which we will be using henceforth. The following theorem characterizes better the relation between Definition 12 and that of a vector spaces with an Hermitian inner product:

Theorem 13. Let (V, S) be a complex vector space V endowed with an Hermitian inner product S . Then

1. there exist a Hilbert space $\mathcal{H}$ together with an Hermitian inner product $(\cdot, \cdot)$ as well as an injective map $J : V \rightarrow \mathcal{H}$ such that $J(V)$ is dense in $\mathcal{H}$ and $(J(v), J(v')) = S(v, v')$ for all $v, v' \in V$.
2. if there exists a second triple $(J_1, \mathcal{H}_1, (\cdot, \cdot)_1)$ such that $J_1 : V \rightarrow \mathcal{H}_1$ is an injective map whose image is dense in $\mathcal{H}_1$ and $(J_1(v), J_1(v'))_1 = S(v, v')$ for all $v, v' \in B$, then there exists a unique unitary operator $U_1 : \mathcal{H} \rightarrow \mathcal{H}_1$ such that $U_1 \circ J = J_1$.

1.3 Reflexivity of Hilbert Spaces

Goal of this subsection is to prove one of the most important properties of Hilbert spaces: They are *reflexive*, namely they are (canonically) isomorphic to their dual space exactly as every finite dimensional vector space. As we will see this is a by-product of Riesz representation theorem. We recall a basic definition of set theory:

Definition 14. Let V be a vector space over the complex numbers and let K be any non empty subset. We say that K is **convex**, if $\forall v, v' \in K$ and $\forall \lambda \in [0, 1]$, $\lambda v + (1 - \lambda)v' \in K$.

Following the notation of [5], we call $\langle K \rangle$ the vector subspace of V generated by all linear combinations of elements in K and we denote with $\overline{K}$ the closure of K in V , whenever V is endowed with a suitable notion of convergence of sequences, *e.g.*, when it has a topology.

Theorem 15. Let $\mathcal{H}$ be a Hilbert space and let $K \subseteq \mathcal{H}$ be a non empty set. Then

1. $K^\perp \doteq \{\psi \in \mathcal{H} \mid (\psi, \phi) = 0, \forall \phi \in K\}$ is a closed subspace of $\mathcal{H}$,
2. $K^\perp = \langle K \rangle^\perp = \overline{\langle K \rangle}^\perp = \overline{\langle K \rangle^\perp}$,
3. If $K \subseteq \mathcal{H}$ is a closed subspace, than $\mathcal{H} = K \oplus K^\perp$,
4. $(K^\perp)^\perp = \overline{\langle K \rangle}$.

Proof. 1. $K^\perp$ is a linear space on account of the bilinearity of the Hermitian inner product. Let $\{\psi_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence of elements in $K^\perp \subseteq \mathcal{H}$. Since $\mathcal{H}$ is complete per hypothesis, there must exist $\psi \in \mathcal{H}$ such that $\lim_{n \rightarrow \infty} \psi_n = \psi$. At the same time, for all $n \in \mathbb{N}$, $(\psi_n, \phi) = 0$ for all $\phi \in K$ and, by the continuity of $(\cdot, \cdot)$ we can take the limit as $n \rightarrow \infty$, obtaining $(\psi, \phi) = 0$, that is $\psi \in K^\perp$.

2. The first identity descends trivially from the bilinear structure of the Hermitian inner product, while the second is nothing but item 1. The third equality follows from this reasoning: Since $\langle K \rangle \subseteq \overline{\langle K \rangle}$, then $\overline{\langle K \rangle}^\perp \subseteq \langle K \rangle^\perp$. At the same time, per definition of closure, it holds $\langle K \rangle^\perp \subseteq \overline{\langle K \rangle}^\perp$, from which it descends $\langle K \rangle^\perp = \overline{\langle K \rangle}^\perp$.

3. Since $K \cap K^\perp = \{0\}$ per construction, we need to show that $\langle K, K^\perp \rangle = \mathcal{H}$, that is $\mathcal{H}$ is generated by linear combinations of elements in K and $K^\perp$. Let $\psi \in \mathcal{H}$ and choose $\phi \in K$ so to minimize $\|\psi - \phi\|$ as $\phi \in K$ varies. Let $\tilde{\phi} \doteq \psi - \phi$. If we show that $\tilde{\phi} \in K^\perp$ we have concluded. Let $\phi_0 \in K$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be

$$f(t) \doteq \|\psi - \phi + t\phi_0\|^2.$$

We choose ϕ_0 so that f is extremal⁴ at $t = 0$. To this end, we compute

$$f'(0) = \lim_{\epsilon \rightarrow 0} \frac{\|\tilde{\phi} + \epsilon\phi_0\|^2 - \|\tilde{\phi}\|^2}{\epsilon} = 2\Re(\tilde{\phi}, \phi_0).$$

Setting $f'(0) = 0$, we obtain that the real part of the inner product between $\tilde{\phi}$ and ϕ_0 vanishes. Repeating the same calculation with $i\phi_0$ in place of ϕ_0 yields that also the imaginary part of the inner product is 0. In other words $\phi_0 \in K^\perp$.

4. Since $K \perp K^\perp$ per construction and since the Hermitian inner product on $\mathcal{H}$ is continuous, we can consider the completion of $\langle K \rangle$ obtaining that $\overline{\langle K \rangle} \subset (K^\perp)^\perp$. Using both the second and the third item of this theorem, we obtain

$$\mathcal{H} = \overline{\langle K \rangle} \oplus \overline{\langle K \rangle}^\perp = \overline{\langle K \rangle} \oplus K^\perp.$$

Suppose that $\psi \in (K^\perp)^\perp \subset \mathcal{H}$. There must exist $\phi_0 \in \overline{\langle K \rangle}$ and $\phi_1 \in K^\perp$ such that $\psi = \phi_0 + \phi_1$. Taking the scalar product with ϕ_1 , we obtain that

$$(\phi_1, \psi) = \|\phi_1\|^2 = 0,$$

where we used that $K^\perp \perp \overline{\langle K \rangle}$ as well as the hypothesis that $\psi \in (K^\perp)^\perp$. As a consequence of $\|\phi_1\| = 0$, $\phi_1 = 0$. Hence $\psi = \phi_0 \in \overline{\langle K \rangle}$, that is $(K^\perp)^\perp \subseteq \overline{\langle K \rangle}$, which concludes the proof. $\square$

Corollary 16. *Let $\mathcal{H}$ be a Hilbert space. A subset $K \subset \mathcal{H}$ is dense in $\mathcal{H}$ if and only if $K^\perp = \{0\}$.*

Proof. This is a straightforward consequence of item 2. and 4. of Theorem 15. $\square$

We can prove the main statement of this subsection:

Theorem 17 (Riesz). *Let $\mathcal{H}$ be a Hilbert space. For any continuous linear functional⁵ $F :$*

⁴Notice that this is actually a minimum.

⁵The notion of continuous functional is the one of Definition 4.

$\mathcal{H} \rightarrow \mathbb{C}$, there exists a unique $\psi_F \in \mathcal{H}$ such that

$$F(\psi) = (\psi_F, \psi).$$

The map $F \mapsto \psi_F$ is a bijection.

Proof. Let F be any continuous linear functional as per hypothesis. Continuity entails that $\ker(F) = \{\psi_0 \in \mathcal{H} \mid F(\psi_0) = 0\}$ is a closed subspace of $\mathcal{H}$. Theorem 15 guarantees us that $\mathcal{H} = \ker(F) \oplus \ker(F)^\perp$. If $\ker(F) = \mathcal{H}$, then we can choose $\psi_F = 0$. If not, then $\dim(\ker(F)^\perp) = 1$. To prove it, let $\phi \in \ker(F)^\perp$. For any $\phi_1 \in \ker(F)^\perp$, $\phi_1 - \frac{F(\phi_1)}{F(\phi)}\phi \in \ker(F)^\perp$ since $\ker(F)^\perp$ is a closed subspace. At the same time this linear combination lies in $\ker(F)$ since

$$F\left(\phi_1 - \frac{F(\phi_1)}{F(\phi)}\phi\right) = F(\phi) - \frac{F(\phi_1)}{F(\phi)}F(\phi) = 0.$$

Yet, we know that $\ker(F) \cap \ker(F)^\perp = \{0\}$. Hence $\phi_1 - \frac{F(\phi_1)}{F(\phi)}\phi = 0$, which means that ϕ generates the whole $\ker(F)^\perp$. We can now make the following ansatz:

$$\psi_F \doteq \frac{\overline{F(\phi)}}{\|\phi\|^2}\phi.$$

Let $\psi \in \mathcal{H}$. We can write it as $\psi = \tilde{\psi} + \tilde{\phi}$ where $\tilde{\psi} \in \ker(F)$ and $\tilde{\phi} \in \ker(F)^\perp$. The previous analysis tells us that $\tilde{\phi} = \frac{F(\tilde{\phi})}{F(\phi)}\phi$. Yet, per linearity of F , $F(\psi) = F(\tilde{\psi}) + F(\tilde{\phi}) = F(\tilde{\phi})$. Hence

$$\tilde{\phi} = \frac{F(\psi)}{F(\phi)}\phi, \implies \psi = \tilde{\psi} + \frac{F(\psi)}{F(\phi)}\phi$$

As a by-product

$$(\psi_F, \psi) = \left(\frac{\overline{F(\phi)}}{\|\phi\|^2}\phi, \tilde{\psi} + \frac{F(\psi)}{F(\phi)}\phi\right) = \left(\frac{\overline{F(\phi)}}{\|\phi\|^2}\phi, \frac{F(\psi)}{F(\phi)}\phi\right) = F(\psi),$$

where we used in the second equality that $(\tilde{\psi}, \phi) = 0$, while, in the third, that the Hermitian inner product is anti-linear in the first entry. The map $F \mapsto \psi_F$ is well-defined and also injective. This is consequence of the fact that the only continuous map F such that $F(\psi) = 0$ for all $\psi \in \mathcal{H}$ can only be the trivial map. $F \mapsto \psi_F$ is also surjective, since, for all $\psi' \in \mathcal{H}$, one can define a continuous linear functional $F_{\psi'} : \mathcal{H} \rightarrow \mathbb{C}$ as $F_{\psi'}(\psi) \doteq (\psi', \psi)$. $\square$

Corollary 18. Every Hilbert space $\mathcal{H}$ is reflexive, that is, if one considers $\mathcal{H}^* \doteq \{F : \mathcal{H} \rightarrow \mathbb{C} \mid F \text{ is continuous and linear}\}$, the linear map $\mathcal{I} : \mathcal{H} \rightarrow (\mathcal{H}^*)^*$, such that

$$\mathcal{I}(\psi)[F] = F(\psi), \quad \forall \psi \in \mathcal{H}, \text{ and } \forall F \in \mathcal{H}^*$$

is a unitary operator, namely a surjective isometry.

Proof. Riesz theorem 17 entails the existence of an isomorphism between $\mathcal{H}$ and its dual $\mathcal{H}^*$. In addition, we can turn this map into an isometry of Hilbert spaces if we endow $\mathcal{H}^*$ with the Hermitian inner product

$$(F', F)_{\mathcal{H}^*} \doteq (\psi_F, \psi_{F'}), \quad (6)$$

where ψ_F and $\psi_{F'}$ are constructed from F and F' as per theorem 17. By applying once more Riesz theorem, we can infer that for every $H \in (\mathcal{H}^*)^*$, there must exist $F_H \in \mathcal{H}^*$ such that $H(F') = (F_H, F')_{\mathcal{H}^*}$ for all $F' \in \mathcal{H}^*$. On the one hand repeating the same line of reasoning as above, we can conclude that there exists an isometry between $\mathcal{H}^*$ and $\mathcal{H}^{**}$ if we endow the latter with the Hermitian inner product

$$(H, H')_{\mathcal{H}^{**}} \doteq (F_{H'}, F_H)_{\mathcal{H}^*}.$$

Combining these results, it descends that there exists an isometry of Hilbert spaces between $\mathcal{H}$ and $\mathcal{H}^{**}$. On the other hand, equation (6) entails that $(F_H, F')_{\mathcal{H}^*} = (\psi_{F'}, \psi_{F_H})_{\mathcal{H}} = F'(\psi_{F_H})$. In other words $H(F') = F'(\psi_{F_H})$, which is nothing but $\mathcal{J}(\psi_{F_H})[F']$. This entails that the map $\mathcal{J}$ is an isomorphism which realizes the isometry between $\mathcal{H}$ and $\mathcal{H}^{**}$, hence $\mathcal{H}$ is reflexive. $\square$

2 Bounded Operators on Hilbert Spaces

In this section we start the study of a notable class of maps between Hilbert spaces whose application to physics are manifold. We shall introduce them gradually, highlighting only the minimal structures which are necessary. Unless stated otherwise we shall work over the field of complex numbers, but all results can be adapted to the real numbers mutatis mutandis.

Definition 19. Let V, V' two complex vector spaces. We call a map $T : V \rightarrow V'$ a **(linear) operator** from V to V' if

$$T(\alpha v + \beta v') = \alpha T(v) + \beta T(v'). \quad \forall v, v' \in V, \text{ and } \forall \alpha, \beta \in \mathbb{C}$$

We call $\mathcal{L}(V, V')$ the collection of all these maps.

The definition can be refined if we assume additionally that we are dealing with Banach spaces:

Definition 20. Let X, Y be two Banach spaces. Then

- We call $\mathcal{B}(X, Y) \subset \mathcal{L}(X, Y)$ the set of all continuous linear operator from X to Y where continuity is defined as per Definition 4. If $Y = X$ then we write $\mathcal{B}(X) \equiv \mathcal{B}(X, X)$ and $\mathcal{L}(X) \equiv \mathcal{L}(X, X)$.
- If $Y = \mathbb{C}$ (or $\mathbb{R}$ if we were working over real vector spaces), we would call $X^* = \mathcal{L}(X, \mathbb{C})$ the **algebraic dual** of X and $X' = \mathcal{B}(X, \mathbb{C})$ the **topological dual** of X .

It is straightforward to see that all these spaces are linear themselves inheriting the vector space structures from X and Y . Nonetheless, the class of all linear operators is certainly huge and it encompasses manifold examples with rather different behaviours. Consider the following simple example:

Example 21. Let $X = Y = l^2(\mathbb{C})$ and let $T : l^2(\mathbb{C}) \rightarrow l^2(\mathbb{C})$ be defined via its action on a generic sequence $z = \{z_n\}_{n \in \mathbb{N}} \in l^2(\mathbb{C})$ as the shift operator

$$T(z_1, z_2, z_3, \dots) = (0, z_1, z_2, \dots).$$

In other words $T(\{z_n\}_{n \in \mathbb{N}}) = \{z'_n\}_{n \in \mathbb{N}}$ where $z'_1 = 0$ while $z'_n = z_{n-1}$ for $n > 1$. If we recall that $\|z\|_{l^2}^2 = \sum_{n=1}^{\infty} |z_n|^2$, a direct computation shows that $\|T(z)\|_{l^2}^2 = \|z\|_{l^2}^2$. Hence, in view of Definition 11, T is an isometry.

As we see T preserves the norm of the Hilbert space element and it is desirable to characterize all the operators which have a similar behaviour.

Theorem 22. Let X, Y be two normed spaces over the complex numbers with norm $\|\cdot\|_X$ and $\|\cdot\|_Y$ respectively. Let $T \in \mathcal{L}(X, Y)$. The following two conditions are equivalent:

1. there exists $K \in \mathbb{R}$ such that $\|T(v)\|_Y \leq K\|v\|_X$ for all $v \in X$,
2. $\sup_{v \in X \setminus \{0\}} \frac{\|T(v)\|_Y}{\|v\|_X} < \infty$

Proof. Suppose 1. holds. Then, for all $v \in V$, $\frac{\|T(v)\|_Y}{\|v\|_X} \leq K < \infty$. Taking the supremum of the left hand side yields the desired result. Suppose now that 2. holds. We can identify K with $\sup_{v \in X \setminus \{0\}} \frac{\|T(v)\|_Y}{\|v\|_X}$ and, per definition of supremum, 1. descends. $\square$

From now on we shall **omit** the subscripts in the norms indicating the space on which they are defined unless there is a risk of confusion. In this case we will reinstate them. The above theorem justifies the following definition:

Definition 23. Let X, Y be normed spaces over the complex numbers. We say that $T \in \mathcal{L}(X, Y)$ is **bounded** if one of the conditions of Theorem 22 are met. We call **operator norm** of T

$$\|T\| = \sup_{v \neq 0} \frac{\|T(v)\|}{\|v\|}.$$

Using the linearity of T is often convenient to rewrite the definition of operator norm in this fully equivalent way:

$$\|T\| = \sup_{v \mid \|v\|=1} \|T(v)\|. \quad (7)$$

Observe that a direct consequence of this definition is that for every bounded operator T between two normed spaces X, Y , it holds that, for every $v \in X$,

$$\|T(v)\| \leq \|T\| \|v\|.$$

It is clear that bounded operators are a nice class of maps between normed spaces, but, as we have seen in Definition 4, we can also require an operator to be continuous, a property which is often desirable as one knows for example from classical mechanics. Luckily enough it turns out that boundedness and continuity of an operator between two normed spaces are two sides of the same coin:

Theorem 24. *Let $T : X \rightarrow Y$ be an operator between two normed spaces over the complex numbers. The following statements are equivalent:*

1. T is continuous at $0 \in X$,
2. T is continuous,
3. T is bounded.

Proof. $3 \implies 1$. This is the simplest implication since, if T is bounded, per linearity, it entails that there exists $K \in \mathbb{R}$ such that $\|T(v - v')\| \leq K\|v - v'\|$, for all $v, v' \in X$. Following Definition 4, for every $\epsilon > 0$, it suffices to take $\delta = \frac{\epsilon}{K}$.

$1 \implies 2$. Suppose that T is continuous at $0 \in X$ and consider $\lim_{v \rightarrow v'} T(v)$. It holds that since the limit is taken with respect to v ,

$$\lim_{v \rightarrow v'} T(v) - T(v') = \lim_{v \rightarrow v'} T(v - v') = \lim_{w \rightarrow 0} T(w) = 0,$$

where $w \doteq v - v'$. Notice that, beside the assumption, we used that T is linear. The second equality can be seen writing the limits in terms of norms.

$2. \implies 3$. Suppose that T is continuous. This is true in particular at $0 \in X$. Hence we can find $\delta > 0$ such that $\|v\| < \delta$ and $\|T(v)\| < 1$ with $v \in X$. Let δ' be such that $0 < \delta' < \delta$ and let $w \in X$. We define $v' \doteq \delta' \frac{w}{\|w\|}$ and we observe that $\|v'\| = \delta' < \delta$. This entails that $\|T(v')\| = \delta' \frac{\|T(w)\|}{\|w\|} < 1$ which implies $\|T(w)\| < \frac{\|w\|}{\delta'}$. This is nothing but condition 1. in Theorem 22 with $K = \frac{1}{\delta'}$. Hence T is bounded. $\square$

To conclude this small excursus, we explain without proving it that there is a reason why the name operator norm is used - see [5, Th. 2.41]:

Proposition 25. *Let X, Y be two normed spaces over the complex numbers. Then the following statements hold true:*

- The operator norm defines a norm over $\mathcal{B}(X, Y)$,
- If Y is a Banach space, then $\mathcal{B}(X, Y)$ is a Banach space itself.
- if Z is a third normed space over the complex numbers than it holds

$$\|\mathbb{I}\| = 1, \quad \|ST\| \leq \|S\|\|T\|, \quad \forall S \in \mathcal{B}(Y, Z) \text{ and } \forall T \in \mathcal{B}(X, Y),$$

where $\mathbb{I}$ is the identity operator on any normed space⁶.

⁶Observe that, if $X = Y = Z$ Proposition 25 entails that $\mathcal{B}(X) \equiv \mathcal{B}(X; X)$ possesses the structure of an algebra.

An additional important consequence which arises from the hypothesis that the target space is a Banach space is the following:

Proposition 26 (Extension of bounded operators). *Let X a normed space over the complex numbers and let Y be a complex Banach space. Let $W \subset X$ be dense and let $T \in \mathcal{B}(W, Y)$, then there exists a unique $\tilde{T} \in \mathcal{B}(X, Y)$ such that $\tilde{T}|_W = T$ and $\|\tilde{T}\| = \|T\|$.*

Proof. Let us show existence. For any $x \in X$, being W dense in X , there exists a sequence $x_n \in W$, $n \in \mathbb{N}$ such that $\lim_{n \rightarrow \infty} x_n = x$. Since T is bounded, it holds that there exists $K > 0$ finite

$$\|T(x_n) - T(x_m)\| \leq K\|x_n - x_m\|.$$

This inequality shows that since $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence, so it must be $\{T(x_n)\}_{n \in \mathbb{N}}$. Being Y a Banach space per hypothesis, its completeness entails that there exists $\lim_{n \rightarrow \infty} T(x_n)$ which we call $\tilde{T}x$. It is immediate to show that, since $\|\cdot\|$ is a continuous map from Y to $\mathbb{R}$, then the limit does not depend on the approximating sequence. Furthermore, per construction (choose the constant sequence), it holds that $\tilde{T}|_W = T$ and that $\tilde{T}$ is linear. Since $\|T(x_n)\| \leq K\|x_n\|$, by taking the limit, one obtains that $\|\tilde{T}\| \leq K$, hence $\tilde{T} \in \mathcal{B}(X, Y)$.

Let us prove uniqueness. Suppose there exists a second extension U with the same properties of $\tilde{T}$. Take any sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements of W converging to $x \in X$. The restriction constraint entails that $\tilde{T}(x_n) = T(x_n) = U(x_n)$. Hence the sequence $(\tilde{T} - U)(x_n)$ is identically 0 for every finite n . It converges to $0 \in Y$, which entails $\tilde{T}(x) = U(x)$ for all $x \in X$.

To conclude we need to control the norm of $\tilde{T}$. Let $\{x_n\}_{n \in \mathbb{N}}$ be a sequence of elements in W converging to x . It holds

$$\|\tilde{T}(x)\| = \lim_{n \rightarrow \infty} \|T(x_n)\| \leq \|T\| \lim_{n \rightarrow \infty} \|x_n\| = \|T\| \|x\|,$$

from which it descends $\|\tilde{T}\| \leq \|T\|$. At the same time

$$\|\tilde{T}\| = \sup_{x \in X, x \neq 0} \frac{\|\tilde{T}(x)\|}{\|x\|} \geq \sup_{x \in W, x \neq 0} \frac{\|\tilde{T}(x)\|}{\|x\|} = \|T\|,$$

where in the last equality we used that $\tilde{T}|_W = T$ and where we used the definition of operator norm as in item 2. of Theorem 22. $\square$

Although it turns out that most of the physically interesting instances of operators, *e.g.*, the Hamiltonian, are not bounded, there are a few examples which are quite important and enlightening:

1. Every unitary operator U on a Hilbert space $\mathcal{H}$ is bounded. As a matter of fact, since U preserves the norm, for every $\psi \in \mathcal{H}$, it holds $\|U(\psi)\| = \|\psi\|$. By taking the supremum it holds that $\|U\| = 1$. The relevance of this observation stems from the consideration that the time evolution of a dynamical system encoded in the Schrödinger equation or

the action of a continuous symmetry (rotations, translations and so on and so forth) on a Hilbert space are all described via a unitary operator. Although at this stage, it might sound even confusing, bear in mind, that this statement relies strongly on Proposition 26 since the time evolution as well as the action of continuous isometries are almost always defined on dense subspaces of $\mathcal{H}$, rather than on the whole Hilbert space.

2. Consider $\mathcal{H} = L^2(0, 1)$ as well as the so-called **multiplication operator** $\hat{x} : L^2(0, 1) \rightarrow L^2(0, 1)$ such that $\hat{x}(\psi) = x\psi$ for all $\psi \in L^2(0, 1)$. The following inequality holds

$$\|\hat{x}(\psi)\|^2 = \int_0^1 dx |x\psi|^2 \leq \int_0^1 dx |\psi|^2 = \|\psi\|^2.$$

As a consequence, not only $\hat{x}$ is well defined map from $L^2(0, 1)$ into itself, but it is also a bounded operator. By taking the supremum over all unit-norm elements of the Hilbert space, it descends $\|\hat{x}\| \leq 1$. Observe that a straightforward generalization yields that the multiplication operator on $L^2(a, b)$ is bounded whenever $-\infty < a < b < \infty$. In addition $\|\hat{x}\| \leq b - a$. It is also immediate to realize that none of these conclusions, including well-posedness, holds true on $L^2(\mathbb{R})$.

3. Consider $\mathcal{H} = L^2(\mathbb{R}^3)$ as well as the standard Cartesian coordinates on $\mathbb{R}^3$. The **parity transformation** is the map $P : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $P(x, y, z) = (-x, -y, -z)$. It induces a counterpart on $\mathcal{H}$, namely the **parity operator** $\hat{P} : L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}^3)$ such that $\hat{P}\psi = \psi(P(x, y, z)) = \psi(-x, -y, -z)$ where $\psi(x, y, z)$ stands for any representative of the equivalence class $\psi \in L^2(\mathbb{R}^3)$. It holds

$$\|\hat{P}(\psi)\|^2 = \int_{\mathbb{R}^3} d^3x |\psi(-x, -y, -z)|^2 = \|\psi\|^2,$$

where the last equality descends after the obvious change of coordinates in the integral. As a by-product $\hat{P}$ is a unitary, hence bounded operator. It plays a special role since it is associated in quantum mechanics and, more generally in quantum field theory, to a non continuous symmetry of the underlying quantum system.

Exercise 7. Verify that $\mathcal{B}(X, Y)$ is a normed space whenever endowed with the operator norm.

We conclude quoting some of the most important theorems concerning operators between Banach spaces – see [5, Th. 2.91]:

Theorem 27 (of Banach-Schauder). *Let X, Y be two Banach spaces over the complex numbers and let $T \in \mathcal{B}(X, Y)$ be surjective. Then T is an open map.*

An important corollary of this theorem is the following

Theorem 28 (Banach inverse operator theorem). *Let X, Y be two Banach spaces over the complex numbers and let $T \in \mathcal{B}(X, Y)$ be bijective. Then*

1. $T^{-1} \in \mathcal{B}(Y, X)$,
2. there exists $K' > 0$ such that, for all $v \in X$ $\|T(v)\| \geq K'\|v\|$.

Proof. 1. T^{-1} inherits linearity from that of T , while the Banach-Schauder theorem guarantees us that T is an open map and thus T^{-1} must be continuous since the pre-image of any open set is itself open. Theorem 24 entails that T^{-1} is bounded.

2. Since T^{-1} is bounded, there exists $K > 0$ such that, for all $w \in Y$, $\|T^{-1}(w)\| \leq K\|w\|$. Notice that we have excluded $K = 0$ since it would be in contradiction to the hypothesis of T being invertible. Since T is a bijection, for all $w \in Y$, there exists $v \in X$ such that $w = T(v)$. Hence $\|v\| = \|T^{-1}(T(v))\| \leq K\|T(v)\|$, from which descends the sought conclusion by fixing $K' = K^{-1}$. $\square$

2.0.1 The Closed Graph Theorem

We focus now on a construction related to normed spaces, which will play a role especially in the section devoted to unbounded operators. Let thus X, Y be normed spaces whose norms are $\|\cdot\|_X$ and $\|\cdot\|_Y$. Since these are vector spaces, we can construct their direct sum $X \oplus Y$ and we can endow it naturally with a norm $\|\cdot\|$ defined as follows:

$$\|(x, y)\| \doteq \max\{\|x\|_X, \|y\|_Y\}. \quad \forall (x, y) \in X \oplus Y \quad (8)$$

Exercise 8. Show that the topology induced on $X \oplus Y$ by $\|\cdot\|$ is equivalent to the product topology induced by $\|\cdot\|_X$ and $\|\cdot\|_Y$.

It is also immediate to infer, that, if $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach spaces, so is $(X \oplus Y, \|\cdot\|)$.

Definition 29. Let X, Y be normed spaces on the complex numbers. Let $T \in \mathcal{L}(X, Y)$ and let

$$G(T) \doteq \{(x, T(x)) \in X \oplus Y \mid x \in X\},$$

be the **graph** of T . We call T **closed** if $G(T)$ is closed in the product topology.

Exercise 9. Prove that $G(T)$ is closed if and only if for any sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in X such that $\lim_{n \rightarrow \infty} x_n = x \in X$ and such that the sequence $\{T(x_n)\}_{n \in \mathbb{N}}$ converges in Y , we have $\lim_{n \rightarrow \infty} T(x_n) = T(x)$.

We can now state the main theorem of this subsection:

Theorem 30 (Closed graph theorem). Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces over the complex numbers. Then $T \in \mathcal{L}(X, Y)$ is bounded if and only if it is closed.

Proof. Suppose $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach space and consider $T \in \mathcal{B}(X, Y)$. Then, if $\{x_n\}_{n \in \mathbb{N}}$ is a sequence of elements in X converging to $x \in X$, then $\|T(x_n)\| \leq \|T\| \|x_n\|$, which implies that $\lim_{n \rightarrow \infty} T(x_n) = T(x)$. This entail that $G(T)$ is closed in the product topology, hence T is a closed operator.

Suppose now that T is closed and consider the bilinear map $\pi_1 : G(T) \rightarrow X$ which projects on the first entry, *i.e.*, $\pi_1(x, T(x)) = x$. The projection is a continuous map and, per definition, $\|\pi_1(x, T(x))\| = \|x\|_X \leq \|(x, T(x))\|$, where the inequality is a consequence of (8). In other words π_1 is a bounded map and Banach inverse operator theorem 28 guarantees us that $\pi_1^{-1} : X \rightarrow G(T) \subset X \oplus Y$ is bounded. Consider now the projection on the second element, $\pi_2 : X \oplus Y \rightarrow G(T)$ such that $\pi_2(x, T(x)) = T(x)$. Using the same line of reasoning as for π_1 , we can conclude that π_2 is a continuous, hence bounded, map. At last we observe that we have written T as $\pi_2 \circ \pi_1^{-1}$. Since these are the composition of bounded maps, T is bounded itself. $\square$

2.1 The Adjoint Operator

We examine now one of the main consequences of Riesz theorem leading to the definition of the adjunction of an operator. The relevance of this new operation cannot be emphasized enough since it is at the hearth of the formulation of quantum mechanics and of all quantum theories. In this section we focus only on the case of bounded operators, which do not encompass all physically or mathematically interesting scenarios, but it can be easily tamed and hence it is well-suited for a first approach. Notice that, unless otherwise specified *all Hilbert spaces are indicated as* $\mathcal{H} \equiv (\mathcal{H}, (\cdot, \cdot))$. We will leave the scalar product understood in the definition unless a disambiguation becomes necessary. We recall also that, unless stated otherwise, all Hilbert spaces can be non separable. For all practical purposes, all results are valid without any requirement of separability unless in the proof one has to choose a countable basis or one has to rely on a theorem where such choice has been made.

Proposition 31. *Let $(\mathcal{H}_1, (\cdot, \cdot)_1)$ and $(\mathcal{H}_2, (\cdot, \cdot)_2)$ be two Hilbert spaces and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. There exists a unique operator $T^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ such that*

$$(\phi, T\psi)_2 = (T^*\phi, \psi)_1, \quad \forall \psi \in \mathcal{H}_2, \text{ and } \forall \phi \in \mathcal{H}_1.$$

The operator $T^ \in \mathcal{L}(\mathcal{H}_2, \mathcal{H}_1)$ is called the (Hermitian) adjoint (or sometimes Hermitian conjugate) of T .*

Proof. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be as per hypothesis and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. Fix $\phi \in \mathcal{H}_2$ and consider the map $f_\phi : \mathcal{H}_1 \rightarrow \mathbb{C}$ such that $f_\phi(\psi) \doteq (\phi, T\psi)_2$. We observe that $f_\phi \in \mathcal{B}(\mathcal{H}_1)$. As a matter of facts f_ϕ inherits linearity in its argument from the bilinearity of the Hermitian inner product on $\mathcal{H}_2$. The Cauchy-Schwarz inequality – see Theorem 10 – entails that

$$|f_\phi(\psi)| = |(\phi, T\psi)| \leq \|\phi\|_2 \|T\psi\|_2 \leq \|\phi\|_2 \|T\| \|\psi\|_1,$$

where we used that T is per assumption bounded. Hence dividing each side of the inequality by $\|\psi\|_1$, one obtains

$$\frac{|f_\phi(\psi)|}{\|\psi\|_1} \leq \|T\| \|\phi\|_2.$$

Since ϕ is arbitrary but fixed, we can take the supremum over all $\psi \in \mathcal{H}_1$ to conclude via 2. in Theorem 22 that f_ϕ is a bounded operator. Theorem 24 entails that f_ϕ is continuous. We can thus apply Riesz's Theorem to infer the existence of an element $\Psi_\phi \in \mathcal{H}_1$ such that

$$(\phi, T\psi)_2 = (\Psi_\phi, \psi)_1 \quad \forall \psi \in \mathcal{H}_1$$

In addition the bilinearity of $(\cdot, \cdot)_2$ ensures that the map $\mathcal{H}_2 \ni \phi \mapsto f_\phi : \mathcal{H}_1 \rightarrow \mathbb{C}$ is linear and, consequently, also the map $\mathcal{H}_2 \ni \phi \mapsto \Psi_\phi \in \mathcal{H}_1$. This justifies introducing the linear operator $T^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ defined as

$$T^*(\phi) \doteq \Psi_\phi.$$

It remains to be seen that T^* enjoys the wanted properties. First of all, for every $\psi \in \mathcal{H}_1$, it holds:

$$(T^*(\phi), \psi)_1 = (\Psi_\phi, \psi)_1 = (\phi, T\psi)_2.$$

We are left with the proof that T^* is unique. We proceed by contradiction, assuming that there exists $K \in \mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$ different from T^* such that, for all $\phi \in \mathcal{H}_2$ and for all $\psi \in \mathcal{H}_1$, $(K(\phi), \psi)_1 = (\phi, T\psi)_2$. This implies also that $(K(\phi), \psi)_1 = (T^*(\phi), \psi)_1$ or, in other words, $((K - T^*)(\phi), \psi)_1 = 0$. Since ψ is arbitrary we can choose $\psi = (K - T^*)(\phi)$, hence obtaining

$$((K - T^*)(\phi), (K - T^*)(\phi))_1 = \|(K - T^*)(\phi)\|_1 = 0,$$

where $\|\cdot\|_1$ is the norm induced by $(\cdot, \cdot)_1$. Item 4. in Definition 1 entails that $(K - T^*)(\phi) = 0$. Since ϕ is arbitrary the sought conclusion descends. $\square$

In full analogy with the nomenclature used in the course of linear algebra when discussing maps between finite dimensional vector spaces, given any two vector spaces V, V' and a linear map $T : V \rightarrow V'$, we call

- *Range* of T the set $\text{Ran}(T) \doteq \{v' \in V' \mid \exists v \in V, \text{ for which } v' = T(v)\},$
- *Kernel* of T The set $\ker(T) \doteq \{v \in V \mid T(v) = 0\}.$

The adjoint of a bounded operator between two Hilbert spaces enjoys many additional notable properties besides the already remarkable ones of existence and uniqueness:

Proposition 32. *Let $(\mathcal{H}_1, (\cdot, \cdot)_1)$ and $(\mathcal{H}_2, (\cdot, \cdot)_2)$ be two Hilbert spaces and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. Then:*

1. The **adjunction map** $*$: $\mathcal{B}(\mathcal{H}_1, \mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ defined as $*(T) = T^*$ for all $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ is antilinear and involutive, namely $(T^*)^* = T$ for all $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$,
2. The adjoint operator T^* is bounded. Furthermore

$$\|T^*\| = \|T\| \quad \text{and} \quad \|TT^*\| = \|T\|^2 = \|T^*T\|.$$

3. if $\tilde{\mathcal{H}}$ is a third Hilbert space, then, for every $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ and for every $S \in \mathcal{B}(\tilde{\mathcal{H}}, \mathcal{H}_1)$

$$(TS)^* = S^*T^*.$$

4. $\ker(T) = [\text{Ran}(T^*)]^\perp$ and $\ker(T^*) = [\text{Ran}(T)]^\perp$,

5. T is bijective if and only the same holds true for T^* . In this case $(T^{-1})^* = (T^*)^{-1}$.

Proof. 1. Since an Hermitian inner product is antilinear with respect to first entry, it descends that, for all $\alpha, \beta \in \mathbb{C}$, for all $\psi \in \mathcal{H}_1$, $\phi \in \mathcal{H}_2$ and for all $T, S \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$

$$(\phi, (\alpha T + \beta S)(\psi))_2 = \alpha(T^*(\phi), \psi)_1 + \beta(S^*(\phi), \psi)_2 = ((\bar{\alpha}T^* + \bar{\beta}S^*)(\phi), \psi)_1,$$

which entails $(\alpha T + \beta S)^* = \bar{\alpha}T^* + \bar{\beta}S^*$. In addition the identities

$$(\phi, T(\psi))_2 = (T^*(\phi), \psi)_1 = (\phi, (T^*)^*\psi)_2, \quad \forall \psi \in \mathcal{H}_1, \forall \phi \in \mathcal{H}_2, \text{ and } \forall T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2),$$

combined with the uniqueness of the adjoint as per Proposition 31 entail that $(T^*)^* = T$.

2. For every $\phi \in \mathcal{H}_2$ and for every $\psi \in \mathcal{H}_1$, Theorem 31 together with the Cauchy-Schwarz inequality guarantees that, for every $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$

$$|(T^*(\phi), \psi)_1| = |(\phi, T(\psi))_2| \leq \|\phi\|_2 \|T(\psi)\|_2 \leq \|\phi\|_2 \|T\| \|\psi\|_1,$$

where all norms are those induced from the Hermitian inner products. The arbitrariness of ψ entails that we can fix it as $\psi = T^*(\phi)$. The above inequality becomes

$$\|T^*(\phi)\|_1^2 \leq \|T^*(\phi)\|_1 \|T\| \|\phi\|_2 \implies \|T^*(\phi)\|_2 \leq \|T\| \|\phi\|_2.$$

Dividing each side of the last inequality by $\|\phi\|_2$ and taking the supremum over all $\phi \in \mathcal{H}_2$, we can use 2. in Theorem 22 to conclude that $T^* \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ and $\|T^*\| \leq \|T\|$. Having established that T^* is a bounded operator we can reverse the above line of reasoning exchanging the role played by T and T^* . Mutatis mutandis and using, where necessary, that the adjunction map is involutive, one gets to the inequality $\|T\| \leq \|T^*\|$ which, combined with the one already proven, yields $\|T\| = \|T^*\|$.

If we recall that Proposition 25 guarantees has that $\mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ is a Banach space and that, for all $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$, $\|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2$. At the same time, per Definition 25, the following chain of relations holds true:

$$\begin{aligned} \|T\|^2 &= \left(\sup_{\psi \neq 0} \frac{\|T(\psi)\|_2}{\|\psi\|_1} \right)^2 = \sup_{\psi \neq 0} \frac{(T(\psi), T(\psi))_2}{\|\psi\|_1^2} = \sup_{\psi \neq 0} \frac{(T^*T(\psi), \psi)_1}{\|\psi\|_1^2} \leq \\ &\quad \sup_{\psi \neq 0} \frac{\|T^*T\psi\|_1}{\|\psi\|_1} = \|T^*T\|. \end{aligned}$$

Notice that the inequality is nothing but the application of Theorem 10, while the second equality stems for the realization that the square of the supremum over a set of positive real numbers

coincides with the supremum of the square of the elements of the set. We have thus shown both that $\|T^*T\| \leq \|T\|^2$ and that $\|T\|^2 \leq \|T^*T\|$, hence $\|T\|^2 = \|T^*T\|$. Since $\|T\| = \|T^*\|$, by exchanging the role played by T and T^* and being the adjunction map involutive, one obtains $\|T^*\|^2 = \|TT^*\|$ and thus the sought $\|T\|^2 = \|T^*T\| = \|TT^*\|$.

3. for every $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$, $S \in \mathcal{B}(\tilde{\mathcal{H}}, \mathcal{H}_1)$ and for all $\phi \in \mathcal{H}_2$, $\psi \in \tilde{\mathcal{H}}$, the following chain of equalities holds true:

$$((TS)^*(\phi), \psi)_{\tilde{\mathcal{H}}} = (\phi, TS(\psi))_2 = (T^*(\phi), S(\psi))_1 = (S^*(T^*(\phi)), \psi)_{\tilde{\mathcal{H}}},$$

where $(\cdot, \cdot)_{\tilde{\mathcal{H}}}$ indicates the Hermitian inner product in $\tilde{\mathcal{H}}$. The uniqueness of the adjoint operator, proven in Proposition 31, guarantees that $(TS)^* = S^*T^*$.

4. We prove only the first one since the second follows suit. Suppose that $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ and that $\psi \in \ker(T)$. Then, since $T(\psi) = 0$, for every $\phi \in \mathcal{H}_2$

$$0 = (\phi, T(\psi))_2 = (T^*\phi, \psi)_1.$$

The arbitrariness of ϕ entails that $\psi \in [\text{Ran}(T^*)]^\perp$. Since we have only used identities, we can trace our step back, starting from the assumption that $\psi \in [\text{Ran}(T^*)]^\perp$, proving as a consequence that $\psi \in \ker(T)$.

5. Suppose T is bijective, then Banach inverse operator theorem 28 guarantees that $T^{-1} \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$. Boundedness implies existence of the adjoint as per Proposition 31. Observing that the adjoint of the identity map $\mathbb{I}$ on a given Hilbert space is the identity itself, the following chain of identities holds true in view of point 3. of this proposition:

$$\mathbb{I} = (\mathbb{I})^* = (TT^{-1})^* = (T^{-1})^*T^* \quad \text{and} \quad \mathbb{I} = (\mathbb{I})^* = (T^{-1}T)^* = T^*(T^{-1})^*.$$

This is tantamount to saying that T^* is invertible, hence a bijection. Furthermore, by direct inspection, it descends that $(T^{-1})^* = (T^*)^{-1}$. The same line of reasoning leads to the sought conclusion starting from the hypothesis that T^* is a bijection. The only difference is the necessity of using also point 1. of this proposition, $(T^*)^* = T$ in particular. $\square$

2.1.1 Further Properties of the Adjoint Operators

We have learned that, given any pair of Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ and a bounded operator $T : \mathcal{H}_1 \rightarrow \mathcal{H}_2$, there is a canonical way to associate to it an adjoint $T^* \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$. We can now make a clever use of this information to identify in between the plethora of all possible bounded, linear operators notable subclasses, of paramount relevance especially in quantum mechanics. We recall that, whenever the target and the source Hilbert space coincide, *i.e.*, $\mathcal{H}_1 = \mathcal{H}_2 = \mathcal{H}$, we shall use the notation $\mathcal{B}(\mathcal{H})$ to indicate the collection of bounded, linear operators from $\mathcal{H}$ to itself. From now on we will also omit all subscripts in the norms and in the Hermitian scalar products whenever the space to which they refer will be clear from the context. So... be careful and pay attention!

Definition 33. Let $\mathcal{H}$ and $\mathcal{H}'$ be two Hilbert spaces. Then we call

- **normal** any $T \in \mathcal{B}(\mathcal{H})$ such that $TT^* = T^*T$,
- **self adjoint** any $T \in \mathcal{B}(\mathcal{H})$ such that $T = T^*$
- **isometric** any $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ such that $(T\psi', T\psi)' = (\psi, \psi')$, for all $\psi, \psi' \in \mathcal{H}$
- **unitary** any isometric operator $T \in \mathcal{B}(\mathcal{H}, \mathcal{H}')$, which is moreover surjective.
- **positive** any linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ such that $(\psi, T\psi) \geq 0$ for all $\psi \in \mathcal{H}$. In this case we say that $T \geq 0$.

Many of these definitions have equivalent formulations, which, depending on the context, might be more useful. We propose two exercises in this direction:

Exercise 10. Let $\mathcal{H}$ and $\mathcal{H}'$ be two Hilbert spaces and let $\mathbb{I}$ be the identity operator on $\mathcal{H}$. Show that $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ is isometric if and only if it is bounded and $T^*T = \mathbb{I}$.

Exercise 11. Let $\mathcal{H}$ and $\mathcal{H}'$ be two Hilbert spaces and let $\mathbb{I}, \mathbb{I}'$ be the respective identity operators. Show that $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ is unitary if and only if it is bounded and $T^*T = \mathbb{I}$, while $TT^* = \mathbb{I}'$.

A useful lemma is the following:

Lemma 34. Let $\mathcal{H}$ be a Hilbert space and let T be a bounded, self-adjoint operator thereon. Then

$$\|T\| = \sup\{ |(\psi, T\psi)|, \mid \psi \in \mathcal{H}, \text{ such that } \|\psi\| = 1 \}. \quad (9)$$

More generally if $T \in \mathcal{L}(\mathcal{H})$ is such that $(\psi, T\psi) = (T\psi, \psi)$ for all $\psi \in \mathcal{H}$ and the right-hand side of (9) is finite, then T is bounded.

Proof. Let $K \doteq \sup\{ |(\psi, T\psi)|, \mid \psi \in \mathcal{H}, \text{ such that } \|\psi\| = 1 \}$. Since we consider unit-norm vectors and since T is bounded, $|(\psi, T\psi)| \leq \|\psi\| \|T\| \|\psi\| = \|T\|$. Hence $K \leq \|T\|$. Since the Hermitian inner product is linear on the second entry and antilinear in the first, it holds that, for all $\psi, \psi' \in \mathcal{H}$

$$4(\psi, T\psi') = (\psi + \psi', T(\psi + \psi')) - (\psi - \psi', T(\psi - \psi')) - i(\psi + i\psi', T(\psi + i\psi')) + i(\psi - i\psi', T(\psi - i\psi')).$$

Taking the real part of both sides⁷ we obtain

$$4\Re(\psi, T\psi') = (\psi + \psi', T(\psi + \psi')) - (\psi - \psi', T(\psi - \psi')).$$

Applying the Cauchy-Schwarz inequality together with the parallelogram rule, we obtain

$$4\Re(\psi, T\psi') \leq K(\|\psi + \psi'\|^2) + K(\|\psi - \psi'\|^2) = 2K(\|\psi\|^2 + \|\psi'\|^2). \quad (10)$$

⁷Observe that, for any Hilbert space $\mathcal{H}$ and for any self-adjoint $T \in \mathcal{L}(\mathcal{H})$ it holds $\overline{(\psi, T\psi)} = (T\psi, \psi) = (\psi, T\psi)$ no matter the choice of $\psi \in \mathcal{H}$.

Here we used that $(\psi + \psi', T(\psi + \psi')) \leq K\|\psi + \psi'\|^2$ because, per definition of K , it holds

$$\left| \left(\frac{\psi + \psi'}{\|\psi + \psi'\|}, T \left(\frac{\psi + \psi'}{\|\psi + \psi'\|} \right) \right) \right| = \left(\frac{\psi + \psi'}{\|\psi + \psi'\|}, T \left(\frac{\psi + \psi'}{\|\psi + \psi'\|} \right) \right) \leq K,$$

and similarly using $\psi - \psi'$. Since both ψ and ψ' in (10) are arbitrary, we can choose ψ' so that $\|\psi'\| = 1$, while $\psi = \frac{T\psi'}{\|T\psi'\|}$ provided that $T\psi' \neq 0$. Since $\|\psi\| = 1$, the inequality (10) becomes

$$4\|T\psi'\| \leq 4K.$$

Notice that, if we would have considered the special case in which $T\psi' = 0$, then this last inequality is automatically satisfied. Using (7), we can take the supremum of the left hand side to conclude that $\|T\| \leq K$.

The last part of the lemma is nothing but the second part of the above proof where it is established that $\|T\| \leq K$, hence that T is bounded. $\square$

Before proceeding with our analysis, three short comments are in due course:

- Given two linear operators $T, U : \mathcal{H} \rightarrow \mathcal{H}$, we can write that $T \geq U$ if $T - U$ is a positive operator,
- All self-adjoint and all unitary operators are normal, but the converse is not true,
- Not all isometric operators from an Hilbert space to itself are unitary as one can readily infer by considering the shift operator defined in Example 21. We leave the details of the computation to the reader.

In view of the first comment we can prove a rather interesting lemma:

Lemma 35. *Let $\mathcal{H}$ be a complex Hilbert space. Then the relation $\geq$ defined out of the notion of positive operators establishes a partial order on $\mathcal{L}(\mathcal{H})$.*

Proof. The relation $\geq$ identified in Definition 33 cannot be of total order since, given two generic $A, B \in \mathcal{L}(\mathcal{H})$, neither $A - B$ nor $B - A$ are necessarily positive. In order to see whether it realizes a partial order, we need to verify the three defining properties of reflexivity, antisymmetry and transitivity.

The first is immediate since, for every $A \in \mathcal{L}(\mathcal{H})$, $A - A = 0$; hence $A \geq 0$. Suppose now that $A, B \in \mathcal{L}(\mathcal{H})$ are such that both $A \geq B$ and $B \geq A$. For every $\psi \in \mathcal{H}$, it holds that both $(\psi, (A - B)\psi)$ and $(\psi, (B - A)\psi)$ are greater or equal to 0. The only option is that $(\psi, (A - B)\psi) = 0$. Using the polarization identity – see Exercise 5 – once replacing ψ with $\psi + \psi'$, $\psi' \in \mathcal{H}$ arbitrary, and once with $\psi + i\psi'$, one establishes that $(\psi', (A - B)\psi) = 0$ for all $\psi, \psi' \in \mathcal{H}$. Choose now a sequence of elements ψ_n such that $\lim_{n \rightarrow \infty} \psi_n = (A - B)\psi$. It holds that $(\psi_n, (A - B)\psi) = 0$. Taking the limit as $n \rightarrow \infty$, thanks to the Cauchy-Schwarz inequality we can conclude that

$$0 = \lim_{n \rightarrow \infty} (\psi_n, (A - B)\psi) = ((A - B)\psi, (A - B)\psi) = \|(A - B)\psi\|^2.$$

In other words $(A - B)\psi = 0$ for all $\psi \in \mathcal{H}$, hence $A = B$.

To conclude we need to establish transitivity. Let $A, B, C \in \mathcal{L}(\mathcal{H})$ be such that $A \geq B$ and $B \geq C$. Then, since the Hermitian inner product is bilinear, it holds that, for all $\psi \in \mathcal{H}$

$$(\psi, (A - C)\psi) = (\psi, (A - B + B - C)\psi) = (\psi, (A - B)\psi) + (\psi, (B - C)\psi) \geq 0.$$

In other words $A \geq C$. □

We stress the hypothesis that $\mathcal{H}$ is a complex Hilbert space, since the above lemma fails if $\mathcal{H}$ is built over the reals. There are several reasons for this fact and finding one is a good exercise:

Exercise 12. Establish why $\geq$ cannot define a partial order on a real Hilbert space.

There is also a notable case where unitary and isometric operators coincide:

Exercise 13. Let $\mathcal{H}$ be a finite dimensional Hilbert space. Show that all isometric operators thereon are also unitary.

We introduce now the simplest, albeit the most important concept of the whole course. Its applications are manifold, being linked especially to the role and to the outcome of a measure in a quantum system. At the same time the definition which follows is nothing but the natural and straightforward generalization of the counterpart learned when studying matrices in linear algebra.

Definition 36. Let V be a complex vector space and let $T \in \mathcal{L}(V)$. We say that $\lambda \in \mathbb{C}$ is an **eigenvalue** of T if there exists an **eigenvector**, namely $v \in V \setminus \{0\}$ such that

$$T(v) = \lambda v.$$

We call **eigenspace** of T relative to the eigenvalue λ , the vector subspace $W_\lambda \subseteq V$, built as the (complex) span of all eigenvectors of λ .

We can now prove a few properties descending from Definition 36:

Proposition 37. Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{B}(\mathcal{H})$ be normal. Then

1. For all $\psi \in \mathcal{H}$ it holds $\|T\psi\| = \|T^*\psi\|$, hence $\ker(T) = \ker(T^*)$ and $\overline{\text{Ran}(T)} = \overline{\text{Ran}(T^*)}$, where the bar stands for the closure to a Hilbert space.
2. λ is an eigenvalue of T with eigenvector ψ if $\bar{\lambda}$ is an eigenvalue of T^* with the same eigenvector,
3. Let λ, λ' be two distinct eigenvalues. Then $W_\lambda \perp W_{\lambda'}$,

Proof. 1. Let T be a bounded, normal operator on $\mathcal{H}$. Hence, for all $\psi \in \mathcal{H}$

$$\|T\psi\|^2 = (T\psi, T\psi) = (T^*T\psi, \psi) = (TT^*\psi, \psi) = (T^*\psi, T^*\psi) = \|T^*\psi\|^2.$$

From this chain of identities, it is immediate to see that if $\psi \in \ker(T)$ then it lies also in $\ker(T^*)$ and vice-versa. Item 4. in Proposition 32 entails that $[\text{Ran}(T^*)]^\perp = [\text{Ran}(T)]^\perp$. To conclude it suffices to recall the generic property of Hilbert spaces, according to which, if we consider any linear subspace W of an Hilbert space $\mathcal{H}$ then $(W^\perp)^\perp = \overline{W}$. In other words $\overline{\text{Ran}(T^*)} = \overline{\text{Ran}(T)}$.

2. Let $\lambda \in \mathbb{C}$. In view of the antilinearity proven in Proposition 32, $(T - \lambda\mathbb{I})^* = T^* - \bar{\lambda}\mathbb{I}$. Furthermore, per direct inspection, $(T - \lambda\mathbb{I})$ is normal since also T is. By item 1. of this proposition, we can infer that, for all $\psi \in \mathcal{H}$

$$\|(T - \lambda\mathbb{I})\psi\| = \|(T^* - \bar{\lambda}\mathbb{I})\psi\|,$$

from which it descends the sought statement since, if $(T - \lambda\mathbb{I})\psi = 0$, then also $(T^* - \bar{\lambda}\mathbb{I})\psi$ vanishes.

3. Let $\lambda, \lambda' \in \mathbb{C}$ be eigenvalues relative respectively to the eigenvectors $\psi, \psi' \in \mathcal{H}$. Using item 2. of this proposition, it holds:

$$\lambda(\psi', \psi) = (\psi', T\psi) = (T^*\psi', \psi) = (\bar{\lambda}'\psi', \psi) = \lambda'(\psi', \psi).$$

Since, per hypothesis $\lambda \neq \lambda'$, the only possibility is that $(\psi', \psi) = 0$, from which the sought result descends. $\square$

Corollary 38. *Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{L}(\mathcal{H})$. Then*

1. *If $T \geq 0$, then its eigenvalues are real and non-negative,*
2. *If T is bounded and self-adjoint, then its eigenvalues are real,*
3. *if T is isometric its eigenvalues λ are such that $|\lambda| = 1$.*

Proof. 1. Suppose that T is positive and that $\lambda \in \mathbb{C}$ is an eigenvalue relative to the eigenvector $\psi \in \mathcal{H}$. Then

$$0 \leq (\psi, T\psi) = \lambda(\psi, \psi) = \lambda\|\psi\|^2.$$

This is possible only if λ is real and non-negative.

2. Suppose that T is bounded and self-adjoint and let λ be an eigenvalue relative to the eigenvector $\psi \in \mathcal{H}$. Then, by item 2. of Proposition 37

$$\lambda(\psi, \psi) = (\psi, T\psi) = (T\psi, \psi) = \bar{\lambda}(\psi, \psi).$$

Since $\psi \neq 0$, it holds that $\lambda = \bar{\lambda}$.

3. Suppose that T is isometric and that λ is an eigenvalue relative to the eigenvector $\psi \in \mathcal{H}$. Then

$$|\lambda|^2(\psi, \psi) = (T\psi, T\psi) = (\psi, \psi).$$

Since $\psi \neq 0$, it descends $|\lambda| = 1$. $\square$

This corollary guarantees us that, when working with bounded operators, the self-adjoint and the positive ones have real eigenvalues. In view of the quantum mechanical interpretation of the eigenvalues of an operator as the possible results of a measurement process of a system, it becomes of paramount importance to establish criteria to identify operators whose eigenvalues are real. The next proposition is a first step in this direction, with the net advantage that it tells us that positive bounded operators are just a special class of the self-adjoint ones:

Proposition 39. *Let $\mathcal{H}$ be a Hilbert space. Then the following statements hold true:*

1. *If $T \in \mathcal{L}(\mathcal{H})$ is such that $(\psi', T\psi) = (T\psi', \psi)$ for all $\psi, \psi' \in \mathcal{H}$, then T is bounded and self-adjoint,*
2. *if $T \in \mathcal{B}(\mathcal{H})$ is such that $(\psi, T\psi) = (T\psi, \psi)$ for all $\psi \in \mathcal{H}$, then T is self-adjoint,*
3. *if $T \in \mathcal{B}(\mathcal{H})$ is positive, then it is also self-adjoint.*

Proof. 1. It suffices to show that T is bounded, since the uniqueness of the adjoint proven in Proposition 31 entails that, in this case, $T = T^*$. Boundedness is a by-product of the *closed graph theorem* 30. We focus only on the remaining items.

2. The hypothesis entails that, for any $\psi \in \mathcal{H}$, $((T - T^*)\psi, \psi) = 0$. The same argument used in Lemma 35 guarantees that this entails $\|(T - T^*)\psi\| = 0$, hence $T = T^*$.

3. Suppose now that T is positive. Then, for any $\psi \in \mathcal{H}$ it holds that $(\psi, T\psi) \geq 0$ and in particular it is real. Thus it coincides with its complex conjugate. By the third property in Definition 9, this is tantamount to saying that

$$(\psi, T\psi) = (T\psi, \psi).$$

Hence we are back to the case analysed in item 2. □

2.1.2 Orthogonal Projectors

We need one last elementary notion related to boundary operators before dwelling into more technical results:

Definition 40. *Let $\mathcal{H}$ be a Hilbert space. We say that $P \in \mathcal{B}(\mathcal{H})$ is an **orthogonal projector** if $P^2 = P$ and $P^* = P$.*

Lemma 41. *Let $\mathcal{H}$ be a Hilbert space and let P be an orthogonal projector. Then P is a positive operator.*

Proof. Let $\psi \in \mathcal{H}$ be arbitrary. Then it holds

$$(\psi, P\psi) = (\psi, P^2\psi) = (P^*\psi, P\psi) = (P\psi, P\psi) = \|P\psi\|^2 \geq 0,$$

which entails that $P \geq 0$ since P is bounded per construction. □

One of the nice properties of orthogonal projectors is the possibility of characterizing them with great detail. Since they are of extreme usefulness in analysing self-adjoint operators, this is a welcome feature:

Theorem 42. *Let $\mathcal{H}$ be a Hilbert space and let $P \in \mathcal{B}(\mathcal{H})$ be an orthogonal projector. Let $W \doteq P[\mathcal{H}]$. The following holds:*

1. $Q = \mathbb{I} - P$ is an orthogonal projector, being $\mathbb{I}$ the identity on $\mathcal{H}$,
2. $Q(\mathcal{H}) = W^\perp$ and $\mathcal{H} = W \oplus W^\perp$,
3. For all $\psi \in \mathcal{H}$, $\|\psi - P(\psi)\| = \min\{\|\psi - \psi'\|; \psi' \in W\}$,
4. Let e_i , $i \in I$ be a basis of W with the index i running over a set I . Then

$$P(\cdot) = s - \sum_{i \in I} e_i(e_i, \cdot),$$

where s — entails that the convergence of the sum is meant with respect to the strong topology.

5. $\mathbb{I} \geq P$ and, unless $P = 0$, then $\|P\| = 1$.

Proof. 1. We observe that $Q^2 = (\mathbb{I} - P)^2 = \mathbb{I} + P^2 - 2P = \mathbb{I} - P$ since P is a projector per hypothesis. Furthermore $Q^* = (\mathbb{I} - P)^* = \mathbb{I}^* - P^* = Q$. Hence Q is an orthogonal projector

2. Suppose $\psi' = Q(\tilde{\psi})$, $\tilde{\psi} \in \mathcal{H}$. Then, for every $\psi \in W$ it holds

$$(\psi, \psi') = (\psi, Q\tilde{\psi}) = (Q^*\psi, \tilde{\psi}) = (\psi - P\psi, \tilde{\psi}) = 0,$$

where we used that $Q^* = Q$ and that $P\psi = \psi$ since $\psi \in W$. We have shown that $Q(\mathcal{H}) \subseteq W^\perp$. Suppose now that W is a closed subspace and that $\mathcal{H} = W \oplus Q(\mathcal{H})$. Since the theory of Hilbert spaces guarantees also that $\mathcal{H} = W \oplus W^\perp$, it descends that $Q(\mathcal{H}) = W^\perp$ per the uniqueness of the decomposition. We show first that W is closed. Let $\{\psi_n\}_{n \in \mathbb{N}}$ be such that $P(\psi_n)$ is a Cauchy sequence. Since Hilbert spaces are complete, there must exist $\Psi \in \mathcal{H}$ such that $\lim_{n \rightarrow \infty} P(\psi_n) = \Psi$. Since P is bounded, hence continuous per Proposition 24, we can apply P to the last identity, exchanging it with the limits to obtain $\lim_{n \rightarrow \infty} P(\psi_n) = P(\Psi)$ since $P^2 = P$. Hence $\Psi = P\Psi$ and $\Psi \in W$. We have shown that W is complete, thus a closed subspace. Let $\psi \in \mathcal{H}$ be arbitrary; we can write $\psi = w + v$, where $w = P(\psi) \in W$ and $v = Q(\psi) \in Q(\mathcal{H})$. We need to show that this decomposition is unique which is tantamount to proving that $W \cap Q(\mathcal{H}) = \emptyset$.

Suppose there exists $\phi \in W \cap Q(\mathcal{H})$. Then $\phi = P\phi = Q\phi$ which entails $2P\phi = \phi$. Since $\phi = P\phi$, it descends $P\phi = 0$, hence $\phi = 0$.

3. and 4. These are technical results which we omit. The reader can refer to [5].

5. Let $\psi \in \mathcal{H}$ be arbitrary and let P be an orthogonal project. Then

$$0 \leq (Q\psi, Q\psi) = (\psi, Q^2\psi) = (\psi, Q\psi) = \|\psi\|^2 - (\psi, P\psi).$$

Hence $\mathbb{I} \geq P$ and $(P\psi, P\psi) = (\psi, P\psi) \leq \|\psi\|^2$. By (7), this entails taking the supremum over all unit norm $\psi \in \mathcal{H}$ that $\|P\| \leq 1$. Yet, if $P \neq 0$, there must exist $\psi' \in W$ and without loss of generality we can take it of unit norm. Hence, since $P\psi' = \psi'$, we have found a vector for which $\|P\psi\| = 1$, hence $\|P\| = 1$. $\square$

The next statement shows that the requirement of orthogonality for projectors does not entail any loss of generality:

Proposition 43. *Let $\mathcal{H}$ be a Hilbert space and let W be a closed subspace. Let $P : \mathcal{H} \rightarrow W$ and $Q : \mathcal{H} \rightarrow W^\perp$ be the associated projectors. They are orthogonal.*

Proof. Since W is closed subspace, $\mathcal{H} = W \oplus W^\perp$. Hence $Q = \mathbb{I} - P$ where $\mathbb{I}$ is the identity operator on $\mathcal{H}$. It suffices thus to prove that $P^* = P$. Let $\psi, \psi' \in \mathcal{H}$ be arbitrary and let $w, w' \in W$ and $v, v' \in W^\perp$ be the unique elements for which $\psi^{(t)} = w^{(t)} + v^{(t)}$. Then

$$(\psi', P\psi) = (w' + v', w) = (w', w).$$

At the same time

$$(P\psi', \psi) = (w', w + v) = (w', w),$$

from which it descends that $(P\psi', \psi) = (\psi', P\psi) = (P^*\psi', \psi)$ where we used the definition of adjoint in the last identity. It descends $((P - P^*)\psi', \psi) = 0$ for all $\psi \in \mathcal{H}$, in particular if $\psi = (P - P^*)\psi'$. This entails $\|(P - P^*)\psi'\|^2 = 0$ for all $\psi' \in \mathcal{H}$. Hence $P\psi' = P^*\psi'$, i.e., $P = P^*$. $\square$

2.2 The Square Root of a Positive Operator

Positive, bounded operators enjoy several special properties, the most notable being that they are always self-adjoint and their eigenvalues lie in $[0, \infty)$. Additionally we can perform on them certain operations which are in general ill-defined. We review here the most important one:

Definition 44. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. We say that $B \in \mathcal{B}(\mathcal{H})$ is a **square root** of A if $A = B^2$. Furthermore if $B \geq 0$ we call it a **positive root**.*

Positive roots are often indicated also as $\sqrt{A}$, but it is clear that, in general, roots might not be well-defined. We shall give sufficient conditions for their existence. As a preliminary result we need an ancillary proposition, whose proof we omit, being technical even for the taste of the author [5, Prop. 3.65]:

Proposition 45. *Let $\mathcal{H}$ be a Hilbert space and let $\{A_n\}_{n \in \mathbb{N}}$ be a non decreasing (non-increasing) sequence of bounded, self-adjoint operators thereon. If $\{A_n\}_{n \in \mathbb{N}}$ is bounded from below (above) by $K \in \mathcal{B}(\mathcal{H})$, then there exists a bounded, self-adjoint operator A on $\mathcal{H}$ such that $A \leq K$ ($A \geq K$) and*

$$s - \lim_{n \rightarrow \infty} A_n = A.$$

We can now state the main result of this section:

Theorem 46. *Let $\mathcal{H}$ be an Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$ be positive. Then there exists a unique positive square root $\sqrt{A}$ such that*

- (a) *it commutes with any bounded operator B on $\mathcal{H}$, which, in turn, commutes with A ,*
- (b) *it is bijective is also A is bijective.*

Proof. Just kidding... have a look at [5, Th. 3.66] and you will see what I mean! $\square$

Definition 47. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. We call **modulus** of A the bounded, positive and self adjoint operator*

$$|A| = \sqrt{A^*A}.$$

Corollary 48. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. Then its modulus enjoys the following properties:*

- a) $\ker(|A|) = \ker(A)$,
- b) $\overline{\text{Ran}(|A|)} = (\ker(A))^\perp$

Proof. We start from a). For any $\psi \in \mathcal{H}$ it holds that

$$\| |A|\psi \|^2 = (|A|\psi, |A|\psi) = (\psi, |A|^2\psi) = (\psi, A^*A\psi) = (A\psi, A\psi) = \|A\psi\|^2, \quad (11)$$

where we used that $|A|$ is self-adjoint. From the identities, it is manifest that $|A|\psi = 0$ if and only if $A\psi = 0$.

Let us focus on b). In view of the theory of Hilbert spaces we know that, for any subspace $W \in \mathcal{H}$, $\overline{W} = (W^\perp)^\perp$. Therefore $\overline{\text{Ran}(|A|)} = ((\text{Ran}(|A|))^\perp)^\perp$. At the same time $(\text{Ran}(|A|))^\perp = \ker(|A|)$. As a matter of facts, suppose $\psi \in (\text{Ran}(|A|))^\perp$; for any $\psi' \in \mathcal{H}$

$$0 = (\psi, |A|\psi') = (|A|\psi, \psi').$$

Since ψ' is arbitrary, the only option is $\psi \in \ker(|A|)$. Reading this equality from right to the left, we realize that also the converse holds true, namely if $\psi \in \ker(|A|)$, then $\psi \in (\text{Ran}(|A|))^\perp$. Putting together these results, we obtain $\overline{\text{Ran}(|A|)} = (\ker(|A|))^\perp$. Using item a) of this corollary we obtain the sought conclusion. $\square$

Theorem 49 (Polar Decomposition). *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. Then there exists a unique pair of operators $P, U \in \mathcal{B}(\mathcal{H})$ such that*

- a) $A = UP$,
- b) $P \geq 0$,
- c) U is isometric on $\text{Ran}(P)$,
- d) $U|_{\ker(P)} = 0$.

Furthermore it holds that

- 1. $P = |A|$ and thus $\ker(U) = \ker(A) = \ker(P) = [\text{Ran}(P)]^\perp$,
- 2. if A is bijective, then $U = A|A|^{-1}$.

The splitting $A = UP$ is called the **polar decomposition** of A .

Proof. We start proving uniqueness of the decomposition. Let us focus first of all on P . If any decomposition $A = UP$ with the given properties exists, then $A^* = (UP)^* = PU^*$ since P is positive and bounded, hence self-adjoint per Proposition 39. As a by-product $A^*A = PU^*UP$. Since U is isometric on $\text{Ran}(P)$, it holds that, for all $\psi, \psi' \in \mathcal{H}$

$$(UP\psi, UP\psi') = (P\psi, P\psi') \implies (\psi, PU^*UP\psi') = (\psi, P^2\psi'),$$

which entails $A^*A = P^2$. Hence, for any decomposition as per hypothesis, the positive operator P is the unique positive square root of A^*A , namely $|A|$.

We prove uniqueness of U . As a starting point, since $\mathcal{H} = \ker(P) \oplus (\ker(P))^\perp$ and since P is self-adjoint being positive, we can use item 4. in Proposition 32 to conclude that

$$\mathcal{H} = \ker(P) \oplus [\text{Ran}(P^*)^\perp]^\perp = \ker(P) \oplus \overline{\text{Ran}(P^*)} = \ker(P) \oplus \overline{\text{Ran}(P)}.$$

In view of this decomposition, defining an operator on $\mathcal{H}$ unambiguously is tantamount to specifying its image on $\ker(P)$ and $\overline{\text{Ran}(P)}$. We set $U|_{\ker(P)} = 0$, while $U|_{\text{Ran}(P)}$ is defined via the relation $UP\psi = A\psi$ for every $\psi \in \mathcal{H}$. Since U is bounded per construction, so it is on $\text{Ran}(P)$ and thus it can be extended uniquely to the closure $\overline{\text{Ran}(P)}$ on account of Proposition 26. In other words U is unique. Observe that since $P = |A|$, it holds that, for any $\psi \in \mathcal{H}$, $A\psi = U(|A|\psi)$. If we combine this identity with (11) guaranteeing that $\|A\psi\| = \||A|\psi\|$, we can infer that U is an isometry on $\text{Ran}(P)$. A direct inspection of the above argument entails that, besides uniqueness, we have also shown existence as well as item 1.

We are left with 2. We observe that, if A is injective, item 1. guarantees that also U is injective. At the same time the decomposition $A = UP$ guarantees that $\text{Ran}(A) \subseteq \text{Ran}(U)$ which is tantamount to saying that U is surjective if A is. Hence if A is bijective, U is a bijective isometry, in other words a unitary operator. At last, since $P = |A|$, it holds $A = U|A|$. Since both A and U are bijective, so must be $|A|$. Hence it admits an inverse and it is meaningful to write $U = A|A|^{-1}$.

□

3 Compact Operators

In this section we introduce and we discuss the properties of a notable class of operators. They are of paramount relevance especially on infinite dimensional Hilbert spaces since they represent the closest possible extension of what matrices are on finite dimensional Hilbert spaces. We divide the section in two main parts: The first subsection introduce the main definition and properties, making only use of normed spaces. The reader, who is not familiar with the basic aspects of topology, is strongly advised to revise them. In the second part we will focus mainly on Hilbert spaces and, although, not always necessary, we will make the assumption that all spaces are separable.

3.1 Compact Operators on Normed Spaces

In this section we consider normed spaces which are a special instance of topological spaces with the topology induced by the norm itself, see Appendix A. A preliminary definition is the following:

Definition 50. *Let X be a normed space. A subset $W \subset X$ is said to be **sequentially compact** if and only if every sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in W has a subsequence converging in W .*

Proposition 51. *Let X be a normed space and let $W \subseteq X$. If any sequence of elements in W admits a converging subsequence, then W is relatively compact. The limit might not lie in W .*

Proof. W is relatively compact if $\overline{W}$ is compact. In view of Definition 50 it suffices to prove that for every sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in $\overline{W}$, there exists a subsequence converging in $\overline{W}$. Consider any such sequence and fix n . Per definition of closure there must exist a sequence of elements $\{y_k^{(n)}\}$ in W converging to x_n as $k \rightarrow \infty$. Considering a sufficiently large $\bar{k}$, we can construct a third sequence

$$z_n \doteq y_{\bar{k}}^{(n)}, \quad \|x_n - z_n\| < n^{-1}.$$

Per assumption we can extract a subsequence $\{z_{n_p}\}_{n_p \in \mathbb{N}}$ converging to an element $x \in \overline{W}$. Furthermore

$$\|x_{n_p} - x\| \leq \|x_{n_p} - z_{n_p}\| + \|z_{n_p} - x\|,$$

where the left hand side can be made smaller at leisure since $\|x_{n_p} - z_{n_p}\| < n_p^{-1}$. In other words $x_{n_p} \rightarrow x$ as $n_p \rightarrow \infty$. $\square$

A technical corollary whose proof requires Baire's theorem, which we have not discussed (hence we omit the proof), is the following

Corollary 52. *Let X be an infinite dimensional normed space. Then*

- *if $W \subset X$ is compact, then the interior of W is empty,*
- *if X is a Banach space, then X cannot be obtained as a countable union of compact subsets.*

Before moving to the theory of operators, we remark that an important consequence of these results is the proof of one key difference between finite and infinite dimensional normed spaces. As shown in [5, Prop. 4.5], on any, infinite dimensional, normed space the closure of any open ball of any centre and of any radius cannot be compact. On the contrary, in finite dimension, this is the case as one can show by using the Heine-Borel theorem.

Definition 53. *Let X be a normed space. We say that $W \subset X$ is **bounded** if there exist $x \in X$ and $\delta > 0$ such that $W \subset B_\delta(x) \doteq \{x' \in X \mid \|x - x'\| < \delta\}$.*

Definition 54. *Let X, Y be normed spaces over the complex numbers. We say that $T \in \mathcal{L}(X, Y)$ is a **compact operator** if one of these two conditions is met:*

1. For any bounded subset $W \subset X$, $T(W)$ is relatively compact in Y ,
2. If $\{x_n\}_{n \in \mathbb{N}}$ is a bounded sequence of elements in X , there exists a subsequence $\{x_{n_k}\}_{n_k \in \mathbb{N}}$ such that $T(x_{n_k})$ converges in Y .

The set of all compact operators is indicated with $\mathcal{B}_\infty(X, Y)$ or with $\mathcal{B}_\infty(X)$ if $X = Y$.

Notice that, while the first condition automatically implies the second, the converse is a consequence of Proposition 51. We prove a few notable properties of compact operators, both individually and as a set:

Lemma 55. *Let X, Y be normed spaces over the complex numbers. Then any compact operator $T : X \rightarrow Y$ is also bounded, that is $\mathcal{B}_\infty(X, Y) \subset \mathcal{B}(X, Y)$.*

Proof. Consider $B_1(0)$ the unit ball in X centred at 0. Since T is compact, $T(B_1(0))$ is relatively compact. Its closure can be thus covered with a sufficient number, say N , of balls $B_r(y_i)$, $i = 1, \dots, N$. Hence $\overline{T(B_1(0))} \subset \bigcup_{i=1}^N B_r(y_i) \subset B_{R+r}(0)$ where R is the maximum distance between the centres y_i and the origin 0 in Y . As a by-product, for any $v \in X$ such that $\|v\| = 1$, $\|T(v)\| \leq R + r$, hence $\|T\| \leq r + R$. $\square$

Proposition 56. *Let X, Y be normed spaces over the complex numbers and let $\mathcal{B}_\infty(X, Y)$ be the set of compact operators. Then*

- (a) $\mathcal{B}_\infty(X, Y)$ is a vector subspace of $\mathcal{B}(X, Y)$.
- (b) if Z is normed subspace and $A \in \mathcal{B}_\infty(X, Y)$, then⁸ $AB \in \mathcal{B}_\infty(Z, Y)$ for all $B \in \mathcal{B}(Z, X)$ and $CA \in \mathcal{B}_\infty(X, Z)$ for all $C \in \mathcal{B}(Y, Z)$.
- (c) Let Y be a Banach space and let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of compact operators from X to Y converging uniformly to a bounded operator A . Then $A \in \mathcal{B}_\infty(X, Y)$. In other words $\mathcal{B}_\infty(X, Y)$ is a closed subspace of $\mathcal{B}(X, Y)$ in the norm topology.

The proof of this proposition is very long and we omit it. The interested reader can refer to [5, Prop. 4.9].

Exercise 14. Prove that the identity operator cannot be a compact operator on an infinite dimensional normed space.

Lemma 57 (Banach). *Let x_i with i running over a finite or infinite set of indices be a sequence of linearly independent vectors in a normed space X . Then there exists a corresponding sequence of elements y_j in X such that, for all admissible integers n*

1. $\|y_n\| = 1$ for all n ,
2. $y_n \in X_n \doteq \text{span}_{\mathbb{C}}(x_1, \dots, x_n)$,

⁸In other words we are proving that, for any normed space X , $\mathcal{B}_\infty(X)$ is a (non-unital) algebra.

$$3. d(y_n, X_{n-1}) \doteq \inf_{x \in X_{n-1}} \|x - y_n\| > \frac{1}{2}.$$

Proof. First of all we observe that $d(y_n, X_{n-1})$ exists and it is finite since it is the infimum of a non empty set of real numbers, bounded from below by 0. We build the sought sequence as follows: Let $y_1 \doteq \frac{x_1}{\|x_1\|}$. Since the vectors x_i are linearly independent for all values of i , we know that $x_n \notin X_{n-1}$ and $d(x_n, X_{n-1}) = k > 0$. Choose $x' \in X_{n-1}$ so that $k < \|x_n - x'\| < 2k$. Since $x' \in X_{n-1}$, $k = d(x_n, X_{n-1}) = d(x_n - x', X_{n-1})$. We define $y_n \doteq \frac{x_n - x'}{\|x_n - x'\|}$. Per direct inspection and per construction the new sequence y_n satisfies all three conditions. $\square$

Theorem 58. *Let X be a normed space and let $T \in \mathcal{B}_\infty(X)$. Then*

1. $\forall \delta > 0$, there exists a finite number of eigenspaces of T with eigenvalue λ such that $|\lambda| > \delta$,
2. If $\lambda \neq 0$ is an eigenvalue of T with eigenspace W_λ , then $\dim(W_\lambda) < \infty$,
3. The eigenvalues of T form a bounded, at most countable, set. They can be ordered so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq 0$, where 0 is the only possible limit point.

Proof. 1. and 2. If the normed space is a finite dimensional set the theorem is basically a trivial consequence of standard results from linear algebra. Hence we assume that X is infinite dimensional. Let $\lambda_1, \lambda_2, \dots$ be a sequence of distinct eigenvalues, possibly with multiplicities, such that, for any arbitrary, but fixed $\delta > 0$, there exists $\bar{i} \in \mathbb{N}$ such that $|\lambda_i| > \delta$ for all $i > \bar{i}$. Suppose there exists a corresponding infinite sequence of linearly independent eigenvectors $x_1, x_2, \dots$. Using Banach Lemma 57, we can construct a corresponding sequence $y_1, y_2, \dots$ enjoying all properties proven in the lemma. Consider the associated sequence $\{\frac{y_n}{\lambda_n}\}_{n \in \mathbb{N}}$. It is bounded since $|\lambda_n| > \delta$ for all n . Since T is a compact operator, there must exist a converging subsequence of $\{T(\frac{y_n}{\lambda_n})\}_{n \in \mathbb{N}}$. Notice that per construction, for each index n in the sequence, there exists $\beta_k \in \mathbb{C}$, $k = 1, 2, \dots, n$ such that $y_n = \sum_{k=1}^n \beta_k x_k$. Since each x_k is an eigenvector of T with eigenvalue λ_k

$$T\left(\frac{y_n}{\lambda_n}\right) = \sum_{k=1}^n \beta_k \frac{\lambda_k}{\lambda_n} x_k = y_n + z_n,$$

where, per direct inspection $z_n = \sum_{k=1}^{n-1} \beta_k \left(\frac{\lambda_k}{\lambda_n} - 1\right) x_k$. Let i, j be two indices in the sequences that we are considering and assume that $i > j$. Then

$$\left\|T\left(\frac{y_i}{\lambda_i}\right) - T\left(\frac{y_j}{\lambda_j}\right)\right\| = \|y_i + z_i - y_j - z_j\| > \frac{1}{2},$$

where, using the notation of Banach lemma, $y_j - z_j - z_i \in X_{i-1}$. Item 3. in Lemma 57 entails the above inequality. We have found a contradiction to the hypothesis that $T \in \mathcal{B}_\infty(X)$, since there cannot exist a converging subsequence of $\{T(\frac{y_n}{\lambda_n})\}_{n \in \mathbb{N}}$. Hence there cannot exist infinite linearly independent eigenvectors with eigenvalues greater than any, but fixed, strictly positive number. This proves both 1. and 2.

3. This is a straightforward consequence of item 1., choosing a sequence $\delta_n = n^{-1}$. For each, but fixed $n \in \mathbb{N}$, we have a finite number of eigenvalues, which thus are a bounded set which can be ordered increasingly with respect to their absolute value. By taking $n \rightarrow \infty$, at most we get a countable set, but still bounded. $\square$

3.2 Compact Operators on Hilbert Spaces and their Spectral Representation

In this section, we specialize our analysis of compact operators to the case when the normed space is actually a Hilbert space. A first alternative characterization of compact operators is the following:

Proposition 59. *Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{B}(\mathcal{H})$. Then T is compact if and only if $|T|$ – see Definition 47 – is compact.*

Proof. Suppose T is compact and let $\{x_n\}_{n \in \mathbb{N}}$ be a bounded sequence in $\mathcal{H}$. Since T is compact, we can extract a converging sequence $\{T(x_{n_k})\}_{n_k \in \mathbb{N}}$. Consider now $|T|(x_{n_k})$; (11) guarantees that $\||T|(x_{n_k})\| = \|T(x_{n_k})\|$. Hence also the sequence $|T|(x_{n_k})$ is converging. Exchanging the role of T and $|T|$ we can repeat the same line of reasoning, hence proving the proposition. $\square$

We recall that all properties proven in the previous part of the section are inherited by compact operators on Hilbert spaces, especially that $\mathcal{B}_\infty(\mathcal{H})$ is a vector subspace of $\mathcal{B}(\mathcal{H})$ and that the product between a compact and a bounded operator is still compact – see Proposition 56. In a more refined language this entails that $\mathcal{B}_\infty(\mathcal{H})$ is a two-sided ideal of the algebra of bounded operators. Hence it is legitimate to consider the quotient between these vector spaces and resulting space inherits the algebra structure. Actually we can prove something even stronger which emphasizes the role of compact operators within the class of bounded ones:

Proposition 60. *Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{B}_\infty(\mathcal{H})$. Then $T^* \in \mathcal{B}_\infty(\mathcal{H})$.*

Proof. Let us consider $T \in \mathcal{B}_\infty(\mathcal{H})$. Proposition 59 entails that also $|T| \in \mathcal{B}_\infty(\mathcal{H})$. Using the polar decomposition, seen in Theorem 49, we know that there exists $U \in \mathcal{B}(\mathcal{H})$ such that $T = U|T|$. Per construction the modulus of a bounded operator is self-adjoint. Hence $T^* = |T|U^*$. Yet item 2. of Proposition 32 guarantees us that U^* is bounded. In other words we have written T^* as the product of a compact and of a bounded operator. Hence, in view of Proposition 56, it is a compact operator. $\square$

We can now prove the first and the simplest instance of one of the most important structural aspects of a large class of operators on Hilbert spaces, namely the so-called *spectral decomposition*, often referred to as *spectral theorem* or *spectral calculus*.

Theorem 61 (Hilbert). *Let $\mathcal{H}$ be an Hilbert space and let T be a self-adjoint, compact operator thereon. Then, every eigenspace of T with non-vanishing eigenvalue has finite dimension. Furthermore, if we call $\sigma_p(T)$ the collection of all eigenvalues, it holds:*

1. $\sigma_p(T)$ is a non empty, real set,

2. $\|T\| = \sup\{|\lambda| \mid \lambda \in \sigma_p(T)\},$
3. $T = 0$ if and only if 0 is the only eigenvalue.

Proof. Let us start by proving that every eigenspace of T with non vanishing eigenvalue $\lambda \in \mathbb{C}$ is finite dimensional. The idea is to use [5, Prop. 4.5], according to which, on any normed space the closure of any open ball of any centre and of any radius cannot be compact if the space has infinite dimension. Let $\mathcal{H}_\lambda$ be the eigenspace relative to λ and let $\mathcal{B}_1(0) \subset \mathcal{H}_\lambda$ be the unit ball centred at the origin in $\mathcal{H}_\lambda$. Notice that $\lambda^{-1}\mathcal{B}_1(0)$ is bounded per construction, hence, being T a compact operator $\mathcal{B}_1(0) = T(\lambda^{-1}\mathcal{B}_1(0))$ is relatively compact. This means that $\overline{\mathcal{B}_1(0)}$ is compact, but, in view of [5, Prop. 4.5], this can happen only if $\dim \mathcal{H}_\lambda < \infty$.

1. and 2. We prove that $\sigma_p(T)$ is non empty. If so it is certainly real since T is per assumption self-adjoint and all its eigenvalues are real as per item 2. of Corollary 38. We can rewrite (9) as

$$-\|T\| \leq (\psi, T\psi) \leq \|T\|, \quad \forall \psi \in \mathcal{H} \text{ such that } \|\psi\| = 1$$

which entails that the equality can be obtained either as the infimum or as the supremum of $(\psi, T\psi)$ over all unit norm $\psi \in \mathcal{H}$. Consider the latter case, the former following suit. Assume $\|T\| > 0$ or the theorem becomes trivial as $T = 0$. Fix an eigenvalue λ and a relative eigenvector ψ_λ such that $\|\psi_\lambda\| = 1$. The Cauchy-Schwarz inequality entails that

$$\|T\| \geq |(\psi_\lambda, T\psi_\lambda)| = |\lambda|(\psi_\lambda, \psi_\lambda) = |\lambda|.$$

In other words $\sup_{\lambda \in \sigma_p(T)} \{|\lambda|\} \leq \|T\|$. We exhibit now an eigenvector whose eigenvalue is indeed $\|T\|$. Consider a sequence of elements $\{\psi_n\}_{n \in \mathbb{N}}$ such that $\|\psi_n\| = 1$ for all n and $(\psi_n, T\psi_n) \rightarrow \|T\|$ as $n \rightarrow \infty$. Since $\|T(\psi_n)\| \leq \|T\| \|\psi_n\| = \|T\|$, it holds

$$\|(T(\psi_n) - \|T\| \psi_n)\|^2 = \|T(\psi_n)\|^2 + \|T\|^2 - 2\|T\|(\psi_n, T\psi_n) \leq 2\|T\|^2 - 2\|T\|(\psi_n, T\psi_n),$$

which converges to 0 as $n \rightarrow \infty$. To conclude we need to extract from $\{\psi_n\}_{n \in \mathbb{N}}$ a converging subsequence. Since the sequence is bounded, being $\|\psi_n\| = 1$ for all $n \in \mathbb{N}$, and since T is compact per assumption, there exists $\{T(\psi_{n_k})\}_{n_k \in \mathbb{N}}$, a converging subsequence of $\{T(\psi_n)\}_{n \in \mathbb{N}}$. As a by-product, define

$$\psi_{n_k} = \frac{1}{\|T\|} (T(\psi_{n_k}) - (T(\psi_{n_k}) - \|T\| \psi_{n_k})),$$

which converges to a certain $\psi \in \mathcal{H}$ as $n_k \rightarrow \infty$ since $\lim_{n_k \rightarrow \infty} (T(\psi_{n_k}) - \|T\| \psi_{n_k}) = 0$. In other words we have shown that $T\psi = \|T\|\psi$, which is the sought conclusion.

3. This is straightforward once we established that if 0 is the only eigenvalue, then $\|T\| = 0$, hence $T = 0$. $\square$

The more celebrated theorem proven by Hilbert is actually a continuation of the previous one:

Theorem 62 (Hilbert). *Let $\mathcal{H}$ be an Hilbert space and let T be a self-adjoint, compact operator thereon and let $\sigma_p(T)$ be the collection of its eigenvalues. Then*

1. *If P_λ is the orthogonal projector on the eigenspace of T relative to the eigenvalue λ , we can write*

$$T = \sum_{\lambda \in \sigma_p(T)} \lambda P_\lambda,$$

where the sum is understood in the uniform topology and all eigenvalues are collected in a decreasing order with respect to their absolute value.

2. *$\mathcal{H}$ admits a basis of eigenvectors of T .*

Proof. 1. Consider any eigenvalue $\lambda \in \mathbb{C}$ with eigenspace $\mathcal{H}_\lambda$ and let $P_\lambda : \mathcal{H} \rightarrow \mathcal{H}_\lambda$ be the orthogonal projector. Theorem 42 guarantees us that $\mathcal{H} = \mathcal{H}_\lambda \oplus \mathcal{H}_\lambda^\perp$ and that $Q_\lambda = \mathbb{I} - P_\lambda$ is an orthogonal projector on $\mathcal{H}_\lambda^\perp$. In addition we can show that

$$TP_\lambda = P_\lambda T = \lambda P_\lambda.$$

As a matter of fact, for any $\psi \in \mathcal{H}$, it holds

$$TP_\lambda \psi = \lambda P_\lambda \psi \implies TP_\lambda = \lambda P_\lambda$$

By taking the adjoint and using the hypothesis that $P_\lambda^* = P_\lambda$ and $T^* = T$, we obtain

$$(TP_\lambda)^* = (\lambda P_\lambda)^* \iff P_\lambda T = \lambda P_\lambda,$$

where we used that $\lambda \in \mathbb{R}$ as per Theorem 61. Following slavishly this line of reasoning we draw the same conclusion with respect to Q_λ , namely $Q_\lambda T = TQ_\lambda$. It holds

$$T = T\mathbb{I} = T(P_\lambda + Q_\lambda) = \lambda P_\lambda + Q_\lambda T.$$

In view of the properties of both Q_λ and T we know that $(Q_\lambda T)^* = TQ_\lambda = Q_\lambda T$, that is $Q_\lambda T$ is self-adjoint. Furthermore it is compact since $T \in \mathcal{B}_\infty(\mathcal{H})$ and all projectors Q_λ are bounded. In addition, since $P_\lambda Q_\lambda = Q_\lambda P_\lambda$, it holds

$$P_\lambda(Q_\lambda T) = (Q_\lambda T)P_\lambda.$$

In view of these results, consider $\lambda_0 \in \mathbb{C}$ the eigenvalue of T with highest absolute value. Using the notation $\mathcal{H}_0 \equiv \mathcal{H}_{\lambda_0}$ and similarly for the projection operators, we write $T = \lambda_0 P_0 + T_1$, where $T_1 = Q_0 T$. If $T_1 = 0$ the proof ends; if not, since $T_1 \in \mathcal{B}_\infty(\mathcal{H})$ and since it is self-adjoint, we consider λ_1 , the eigenvalue of T_1 with highest absolute value. Let ψ_1 be an eigenvector relative to λ_1 . This is also an eigenvector of T since

$$T\psi_1 = \lambda_0 P_0 \psi_1 + T_1 \psi_1 = \lambda_1 \psi_1,$$

since $\psi \in \mathcal{H}_0^\perp$. Hence the eigenspace of T and of T_1 relative to λ_1 coincide and we can write $T = \lambda_0 P_0 + \lambda_1 P_1 + T_2$. From Theorem 61 we know, moreover, that $\|T\| = \lambda_0$ and $\|T_1\| = \lambda_1$. We can reiterate the construction for as many steps as necessary obtaining

$$T = \sum_{i=0}^{n-1} \lambda_i P_i + T_n,$$

where $|\lambda_0| \geq |\lambda_1| \geq \dots \geq |\lambda_{n-1}|$. We claim that this sequence of eigenvalues must converge to 0. To show it, consider a vector of unit norm ψ_i in each of the eigenspace $\mathcal{H}_i$. This is a bounded sequence and, thus, being T compact per hypothesis, there must exist a subsequence of $\{T(\psi_i)\}_{i \in \mathbb{N}}$ converging. Suppose by contradiction that there exists $\epsilon > 0$ such that the absolute value of all eigenvalues is greater than ϵ . Then, for all p, m with $p > m$

$$\|T(\psi_p) - T(\psi_m)\|^2 = \|\lambda_p \psi_p - \lambda_m \psi_m\|^2 = |\lambda_p|^2 + |\lambda_m|^2 \geq 2\epsilon,$$

where we used in the last equality that, per construction ψ_p and ψ_m are orthogonal. The above inequality entails that $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$. Hence $\{T(\psi_n)\}_{n \in \mathbb{N}}$ is not a Cauchy sequence and we cannot extract from it a converging subsequence. In other words we have shown that

$$T = \sum_{n=0}^{\infty} \lambda_n P_n,$$

which converges in the uniform topology. From the construction it is also clear that the ordering of the eigenvalues is dictated by convenience and ultimately it plays no role.

We need to prove that the sum encompasses all non-vanishing eigenvalues of T . Suppose there exists $\lambda \neq \lambda_n$ for all possible values of n . Let us call P_λ the orthogonal projective on the associated eigenspace. Since eigenspaces relative to distinct eigenvalues of a self-adjoint, bounded operator are orthogonal, *c.f.* Proposition 37, it holds that $P_\lambda P_{\lambda_n} = P_{\lambda_n} P_\lambda = 0$ for all possible values of n . In addition

$$TP_\lambda = \sum_{n=0}^{\infty} \lambda_n P_{\lambda_n} P_\lambda = 0,$$

which entails that, for every $\psi_\lambda \in \mathcal{H}_\lambda$ $T\psi_\lambda = TP_\lambda \psi_\lambda = 0$, namely ψ_λ is an eigenvector of eigenvalue 0. In other words $\lambda = 0$.

2. Consider each eigenspace $\mathcal{H}_n$ and consider an orthogonal basis on each of these spaces. We need to prove that the union of all these basis is a set of generators for $\mathcal{H}$. It suffices that the orthogonal complement of their span is trivial. Let ψ be in such orthogonal space. Then $T\psi = \sum_{n=0}^{\infty} \lambda_n P_{\lambda_n} \psi = 0$. In other words $\psi \in \ker(T)$, namely it is an eigenvector of eigenvalue 0. At the same time, being orthogonal to each of the eigenspaces built from each P_{λ_n} , $\psi \in \ker(T)^\perp$. To summarize $\psi \in \ker(T) \cap \ker(T)^\perp = \{0\}$. $\square$

Finished? Nope, we have barely begun!!! But we learned something: knowing the eigenvalues of a compact, self-adjoint operator together with their multiplicities is tantamount to knowing the whole operator. Isn't it like with matrices? I told you so! Oh, before I forget, I wanted to tell you (probably "you" is just me ... ah no, ciao Beatrice!) one last thing without proving it:

Theorem 63. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}_\infty(\mathcal{H})$ with $A \neq 0$. Let $\lambda_i, i \in \mathbb{N} \cup \{0\}$ be the non-vanishing eigenvalues of $|A|$ ordered decreasingly. They are called $\text{sing}(A)$, the **singular values of A** . Let $\psi_{\lambda_i, j} \in \mathcal{H}$ be an orthonormal basis of the eigenspace relative to λ_i with j running from 1 to m_i the multiplicity of λ_i . Then*

$$A(\cdot) = \sum_{\lambda_i \in \text{sing}(A)} \sum_{j=1}^{m_i} \lambda_i (\psi_{\lambda_i, j}, \cdot) \tilde{\psi}_{\lambda_i, j},$$

and

$$A^*(\cdot) = \sum_{\lambda_i \in \text{sing}(A)} \sum_{j=1}^{m_i} \lambda_i (\tilde{\psi}_{\lambda_i, j}, \cdot) \psi_{\lambda_i, j},$$

where $\tilde{\psi}_{\lambda_i, j} \doteq \frac{1}{\lambda_i} A(\psi_{\lambda_i, j})$.

3.3 Hilbert-Schmidt Operators

We devote this section to introducing and to studying the main properties of a notable class of compact operators with manifold applications ranging from quantum mechanics to theory of PDEs.

Definition 64. *Let $\mathcal{H}$ be a Hilbert space. We say that $A \in \mathcal{B}(\mathcal{H})$ is a **Hilbert-Schmidt operator** if there is an orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ in $\mathcal{H}$ such that*

$$\sum_{i=1}^{\infty} \|A\psi_i\|^2 < \infty.$$

We call $\mathcal{B}_2(\mathcal{H})$ the collection of all Hilbert-Schmidt operators on $\mathcal{H}$ and we call

$$\|A\|_2 \doteq \sqrt{\sum_{i=1}^{\infty} \|A\psi_i\|^2}. \quad (12)$$

One faces at least two dilemmas: Is the definition independent from the choice of a basis and why is this class of operators so useful? We shall give immediately an answer to both queries (otherwise you might have a sleepless night!). Before proceeding, recall that we are assuming that all Hilbert spaces are separable.

Theorem 65. *Let $\mathcal{H}$ be a Hilbert space and let $\{\psi_i\}_{i \in \mathbb{N}}, \{\phi_j\}_{j \in \mathbb{N}}$ be two orthonormal bases of $\mathcal{H}$. Let $A \in \mathcal{B}(\mathcal{H})$; then*

(a) $\sum_{i=1}^{\infty} \|A\psi_i\|^2$ is finite if and only if such is $\sum_{j=1}^{\infty} \|A\phi_j\|^2$ and the two sums coincide,

(b) $\sum_{i=1}^{\infty} \|A\psi_i\|^2$ is finite if and only if such is $\sum_{j=1}^{\infty} \|A^*\phi_j\|^2$ and the two sums coincide.

Proof. Let us consider an arbitrary $A \in \mathcal{B}(\mathcal{H})$ and two bases of $\mathcal{H}$ as per hypothesis. Assume that $\|A\|_2^2$ is finite; then the following chain of equalities holds in view of Parseval identity:

$$\sum_{i=1}^{\infty} \|A(\psi_i)\|^2 = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |(A(\psi_i), \phi_j)|^2 = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} |(\psi_i, A^*(\phi_j))|^2 = \sum_{j=1}^{\infty} \|A^*(\phi_j)\|^2,$$

where the two sums can be exchanged since all terms are positive. All these identities would hold true having assumed that the right most term would be convergent. In other words we have proven item (b). The first one is now a direct consequence of (b). $\square$

The last theorem guarantees that the notion of an Hilbert-Schmidt operator is well-posed, but it does not tell us why it is useful. Well, the reason is ultimately that $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space. The following proposition clarifies better this statement.

Proposition 66. *Let $\mathcal{H}$ be a Hilbert space. Then $\mathcal{B}_2(\mathcal{H})$ is a subspace of $\mathcal{B}(\mathcal{H})$ such that*

1. $\|A\|_2 = \|A^*\|_2$ for all $A \in \mathcal{B}_2(\mathcal{H})$,
2. $\|AK\|_2 \leq \|K\| \|A\|_2$ and $\|KA\|_2 \leq \|K\| \|A\|_2$ for all $A \in \mathcal{B}_2(\mathcal{H})$ and for all $K \in \mathcal{B}(\mathcal{H})$,
3. $\|A\| \leq \|A\|_2$ for all $A \in \mathcal{B}_2(\mathcal{H})$.

Proof. To start with, we need to prove that $\mathcal{B}_2(\mathcal{H})$ is a linear subspace of $\mathcal{B}(\mathcal{H})$. It is per construction closed under multiplication by a scalar. We need only to verify that, for all $A, B \in \mathcal{B}_2(\mathcal{H})$, $(A + B) \in \mathcal{B}_2(\mathcal{H})$. Let $\{\psi_i\}_{i \in \mathbb{N}}$ be an arbitrary orthonormal basis. Then

$$\sum_{i=1}^{\infty} \|(A + B)\psi_i\|^2 \leq \sum_{i=1}^{\infty} (\|A\psi_i\| + \|B\psi_i\|)^2 \leq 2 \left[\sum_{i=1}^{\infty} \|B\psi_i\|^2 + \sum_{i=1}^{\infty} \|A\psi_i\|^2 \right] < \infty,$$

where we used the triangle inequality.

1. is a direct consequence of the proof of Theorem 65.
2. Let $A \in \mathcal{B}_2(\mathcal{H})$, $K \in \mathcal{B}(\mathcal{H})$ and let $\{\psi_i\}_{i \in \mathbb{N}}$ be an arbitrary orthonormal basis. Then

$$\|KA\|_2^2 = \sum_{i=1}^{\infty} \|(KA)\psi_i\|^2 \leq \sum_{i=1}^{\infty} \|K\|^2 \|A\psi_i\|^2 = \|K\|^2 \|A\|_2^2.$$

In other words $KA \in \mathcal{B}_2(\mathcal{H})$. At the same time, by item 1. we know that

$$\|(AK)^*\|_2 = \|K^*A^*\|_2 \leq \|K^*\| \|A^*\|_2 = \|K\| \|A\|_2,$$

where we used Theorem 65 to conclude that $A^* \in \mathcal{B}_2(\mathcal{H})$ as well as Proposition 32 to infer that $\|K^*\| = \|K\|$.

3. Let $A \in \mathcal{B}_2(\mathcal{H})$ and let $\{\psi_i\}_{i \in \mathbb{N}}$ be an arbitrary orthonormal basis. It holds by (7)

$$\begin{aligned} \|A\| &= \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \|A\phi\| = \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} (\|A\phi\|^2)^{\frac{1}{2}} = \\ &= \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \left(\sum_{i=1}^{\infty} |(\psi_i, A\phi)|^2 \right) = \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \left(\sum_{i=1}^{\infty} |(A^*\psi_i, \phi)|^2 \right)^{\frac{1}{2}} \leq \\ &\leq \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \left(\sum_{i=1}^{\infty} \|A^*\psi_i\|^2 \|\phi\|^2 \right)^{\frac{1}{2}} = \|A^*\|_2 = \|A\|_2, \end{aligned}$$

where we used at different stages both the Parseval equality and item 1. $\square$

From a more technical perspective this proposition tell us that Hilbert-Schmidt operators not only form an algebra but rather an ideal in the algebra of bounded operators.

Proposition 67. *Let $\mathcal{H}$ be an Hilbert space and let $A, B \in \mathcal{B}_2(\mathcal{H})$. Let $\{\psi_i\}_{i \in \mathbb{N}}$ be an orthonormal basis of $\mathcal{H}$. Let $(\cdot, \cdot)_2 : \mathcal{B}_2(\mathcal{H}) \times \mathcal{B}_2(\mathcal{H}) \rightarrow \mathbb{C}$ be*

$$(A, B)_2 \doteq \sum_{i=1}^{\infty} (A\psi_i, B\psi_i). \quad \forall A, B \in \mathcal{B}_2(\mathcal{H}) \quad (13)$$

This is well-defined map which determines a scalar-product on $\mathcal{B}_2(\mathcal{H})$ such that, for all $A \in \mathcal{B}_2(\mathcal{H})$, $\|A\|_2^2 = (A, A)_2$.

Proof. Notice that the map $(\cdot, \cdot)_2$ is bilinear and convergent, since, for any $i \in \mathbb{N}$

$$\sum_{i=1}^{\infty} |(A\psi_i, B\psi_i)| \leq \sum_{i=1}^{\infty} \|A\psi_i\| \|B\psi_i\| \leq \frac{1}{2} \sum_{i=1}^{\infty} (\|A\psi_i\|^2 + \|B\psi_i\|^2) \leq \infty,$$

for any choice of $A, B \in \mathcal{B}_2(\mathcal{H})$. The last chain of inequalities holds true due to the Cauchy-Schwarz inequality. Having established convergence, it is straightforward to verify that we have constructed an Hermitian inner product as per Definition 9 since all properties are inherited from those of $(\cdot, \cdot)$. To conclude we observe that, once it has been established that $(\cdot, \cdot)_2$ is an Hermitian inner product, we have the polarisation identity at our disposal which, in combination with Theorem 65, entails that $(\cdot, \cdot)_2$ does not depend on the choice of the orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$. $\square$

We conclude this section with two additional results, starting from a corollary whose proof we omit.

Corollary 68. *Let $\mathcal{H}$ be a Hilbert space. Then every Hilbert-Schmidt operator is also compact.*

Proposition 69. *Let $\mathcal{H}$ be an Hilbert space. Then $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space with respect to (13). Furthermore, for every $A \in \mathcal{B}_2(\mathcal{H})$*

$$\|A\|_2^2 = \sum_{\lambda \in \text{sing}(A)} m_\lambda \lambda^2,$$

where m_λ is the multiplicity of the eigenvalue λ of $|A|$.

Proof. We divide our analysis in two parts. In the first we show that $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space, in the second we prove the remaining statements of the proposition.

In view of Proposition 67 we need to show only the completeness of $\mathcal{B}_2(\mathcal{H})$. Consider an arbitrary orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ and let $\{A_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence of Hilbert-Schmidt operators. In view of item 3. of Proposition 66 we know that $\{A_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathcal{B}(\mathcal{H})$. The latter is a Banach space, hence complete. There exists $A \in \mathcal{B}(\mathcal{H})$ such that $\lim_{n \rightarrow \infty} \|A_n - A\| = 0$. By definition of Cauchy sequence, we know that for all $\epsilon > 0$, we can find two sufficiently large indices $m, k \in \mathbb{N}$ such that

$$\sum_{i=1}^{\infty} \|(A_k - A_m)\psi_i\|^2 = \|A_k - A_m\|_2^2 \leq \epsilon^2.$$

Hence for every but fixed $q \in \mathbb{N}$

$$\sum_{i=1}^q \|(A_k - A_m)\psi_i\|^2 \leq \epsilon^2,$$

which allows us to take the limit as $m \rightarrow \infty$ to obtain

$$\sum_{i=1}^q \|(A_k - A)\psi_i\|^2 \leq \epsilon^2.$$

In view of the arbitrariness of q , it descends

$$\sum_{i=1}^{\infty} \|(A_k - A)\psi_i\|^2 \leq \epsilon^2,$$

which entails that $(A - A_k) \in \mathcal{B}_2(\mathcal{H})$. Since $A_k \in \mathcal{B}_2(\mathcal{H})$ per hypothesis, we have that $A = (A - A_k) + A_k \in \mathcal{B}_2(\mathcal{H})$ thanks to Proposition 66. We have shown the convergence of the Cauchy sequence to an element of $\mathcal{B}_2(\mathcal{H})$.

Consider now $A \in \mathcal{B}_2(\mathcal{H})$ and $|A|$. Since Corollary 68 entails that $A \in \mathcal{B}_\infty(\mathcal{H})$, then $|A|$ is both positive and compact, see Definition 47 and Proposition 59. Following Theorem 63, we construct first a basis $\{\phi_{\lambda,i}\}_{\lambda \in \text{sing}(A)}$ of $(\ker(|A|))^\perp$ where each eigenvector comes with a

multiplicity i running from 1 to m_λ . We complete the basis of $\mathcal{H}$ with $\{\psi_j\}_{j \in \mathbb{N}}$, a basis of $\ker(|A|)$ which coincides with $\ker(A)$ by Corollary 48. We can thus write

$$\|A\|_2^2 = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} (A\phi_{\lambda,i}, A\phi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} (A^*A\phi_{\lambda,i}, \phi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \lambda^2 m_\lambda,$$

where we used that $|A|\phi_{\lambda,i} = \sqrt{A^*A}\phi_{\lambda,i} = \lambda\phi_{\lambda,i}$. $\square$

3.4 Trace Class Operators

In this section we define and we discuss the main properties of a notable class of bounded operators, which are known either as trace class or as **nuclear operators**. Their relevance is enormous being naturally associated to quantum states in quantum mechanics over finite dimensional Hilbert spaces and being the starting point for the notion of thermal states in quantum statistical mechanics. As in the previous section, we will be assuming that all Hilbert spaces are separable even though we shall not mention it explicitly and, for this reason, all calculations will be done under the assumption that the Hilbert space is infinite dimensional. If this were not to be the case, all results would still be true barring the change of a rather limited (and irrelevant) number of details. As a starting point we need a preliminary result:

Proposition 70. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. The following three statements are equivalent:*

1. *There exists an orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ such that*

$$\sum_{i=1}^{\infty} (\psi_i, |A|\psi_i) < \infty.$$

2. *$\sqrt{|A|}$ is a Hilbert-Schmidt operator,*

3. *$A \in \mathcal{B}_\infty(\mathcal{H})$ and the sum of all elements of the set $\{\lambda m_\lambda\}_{\lambda \in \text{sing}(A)}$ is finite.*

Proof. 1. $\iff$ 2. Since $|A| = \sqrt{|A|}\sqrt{|A|}$ and since both $|A|$ and $\sqrt{|A|}$ are self-adjoint per construction, condition 1. translates as

$$\sum_{i=1}^{\infty} (\psi_i, \sqrt{|A|}\sqrt{|A|}\psi_i) = \sum_{i=1}^{\infty} (\sqrt{|A|}\psi_i, \sqrt{|A|}\psi_i) = \|\sqrt{|A|}\|_2^2 < \infty,$$

which entails that $\sqrt{|A|} \in \mathcal{B}_2(\mathcal{H})$. The converse follows by tracing backwards this reasoning.

2. $\iff$ 3. Corollary 68 entails that $\sqrt{|A|}$ is a compact operator. Proposition 56 guarantees that the product of compact operators is still compact, hence $|A| \in \mathcal{B}_\infty(\mathcal{H})$. Let us consider

the basis made out of unit norm eigenvectors of $|A|$, namely $\{\psi_{\lambda,i}\}$, i running from 1 to m_λ , the multiplicity of λ . Since $|A|\psi_{\lambda,i} = \lambda\psi_{\lambda,i}$ for any value of i , it holds

$$\|\sqrt{A}\|_2^2 = \sum_{\lambda} \sum_{i=1}^{m_\lambda} \left(\sqrt{|A|}\psi_{\lambda,i}, \sqrt{|A|}\psi_{\lambda,i} \right) = \sum_{\lambda} \sum_{i=1}^{m_\lambda} (\psi_{\lambda,i}, |A|\psi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \lambda m_\lambda,$$

where in the second equality we used that $\sqrt{|A|}$ is self-adjoint and in the last we observed that all eigenvectors of the eigenvalue 0 do not contribute to the sum. Tracing backwards the proof shows that also the converse implication holds true. $\square$

In view of this last proposition,

Definition 71. Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. We call it a **trace class operator** if $\sqrt{|A|}$ is a Hilbert-Schmidt operator. The set of all these operators is indicated as $\mathcal{B}_1(\mathcal{H})$ and, for all $A \in \mathcal{B}_1(\mathcal{H})$,

$$\|A\|_1 \doteq \|\sqrt{|A|}\|_2^2 = \sum_{\lambda \in \text{sing}(A)} \lambda m_\lambda.$$

We have now defined different classes of operators on Hilbert spaces, but we have not yet completely investigated their relation. The following corollary goes one step in this direction:

Corollary 72. Let $\mathcal{H}$ be a Hilbert space. The following inclusions hold

$$\mathcal{B}_1(\mathcal{H}) \subset \mathcal{B}_2(\mathcal{H}) \subset \mathcal{B}_\infty(\mathcal{H}) \subset \mathcal{B}(\mathcal{H}).$$

Proof. We have already shown all the inclusions except for the first one. Suppose $A \in \mathcal{B}_1(\mathcal{H})$. Per Definition 71, this means that $\sqrt{|A|} \in \mathcal{B}_2(\mathcal{H})$, but item 2. of Proposition 66 entails that the product of Hilbert-Schmidt operators lies still in this class. Hence $|A| = \sqrt{|A|}\sqrt{|A|} \in \mathcal{B}_2(\mathcal{H})$. Since A is bounded per assumption, we can use the polar decomposition of Theorem 49 to write $A = U|A|$ where $U \in \mathcal{B}(\mathcal{H})$. Still item 2. of Proposition 66 guarantees that $A \in \mathcal{B}_2(\mathcal{H})$. $\square$

I am sure that, at the fifth semester of your studies, you have learned that no name is give by chance to an important mathematical object. Hence you must (and I stress that you must, especially you Simone!) be wondering why the name trace class and how an operation, which is intrinsically well-defined on a finite dimensional Hilbert space, can be generalized to the infinite dimensional scenario. Before answering to this question, time for a filler episode... well for another theorem which, in a mathematically advanced language, is telling us that trace-class operators form an ideal in the space of bounded operators. As we have already proven several similar results, we shall omit the proof, which can be found in [5, Theorem 4.32]:

Theorem 73. Let $\mathcal{H}$ be a Hilbert space. The following statements hold true:

- (a) For every $A \in \mathcal{B}_1(\mathcal{H})$, there exist $B, C \in \mathcal{B}_2(\mathcal{H})$ such that $A = BC$. Conversely for every pair of Hilbert-Schmidt operators B, C on $\mathcal{H}$, $BC \in \mathcal{B}_1(\mathcal{H})$ and $\|BC\|_1 \leq \|B\|_2 \|C\|_2$.

(b) $\mathcal{B}_1(\mathcal{H})$ is a linear subspace of $\mathcal{B}(\mathcal{H})$ and, moreover, $\forall A \in \mathcal{B}_1(\mathcal{H})$ and $\forall K \in \mathcal{B}(\mathcal{H})$, $KA, AK \in \mathcal{B}_1(\mathcal{H})$ and

$$\|KA\|_1 \leq \|K\| \|A\|_1 \quad \text{and} \quad \|AK\|_1 \leq \|K\| \|A\|_1,$$

(c) $\|\cdot\|_1$ is a norm on $\mathcal{B}_1(\mathcal{H})$.

One could go one step further actually showing that $\mathcal{B}_1(\mathcal{H})$ is complete with respect to the norm $\|\cdot\|_1$, hence it is a Banach space.

We can now answer to question asked above:

Proposition 74. *Let $\mathcal{H}$ be a Hilbert space. If $\{\psi_i\}_{i \in \mathbb{N}}$ is an orthonormal basis of $\mathcal{H}$, we call **trace** $Tr : \mathcal{B}_1(\mathcal{H}) \rightarrow \mathbb{C}$ the map*

$$Tr(A) = \sum_{i=1}^{\infty} (\psi_i, A\psi_i). \quad \forall A \in \mathcal{B}_1(\mathcal{H}) \quad (14)$$

In addition it holds that

1. The image of Tr does not depend on the choice of the basis,
2. For every $B, C \in \mathcal{B}_2(\mathcal{H})$, it holds

$$Tr(BC) = (B^*, C)_2,$$

3. for every $A \in \mathcal{B}_1(\mathcal{H})$, $|A| \in \mathcal{B}_1(\mathcal{H})$ and

$$\|A\|_1 = Tr(|A|).$$

Proof. 1. and 2. In view of Theorem 73 we can write any $A \in \mathcal{B}_1(\mathcal{H})$ as the product $A = BC$, $B, C \in \mathcal{B}_2(\mathcal{H})$. In view of (13), we can write

$$(B^*, C)_2 = \sum_{i=1}^{\infty} (B^* \psi_i, C \psi_i) = \sum_{i=1}^{\infty} (\psi_i, BC \psi_i) = \sum_{i=1}^{\infty} (\psi_i, A \psi_i) = Tr(A),$$

where $\{\psi_i\}_{i \in \mathbb{N}}$ is an orthonormal basis of $\mathcal{H}$. Since item 1. of Proposition 73 guarantees us also the the product of two Hilbert-Schmidt operators is trace class, this last chain of identities tell us not only that 2. but also that 1. is true, since the Hilbert-Schmidt Hermitian inner product does not depend on the choice of the basis.

3. Since $|A| = (|A|)$ we know that from (12) $\| |A| \|_1 = \|A\|_1$ which is finite since $A \in \mathcal{B}_1(\mathcal{H})$ per assumption. If we choose $\{\psi_{\lambda,i}\}$ a basis of eigenvectors of $|A|$, with the index i running from 1 to m_λ , the multiplicity of the eigenvalue λ , it holds per Definition 71 that

$$Tr(|A|) = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} (\psi_{\lambda,i}, |A| \psi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} \lambda = \sum_{\lambda \in \text{sing}(A)} \lambda m_\lambda = \|A\|_1,$$

which is the sought identity. $\square$

This last proposition is indeed a justification to the use of the word trace, but one might still be wondering whether the operator Tr inherits also all the properties of the trace of a matrix. The answer is essentially yes, but it does not hurt (me, you... maybe!) to prove some of them in this context:

Proposition 75. *Let $\mathcal{H}$ be a Hilbert space. Then the trace operator enjoys the following properties:*

- For every $A, B \in \mathcal{B}_1(\mathcal{H})$ and for all $\alpha, \beta \in \mathbb{C}$

$$Tr(\alpha A + \beta B) = \alpha Tr(A) + \beta Tr(B), \quad Tr(A^*) = \overline{Tr(A)},$$

- it is cyclic, namely, for every $A \in \mathcal{B}_1(\mathcal{H})$ and for every $K \in \mathcal{B}(\mathcal{H})$, it holds

$$Tr(AK) = Tr(KA).$$

Proof. 1. It is an immediate consequence of (14) and of the bi-linearity of the Hermitian inner product on $\mathcal{H}$.

2. To start with, suppose that both A and K lie in $\mathcal{B}_2(\mathcal{H})$. Then item 2. of Proposition 74 guarantees that

$$Tr(AK) = (A^*, K)_2 \quad \text{and} \quad Tr(KA) = (K^*, A)_2.$$

Since $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space with respect to $(\cdot, \cdot)_2$, we can use the polarisation identity to write

$$4(A^*, K)_2 = \|A^* + K\|_2^2 + \|A^* - K\|_2^2 - i\|A^* + iK\|_2^2 + i\|A^* - iK\|_2^2.$$

Since item 1. in Proposition 66 guarantees us, that for every $C \in \mathcal{B}_2(\mathcal{H})$, $\|C^*\|_2 = \|C\|_2$, we can rewrite the above identity as

$$4(A^*, K)_2 = \|A + K^*\|_2^2 + \|A - K^*\|_2^2 - i\|K^* + iA\|_2^2 + i\|K^* - iA\|_2^2 = 4(K^*, A).$$

Suppose now that $K \in \mathcal{B}(\mathcal{H})$, but not necessarily compact. In view of item 1. of Theorem 73, we can decompose $A = CD$, with $C, D \in \mathcal{B}_2(\mathcal{H})$. Hence

$$Tr(AK) = Tr(CDK) = Tr(DKC) = Tr(KCD) = Tr(KA),$$

which concludes the proof. □

4 Unbounded Operators

In the previous section we have studied bounded operators and we have seen that their behaviour is well-under control. So where is the problem? Well... most of the physically interesting operators do not fall in that class. The prime examples are the position and the momentum operators, not to mention the various Hamiltonians which are at the hearth of manifold models.

Depending on the depth of your knowledge of quantum mechanics, the above examples might or might not make a lot of sense. In any case let me offer you, my dear reader(s), an example of what I mean.

Consider $\mathcal{H} = L^2(I)$ where $I = (1, \infty)$ and the measure is the standard Lebesgue one. A natural operation that we want to be able to perform on such Hilbert space is the multiplication by x the coordinate along $I \subset \mathbb{R}$. This is a special example of the position operator that we will be studying at a later stage. Yet the question is whether what we are trying to define is meaningful. Consider two cases:

- Let $\psi(x) = e^{-\frac{x}{2}}$. This lies in $\mathcal{H}$ since

$$\int_1^\infty dx |\psi(x)|^2 = \int_1^\infty dx e^{-x} = -e^{-x}|_1^\infty = \frac{1}{e}.$$

Multiplying $\psi(x)$ by x yields $xe^{-\frac{x}{2}}$ and

$$\int_1^\infty dx x^2 e^{-x} = \frac{5}{e},$$

which guarantees us that $x\psi(x) \in \mathcal{H}$.

- Let $\psi(x) = x^{-1}$. This lies in $\mathcal{H}$ since

$$\int_1^\infty dx x^{-2} = -x^{-1}|_1^\infty = 1.$$

Multiplying by x yields the constant function 1, whose integral from 1 to infinity clearly diverges.

In other words it is easy to define a large class of operators which are well-defined only on certain elements of a given Hilbert spaces. As we will see in this section, especially via the Hillinger-Toeplitz theorem, these operators are characterized by one notable characteristic: They are unbounded, namely their operator norm as per (7) diverges. As before, we shall try to keep the minimum number of assumptions and structures.

Hence as a starting point we consider only complex vector spaces V . As a rule, from now on, we have in mind the infinite dimensional scenario:

Definition 76. *Let V be a complex vector space. We call a linear map*

$$T : D(T) \subset V \rightarrow V,$$

*an **operator** on the **domain** $D(T)$. The **graph** of T is defined as*

$$G(T) = \{(v, T(v)) \in V \oplus V \mid v \in D(T)\}.$$

As a straightforward consequence of this definition, we can infer that the standard operations between operators are still meaningful provided we pay attention at the correct domain of definition. As a matter of fact:

- if T, S are two operators on V , then also TS is an operator on V with the so-called **standard domain**

$$D(TS) = \{v \in D(S) \mid S(v) \in D(T)\},$$

- if T is an operator on V and $\alpha \in \mathbb{C}$, also αT is an operator on V with **standard domain** $D(\alpha T) = D(T)$, if $\alpha \neq 0$. If $\alpha = 0$, then the operator is everywhere defined coinciding with the trivial operator.

Notice that, in view of these remarks, the associative and distributive properties respectively of the sum and of the product of operators are valid. Furthermore it happens in many concrete cases that we are able to define easily an operator on a given domain, to realize at a later stage that such domain could have been enlarged. This leads to the following rather intuitive definition:

Definition 77. Let V be a vector space and let T be an operator thereon. A second operator T' is called an **extension** of T if $G(T) \subset G(T')$. In this case we write $T \subset T'$.

We consider now a normed space X in place of a simple vector space. Observe that the Cartesian product $X \times X$ is at the same time the vector space $X \oplus X$ and a topological space endowed with the product topology as per Definition 212. This is important since it makes precise the notion of a closed set and of a closure in $X \times X$, hence justifying the following definition which introduce a key concept in the theory of operators:

Definition 78. Let X be a normed space and let $A : D(A) \subseteq X \rightarrow X$ be an operator. We say that

- A is **closed** if $G(A)$ is closed in the product topology – see Definition 212,
- A is **closable** if $\overline{G(A)}$ is the graph of an operator, which we denote as $\bar{A}$, the **closure** of A .

Exercise 15. Show that an equivalent definition of a closed operator is the following: A is closed if and only if for every sequence $\{x_n\}_{n \in \mathbb{N}}$, $x_n \in D(A)$ for all n , such that $\lim_{n \rightarrow \infty} x_n = x \in X$ and $\lim_{n \rightarrow \infty} A(x_n) = y \in X$, then $x \in D(A)$ and $y = A(x)$.

We can characterize closable operators in terms of their graph:

Proposition 79. Let $A : D(A) \subset X \rightarrow X$. The following statements are equivalent:

1. A is closable,
2. $\overline{G(A)}$ does not contain points of the form $(0, z)$, $z \neq 0$,
3. A admits closed extensions.

Proof. 1. $\iff$ 2. A is not closable if and only if there exist two sequences $\{x_n\}_{n \in \mathbb{N}}$ and $\{x'_n\}_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x'_n = x$, but $\lim_{n \rightarrow \infty} A(x_n) = y$ and $\lim_{n \rightarrow \infty} A(x'_n) = y'$ with $y \neq y'$. By linearity this is tantamount to saying that the sequence whose elements are $\tilde{x}_n \doteq x_n - x'_n$ converges to 0 as $n \rightarrow \infty$, while the sequence $A(\tilde{x}_n)$ converges to $y - y' = z \neq 0$. In turn, this amounts to saying that $\overline{G(A)}$ contains points of the form $(0, z)$ with $z \neq 0$. By negating this statement, the conclusion follows. It is also clear that the argument can be traced back to prove the reverse implication.

1. $\iff$ 3. If A is closable than $\bar{A}$ is a closed extension. Conversely, if A admits a closed extension, say B , per Definition 77 and 78, there cannot be in $\overline{G(A)}$ elements of the form $(0, z)$ with $z \neq 0$ since, otherwise, $(0, z) \in \overline{G(A)} \subset \overline{G(B)} = G(B)$. This is a consequence of $A \subset B$ and of B , being closed. Yet, this is also a contradiction since it would entail that, at the same time, the closed operator B is linear and $B(0) = z \neq 0$. $\square$

The situation improves if we assume that X is complete:

Lemma 80. *Let X, Y be Banach spaces, let $T \in \mathcal{B}(X, Y)$ and let $A : D(A) \subset Y \rightarrow Y$ be an operator on Y . If A is closable and $\text{Ran}(T) \subset D(A)$, then $AT \in \mathcal{B}(X, Y)$.*

Proof. Since A is closable, $AT = \bar{A}T$ on $D(A)$ and $\text{Ran}(T) \subset D(A)$. We can show that $\bar{A}T$ is closed. Consider a sequence $\{x_n\}_{n \in \mathbb{N}}$ such that both $\lim_{n \rightarrow \infty} x_n = x$, $x \in X$ and $\lim_{n \rightarrow \infty} \bar{A}T(x_n) = y$, $y \in Y$. Proposition 24 guarantees us that T is continuous, hence there must exist $z \in Y$ such that $\lim_{n \rightarrow \infty} T(x_n) = z$. Since $\bar{A}$ is closed per assumption, $z \in D(\bar{A})$ and $\bar{A}z = y$. Hence $(\bar{A}T)(x) = y$, which implies that $\bar{A}T$ is closed. We can now invoke the closed graph theorem 30 to conclude the proof. $\square$

One last remark is in due course. We have considered only Banach spaces since we know that the direct sum of two of them is still a Banach space, if endowed with the product topology. If we were to consider $\mathcal{H}, \mathcal{H}'$, two Hilbert spaces with Hermitian inner product $(\cdot, \cdot)$ and $(\cdot, \cdot)'$ respectively, we could transfer all results that we have obtained. In addition we also know that $\mathcal{H} \oplus \mathcal{H}'$ is again a Hilbert space if endowed with the Hermitian inner product

$$((x, y), (x', y'))_{\mathcal{H} \oplus \mathcal{H}'} \doteq (x, x') + (y, y'). \quad \forall (x, y), (x', y') \in \mathcal{H} \oplus \mathcal{H}' \quad (15)$$

We conclude this short section with a simple exercise

Exercise 16. Let $\mathcal{H}, \mathcal{H}'$ be two Hilbert spaces with Hermitian inner product $(\cdot, \cdot)$ and $(\cdot, \cdot)'$. Prove that $\mathcal{H} \oplus \mathcal{H}'$ is a Hilbert space (note that only the property of being complete is non trivial) and that the inner product (15) induces the product topology.

4.1 The Hermitian Adjoint Operator

As we have seen in the previous section, one of the key operations on bounded operators is that of adjunction. On the one hand, the adjoint operator exists as a by-product of Riesz's theorem,

while, on the other hand, one can identify the special class of self-adjoint operator. These are noteworthy since their eigenvalues are all real and, thus, they are the natural candidates for playing a pivotal role in quantum theories.

Unfortunately, the moment we drop the hypothesis of boundedness, we are back to square one since already Riesz's theorem cannot be used. We analyse now an alternative approach. Unless stated otherwise, we consider always a linear operator $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$ with a *dense* domain $D(T)$ in a Hilbert space $\mathcal{H}$. In principle we do not need to assume that it is separable, but it will not hurt doing so.

We recall that our guiding principle is looking for an operator T^* together with its domain of definition $D(T^*)$ so that the relation $(\phi, T\psi) = (T^*\phi, \psi)$ holds true for all $\psi \in D(T)$ and for all $\phi \in D(T^*)$. Bearing in mind the construction of the adjoint for bounded operators, we start from the following ansatz as far as the domain of the adjoint is concerned:

$$D(T^*) = \{\phi \in \mathcal{H} \mid \exists \Psi_\phi \in \mathcal{H} \text{ for which } (\phi, T\psi) = (\Psi_\phi, \psi), \forall \psi \in D(T)\}. \quad (16)$$

We observe that the definition is meaningful for the following reasons:

- $D(T^*)$ is never empty since at least $0 \in D(T^*)$ by setting $\Psi_0 = 0$. Furthermore it is a linear space since this property is inherited from the one of the Hermitian inner product,
- In general the assignment $D(T^*) \ni \phi \mapsto \Psi_\phi \in \mathcal{H}$ defines a linear functional if the image is unique. In general this is not the case unless $D(T)$ is dense. Suppose that, for a given $\phi \in D(T^*)$, there exists Ψ_ϕ and Ψ'_ϕ in $\mathcal{H}$ such that $(\Psi_\phi, \psi) = (\phi, T\psi) = (\Psi'_\phi, \psi)$, for all $\psi \in D(T)$. In other words $(\Psi_\phi - \Psi'_\phi, \psi) = 0$ if $\psi \in D(T)$. This identity yields no contradiction unless $D(T)$ is dense in $\mathcal{H}$. In this case, we consider a sequence $\{\psi_n\}_{n \in \mathbb{N}}$ such that $\psi_n \rightarrow \Psi_\phi - \Psi'_\phi$ as $n \rightarrow \infty$. Since, for all $n \in \mathbb{N}$, $(\Psi_\phi - \Psi'_\phi, \psi_n) = 0$ and since the Hermitian inner product is continuous, we can take the limit as $n \rightarrow \infty$, to obtain $\|\Psi_\phi - \Psi'_\phi\|^2 = 0$. This entails $\Psi_\phi - \Psi'_\phi = 0$.

These observations justify the following line of reasoning:

Definition 81. Let $\mathcal{H}$ be an Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a dense operator thereon. We call **adjoint** the operator $T^* : D(T^*) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ whose domain is (16) and whose image is $T^*(\phi) = \Psi_\phi \in \mathcal{H}$ for all $\phi \in D(T^*)$.

Notice that, even though T is densely defined there is no reason why T^* should also enjoy this property. Hence a priori $(T^*)^*$ might not be defined. On the bright side the adjoint enjoys some notable properties which are an almost straightforward consequence of the definition:

Exercise 17. Let T, T' be two densely defined operators on $\mathcal{H}$. Prove that

1. If $T \subset T'$ – see Definition 77, then $(T')^* \subset T^*$,
2. $(T')^*T^* \subseteq (TT')^*$ where the equality holds if $T \in \mathcal{B}(\mathcal{H})$.

Theorem 82. Let T be a densely defined operator on a Hilbert space $\mathcal{H}$. Then

1. T^* is closed and $G(T^*) = [\tau(G(T))]^\perp$ where

$$\tau : \mathcal{H} \oplus \mathcal{H} \rightarrow \mathcal{H} \oplus \mathcal{H}, \quad (\phi, \psi) \mapsto \tau((\phi, \psi)) = (-\psi, \phi). \quad (17)$$

2. T is closable if and only if $D(T^*)$ is dense and, in this case,

$$T \subseteq \overline{T} = (T^*)^*.$$

Proof. 1. Using (17), it holds

$$[\tau(G(T))]^\perp = \{(\phi, \psi) \in \mathcal{H} \oplus \mathcal{H} \mid -(\phi, T(\tilde{\psi}))_{\mathcal{H}} + (\psi, \tilde{\psi})_{\mathcal{H}} = 0, \forall \tilde{\psi} \in D(T)\}.$$

In other words we can identify ψ with $T^*(\phi)$, which means that $[\tau(G(T))]^\perp$ coincides with $G(T^*)$. In addition, since $[\tau(G(T))]^\perp$ is the orthogonal complement of a set in a Hilbert space, this is always closed. Hence the graph of T^* is closed and so is, per definition, the operator itself.

2. Let T be as according to the hypothesis and, by the theory of Hilbert spaces, it holds $\overline{G(T)} = [[G(T)]^\perp]^\perp$. In addition, since $\tau \circ \tau = -\mathbb{I}$, where $\mathbb{I}$ is here the identity operator on $\mathcal{H} \oplus \mathcal{H}$, and since, for any set $W \in \mathcal{H}$, $W^\perp = (-W)^\perp = -W^\perp$, we can conclude

$$\overline{G(T)} = -\tau \left(\left[\tau \left([G(T)]^\perp \right) \right]^\perp \right) = -\tau \left[G(T^*)^\perp \right] = \tau \left[G(T^*)^\perp \right],$$

where we used at different stages the results of item 1. and that τ commutes with the operation $\perp$ of taking the orthogonal complement. Proposition 79 guarantees us that $\overline{G(T)}$ is the graph of an operator, $\overline{T}$, if and only if it does not contain points of the form $(0, \psi) \in \mathcal{H} \oplus \mathcal{H}$. The above chain of identities tells as that this is equivalent to saying that there cannot exist $\psi \in \mathcal{H}$ such that $(0, \psi) \perp (-T^*\phi, \phi)$ for any $\phi \in D(T^*)$. Yet this holds true if and only if $\psi \in [D(T^*)]^\perp$. In other words $\overline{G(T)}$ is the graph of an operator if and only if the orthogonal complement of $D(T^*)$ is trivial, which is equivalent to saying that T^* is densely defined.

If this is the case, then $(T^*)^*$ is well-defined and, by item 1. of this theorem,

$$\overline{G(T)} = \tau \left[(G(T)^*)^\perp \right] = G((T^*)^*).$$

To conclude it suffices to observe that, per definition of closure $\overline{G(T)} = G(\overline{T})$. $\square$

Corollary 83. *Let T be a densely defined operator on a Hilbert space $\mathcal{H}$. Then*

1. $\ker(T^*) = [\text{Ran}(T)]^\perp$ and $\ker(T) \subseteq [\text{Ran}(T^*)]^\perp$ where the equality holds if $D(T^*)$ is dense in $\mathcal{H}$ and T is closed

2. if T is closed, then

$$\mathcal{H} \oplus \mathcal{H} = \tau(G(T)) \oplus G(T^*).$$

Proof. 1. This descends from the identity $(T^*\phi, \psi) = (\phi, T\psi)$ for all $(\phi, \psi) \in D(T^*) \oplus D(T)$, together with item 2. in Theorem 82.

2. If T is closed, then $\tau(G(T))$ is closed since τ is a unitary operator on $\mathcal{H} \oplus \mathcal{H}$. The statement descends from item 1. in Theorem 82 together with the standard properties of the orthogonal complement studied in the course of mathematical methods for physicists. $\square$

Exercise 18. Let T be a densely defined operator on a Hilbert space $\mathcal{H}$. Prove that

$$\ker(T^* - \lambda\mathbb{I}) = [\text{Ran}(T - \bar{\lambda}\mathbb{I})]^\perp, \quad \text{and} \quad \text{Ker}(T - \bar{\lambda}\mathbb{I}) \subseteq [\text{Ran}(T^* - \bar{\lambda}\mathbb{I})]^\perp,$$

where $\lambda \in \mathbb{C}$ and $\mathbb{I}$ is the identity operator on $\mathcal{H}$.

4.2 Essentially Self-Adjoint Operators

In view of the analysis of the previous part of this section, we are able to extend the notion of self-adjointness to operators which are not necessarily everywhere defined. More importantly we will be able to enlarge the class of potential physical observables by identifying a class of operators which are not self-adjoint, but admit a *unique* extension with such property. As we will emphasize, uniqueness is the key point and, in many cases, this is not available. The prime example will be the Laplace operator in a bounded region, say a box in $\mathbb{R}^n$, which admits multiple self-adjoint extensions, each of which can be interpreted as the choice of a boundary condition (Dirichlet, Neumann or Robin). We start with some nomenclature.

Definition 84. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$. We say that T is

- **Hermitian** if $(\phi, T\psi) = (T\phi, \psi)$ for all $\phi, \psi \in D(T)$,
- **symmetric** if T is Hermitian and $D(T)$ is dense in $\mathcal{H}$,
- **self-adjoint** if $D(T)$ is dense and $T = T^*$,
- **essentially self-adjoint** if both $D(T)$ and $D(T^*)$ are dense and $T^* = (T^*)^*$.
- **normal** if $T^*T = TT^*$ where both sides are defined on the standard domain.

Observe that we have defined a new class of operators, those essentially self-adjoint which has no counterpart in the previous sections. In addition there are different, yet equivalent definitions of the above classes of operators. What a better occasion to offer you a new exercise? Actually, while reading this script, I realized that I am offering two exercises! This must be your lucky day (think how an unlucky day could turn out to be!)

Exercise 19. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$. Prove that T is symmetric if and only if $\overline{D(T)} = \mathcal{H}$ and $T \subset T^*$.

Exercise 20. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$. Prove that T is essentially self-adjoint if and only if $D(T)$ is dense in $\mathcal{H}$, T is closable and $T^* = \bar{T}$.

Before proceeding to a deeper analysis of essentially self-adjoint operators, we are falling short of one important answer, namely why we are calling the whole section "unbounded operators" when, up to now, we have been discussing mainly operators which are not everywhere defined. The following celebrated theorem gives (partly) a reason:

Theorem 85 (Hellinger-Toeplitz). *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator. If T is Hermitian and $D(T) = \mathcal{H}$ then it is bounded and self-adjoint.*

Proof. This is a consequence of item 1. of Proposition 39. □

Exercise 21. Let $\mathcal{H}, \mathcal{H}'$ be two Hilbert spaces and let $U : \mathcal{H} \rightarrow \mathcal{H}'$ be a unitary operator. If $T : D(T) \rightarrow \mathcal{H}$ and if $T' : D(T') \rightarrow \mathcal{H}'$ is such that

$$T' = UTU^{-1} \quad \text{and} \quad D(T') = UD(T),$$

then T' enjoys any of the following properties, if enjoyed also by T : closable, closed, Hermitian, symmetric, normal, self-adjoint and essentially self-adjoint.

IMPORTANT: Since we will be acting with the adjoint map several times on each operator, we shall adopt the notation T^{**} in place of $(T^*)^*$ and so on and so forth.

We start identifying criteria to distinguish essentially self-adjoint operators and to unveil their properties:

Proposition 86. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator. Then*

1. *if $D(T), D(T^*)$ and $D(T^{**})$ are dense, then*

$$T^* = \overline{T^*} = \overline{T^*}^* = T^{***}.$$

2. *T is essentially self-adjoint if and only if $\overline{T}$ is self-adjoint,*
3. *If T is self-adjoint, it is maximally symmetric, that is it does not admit a proper extension to a symmetric operator,*
4. *if T is essentially self-adjoint, it admits one and only one self-adjoint extension, namely $\overline{T}$ which coincides with T^* .*

Proof. 1. Since $D(T), D(T^*)$ and $D(T^{**})$ are dense per hypothesis, we know that T^*, T^{**} and T^{***} exist. In view of item 2. of Theorem 82, we can conclude that $\overline{T^*} = T^{***} = (T^{**})^* = \overline{T^*}$. Item 1. of Theorem 82 entails that $T^* = \overline{T^*}$.

2. Suppose that T is essentially self-adjoint. As shown in Exercise 20, this entails that $\overline{T} = T^*$ and thus $D(\overline{T}) = D(T^*)$. By taking the adjoint, we obtain $\overline{T^*} = T^{**}$, but item 2. of Theorem 82 guarantees us that $\overline{T} = T^{**}$. Hence $\overline{T} = \overline{T^*}$ and, since the domains are dense, $\overline{T}$ is self-adjoint. Conversely suppose that $\overline{T}$ is self-adjoint. Then there exists its adjoint and $\overline{T^*} = \overline{T}$,

which implies that $D(T)$, $D(T^*)$ and $D(T^{**})$ are dense. In view of item 1. $T^* = \overline{T^*} = \overline{T^*}^*$. Hence $T^* = \overline{T}$ and T is essentially self-adjoint.

3. Suppose that T is self-adjoint and that there exists a symmetric operator K such that $T \subseteq K$. By taking the adjoint, we obtain $K^* \subseteq T^*$. Yet, since K is symmetric $K \subseteq K^*$. We have obtained the following chain of inclusions

$$T \subseteq K \subseteq K^* \subseteq T^*,$$

which collapses to equalities since $T = T^*$ by the hypothesis that T is self-adjoint.

4. Since T is essentially self-adjoint, we know from Exercise 20 that $\overline{T}$ is a self-adjoint extension. Suppose there exists a second one K . Forcefully $T \subset K$, which entails by taking the adjoint that $K = K^* \subseteq T^*$. Repeating the operation once more we obtain $T^{**} \subseteq K^{**} = K$. Yet, per definition of an essentially self-adjoint operator $T^* = T^{**}$. Hence we obtain that $K = K^* \subset T^* = T^{**} \subseteq K^{**} = K$. In other words $K = T^{**}$, but $T^{**} = \overline{T}$ in view of item 2. of Theorem 82. $\square$

Another relevant characteristic of an essentially self-adjoint operator is the following:

Theorem 87. *Let $\mathcal{H}$ be a Hilbert space and let T be a symmetric operator thereon. The following statements are equivalent:*

1. T is self-adjoint,
2. T is closed and $\ker(T^* \pm i\mathbb{I}) = \{0\}$ where $\mathbb{I}$ is the identity operator on $\mathcal{H}$,
3. $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$.

Proof. 1. $\implies$ 2. Suppose $T = T^*$. By item 1. of Proposition 82 we know that T^* is closed, hence T is closed. Suppose that $\psi \in \ker(T^* \pm i\mathbb{I})$. Then

$$T^*\psi = T\psi = \mp i\psi \implies \mp i(\psi, \psi) = (\psi, T\psi) = (T\psi, \psi) = (\mp i\psi, \psi) = \pm i(\psi, \psi),$$

which can hold true only if $(\psi, \psi) = 0$, i.e., $\psi = 0$.

2. $\implies$ 3. On account of Exercise 18, we know that $\ker(T^* \pm i\mathbb{I}) = [\text{Ran}(T \mp i\mathbb{I})]^\perp = \{0\}$. In other words $\text{Ran}(T \mp i\mathbb{I})$ is dense in $\mathcal{H}$. Since T and thus $T \pm i\mathbb{I}$ are closed and since $D(T)$ is dense, for all $\psi \in \mathcal{H}$ we can find a sequence $\{\psi_n\}_{n \in \mathbb{N}}$, with $\psi_n \in D(T)$ for all $n \in \mathbb{N}$, such that $\lim_{n \rightarrow \infty} (T\psi_n \mp i\psi_n) = \psi$. Notice that, for all $n \in \mathbb{N}$,

$$\|(T \mp i\mathbb{I})\psi_n\|^2 = \|T(\psi_n)\|^2 + \|\psi_n\|^2 \geq \|\psi_n\|^2.$$

This entails that since the left hand side converges as $n \rightarrow \infty$, so it does the right hand side. In other words there exists $\tilde{\psi} \in \mathcal{H}$ such that $\lim_{n \rightarrow \infty} \psi_n = \tilde{\psi}$. Since $T \mp i\mathbb{I}$ is closed $(T \mp i\mathbb{I})\tilde{\psi} = \psi$, that is $\text{Ran}(T \pm i\mathbb{I}) = \overline{\text{Ran}(T \pm i\mathbb{I})} = \mathcal{H}$.

3. $\implies$ 1. Since $T \subset T^*$, we need to show that $D(T^*) \subset D(T)$. Let $\psi \in D(T^*)$. Since $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$, there must exist $\phi \in D(T)$ such that

$$(T \pm i\mathbb{I})\phi = (T^* \pm i\mathbb{I})\psi.$$

Yet, $T = T^*$ on $D(T)$ and, thus, $(T^* \pm i\mathbb{I})(\psi - \phi) = 0$. The hypothesis entails also that $\ker(T^* \pm i\mathbb{I}) = [\text{Ran}(T \mp i\mathbb{I})]^\perp = \{0\}$, which yields in turn $\psi = \phi$ with $\phi \in D(T)$ per assumption and thus $\psi \in D(T)$. $\square$

Theorem 88. *Let $\mathcal{H}$ be a Hilbert space and let T be a symmetric operator thereon. The following statements are equivalent:*

1. T is essentially self-adjoint,
2. $\ker(T^* \pm i\mathbb{I}) = \{0\}$ where $\mathbb{I}$ is the identity operator on $\mathcal{H}$,
3. $\overline{\text{Ran}(T \pm i\mathbb{I})} = \mathcal{H}$.

Proof. 1. $\iff$ 2. If T is essentially self-adjoint, then T^* is self-adjoint and closed. We can apply to it the second criterion in Theorem 87 to conclude that $\ker(T^{**} \pm i\mathbb{I}) = \{0\}$. Since $T^{**} = T^*$ the sought statement is proven.

Suppose now that $\ker(T^* \pm i\mathbb{I}) = \{0\}$. Since T is symmetric, we know that $T \subseteq T^*$ and, by taking the adjoint $T^{**} \subseteq T^*$. Hence $D(T^*)$ is dense, being such $D(T)$. Once more item 2. of Theorem 82 entails that $T \subseteq \overline{T} = T^{**}$, which yields $\overline{T} \subseteq T^*$. The first item of Proposition 86 guarantees as a consequence that $T^* = \overline{T}^*$. In other words we have shown that $\overline{T}$ is symmetric. We can now apply item 2. of Theorem 87 to conclude that $\overline{T}$ is self-adjoint, but item 2. of Proposition 86 tells us that this is equivalent to having T essentially self-adjoint.

2. $\iff$ 3. This is a direct consequence of $[\text{Ran}(T \pm i\mathbb{I})]^\perp = \ker(T^* \mp i\mathbb{I})$ together with

$$\overline{\text{Ran}(T \pm i\mathbb{I})} \oplus [\text{Ran}(T \pm i\mathbb{I})]^\perp = \mathcal{H}.$$

The theorem stands proven. $\square$

We conclude this subsection with a key definition:

Definition 89. *Let $\mathcal{H}$ be a Hilbert space and let T be a closable, densely defined operator on $\mathcal{H}$. A dense subspace $W \subset D(T)$ is called a **core** for T if*

$$\overline{T|_W} = \overline{T}.$$

Proposition 90. *Let $\mathcal{H}$ be a Hilbert space and let T be a self-adjoint operator thereon. A subspace W is a core for T if and only if $T|_W$ is essentially self-adjoint.*

Proof. If $T|_W$ is essentially self-adjoint, then it admits a unique self-adjoint extension which must coincide with its closure. Hence $T|_W$ is a core. Conversely if $T|_W$ is a core, the closure is self-adjoint since it coincides with T itself. $\square$

4.2.1 The Position and the Momentum Operator – Definitions

In quantum mechanics there are two key examples which rely on the theory that we have developed so far: the momentum and the position operator. We will omit any proof since, for the first one, we need to get acquainted with the notion of Fourier transform and, for both, we need to study more in detail the smooth and compactly supported functions. We will discuss it in the second part of the course, hence we will postpone the analysis especially of the momentum operator to that moment. (ah ah ah moment and momentum!!!... The author needs holidays!)

Let $\mathcal{H} = L^2(\mathbb{R}^n, d^n x)$ where $d^n x$ indicates that we are considering the Lebesgue measure. We will omit henceforth indicating the measure which is left understood. For every $i = 1, \dots, n$ define the **position operators**

$$X_i : D(X_i) \subset \mathcal{H} \rightarrow \mathcal{H}, \quad \psi \mapsto x_i \psi, \quad (18)$$

where x_i represents the multiplication by the i -th coordinate over $\mathbb{R}^n$ endowed with the standard Euclidean coordinate system. The domain of definition is

$$D(X_i) \doteq \{\psi \in L^2(\mathbb{R}^n) \mid x_i \psi \in L^2(\mathbb{R}^n)\}. \quad (19)$$

Proposition 91. *Each position operator X_i , $i = 1, \dots, n$ on $D(X_i) \subset L^2(\mathbb{R}^n)$ is self-adjoint and $\mathcal{D}(\mathbb{R}^n) \equiv C_0^\infty(\mathbb{R}^n)$ is a core for X_i .*

As already mentioned all proofs are postponed to later sections of these notes. To define the momentum operator, we need a preliminary definition:

Definition 92. *A function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is called **locally integrable** if $fg \in L^1(\mathbb{R}^n)$ for all $g \in C_0^\infty(\mathbb{R}^n)$. We call $h = w - \partial_{x_j} f$ the **weak derivative** of f along the direction x_j , if both f and h are locally integrable and*

$$\int_{\mathbb{R}^n} d^n x h(x) g(x) = - \int_{\mathbb{R}^n} d^n x f(x) \frac{\partial g}{\partial x_j}. \quad \forall g \in C_0^\infty(\mathbb{R}^n)$$

If we observe that any element of $L^2(\mathbb{R}^n)$ is also locally integrable, we can call accordingly **momentum operator**

$$P_j : D(P_j) \subset L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n), \quad \psi \mapsto -i w - \partial_{x_j} \psi \quad (20)$$

with domain of definition

$$D(P_j) \doteq \{\psi \in L^2(\mathbb{R}^n) \mid \exists w - \frac{\partial \psi}{\partial x_j} \in L^2(\mathbb{R}^n)\}. \quad (21)$$

Proposition 93. *Each momentum operator P_j , $j = 1, \dots, n$ on $D(P_j) \subset L^2(\mathbb{R}^n)$ is self-adjoint and $\mathcal{D}(\mathbb{R}^n) \equiv C_0^\infty(\mathbb{R}^n)$ is a core for X_i .*

4.3 The Deficiency Indices

Up to now we have investigated those operators admitting a unique self-adjoint extension. In many interesting cases, such as for example the Laplace operator on a bounded domain, we have to give up on the request of uniqueness. We need therefore to establish useful criteria to decide whether and how many self-adjoint extensions exist. To this avail the first step is the extension of the concept of isometry to unbounded operators and the definition of the so-called Cayley transform:

Definition 94. Let $\mathcal{H}$ be a Hilbert space and let $U : D(U) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator. It is called an **isometry** if

$$(U\psi, U\psi') = (\psi, \psi'), \quad \forall \psi, \psi' \in D(U)$$

which is equivalent to asking $\|U(\psi)\| = \|\psi\|$ for all $\psi \in D(U)$.

We recall that the map $\mathbb{R} \ni x \mapsto \frac{x-i}{x+i}$ is a bijection to $\mathbb{S}^1 \setminus \{1\} \subset \mathbb{C}$. This inspires the notion of Cayley transform

$$V(T) = (T - i\mathbb{I})(T + i\mathbb{I})^{-1}, \quad (22)$$

where T is an operator on a Hilbert space $\mathcal{H}$. In general the Cayley transform is not even well-defined, but the following proposition clarifies a lot its usefulness:

Proposition 95. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a symmetric operator. Then

- $T + i\mathbb{I}$ is injective,
- the Cayley transform (22) is well defined,
- $V(T)$ is an isometry and $\text{Ran}(V) = \text{Ran}(T - i\mathbb{I})$.

Proof. Since T is a symmetric operator, for every $\psi \in D(T)$, it holds that

$$\|(T + i\mathbb{I})\psi\|^2 = \|(T - i\mathbb{I})\psi\|^2 = \|T(\psi)\|^2 + \|\psi\|^2 \geq \|\psi\|^2. \quad (23)$$

Suppose $\psi \in \ker(T \pm i\mathbb{I})$. Hence $\|(T \pm i\mathbb{I})\psi\|^2 = 0$ implies $\|\psi\|^2 = 0$, which is tantamount to saying $\psi = 0$. In other words both $T \pm i\mathbb{I}$ are injective. Hence V is well defined on $\text{Ran}(T + i\mathbb{I})$ where $(T + i\mathbb{I})^{-1}$ exists. Notice also that $T - i\mathbb{I} = T + i\mathbb{I} - 2i\mathbb{I}$ and the standard domain of definition is $D(T + i\mathbb{I}) \cap D(\mathbb{I}) = D(T + i\mathbb{I})$ since $D(\mathbb{I}) = \mathcal{H}$. The same holds true exchanging i with $-i$. We need only to show that $V(T)$ is an isometry. Yet, for any $\phi \in D(V)$, $\tilde{\psi} \doteq (T + i\mathbb{I})^{-1}\phi \in D(T - i\mathbb{I})$. Hence by (23), we can conclude

$$\|V(T)\phi\| = \|(T - i\mathbb{I})\tilde{\psi}\| = \|(T + i\mathbb{I})\tilde{\psi}\| = \|\phi\|,$$

which the sought result. □

More generally we can prove

Proposition 96. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. If the Cayley transform (22) is well-defined, then*

1. $\mathbb{I} - V$ is injective,
2. $\text{Ran}(\mathbb{I} - V) = D(T)$ and $T = i(\mathbb{I} + V)(\mathbb{I} - V)^{-1}$.

Proof. We observe that (22) is defined on $D(V) = \{\psi \in \mathcal{H} \mid \exists \phi \in D(T) \text{ such that } \psi = (T + i\mathbb{I})\phi\}$. For all $\psi \in D(V)$, $V\psi = (T - i\mathbb{I})\phi$, from which it descends

$$(\mathbb{I} + V)\psi = 2T\phi, \quad \text{and} \quad (\mathbb{I} - V)\psi = 2i\phi.$$

It follows that if $\psi \in \ker(\mathbb{I} - V)$, then $\phi = 0$, but, since $\psi = (T + i\mathbb{I})\phi$, also $\psi = 0$. At the same time, since $\phi \in D(T)$, $\text{Ran}(\mathbb{I} - V) = D(T)$ and, combining together the above two equalities, we obtain $T = i(\mathbb{I} + V)(\mathbb{I} - V)^{-1}$. $\square$

The most important result is the following:

Theorem 97. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. Then*

- (a) *if T is symmetric, then it is self-adjoint if and only if $V(T)$ is unitary on $\mathcal{H}$,*
- (b) *if $V : \mathcal{H} \rightarrow \mathcal{H}$ is a unitary operator on $\mathcal{H}$ and $\mathbb{I} - V$ is injective, then it is the Cayley transform of a self-adjoint operator on $\mathcal{H}$.*

Proof. (a) Suppose that $T = T^*$. Then item 1. of Theorem 87 guarantees us that $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$. Item 3. of Proposition 95 entails that on these domain $V(T)$ is an isometry. Since $D(V) = \mathcal{H}$, V is also surjective, hence unitary.

Conversely, suppose that $V(T)$ is a unitary operator. Proposition 95 guarantees us that $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$ and item 3. of Theorem 87 that $T = T^*$.

(b) It suffices to show that V is the Cayley transform of a symmetric operator. Then item 1. of this Theorem allows us to draw the conclusion, we want. Per assumption, the map $\mathbb{I} - V$ is bijective from $D(V) = \mathcal{H}$ to $\text{Ran}(\mathbb{I} - V)$. Let us define the auxiliary operator A such that

$$A\psi = i(\phi + V\phi), \quad \forall \psi \in \mathcal{H} \text{ such that } \psi = \phi - V\phi. \quad (24)$$

Since ϕ is unique the operator A is well defined. Suppose $\psi, \psi' \in D(A) = \text{Ran}(\mathbb{I} - V)$ with $\psi = \phi - V\phi$ and $\psi' = \phi' - V\phi'$, with $\phi, \phi' \in D(V)$. Since V is per hypothesis an isometry

$$(A\psi, \psi') = -i(\phi + V\phi, \phi' - V\phi') = -i(V\phi, \phi') + i(\phi, V\phi') = (\phi - V\phi, i\phi' + iV\phi') = (\psi, A\psi').$$

In other words A is Hermitian. To show that A is symmetric, we must prove that its domain is dense. Yet $D(A) = \text{Ran}(\mathbb{I} - V)$ and $[\text{Ran}(\mathbb{I} - V)]^\perp = \ker(\mathbb{I} - V^*)$. The latter must be trivial, since, if there would exists $0 \neq \tilde{\psi} \in \ker(\mathbb{I} - V^*)$, then $V^*\tilde{\psi} = \tilde{\psi}$. Applying V and Exercise 11,

we obtain $V\tilde{\psi} = VV^*\tilde{\psi} = \tilde{\psi}$, namely $\tilde{\psi} \in \ker(V - \mathbb{I})$ which is impossible since $V - \mathbb{I}$ is injective. In other words $\text{Ran}(\mathbb{I} - V)$ and thus $D(A)$ are densely defined.

To conclude, we show that V is the Cayley transform of A as in (24). Notice that this equation entails that, for all $\psi \in \mathcal{H}$,

$$2iV\phi = A\psi - i\psi \quad \text{and} \quad 2i\phi = A\psi + i\psi$$

where $\phi \in D(A)$. As a by-product we observe that, since $\mathcal{H} = D(V) = \text{Ran}(A + i\mathbb{I})$, we have that

$$V(A + i\mathbb{I}) = (A - i\mathbb{I}),$$

namely V is the Cayley transform of A , as sought. $\square$

All these theorems suggest that the following objects play a key role:

Definition 98. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a symmetric operator thereon. Then we call **deficiency indices** of T

$$d_{\pm}(T) \doteq \dim(\ker(T^* \pm i\mathbb{I})). \quad (25)$$

We state now a theorem unveiling the key application of deficiency indices. Yet, since its proof is very long and technical, we omit it – the brave ones could refer to [5, Th. 5.36]

Theorem 99. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. Then

1. T admits self-adjoint extensions if and only if $d_+(T) = d_-(T)$.
2. If $d_+(T) = d_-(T)$, there is a bijection between the self-adjoint extensions of T and the surjective isometries from $\ker(T^* - i\mathbb{I})$ to $\ker(T^* + i\mathbb{I})$.

Proof. 1. Assume T admits a self-adjoint extension S , i.e. $T \subset S = S^*$. If we call $V(T)$ and $V(S)$ the Cayley transform of T and S respectively, item b) of Theorem 97 entails that $V(S) : \mathcal{H} \rightarrow \mathcal{H}$ is a unitary operator. In addition (22) guarantees that $V(T) \subset V(S)$. This is a by product of the fact that, since $T \subset S$, then $T - i\mathbb{I} \subset S - i\mathbb{I}$ and $(T + i\mathbb{I})^{-1} \subset (S + i\mathbb{I})^{-1}$. This entails also that $V(S)[\text{Ran}(T + i\mathbb{I})] \subset \text{Ran}(T - i\mathbb{I})$. Since $V(S)$ is a unitary operator, if $\psi \in \mathcal{H}$ is such that $\psi \in \text{Ran}(T + i\mathbb{I})^{\perp}$ then $V(S)\psi \in [(V(S)(\text{Ran}(T + i\mathbb{I})))]^{\perp} = \text{Ran}(T - i\mathbb{I})^{\perp}$. Here equality holds true since $V(S)$ is surjective, being a unitary operator. Using Exercise 18 we know that $\text{Ran}(T \pm i\mathbb{I})^{\perp} = \ker(T^* \mp i\mathbb{I})$ and thus $[(V(S)(\text{Ran}(T + i\mathbb{I})))]^{\perp} = \text{Ran}(T - i\mathbb{I})^{\perp}$ translates to $V(S)[\ker(T^* - i\mathbb{I})] = \ker(T^* + i\mathbb{I})$. This entails that the deficiency subspaces are in 1:1 correspondence, that is $d_+(T) = d_-(T)$.

Suppose now that T has equal deficiency indices and let $U_0 : \ker(T^* + i\mathbb{I}) \rightarrow \ker(T^* - i\mathbb{I})$ be a unitary operator. Using once more Exercise 18, we can decompose $\mathcal{H} = \overline{\text{Ran}(T + i\mathbb{I})} \oplus \ker(T^* - i\mathbb{I}) = \overline{\text{Ran}(T - i\mathbb{I})} \oplus \ker(T^* + i\mathbb{I})$. As a consequence we can construct a unitary operator $U : \mathcal{H} \rightarrow \mathcal{H}$ such that $U = V(T) \oplus U_0$ where $V(T)$ is the Cayley transform of T and where, for

every $\psi \in \mathcal{H}$, $U\psi = V(T)\phi + U_0\psi_0$ with $\psi = \phi + \psi_0$, $\phi \in \overline{\text{Ran}(T + i\mathbb{I})}$ and $\psi_0 \in \ker(T^* - i\mathbb{I})$. Observe that, since $V(T)$ is an isometry on $\text{Ran}(T + i\mathbb{I})$, it can be extended uniquely to an isometry on $\overline{\text{Ran}(T + i\mathbb{I})}$, cf. Proposition 26. It is worth mentioning that U is a bijection since injectivity is inherited from that of $V(T)$ and of U_0 , while surjectivity is a consequence of U_0 being bijective per assumption, while the (extension of the) Cayley transform of a symmetric operator establishes a 1:1 correspondence between $\overline{\text{Ran}(T - i\mathbb{I})}$ and $\overline{\text{Ran}(T + i\mathbb{I})}$. The operator $\mathbb{I} - U$ is injective. As a matter of fact, suppose $\chi \in \ker(\mathbb{I} - U)$. Writing $\chi = \chi_1 + \chi_0$ with $\chi_1 \in \overline{\text{Ran}(T + i\mathbb{I})}$ and $\chi_0 \in \ker(T^* - i\mathbb{I})$, $(\mathbb{I} - U)\chi = 0$ is equivalent to $V(T)\chi_1 = \chi_1$ and $\chi_0 = U_0\chi_0$. The second identity entails $\chi_0 \in \ker(T^* + i\mathbb{I}) \cap \ker(T^* - i\mathbb{I}) = \{0\}$. The first one can be solved only by $\chi_1 = 0$ on account of item 1. of Proposition 96. We can apply item b) of Theorem 97 to conclude that U is the Cayley transform of a self-adjoint operator T_{U_0} . Since $V(T) \subset U$ per construction, $T \subset T_{U_0}$. The number of self-adjoint extensions of T is ruled by the possible choices for U_0 .

2. We have already proven that for each surjective isometry between the deficiency subspaces there exists a self-adjoint extension of T . We need to prove the converse. Let S be such that $T \subset S = S^*$ and let $V(T)$ and $V(S)$ be the respective Cayley transforms. Per construction $V(T) \subset V(S)$ and $V(S)$ is a unitary operator on $\mathcal{H}$. Since $V(T)$ is a surjective isometry from $\overline{\text{Ran}(T - i\mathbb{I})}$ to $\overline{\text{Ran}(T + i\mathbb{I})}$ and since $\mathcal{H} = \overline{\text{Ran}(T + i\mathbb{I})} \oplus \ker(T^* - i\mathbb{I}) = \overline{\text{Ran}(T - i\mathbb{I})} \oplus \ker(T^* + i\mathbb{I})$, $V(S)$ must coincide with $V(T)$ on the first factor of the direct sum. In addition $V(S)$ must identify a unitary operator from $\ker(T^* - i\mathbb{I})$ to $\ker(T^* + i\mathbb{I})$. Furthermore, if S' were a second self-adjoint extension of T , its Cayley transform $V(S')$ would differ from $V(S)$ only for the unitary operator between the deficiency subspaces. This concludes the proof. $\square$

To conclude the section, we prove one last result:

Theorem 100. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. If T is symmetric, it is essentially self-adjoint if and only if it admits a unique self-adjoint extension.*

Proof. Suppose T is essentially self-adjoint. Then it has a unique self-adjoint extension by item 4. of Proposition 86. Theorem 99 guarantees instead that a symmetric operator has a self-adjoint extension only if it has the same deficiency indices, i.e., $d_+ = d_-$. Uniqueness is obtained if $d_+ = d_- = 0$. Yet the first point of Theorem 88 entails that A must be essentially self-adjoint. $\square$

To conclude the section, we explain how to use Theorem 99 to construct explicitly a self-adjoint extension of a symmetric operator such that $d_+(T) = d_-(T)$. Let us consider thus a surjective isometry $U : \ker(T^* - i\mathbb{I}) \rightarrow \ker(T^* + i\mathbb{I})$, then we indicate the extension of the operator T as $T_U : D(T_U) \rightarrow \mathcal{H}$ where

$$D(T_U) \doteq \{\chi \in \mathcal{H} \mid \exists! \phi \in D(\overline{T}) \text{ and } \psi \in \ker(T^* - i\mathbb{I}) \text{ such that } \chi = \phi + (\mathbb{I} + U)\psi\}, \quad (26)$$

from which we can infer that

$$T_U\chi = T\phi + i(\mathbb{I} - U)\psi. \quad \forall \chi \in D(T_U) \quad (27)$$

4.3.1 An example: the momentum operator

Our goal is to show that the concept of deficiency indices is not just abstract, but it represents an effective, manageable, concrete (more lovely adjectives, folks?) tool. To be precise let us consider the closed interval $I = [0, 1]$ and let $AC(I)$ be the collection of all absolutely continuous functions over I ... wait... you don't know what an absolutely continuous function is, isn't it? Not a big deal:

Definition 101. Let $J \subseteq \mathbb{R}$. A function $f : J \rightarrow \mathbb{C}$ is said to be **absolutely continuous** on J if $\forall \epsilon > 0$, there exists $\delta > 0$ such that, whenever we consider a finite sequence of intervals $[x_k, y_k] \in J$, $k = 1, \dots, N$ which are disjoint and enjoying the property

$$\sum_{k=1}^N (y_k - x_k) < \delta,$$

then

$$\sum_{k=1}^N |f(y_k) - f(x_k)| < \epsilon.$$

Observe that, on every compact subset of $\mathbb{R}$, absolutely continuous functions are also continuous, hence, in addition square integrable. This is a prerequisite to make sense of the following definition:

$$A = i \frac{d}{dx} : D(A) \rightarrow L^2(I), \quad \psi \mapsto A(\psi) = i \frac{d\psi}{dx} \quad (28)$$

where we choose

$$D(A) \doteq \{f \in AC(I) \mid f(0) = f(1) = 0\}.$$

Lemma 102. The operator A in (28) is symmetric.

Proof. The domain $D(A)$ is dense in $L^2([0, 1])$ since $C_0^\infty(0, 1) \subset D(A)$. It remains to be shown that A is Hermitian. This descends from the following chain of equalities: For all $\phi, \psi \in D(A)$,

$$(\psi, A\phi) = \int_0^1 dx \, i\bar{\psi}(x) \frac{d\phi(x)}{dx} = i\bar{\psi}\phi|_0^1 - \int_0^1 dx \, i \frac{d\bar{\psi}}{dx} \phi(x) = (A\psi, \phi),$$

where we integrated by parts and in the last step we used both the sesquilinearity of the inner product and the boundary conditions. $\square$

The second step consists in verifying that A is not essentially self-adjoint. To compute the domain of the adjoint operator, it suffices to notice the integration by parts is the only operation that we have to make sure that it makes sense. For this reason it descends that

$$D(A^*) = \{\psi \in L^2(I) \mid \frac{d\psi}{dx} \in AC(I)\}.$$

By repeating the same procedure we can compute the domain of A^{**} , though, in this case, we need to account also for the boundary conditions at $x = 0$ and $x = 1$, else the boundary terms in the integration by parts would not vanish. Hence

$$D(A^{**}) = \{\psi \in AC(I) \mid \frac{d\psi}{dx} \in L^2(I) \text{ and } \psi(0) = \psi(1) = 0\},$$

where the specific choice of the boundary conditions are made so that $A \subset A^{**}$. By direct inspection we realize that $A^* \neq A^{**}$. Hence A is not essentially self-adjoint. Yet, since A is symmetric we can at least ask ourselves whether it admits self-adjoint extensions.

Proposition 103. *The operator A in (28) has deficiency indices $d_+(A) = d_-(A) = 1$.*

Proof. In order to compute the deficiency indices we need to evaluate the dimension of $\ker(A^* \pm i\mathbb{I})$. This is tantamount to finding $\psi \in D(A^*)$ such that $A^*\psi = \mp i\psi$, which translates to $\frac{d\psi}{dx} = \mp \psi$. Since $\psi \in D(A^*)$, then $\frac{d\psi}{dx} \in AC(I)$ and the equation entails that, per consistency, we have to look for solutions $\psi \in AC(I)$. We can then solve directly the ordinary differential equations obtaining $\psi(x) = ae^{\mp x}$, with $a \in \mathbb{C}$. Since the exponential function is smooth on I , it is in particular absolutely continuous. The outcome of this calculation is that the kernel of both $A^* \pm \mathbb{I}$ is one dimensional, being spanned by the choice of a . The deficiency indices are $(1, 1)$. $\square$

Theorem 99 guarantees us that A admits self-adjoint extensions. We want to understand their significance and this descends from item 2. of Theorem 99. To compute all possible surjective isometries from $\mathcal{H}_+ = \ker(A^* + i\mathbb{I})$ and $\mathcal{H}_- = \ker(A^* - i\mathbb{I})$, which are 1-dimensional vector spaces, it suffices to consider their unit-norm generators

$$\varphi_+ = \sqrt{\frac{2}{e^2 - 1}}e^x, \quad \varphi_- = \sqrt{2(e^2 - 1)}e^{-x}.$$

Since we are dealing with finite dimensional Hilbert subspaces, all surjective isometries are actually unitary operators and they are all of the form

$$U_\alpha : \mathcal{H}_+ \rightarrow \mathcal{H}_-, \quad \varphi_+ \mapsto e^{i\alpha}\varphi_-,$$

where $\alpha \in [0, 2\pi)$ is a parameter. In this case it is also easy to construct explicitly the self-adjoint extension. While the operator A is still defined as $i\frac{d}{dx}$, we obtain a 1-parameter family of possible domains of definition for the self-adjoint extensions:

$$D_\alpha = \{f + z\varphi_+ + ze^{i\alpha}\varphi_- \mid f \in D(A) \text{ and } z \in \mathbb{C}\},$$

which can be seen to be nothing but

$$D_\alpha \doteq \{f' \in AC(I) \mid f'(1) = e^{i\beta}f'(0)\},$$

where β is not necessarily equal to α .

5 Spectral Theory

This last chapter on the theory of operators represents a bridge towards more mathematical concepts which are at the hearth of modern functional analysis applied to physical models. Our goal is in particular to generalize the concept of eigenvalue which we used in the previous section and to contextualize it using the more modern language of projective valued measure. Since we have also to learn the theory of distributions we will not enter into many details and we shall focus only on a preliminary set of results.

5.1 The Resolvent and the Spectrum

In this subsection we will generalize concepts and structures which have their roots in linear algebras and in the study of matrices. In particular, if you think about the question of what is an eigenvalue of a complex valued matrix, you can give many equivalent answers, but, if one would ask you, how you compute the possible eigenvalues, the answer would always be to look for the zeros of the characteristic polynomials. If we focus on this last object and we think about the ingredients necessary to construct it, we can realize almost immediately that we need the linearity of the space of matrices, its algebra structure (multiplication of matrices) and the identity. The last missing bit of information is the determinant function. If we think of bounded operators on a normed space, this is actually the only tool that we cannot transport slavishly from the finite dimensional case. Yet we can replace it with a more abstract, albeit equivalent concept. The existence of a non trivial zero of the determinant of a matrix is tantamount to saying that such matrix is not injective, *i.e.* not invertible.

Bearing in mind these considerations, let us introduce a preliminary concept which generalizes the standard one coming from complex analysis:

Definition 104. Let X be a Banach space over the complex numbers and let $\Omega \subset \mathbb{C}$ be a non empty open set. We say that $f : \Omega \rightarrow X$ is **analytic** at $z_0 \in \Omega$ if $\exists \delta > 0$ and $\{a_n\}_{n \in \mathbb{N}}$, $a_n \in X$ for all n , such that

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad \forall z \in B_\delta(z_0)$$

where convergence is meant with respect to the norm topology.

Although this last definition requires a Banach space to make sure that the limit of the converging sequence lies still in X , in the next steps, we will only need normed spaces. The reader is warned and cordially invited to pay attention to this detail.

Definition 105. Let X be a normed space and let $A : D(A) \subseteq X \rightarrow X$ be an operator thereon. We call **resolvent set** of A the collection $\rho(A) \subseteq \mathbb{C}$ of complex numbers λ such that

1. $\overline{\text{Ran}(A - \lambda \mathbb{I})} = X$,
2. $(A - \lambda \mathbb{I}) : D(A) \rightarrow X$ is injective,

3. $(A - \lambda\mathbb{I})^{-1} : \text{Ran}(A - \lambda\mathbb{I}) \rightarrow X$ is bounded.

For all $\lambda \in \rho(A)$ we call **resolvent** the operator

$$R_\lambda(A) \doteq (A - \lambda\mathbb{I})^{-1} : \text{Ran}(A - \lambda\mathbb{I}) \rightarrow D(A).$$

As it happens often in functional analysis, when it is easy and natural to introduce and to characterize a set, its complement is the one actually containing more information:

Definition 106. Let X be a normed space and let $A : D(A) \subseteq X \rightarrow X$ be an operator thereon. We call **spectrum** of A the set $\sigma(A) \doteq \mathbb{C} \setminus \{\rho(A)\}$, where $\rho(A)$ is the resolvent set. The spectrum is divided in three disjoint unions:

- $\sigma_p(A)$, the **point spectrum** of A , made of all $\lambda \in \mathbb{C}$ such that $\ker(A - \lambda\mathbb{I}) \neq \{0\}$,
- $\overline{\sigma_c(A)}$, the **continuous spectrum** of A , made of all $\lambda \in \mathbb{C}$ such that $\ker(A - \lambda\mathbb{I}) = \{0\}$, $\text{Ran}(A - \lambda\mathbb{I}) = X$ but $(A - \lambda\mathbb{I})^{-1}$ is not bounded,
- $\overline{\sigma_r(A)}$, the **residual spectrum** of A , made of all $\lambda \in \mathbb{C}$ such that $\ker(A - \lambda\mathbb{I}) = \{0\}$, but $\text{Ran}(A - \lambda\mathbb{I}) \neq X$.

Remark 3. Notice that $\sigma_p(A)$ consists of the eigenvalues of A , but the spectrum is in general larger. It is interesting the special case when $X = \mathcal{H}$, i.e., it is a Hilbert space. In this case, if the collection of eigenvectors of A forms a basis of $\mathcal{H}$, we say that A has a **purely point spectrum**. Also in this case we should expect in general that $\sigma_p(A) \neq \sigma(A)$. A good example are compact self-adjoint operators which have purely point spectrum as we have seen in the previous section, but 0 can nonetheless lie in the continuous spectrum

We start analysing more in detail the resolvent and the spectrum of notable classes of operators. Although we will not specify it explicitly, we assume always that all our Banach and Hilbert spaces are non trivial.

Theorem 107. Let X be a Banach space and let A be a closed operator thereon. Then

(a) $\lambda \in \rho(A)$ if and only if $A - \lambda\mathbb{I}$ is a bijection from $D(A)$ to X ,

(b) it holds that

- (i) $\rho(A)$ is open,
- (ii) $\sigma(A)$ is closed,
- (iii) if $\rho(A) \neq \emptyset$, the map $\rho(A) \ni \lambda \mapsto R_\lambda(A) \in \mathcal{B}(X)$ is analytic, being $R_\lambda(A)$ the resolvent of A .

Proof. (a) $\Leftarrow$: Suppose that $A - \lambda\mathbb{I}$ is bijective on the whole X . Condition 1. and 2. in Definition 105 are automatically met and we need only to show that $(A - \lambda\mathbb{I})^{-1}$ is bounded. Yet since A is closed per hypothesis, so is $A - \lambda\mathbb{I}$ and its graph in particular. We can apply thus the closed graph theorem 30 to draw the conclusion sought.

$\implies$: Suppose $\lambda \in \rho(A)$. Per Definition 105 we know that $A - \lambda\mathbb{I}$ is injective. We need to prove that $\text{Ran}(A - \lambda\mathbb{I}) = X$. Since $(A - \lambda\mathbb{I})^{-1}$ is a bounded operator, it is also continuous on its domain, per Theorem 24. Hence, for any $\tilde{x} \in \text{Ran}(A - \lambda\mathbb{I})$, there exists $K > 0$ and $\tilde{y} \in D(A)$ such that

$$\|\tilde{y}\| = \|(A - \lambda\mathbb{I})^{-1}(\tilde{x})\| \leq \|K\| \|\tilde{x}\| = \|K\| \|(A - \lambda\mathbb{I})(\tilde{y})\|.$$

Yet, since item 3. of Definition 105 tells us that $\overline{\text{Ran}(A - \lambda\mathbb{I})} = X$ and since X is a Banach space, hence complete, for every $x \in X$, there exists a sequence $\{y_n\}_{n \in \mathbb{N}}$, $y_n \in D(A)$ for all n such that $\lim_{n \rightarrow \infty} (A - \lambda\mathbb{I})(y_n) = x$. Nonetheless the above estimate on the norm of $\tilde{y}$ entails that the sequence $\{y_n\}_{n \in \mathbb{N}}$ must converge to $y \in X$. Since A is a closed operator per hypothesis, we have $y \in D(A)$ and $(A - \lambda\mathbb{I})(y) = x$. In other words $x \in \text{Ran}(A - \lambda\mathbb{I})$ and thus $\text{Ran}(A - \lambda\mathbb{I}) = X$.

(b) Let $\mu \in \rho(A)$ and let $S : \mathbb{C} \rightarrow \mathcal{L}(X)$ be defined as

$$S(\lambda) \doteq \sum_{n=0}^{\infty} (\lambda - \mu)^n R_{\mu}(A)^{n+1},$$

where R_{μ} is the resolvent. Let us check the convergence of S by looking at

$$\sum_{n=0}^{\infty} |(\lambda - \mu)|^n \|R_{\mu}(A)^{n+1}\| \leq \sum_{n=0}^{\infty} |(\lambda - \mu)|^n \|R_{\mu}(A)\|^{n+1} = \|R_{\mu}(A)\| \sum_{n=0}^{\infty} \|(\lambda - \mu)R_{\mu}(A)\|^n.$$

The sum on the right hand side is a geometric series whose convergence radius is $|\lambda - \mu| \|R_{\mu}(A)\| < 1$. Let us now apply both to the right and to the left of $S(\lambda)$ the operator $A - \lambda\mathbb{I} = A - \mu\mathbb{I} + (\mu - \lambda)\mathbb{I}$. The result is

$$(A - \lambda\mathbb{I})S(\lambda) = \mathbb{I}_X \quad \text{and} \quad S(\lambda)(A - \lambda\mathbb{I}) = \mathbb{I}|_{D(A)}.$$

Hence, if $\mu \in \rho(A)$ we know that there exists an open neighbourhood of μ where the left and right inverses of $A - \lambda\mathbb{I}$ exists as operators from X to $D(A)$. By item (a) of this theorem, we can conclude that this neighbourhood is contained in $\rho(A)$. Hence $\rho(A)$ is open and its complement $\sigma(A)$ must be closed. If the resolvent set is not empty, then $R_{\lambda}(A)$ has a Taylor series around any point of $\rho(A)$ with respect to the uniform (norm) topology. Per definition the map $\lambda \mapsto R_{\lambda}(A)$ is analytic and it maps the resolvent in $\mathcal{B}(X)$. $\square$

Further consequences of this result are the following:

Proposition 108. *Let X be a Banach space and let A be a closed operator with $D(A) = X$. Then*

1. $\rho(A) \neq \emptyset$,
2. $\sigma(A) \neq \emptyset$ and it is compact,
3. $|\lambda| \leq \|A\|$ for all $\lambda \in \sigma(A)$.

Proof. In comparison to Theorem 107, we know that X is a Banach space and $D(A) = X$. Hence, since A is closed, the closed graph theorem 30 guarantees us that A is bounded. Suppose now that $\lambda \in \mathbb{C}$ is such that $|\lambda| > \|A\|$. Let $S' : \mathbb{C} \rightarrow \mathcal{L}(\mathcal{H})$ be

$$S'(\lambda) \doteq \sum_{n=0}^{\infty} (-\lambda)^{n+1} A^n.$$

Observe that

$$\|S'(\lambda)\| \leq \sum_{n=0}^{\infty} \|\lambda^{-(n+1)} A\|^n = \|\lambda\|^{-1} \sum_{n=0}^{\infty} \left\| \frac{A}{\lambda} \right\|^n,$$

which is convergent whenever $|\lambda| > \|A\|$. Furthermore it converges to a bounded operator. In addition, per direct inspection, one can prove that

$$(A - \lambda \mathbb{I})S'(\lambda) = S'(\lambda)(A - \lambda \mathbb{I}) = \mathbb{I} \implies S'(\lambda) = R_\lambda(A),$$

where we used item (a) of Theorem 107. Hence every λ such that $|\lambda| > \|A\|$ lies in $\rho(A)$, that is $\rho(A) \neq \emptyset$. The first item is proven. Furthermore if we prove that $\sigma(A)$ is not empty, item (b) of Theorem 107 guarantees that it is a closed set. Yet, the above argument also tells us that for every $z \in \sigma(A)$, $|z| < \|A\|$, namely $\sigma(A)$ is a bounded set in $\mathbb{C}$. Heine-Borel theorem entails compactness. We need only to prove that the spectrum is not empty. By contradiction assume it is. Hence the function $\lambda \mapsto R_\lambda(A)$ is defined on the whole $\mathbb{C}$.

— **WARNING - VERY TECHNICAL — DANGER!!!**

Fix any continuous linear functional $f : X \rightarrow \mathbb{C}$ and fix any $x \in X$. Consider the complex function $g : \mathbb{C} \rightarrow \mathbb{C}$ such that

$$g(\lambda) \doteq f(R_\lambda(A)x).$$

We show that g is analytic over the whole complex plane. Choose $\mu \in \rho(A)$ which is a non empty set, as proven above. In a neighbourhood of μ contained in the resolvent, we can write the following Taylor expansion

$$g(\lambda) = \sum_{n=0}^{\infty} (\lambda - \mu)^n f(R_\mu(A)^{n+1}x),$$

where we used the linearity of f . Absolute convergence follows repeating the proof of item (b) in Theorem 107. Yet, since we have assumed that $\sigma(A) = \emptyset$, the convergence is valid on the whole complex plane. In addition we can also expand

$$g(\lambda) = \sum_{n=0}^{\infty} (-\lambda)^{-(n+1)} f(A^n x),$$

which is absolutely convergent by Abel's theorem⁹ on power series. We can thus write for $|\lambda| \geq 1 + \|A\|$

$$|g(\lambda)| \leq |\lambda|^{-1} \sum_{n=0}^{\infty} \frac{|f(A^n x)|}{|\lambda|^n} \leq \frac{\|f\| \|x\|}{|\lambda|} \sum_{n=0}^{\infty} \left(\frac{\|A\|}{|\lambda|} \right)^n = \frac{\|f\| \|x\|}{|\lambda| - \|A\|} \leq \frac{K}{|\lambda|},$$

where $K > 0$. Since $|g|$ is everywhere continuous and bounded on the entire complex plane, by Liouville's theorem it must be constant on the whole $\mathbb{C}$. Yet $|g(\lambda)|$ vanishes as $\lambda \rightarrow \infty$. Hence $|g(\lambda)| = 0$. This entails that $f(R_\lambda(A)) = 0$ which, by Hahn-Banach theorem, is tantamount to saying that $R_\lambda(A) = 0$ for any $\lambda \in \rho(A)$. This is a contradiction since $R_\lambda(A)$ must invert $A - \lambda\mathbb{I}$. $\square$

Corollary 109. *Let X be a Banach space and let A be a closed operator thereon. Then for any $\lambda, \mu \in \rho(A)$, it holds*

$$R_\lambda(A) - R_\mu(A) = (\lambda - \mu)R_\lambda(A)R_\mu(A). \quad (29)$$

Proof. Suppose that $\lambda, \mu \in \rho(A)$. Per Definition 105, we know that

$$(A - \lambda\mathbb{I})R_\lambda(A) = \mathbb{I} \quad \text{and} \quad (A - \mu\mathbb{I})R_\mu(A) = \mathbb{I}.$$

Multiply the first identity by $R_\mu(A)$ on the left and the second by $R_\lambda(A)$ on the right. Subtract the results and, using the identities $R_\mu(A)R_\lambda(A) = R_\lambda(A)R_\mu(A)$ and¹⁰ $R_\mu(A)A R_\lambda(A) = R_\lambda(A)A R_\mu(A)$, we obtain (29). $\square$

To conclude this preliminary section, we specialize to Hilbert spaces:

Theorem 110. *Let $\mathcal{H}$ be a Hilbert space. Then*

1. *If $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$ is a self-adjoint operator*

- (i) $\sigma(T) \subseteq \mathbb{R}$,
- (ii) $\sigma_r(T) = \emptyset$,
- (iii) *the eigenspaces of T with distinct eigenvalues are orthogonal.*

2. *if $T \in \mathcal{B}(\mathcal{H})$ is unitary*

- (i) $\sigma(T)$ *is a non empty, compact subset of the unit circle $\mathbb{S}^1 \subset \mathbb{C}$,*
- (ii) $\sigma_r(T) = \emptyset$.

3. *if $T \in \mathcal{B}(\mathcal{H})$ is normal,*

- (i) $\sigma_r(T) = \sigma_r(T^*) = \emptyset$,

⁹Let $H(z) = \sum_{k=0}^{\infty} a_k z^k$ be a series such that $\sum_{k=0}^{\infty} a_k$ converges. Then $\lim_{|z| \rightarrow 1^-} H(z) = \sum_{k=0}^{\infty} a_k$ assuming that

H is defined on the unit disk.

¹⁰To prove the second identity, use that $A = (A - \lambda\mathbb{I}) + \lambda\mathbb{I}$.

$$\begin{aligned} (ii) \quad \sigma_p(T^*) &= \overline{\sigma_p(T)}, \\ (iii) \quad \sigma_c(T^*) &= \overline{\sigma_c(T)}. \end{aligned}$$

Proof. Since the proofs of all three points are similar in spirit and in practice, we show only the first one, dividing it in three parts.

(i): Let $\lambda = \mu + i\nu$, with $\mu, \nu \in \mathbb{R}$ and $\nu \neq 0$. We need to show that $\lambda \in \rho(T)$. Let $\psi \in D(T)$ and since T is per assumption self-adjoint,

$$\|(T - \lambda\mathbb{I})\psi\|^2 = \|(T - \mu\mathbb{I})\psi\|^2 + |\nu|^2\|\psi\|^2,$$

from which we infer

$$\|(T - \lambda\mathbb{I})\psi\| \geq |\nu| \|\psi\|$$

and similarly

$$\|(T - \bar{\lambda}\mathbb{I})\psi\| \geq |\nu| \|\psi\|.$$

The operators $T - \lambda\mathbb{I}$ and $T - \bar{\lambda}\mathbb{I}$ are 1 : 1 and $\|(T - \lambda\mathbb{I})^{-1}\| \leq |\nu|^{-1}$, where $(T - \lambda\mathbb{I})^{-1} : \text{Ran}(T - \lambda\mathbb{I}) \rightarrow D(T)$. Observe that

$$\overline{[\text{Ran}(T - \lambda\mathbb{I})]^\perp} = [\text{Ran}(T - \lambda\mathbb{I})]^\perp = \ker(T^* - \bar{\lambda}\mathbb{I}) = \ker(T - \bar{\lambda}\mathbb{I}) = \{0\},$$

where we used that $T - \bar{\lambda}\mathbb{I}$ is injective. In other words we have $\text{Ran}(T - \lambda\mathbb{I})$ dense in $\mathcal{H}$, hence $\lambda \in \rho(T)$.

(ii): Suppose $\lambda \in \sigma(T)$, but $\lambda \notin \sigma_p(T)$. Then $T - \lambda\mathbb{I}$ is bijective and $\ker(T - \lambda\mathbb{I}) = \{0\}$. Since T is self-adjoint and $\lambda \in \mathbb{R}$, then

$$\ker(T^* - \bar{\lambda}\mathbb{I}) = \{0\} \implies [\text{Ran}(T - \lambda\mathbb{I})]^\perp = \ker(T^* - \bar{\lambda}\mathbb{I}) = \{0\},$$

from which it descends that $\overline{\text{Ran}(T - \lambda\mathbb{I})} = \mathcal{H}$. Hence $\lambda \in \sigma_c(T)$.

(iii): Suppose that $\lambda \neq \mu$ and $T\psi = \lambda\psi$, $T\psi' = \mu\psi'$, with $\psi, \psi' \in \mathcal{H}$. Then

$$(\lambda - \mu)(\psi, \psi') = (T\psi, \psi') - (\psi, T\psi') = (\psi, T\psi') - (\psi, T\psi') = 0,$$

where we used that T is per hypothesis self-adjoint and that $\lambda, \mu \in \mathbb{R}$. The only possibility for the above chain of identities to hold true is that $(\psi, \psi') = 0$. $\square$

6 Theory of Distributions

In the second part of the course we shall make the reader (which reader? Ciao Beatrice!) acquainted with the concept of distributions introduced by Laurent Schwartz, in an attempt to make rigorous several objects which were used in theoretical physics as a magic box. The idea is rather simple: Think of singular objects like $(x)^{-\frac{1}{2}}$ as $x \rightarrow 0$ or functions which fail to be continuous or differentiable everywhere such as, $|x|$ and $\sqrt{|x|}$ at $x = 0$ or the Heaviside

function $\Theta(x)$ which is 1 for $x > 0$ and 0 everywhere else. All these functions are globally rather ill-defined, but suppose that you integrate them after multiplying them by a sufficiently regular function, say smooth and compactly supported. Even if the integrand fails to be continuous everywhere, it is integrable over the whole $\mathbb{R}$ and the result is finite.

In quantum theories this situation is basically a daily routine. There are several quantities that we wish to compute as suitable integrals, whose integrands are either ill defined or, in the best case scenario, not even a smooth function. The most famous and, to an extent which we will make precise, the extreme case of this behaviour is the Dirac delta.

Goal of these notes from now on is to teach you the theory of these singular objects. We will follow mainly [1], but a reader is strongly encouraged to learn from the best of the best [2].

6.1 Test Functions

In the whole section we will be working on $\mathbb{R}^n$ and with its open subsets $\Omega \subseteq \mathbb{R}^n$. We will be considering complex valued functions, although we will omit the symbol $\mathbb{C}$ when it is the target space, *i.e.*, we shall write $C^\infty(\mathbb{R}) \equiv C^\infty(\mathbb{R}; \mathbb{C})$. As a starting point we must define a notable class of functions, which is of paramount relevance since it enjoys basically all nice properties that one can imagine and desire (please keep your lust at bay! Be a little bit slothful!). To this end we recall that for every map $f : \mathbb{R}^n \rightarrow \mathbb{C}$ we call **support** of f

$$\text{supp}(f) \doteq \overline{\{x \in \mathbb{R}^n \mid f(x) \neq 0\}}. \quad (30)$$

Definition 111. We call *space of smooth and compactly supported functions (or test function space)* on $\mathbb{R}^n$,

$$\mathcal{D}(\mathbb{R}^n) = C_0^\infty(\mathbb{R}^n) = \{f \in C^\infty(\mathbb{R}^n) \mid \text{supp}(f) \text{ is compact}\}.$$

A question, which I asked myself, when I first read this definition many years ago was whether any such function really exists. Why not assigning on this topic an exercise?

Exercise 22. Prove that $C_0^\infty(\mathbb{R})$ is not empty. (HINT: Study the behaviour of the function

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & x < 0 \\ 0 & x \geq 0 \end{cases},$$

and of the function $g(x) = f(|x|^2 - 1)$.)

It is straightforward to generalize the result of this exercise from $\mathbb{R}$ to $\mathbb{R}^n$ by multiplying several smooth and compactly supported functions depending each on a different variable. The next question, which raises naturally is why are the functions in Definition 111 so special. There are several different reasons, but one of the more important ones is the following. It has in addition far reaching consequences in physics, but we will not discuss them here: Let $C^\infty(\mathbb{R}^n)$ be the space of smooth functions over $\mathbb{R}^n$. This possesses a natural pairing with $C_0^\infty(\mathbb{R}^n)$ as follows:

$$(\cdot, \cdot) : C^\infty(\mathbb{R}^n) \times C_0^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad (\phi, f) \mapsto (\phi, f) = \int_{\mathbb{R}^n} d^n x \phi(x) f(x). \quad (31)$$

In addition the following notable property holds true:

Proposition 112. *The pairing (31) is **separating**, that is, if $\phi, \phi' \in C^\infty(\mathbb{R}^n)$ are such that*

$$(\phi, f) = (\phi', f), \quad \forall f \in C_0^\infty(\mathbb{R}^n)$$

then $\phi = \phi'$.

Proof. Suppose $\phi \neq \phi'$ at a point $x \in \mathbb{R}^n$. Being smooth functions, they must differ in a neighbourhood Ω of x . Suppose that $\phi - \phi'$ is strictly positive in Ω , else exchange them. Pick any f which is a strictly positive, smooth and compactly supported in Ω . Then $\int_{\mathbb{R}^n} d^n x (\phi - \phi')(x) f(x) > \inf_{x \in \Omega} ((\phi - \phi') f) \text{Vol}(\Omega)$, hence a contradiction. $\square$

Remark 4. *Notice that, in the last proposition, we have used smooth and compactly supported functions in an open set $\Omega \subseteq \mathbb{R}^n$. Notice that $C_0^\infty(\Omega)$ is defined as in (111) replacing $\mathbb{R}^n$ with Ω . Yet $C_0^\infty(\Omega)$ does not coincide with the restriction of $C_0^\infty(\mathbb{R}^n)$ to Ω since the support of a given element might be strictly larger than Ω . Hence the restriction would not yield a compactly supported function.*

Proposition 113. *Let $f \in C_0^k(\mathbb{R}^n)$, $0 \leq k \leq \infty$ and let $\rho \in C_0^\infty(\mathbb{R}^n)$ be such that*

$$\rho \geq 0, \quad \text{supp}(\rho) \subseteq \{x \in \mathbb{R}^n \mid |x| \leq 1\}, \quad \text{and} \quad \int_{\mathbb{R}^n} d^n x \rho(x) = 1,$$

where $|x|$ stands for the Euclidean absolute value on $\mathbb{R}^n$. Let $\epsilon > 0$ and let

$$f_\epsilon(x) = \epsilon^{-n} \int_{\mathbb{R}^n} d^n y f(y) \rho\left(\frac{x-y}{\epsilon}\right). \quad (32)$$

Then $f_\epsilon \in C_0^\infty(\mathbb{R}^n)$ and $\text{supp}(f_\epsilon)$ is contained in an ϵ -neighbourhood of $\text{supp}(f)$. In addition¹¹, for all possible $|\alpha|$, $\partial^\alpha f_\epsilon$ converges uniformly to $\partial^\alpha f$ as $\epsilon \rightarrow 0$.

Proof. We observe that, per construction of f_ϵ and per definition of ρ , the integrand of f_ϵ vanishes whenever the distance between x and $\text{supp}(f)$ is larger than ϵ . In addition, since one integrates over y , we can exchange the integration with any derivative with respect to x . Together with the observation that derivatives of a function at most shrink their support, this allows to draw the same conclusion on the support of any derivative of f_ϵ .

To prove the second part, change coordinates from y to $z \doteq \frac{x-y}{\epsilon}$. Then

$$f_\epsilon(x) = \int_{\mathbb{R}^n} d^n z f(x - \epsilon z) \rho(z).$$

¹¹Here $\alpha = (\alpha_1, \dots, \alpha_n)$ is a multi-index and ∂^α stands for $\frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}$.

To verify uniform convergence as $\epsilon \rightarrow 0$,

$$|f_\epsilon(x) - f(x)| \leq \int_{\mathbb{R}^n} d^n z |f(x - \epsilon z) - f(x)| \rho(z),$$

where we used that ρ is a strictly positive and normalized function. In view of the support properties of ρ

$$|f(x - \epsilon z) - f(x)| \rho(z) \leq \sup\{|f(x + x') - f(x)|, |x'| \leq \epsilon\},$$

where x' plays the role of $-\epsilon z$. Since f is k -times differentiable with $k \geq 0$ and uniformly continuous, then the right hand side of the inequality tends to 0 as $\epsilon \rightarrow 0$, proving uniform convergence of f_ϵ . The same argument applies slavishly to any derivative of f_ϵ just by exchanging the derivative and the integration at the level of

$$(\partial^\alpha f_\epsilon)(x) = \int_{\mathbb{R}^n} d^n z (\partial^\alpha f)(x - \epsilon z) \rho(z),$$

which concludes the proof. $\square$

Remark 5. Notice that the above proposition is valid replacing $\mathbb{R}^n$ with any open subset $\Omega \subseteq \mathbb{R}^n$. In addition, observe that every $f \in C_0^\infty(\Omega)$ can be extended canonically to an element of $C_0^\infty(\mathbb{R}^n)$ by setting $f(x) = 0$ for all $x \in \mathbb{R}^n \setminus \Omega$. With a slight abuse of notation we indicate the extended function still with the symbol f .

As a last step, we need to endow $\mathcal{D}(\mathbb{R}^n) \equiv C_0^\infty(\mathbb{R}^n)$ with additional structures. In particular it is desirable to have a notion of topology or of convergence of sequences endowed to it. Without entering into many technical details, a reader should be warned of a key difference between smooth functions over $\mathbb{R}^n$ and those which are k -times differentiable, $k \in \mathbb{N}$ finite, since the latter can be made into a Banach space inducing on $C^k(\mathbb{R}^n)$ a norm structure, while $C^\infty(\mathbb{R}^n)$ do not enjoy this property. Unfortunately restricting our attention to those smooth functions with compact support does not change the outcome.

We start with a notion of convergence of sequences:

Definition 114. Let $\Omega \subseteq \mathbb{R}^n$ be an open subset of $\mathbb{R}^n$. A sequence $\{f_j\}_{j \in \mathbb{N}}$, $f_j \in \mathcal{D}(\Omega)$ for all j , is said to converge to $f \in \mathcal{D}(\Omega)$, i.e. $\lim_{j \rightarrow \infty} f_j = f$ if

1. there exists a compact set $K \subset \Omega$ such that $\text{supp}(f_j) \subseteq K$ for all $j \in \mathbb{N}$,
2. for every multi-index α , $\partial^\alpha f_j$ converges uniformly to $\partial^\alpha f$.

Starting from Definition 114 it is possible to endow $\mathcal{D}(\Omega)$ with a topology. Yet this is a very technically advanced construction, leading to the notion of a locally convex, topological, Fréchet vector space. We prefer omitting it, inviting an interested reader to check [6] or the appendix of [1].

6.2 Distributions

The idea underlying the concept of distribution originates from the following interpretation of (31): If we fix $\phi \in C^\infty(\mathbb{R})$, we can read $(\phi, \cdot)$ as a map from $\mathcal{D}(\mathbb{R}^n)$ to the real numbers. Proposition 112 together with Definition 114 guarantee us that such a map is well-behaved and in addition

$$\lim_{j \rightarrow \infty} (\phi, f_j) \rightarrow (\phi, f), \quad \text{whenever } \lim_{j \rightarrow \infty} f_j = f. \quad (33)$$

Together with this pros there is a notable con, namely a lack of generality, since the starting point is a smooth function. Distributions extend the range of applicability of the idea of mapping compactly supported functions into the real numbers, by noticing that, if we read $\mathcal{D}(\mathbb{R}^n)$ as a vector space, then $(\phi, \cdot)$ can be naturally interpreted as an element of its dual. On finite dimensional vector spaces, dual elements are characterized only by their linearity, yielding a space which is isomorphic to the initial vector space via a Riesz representation theorem. Already, at a level of Hilbert spaces we have seen that an additional requirement on the dual elements has to be imposed. Riesz theorem 17 guarantees that a (complex) Hilbert space $\mathcal{H}$ is isomorphic to the space of continuous linear functionals from $\mathcal{H}$ into $\mathbb{C}$. The following exercise is aimed at showing that the existence of a correspondence with (33):

Exercise 23. Let $\mathcal{H} = L^2(\mathbb{R}^n)$ and, for any but fixed $\psi \in \mathcal{H}$, let $F_\psi : \mathcal{D}(\mathbb{R}^n) \rightarrow \mathbb{R}$ be $F_\psi(f) = (\psi, f)_{L^2(\mathbb{R}^n)}$, where $(\cdot, \cdot)_{L^2(\mathbb{R}^n)}$ is the inner product on $L^2(\mathbb{R}^n)$. Prove that $(\cdot, \cdot)_{L^2(\mathbb{R}^n)}$ coincides with $(\cdot, \cdot)$ in (31) whenever $\psi \in L^2(\mathbb{R}^n) \cap C^\infty(\mathbb{R}^n)$. Show also that F_ψ is continuous in the sense of Definition 4 and that, whenever $\psi \in L^2(\mathbb{R}^n) \cap C^\infty(\mathbb{R}^n)$, continuity is equivalent to requiring (33).

The merit of this exercise is to show the possibility of extending (31) to more singular objects. Yet, this is still not exhaustive as we would like also to include in the set that we consider, objects like the Dirac delta which is certainly not square-integrable. This leads to the following definition, which is probably not the one expected from the above reasoning. Rest assured that you are neither crazy nor wrong. We follow a more historical path in defining formally distributions proving later the equivalence with a more natural characterization, especially in view of our previous comments.

Definition 115. Let $\Omega \subseteq \mathbb{R}^n$. A linear map $u : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ is called a **distribution** if, for every compact set $K \subset \Omega$, there exists $C_K \in [0, \infty)$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$|u(f)| \equiv |(u, f)| \leq C_K \sum_{|\alpha| \leq N} \sup |\partial^\alpha f|, \quad \forall f \in \mathcal{D}(\Omega), \text{ such that } \text{supp}(f) \subseteq K \quad (34)$$

The vector space of all distributions is called $\mathcal{D}'(\Omega)$.

Using Proposition 113 it is not difficult to realize that all the examples discussed above define elements of the space of distributions. Actually an almost immediate extension can be obtained replacing $C^\infty(\Omega)$ with $C^0(\Omega)$. Most notably also Proposition 112 remains valid. Before continuing our analysis, it is worth sparing a few lines for two additional, enlightening examples:

1. Let $L_{loc}^1(\Omega)$ be the space of *locally integrable* functions on $\Omega \subseteq \mathbb{R}^n$, that it is $\psi \in L_{loc}^1(\Omega)$ if ψ is measurable and $\int_K d^n x |\psi(x)| < \infty$ for all compact sets $K \subset \Omega$. Every $\psi \in L_{loc}^1(\Omega)$ identifies a distribution $u_\psi \in \mathcal{D}'(\Omega)$ defined as

$$u_\psi(f) \doteq \int_{\mathbb{R}^n} d^n x \psi(x) f(x). \quad \forall f \in \mathcal{D}(\Omega)$$

Notice that, on the left hand side, we are not writing $\psi(x)$ on purpose since ψ is actually an equivalence class of functions which coincide almost everywhere. The right hand side of the integral does not depend on the choice of the representative and, whenever we choose one, then we are identifying a concrete function. In this case it is meaningful to write $\psi(x)$. It also noteworthy that the map $L_{loc}^1(\Omega) \rightarrow \mathcal{D}'(\Omega)$ is injective since $u_\psi(f) = 0$ for all $f \in \mathcal{D}(\Omega)$ entails $\psi = 0$. This is a direct consequence of measure theory (for example via Proposition 113), a topic probably obscure to the reader. Hence we omit it.

2. Let $\Omega \subseteq \mathbb{R}^n$ and let $x_0 \in \Omega$. We call **delta distribution** (or *Dirac delta*) the linear functional

$$\delta_{x_0} : \mathcal{D}(\Omega) \rightarrow \mathbb{C}, \quad f \mapsto \delta_{x_0}(f) \doteq f(x_0).$$

It is not difficult to see that we have identified an element of $\mathcal{D}'(\Omega)$. As a matter of fact (34) can be split in two sub-cases. Let K be any compact subset of Ω . Suppose $x_0 \notin K$. Then (34) is automatically satisfied, since the left hand side vanishes once f is extended to 0 outside its support. Conversely, if $x_0 \in K$, (34) reads $|f(x_0)| \leq C_K \sum_{|\alpha| \leq N} \sup |\partial^\alpha f|$.

This is automatically verified by choosing $N = 0$ and $C_K = 1$. Notice that it is a typical Viking custom to write $\delta(x - x_0)$ in place of δ_{x_0} .

Having in mind quite a number of examples, we are falling short of one answer to our beloved readers, despite they might coincide with the empty set. To wit, we have not yet explained the relation between Definition 115 and (33). The next theorem solves this quandary.

Theorem 116. *Let $\Omega \subseteq \mathbb{R}^n$ and let $u : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ be a linear map. This is a distribution if and only if it is **sequentially continuous**, namely $\lim_{j \rightarrow \infty} u(f_j) = 0$ for all sequences of $f_j \in \mathcal{D}(\Omega)$ such that $\lim_{j \rightarrow \infty} f_j = 0$ in the sense of Definition 114.*

Proof. $\implies$ Suppose $u \in \mathcal{D}'(\Omega)$. Then sequential continuity descends from Definition 114 and (34). The latter entails that, if f_j is a sequence of elements of $\mathcal{D}(\Omega)$ and $f_j \rightarrow 0$ as $j \rightarrow \infty$, then the right hand side of (34) tends to 0 as $j \rightarrow \infty$. Consequently the same holds true for the left hand side, that is u is sequentially continuous.

$\impliedby$ Let u be sequentially continuous. By contradiction, suppose that $u \notin \mathcal{D}'(\Omega)$. Hence (34) does not hold true and there exists a compact set K such that, for all $N \in \mathbb{N} \cup \{0\}$, the ratio

$\frac{|(u,f)|}{\sum_{|\alpha| \leq N} \sup(|\partial^\alpha f|)}$ is unbounded. In other words, for each $N \in \mathbb{N} \cup \{0\}$, we can find $f_N \in \mathcal{D}(\Omega)$ with $\text{supp}(f_N) \subseteq K$ such that

$$|(u, f_N)| \geq N \sum_{|\alpha| \leq N} \sup(|\partial^\alpha f_N|). \quad (35)$$

Consider the auxiliary function

$$g_N(x) \doteq \frac{f_N(x)}{N \sum_{|\alpha| \leq N} \sup(|\partial^\alpha f_N|)}.$$

Per construction $\text{supp}(g_N) \subseteq K$ and $g_N \in \mathcal{D}(\Omega)$. In addition $|\partial^\beta g_N| \leq N^{-1}$, where β is a multi-index such that $|\beta| \leq N$. This entails that $\lim_{N \rightarrow \infty} g_N(x) = 0$, but (35) yields $|(u, g_N)| \geq 1$ which contradicts sequential continuity. $\square$

Having under control the notion of distribution essentially via the test-functions, it is natural to wonder whether one can introduce on $\mathcal{D}'(\mathbb{R}^n)$ a criterion of convergence similar to Definition 114. In the theory of distributions the trick is always to remember that $\mathcal{D}'(\mathbb{R}^n)$ is constructed from $\mathcal{D}(\mathbb{R}^n)$ per duality.

Definition 117. Let $\Omega \subset \mathbb{R}^n$ and let $\{u_j\}_{j \in \mathbb{N}}$ be a sequence of distributions on Ω . We say that the sequence **converges** to $u \in \mathcal{D}'(\Omega)$, namely $\lim_{j \rightarrow \infty} u_j = u$ if

$$\lim_{j \rightarrow \infty} (u_j, f) = (u, f) \quad \forall f \in \mathcal{D}(\Omega).$$

It is rather simple to infer from this last definition that the convergence of a sequence of distributions is rather different from many other notions of convergence of functions, e.g. pointwise convergence even if each u_j is for example a smooth function. This exercise will strengthen your grasp of this idea

Exercise 24. Consider the sequence of functions $u_j \in C^\infty(\mathbb{R})$ defined as $u_j(x) = e^{ijx}$. Prove that the sequence converges pointwisely if and only if $x = 2k\pi$ with k integer, whereas it is always convergent in $\mathcal{D}'(\mathbb{R})$.

At the same time, there are other mathematical tools which allow to control the convergence of special sequences of distributions. For example a direct application of the Lebesgue theorem of dominated convergence ensures the following:

Theorem 118. Let $\Omega \subset \mathbb{R}^n$ and let $\{u_j\}_{j \in \mathbb{N}}$ be a sequence of elements in $L^1_{loc}(\Omega)$ converging to a function u . If there exists $g \in L^1_{loc}(\Omega)$ such that $|u_j| \leq g$ for all $j \in \mathbb{N}$, then $u \in L^1_{loc}(\Omega)$ and $\lim_{j \rightarrow \infty} u_j = u$ also in $\mathcal{D}'(\Omega)$.

A second criterion is the following:

Theorem 119. *Let $\Omega \subset \mathbb{R}^n$ and let $\{u_j\}_{j \in \mathbb{N}}$ be a sequence of elements in $\mathcal{D}'(\Omega)$, such that, for all $f \in \mathcal{D}(\Omega)$, there exists and it is finite $\lim_{j \rightarrow \infty} (u_j, f)$. Then the linear functional $u : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ defined so that*

$$u(f) \doteq \lim_{j \rightarrow \infty} (u_j, f), \quad \forall f \in \mathcal{D}(\Omega)$$

identifies an element of $\mathcal{D}'(\Omega)$.

The proof is rather technical, but the brave and wise student can read it in [2, Th 2.1.8]. Observe in addition that Theorem 119 guarantees that $\mathcal{D}'(\Omega)$ is complete.

6.2.1 Localization of Distributions

One of the most important characteristics of the theory of distributions is that it shows common sense. Things that you would expect a distribution to do, are done; properties that you would expect a distribution to fulfil are fulfilled. Let us now focus on a key example of this very philosophical statement.

Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \Omega$. From the definition of support of a function it is immediate to realize the existence of a canonical embedding $\iota_{\Omega' \Omega} : \mathcal{D}(\Omega') \rightarrow \mathcal{D}(\Omega)$ which is obtained by extending any smooth and compactly supported function in Ω' to one on Ω by defining it to be 0 on all points $x \in \Omega \setminus \Omega'$. The embedding map is usually left-understood. While test functions are extended from a smaller to a bigger set, distributions behave dually, hence they can be restricted from a larger to a smaller set. Most notably, if $u \in \mathcal{D}'(\Omega)$, we can set $v \doteq u|_{\Omega'} \in \mathcal{D}'(\Omega')$ via the defining relation

$$v(f) \doteq u(f) \quad \forall f \in \mathcal{D}(\Omega').$$

It is immediate to show that the restriction preserves sequential continuity. At the same time it is not difficult to realize that the restriction is not an injective map. As a matter of fact we say that $u, u' \in \mathcal{D}'(\Omega)$ *coincide* on Ω' if and only if

$$u(f) = u'(f). \quad \forall f \in \mathcal{D}(\Omega')$$

Remark 6. *It is easy to construct examples using distributions. As a matter of fact, consider $\Omega' \subset \Omega$ and consider any $h \in \mathcal{D}(\Omega)$. Since $\mathcal{D}(\Omega) \subset C^\infty(\Omega) \subset \mathcal{D}'(\Omega)$, h identifies a distribution $u_h \in \mathcal{D}'(\Omega)$. Consider now any point $x_0 \in \Omega \setminus \Omega'$. Since the space of distributions is linear (it is a vector space), we can construct $u_{h, x_0} \doteq u_h + \delta_{x_0}$ where δ_{x_0} is the Dirac- δ centred at x_0 . Since $\delta_{x_0}(f) = 0$ for all $f \in \mathcal{D}(\Omega')$, it is clear that u_h and u_{h, x_0} coincide on Ω' .*

An important question, especially if one wants to generalize distributions to arenas more complicated than $\mathbb{R}^n$, is whether it is possible to reconstruct completely a distribution from a sufficient number of its restrictions. This is known as the **localization** problem. To solve it, we need an important tool, known as *partition of unity*.

Proposition 120. *Let K be a compact subset of $\mathbb{R}^n$ and let $\{U_i\}_{i=1, \dots, m < \infty}$ be a collection of open subsets such that $K \subset \bigcup_{i=1}^m U_i$. Then there exists $\psi_i \in \mathcal{D}(U_i)$, $i = 1, \dots, m$ such that*

$$a) \ 0 \leq \psi_i \leq 1, \quad \forall i = 1, \dots, m$$

$$b) \ \sum_{i=1}^m \psi_i \leq 1 \text{ and } \sum_{i=1}^m \psi_i = 1 \text{ on } K.$$

The collection $\{\psi_i\}_{i=1, \dots, m}$ is called a **partition of unity** (of K subordinated to the cover $\{U_i\}$).

Proof. Consider first the case $m = 1$ and recall, that, for every $x \in \mathbb{R}^n$ and for every open subset $\Omega \subseteq \mathbb{R}^n$, there is a notion of distance

$$d(x, \Omega) \doteq \inf_{y \in \Omega} \{|x - y|\},$$

which can be read as $d_\Omega \in C^0(\mathbb{R}^n)$, where $d_\Omega(x) = d(x, \Omega)$. Let now $\delta > 0$ and let $K_\delta \doteq \{x \in \mathbb{R}^n \mid d(x, K) < \delta\}$ where K is taken as per hypothesis. In other words we have constructed a δ -neighbourhood of K . Let now $\epsilon > 0$ so that

$$4\epsilon < \inf\{|x - y| \mid x \in K \text{ and } y \in \mathbb{R}^n \setminus U_1, \},$$

where X_1 is taken as per hypothesis. Set

$$f(x) \doteq \begin{cases} 1 - \frac{d(x, K_{2\epsilon})}{\epsilon} & x \in K_{3\epsilon} \\ 0 & x \in \mathbb{R}^n \setminus K_{3\epsilon} \end{cases}.$$

Plugging this function in (32) constructed in Proposition 6, we obtain that the sought $\psi(x)$ is nothing but $f_\epsilon(x)$.

Suppose now that $m > 1$. For every point $x \in K$, there exists $r = r(x) > 0$ such that the closure of the Euclidean ball $B_r(x)$ is contained in one U_i . In this way we obtain a collection of open balls covering K and containing a finite subcover since K is compact per hypothesis. For every $i = 1, \dots, m$, consider the union of all closed balls contained in U_i . We call this set K_i and it is compact. In turn K is contained in the union of all K_i . Since we have already addressed the case $m = 1$, we can find functions $\phi_i \in C_0^\infty(U_i)$, $i = 1, \dots, m$ fulfilling all the requests of the statement of the proposition for K_i in place of K . To conclude define

$$\psi_1 = \phi_1, \quad \psi_2 = \phi_2(1 - \phi_1), \quad \dots \quad \psi_m = \phi_m \prod_{i=1}^{m-1} (1 - \phi_i).$$

Per construction, for all $i = 1, \dots, m$, $\text{supp}(\psi_i) \subset U_i$, $0 \leq \psi \leq 1$ and $\sum_{i=1}^m \psi_i(x) = 1$, which are the sought requirements. $\square$

Remark 7. Notice that the existence of a partition of unity is often exploited in physics. As a matter of fact Proposition 120 entails that, if K is a compact subset of $\Omega \subseteq \mathbb{R}^n$, we can always find $\Lambda \in \mathcal{D}(\mathbb{R}^n)$ such that $\text{supp}(\Lambda) \subset \Omega$, $0 \leq \Lambda \leq 1$ and $\Lambda = 1$ on K . Such Λ is called a **cut-off** (or bump) function.

Remark 8. Observe that, since we are interested in a compact set K , by Heine-Borel theorem, we know that it is a closed and bounded set. Hence we need only a finite number of open sets to cover it. More generally a partition of unity can be found also for sets requiring in turn an infinite, but countable number of open sets to cover them. This generalization calls for constraints on the topology of the underlying space, which are of course automatically satisfied by $\mathbb{R}^n$. In other words we are lead to introducing the concept of **second countable**. Without this assumption, integration beyond $\mathbb{R}^n$ would not be possible.

Definition 121. Let $\Omega \subset \mathbb{R}^n$ and let $u \in \mathcal{D}'(\Omega)$. We call **support** of u the set

$$\text{supp}(u) = \Omega \setminus \{x \in \Omega \mid \exists \mathcal{O} \subset \mathbb{R}^n, \text{ such that } x \in \mathcal{O} \text{ and } u|_{\mathcal{O}} = 0\}.$$

Corollary 122. Let $\Omega \subseteq \mathbb{R}^n$, let $u \in \mathcal{D}'(\Omega)$ and let $f \in \mathcal{D}(\Omega)$. Then $u(f) = 0$ if $\text{supp}(u) \cap \text{supp}(f) = \emptyset$.

Proof. Consider an open cover of K and a finite collection of opens subsets $\{U_i\}_{i=1, \dots, m < \infty}$ such that $K \subset \bigcup_{i=1}^m U_i$. Choose a partition of unity $\{\psi_i\}_{i=1, \dots, m}$ subordinated to this cover as per Proposition 120. It turns out that

$$(u, f) = (u, \sum_{i=1}^m \psi_i f) = \sum_{i=1}^m (u, \psi_i f),$$

where, in the last step, we used that m is finite. Since both K and $\text{supp}(u)$ are compact, without loss of generality and as a consequence of $\mathbb{R}^n$ being Hausdorff, we can choose U_i in such a way that $U_i \cap \text{supp}(u) = \emptyset$ for all $i = 1, \dots, m$. Consequently, since $\psi_i f \in C_0^\infty(U_i)$, $i = 1, \dots, m$ and since u vanishes thereon on account of Definition 121, we can conclude that

$$0 = \sum_{i=1}^m (u, \psi_i f) = (u, f),$$

which is the sought result. □

Observe that, as a consequence of Corollary 122, if $u \in \mathcal{D}'(\Omega)$ and if $\{U_i\}_{i=1, \dots, m < \infty}$ is a collection of open subsets covering Ω , then, if $u|_{U_i} = 0$ for all $i = 1, \dots, m$, $u = 0$.

Theorem 123. Let $\Omega \subset \mathbb{R}^n$ and let $\{\Omega_i\}_{i \in I}$, I being an index set, be an open cover of Ω . For any $i \in I$, let $u_i \in \mathcal{D}'(\Omega_i)$ be such that

$$u_i = u_j \text{ on } \Omega_i \cap \Omega_j \neq \emptyset.$$

Then, $\exists! u \in \mathcal{D}'(\Omega)$ such that $u|_{\Omega_i} = u_i$.

Proof. Uniqueness is a direct consequence of Corollary 122. Hence we need to show only the existence. Let $f \in C_0^\infty(\Omega)$ and let Ω_j be a finite collection of open subsets covering $\text{supp}(f)$, that

is $j = 1, \dots, m \in I$. By Theorem 120 we can find a partition of unity $\{\psi_j\}_{j=1, \dots, m}$ subordinated to this cover and we can define $u \in \mathcal{D}'(\Omega)$ via

$$(u, f) \doteq \sum_{j=1}^m (u_j, f\psi_j).$$

For the definition to be well-posed, u must not depend on the choice of the partition of unity or of the open cover. Consider a second cover Ω'_k , $k = 1, \dots, l < \infty$ and a partition of unity $\{\phi_k\}_{k=1, \dots, l}$ subordinated to it. Then

$$(u, f) = \sum_{j=1}^m (u_j, f\psi_j) = \sum_{j=1}^m \sum_{k=1}^l (u_j, \psi_j \phi_k f) = \sum_{j=1}^m \sum_{k=1}^l (u_k, \psi_j \phi_k f) = \sum_{k=1}^l (u_k, \phi_k f),$$

where, in the second and in the forth equality we used that $\sum_{k=1}^l \phi_k = \sum_{j=1}^m \psi_j = 1$, in the third, that the sums can be exchanged running both on finite indices and that, per assumption, $u_j = u_k$ whenever $\Omega_j \cap \Omega'_k \neq \emptyset$. We have shown that u is a well-defined linear map from $\mathcal{D}(\Omega)$ to $\mathbb{C}$. It remains to be seen that u is sequentially continuous... a very nice exercise. $\square$

Exercise 25. Complete the proof of the previous theorem by showing that u is a distribution. (Hint: Do not use sequential continuity, but rather the inequality in Definition 115.)

7 Elementary Operations on Distributions

In the previous section we have defined the concept of distribution and we have seen that their action is a natural generalization of (31) which is built starting from a smooth function over $\mathbb{R}^n$. As a matter of fact we have seen that, for every open subset $\Omega \subseteq \mathbb{R}^n$, $C^\infty(\Omega) \subset \mathcal{D}'(\Omega)$. Goal of this section instead is to show that there exist well-defined, useful operations on $\mathcal{D}'(\Omega)$ which have their roots in more familiar counterparts defined for smooth functions. For this reason we discuss these operations first of all at a level of $C^\infty(\Omega)$.

1. Let $\phi \in C^\infty(\Omega)$ and let $u_\phi : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ the distribution defined via (31). For the sake of simplicity we also assume that $n = 1$ and that $\Omega = \mathbb{R}$. The first operation which is well defined on u_ϕ is that of derivation. Consider $\phi'(x) \doteq \frac{d\phi}{dx} \in C^\infty(\mathbb{R})$. This defines a distribution $u_{\phi'}$ such that, for all $f \in \mathcal{D}(\mathbb{R})$

$$(u_{\phi'}, f) = \int_{\mathbb{R}} dx \frac{d\phi}{dx} f(x) = \phi f|_{-\infty}^{\infty} - \int_{\mathbb{R}} dx \phi(x) \frac{df}{dx} = - \int_{\mathbb{R}} dx \phi(x) \frac{df}{dx}, \quad (36)$$

where we used that, being f compactly supported, also $\phi f \in \mathcal{D}(\mathbb{R})$ and $\lim_{x \rightarrow \pm\infty} \phi f = 0$. If we observe that the derivative shrinks at most the support of a function, we have that $\frac{df}{dx} \in \mathcal{D}(\mathbb{R})$.

Thus the distribution $u_{\phi'}$ can be equivalently defined in terms of u_{ϕ} via the defining relation

$$(u_{\phi'}, f) = - \left(u_{\phi}, \frac{df}{dx} \right). \quad \forall f \in \mathcal{D}(\mathbb{R})$$

It is not hard to realize that this last relation relies only on the fact that $\frac{df}{dx} \in \mathcal{D}(\mathbb{R})$ and, thus, it can be naturally taken as the defining identity for the left hand side whenever we consider a generic distribution and not one built out of $\phi \in C^{\infty}(\mathbb{R})$.

2. Let $\phi \in C^{\infty}(\Omega)$ and let $u_{\phi} : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ the distribution defined via (31). For the sake of simplicity we assume again that $n = 1$ and that $\Omega = \mathbb{R}$. The second operation which is well defined on u_{ϕ} is that of multiplication by a second arbitrary but fixed function $h \in C^{\infty}(\mathbb{R})$. Consider now $h\phi \in C^{\infty}(\mathbb{R})$ defined via pointwise multiplication. It identifies a distribution $u_{h\phi}$ such that, for all $f \in \mathcal{D}(\mathbb{R})$,

$$u_{h\phi}(f) = \int_{\mathbb{R}} dx (h\phi)(x) f(x) = \int_{\mathbb{R}} dx \phi(x) (hf)(x) = u_{\phi}(hf), \quad (37)$$

where we used that, being f compactly supported, also $hf \in \mathcal{D}(\mathbb{R})$. In other words the distribution $u_{h\phi}$ can be equivalently defined in terms of u_{ϕ} via the defining relation

$$u_{h\phi}(f) = u_{\phi}(hf), \quad \forall f \in \mathcal{D}(\mathbb{R}).$$

Also in this case, it is not hard to realize that this last relation relies only on the fact that $hf \in \mathcal{D}(\mathbb{R})$ and, thus, it can be naturally taken as the defining identity for the left hand side whenever we consider a generic distribution and not one built out of $\phi \in C^{\infty}(\mathbb{R})$. In this more general scenario, the left hand side will be written as hu , namely the multiplication of $u \in \mathcal{D}'(\mathbb{R})$ by $h \in C^{\infty}(\mathbb{R})$.

7.1 The Derivative of a Distribution

Consider now a generic scenario. In view of the previous discussion we define:

Definition 124. Let $\Omega \subseteq \mathbb{R}^n$ and let $u \in \mathcal{D}'(\Omega)$. We call **distributional derivative** of u along the (Euclidean) x_i -direction, $\frac{\partial u}{\partial x_i} \equiv \partial_i u \in \mathcal{D}'(\Omega)$ defined as

$$(\partial_i u, f) = -(u, \partial_i f). \quad \forall f \in \mathcal{D}(\Omega)$$

Equivalently, we define higher order derivatives as

$$(\partial^{\alpha} u, f) = (-1)^{|\alpha|} (u, \partial^{\alpha} f), \quad \forall f \in \mathcal{D}(\Omega)$$

where α is arbitrary multi-index.

Proposition 125. Let $\{u_k\}_{k \in \mathbb{N}}$ be a sequence of distributions of $\mathcal{D}'(\Omega)$ such that $\lim_{k \rightarrow \infty} u_k = u \in \mathcal{D}'(\Omega)$. Then, for every multi-index α , $\lim_{k \rightarrow \infty} \partial^{\alpha} u_k = \partial^{\alpha} u \in \mathcal{D}'(\Omega)$.

Proof. Let $f \in \mathcal{D}(\Omega)$. Then, by Definition 124,

$$(\partial^\alpha u_k, f) = (-1)^{|\alpha|}(u_k, \partial^\alpha f) \longrightarrow (-1)^{|\alpha|}(u, \partial^\alpha f) = (\partial^\alpha u, f),$$

where we have taken the limit as $k \rightarrow \infty$, since the sequence $\{u_k\}_{k \in \mathbb{N}}$ is per hypothesis convergent to $u \in \mathcal{D}'(\Omega)$. In the last step we use once more the definition of distributional derivative. $\square$

Remark 9. As a consequence of Definition 124, one can observe that, if $f, h \in C^0(\Omega)$ and the distributional equation $\partial_i f = h$ holds true, then $\partial_i f = \frac{\partial f}{\partial x_i}$ exists in the usual sense of a derivative of a function.

Having understood the definition of derivative of a distribution and its main properties, it is interesting to see two notable examples:

Example 126. Let us consider the Heaviside Θ -function defined on $\mathbb{R}$ as

$$\Theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Although it is not a continuous function, $\Theta \in L^1_{loc}(\mathbb{R}) \subseteq \mathcal{D}'(\mathbb{R})$. Let us compute its derivative using Definition 124. Consider any $f \in \mathcal{D}(\mathbb{R})$ and let $\Theta'(x) = \frac{d\Theta}{dx}$. Then

$$(\Theta'(x), f) = - \left(\Theta, \frac{df}{dx} \right) = - \int_0^\infty \frac{df}{dx} = -f|_0^\infty = f(0) = (\delta, f),$$

where we used in the integral that the Heaviside function is 0 on $(-\infty, 0]$ and that, being compactly supported, f vanishes at infinity. In other words we have proven that

$$\Theta' = \delta.$$

A simple variation of the above reasoning shows that $\Theta'(x - a) = \delta_a$ for all $a \in \mathbb{R}$.

The second example that we consider originates from the observation that the function x^{-1} on $\mathbb{R}$ is not an element of $L^1_{loc}(\mathbb{R})$ on account of the pathological divergence at 0. More generally, even to make sense of this function as a distribution is more complicated. A wise reader should learn from [2, §3.2], but here we will present a limited analysis following [1]:

Example 127. In order to define the distribution x^{-1} we start by observing that this is function is $L^1_{loc}(I)$ where $I \subset \mathbb{R}$ is any open set which does not contain $x = 0$. Furthermore, by elementary analysis, for every $f \in \mathcal{D}(I)$, it turns out that

$$(x^{-1}, f) = \left(\frac{d \ln(|x|)}{dx}, f \right),$$

or in other words x^{-1} is the distributional derivative of $\ln(|x|)$ which is actually an element of $L^1_{loc}(\mathbb{R}) \subset \mathcal{D}'(\mathbb{R})$. Notice that the above identity does not hold true for f such that $0 \in \text{supp}(f)$. To understand what happens at the origin, we observe that, if we allow for any $f \in \mathcal{D}(\mathbb{R})$

$$\begin{aligned} \left(\frac{d \ln(|x|)}{dx}, f \right) &= - \left(\ln |x|, \frac{df}{dx} \right) = - \lim_{\epsilon \rightarrow 0^+} \left(\int_{-\infty}^{-\epsilon} dx \ln(-x) \frac{df}{dx} + \int_{\epsilon}^{\infty} dx \ln(x) \frac{df}{dx} \right) = \\ &= \lim_{\epsilon \rightarrow 0^+} \left((f(\epsilon) - f(-\epsilon)) \ln \epsilon + \int_{-\infty}^{-\epsilon} dx \frac{f(x)}{x} + \int_{\epsilon}^{\infty} dx \frac{f(x)}{x} \right). \end{aligned}$$

Since $f \in \mathcal{D}(\mathbb{R})$, by taking a Taylor expansion around 0, we can realize that the term proportion to the logarithm in the above expression is $O(\epsilon \ln \epsilon)$, hence it tends to 0 as $\epsilon \rightarrow 0^+$. The remaining integrals are finite and well-defined whenever $0 \notin \text{supp}(f)$. In order to compute the limit when 0 lies in the support, we adapt Proposition 125 to this scenario. Consider a complex variable $z = x + iy$ with $y > 0$ and define

$$\ln(z) = \ln |z| + i \arg(z), \quad 0 < \arg(z) < \pi$$

so that the map $x \mapsto \ln(z)$ is a distribution which depends on a positive parameter y . For $x \neq 0$, the limit for $y \rightarrow 0^+$ exists and it is

$$\lim_{y \rightarrow 0^+} \ln(z) = \ln(|x|) + i\pi(1 - \Theta(x)), \quad x \neq 0$$

It is not difficult to see that we can now apply Theorem 118 concluding that this last identity holds in $\mathcal{D}'(\mathbb{R})$. Differentiating we obtain

$$\frac{1}{x + i0^+} \doteq \lim_{y \rightarrow 0^+} \frac{1}{x + iy} = \text{P.V.} \left(\frac{1}{x} \right) - i\pi\delta,$$

where $\text{P.V.} \left(\frac{1}{x} \right)$ stand for the *principal value* of the distribution x^{-1} , namely, for any $f \in \mathcal{D}(\mathbb{R})$,

$$\text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x} \doteq \lim_{\epsilon \rightarrow 0^+} \left(\int_{-\infty}^{-\epsilon} dx \frac{f(x)}{x} + \int_{\epsilon}^{\infty} dx \frac{f(x)}{x} \right).$$

7.1.1 The Primitive of a Distribution

An additional advantage of making sense of derivatives at a distributional level arises from the possibility of defining partial differential equations on $\mathcal{D}'(\Omega)$, $\Omega \subseteq \mathbb{R}^n$. We review here the first and the simplest example. We consider $\Omega = \mathbb{R}$, and for a given $v \in \mathcal{D}'(\mathbb{R})$ we look for distributional solutions to the equation

$$u' = v, \quad u' \equiv \frac{du}{dx}.$$

Any solution identifies a so-called **primitive** of v . Let us construct such u . Using Definition 124, we can write for any $f \in \mathcal{D}(\mathbb{R})$ that

$$(u', f) = -(u, f') = (v, f),$$

where the prime symbol will denote henceforth the derivative with respect to x . Since v is assigned, notice that this identity defines unambiguously u on the following class of functions

$$X \doteq \{\phi \in \mathcal{D}(\mathbb{R}) \mid \exists f \in \mathcal{D}(\mathbb{R}) \text{ and } \phi = f'\}.$$

We propose an enlightening exercise:

Exercise 26. Show that X is isomorphic to the collection of all $\phi \in \mathcal{D}(\mathbb{R})$ such that $\int_{\mathbb{R}} dx \phi(x) = 0$. Show also that this last integral can be read as $(1, \phi) = 0$, where $1 \in C^\infty(\mathbb{R}) \subset \mathcal{D}'(\mathbb{R})$ is the constant function.

Resuming our investigation, let us choose $\phi_0(x) \in \mathcal{D}(\mathbb{R})$ such that $\int_{\mathbb{R}} dx \phi_0(x) = 1$. For any $f \in \mathcal{D}(\mathbb{R})$

$$f(x) = f(x) - (1, f)\phi_0(x) + (1, f)\phi_0(x) = \tilde{\phi}(x) + (1, f)\phi_0(x).$$

By direct inspection $\tilde{\phi} \in X$. Hence, in order to define on the whole $\mathcal{D}(\mathbb{R})$ the distribution u , we need to define it beyond X only on the one-dimensional subspace generated by ϕ_0 . In other words we have written

$$\mathcal{D}(\mathbb{R}) = X \oplus \text{span}_{\mathbb{C}}(\phi_0).$$

Let $\pi_1 : \mathcal{D}(\mathbb{R}) \rightarrow X$ be the projection on the first component and let

$$\mu(f) = \int_x^\infty dy (\pi_1(f))(y).$$

It turns out that, for every $f \in \mathcal{D}(\mathbb{R})$, we have written

$$(u, f) = (v, \mu(f)) + (C, f), \tag{38}$$

where $C \doteq (u, \phi_0)$ is an arbitrary constant function. To conclude we need to prove one last bit of information:

Proposition 128. *Let $v \in \mathcal{D}'(\mathbb{R})$ and let u be as in (38) where C is an arbitrary constant function. Then u is a primitive of v and all primitives have this form.*

Proof. As a first step we need only to prove that $u \in \mathcal{D}'(\mathbb{R})$ since, in this case, it is a primitive of v per construction. Using Definition 115, it suffices to show that (34) holds true on every interval of the form $[-a, a]$, $a > 0$. Let $f \in C_0^\infty([-a, a])$. Per construction,

$$\begin{aligned} \text{supp}(\mu(f)) &\subseteq [-a, a] \cup \text{supp}(\phi_0), \\ \text{sup}(|\mu(f)|) &\leq 2a (1 + \text{sup} |\phi_0|) \text{sup} |f|, \\ \text{sup}(|(\mu(f))^{(k)}|) &\leq \text{sup}(|f^{(k-1)}|) + 2a \text{sup} |\phi_0^{(k-1)}| \text{sup} |f|, \quad k \geq 1, \end{aligned}$$

where the superscript (k) indicates the number of derivative with respect to x . In other words (34) for u can be derived from the same inequality for v . Hence $u \in \mathcal{D}'(\mathbb{R})$.

Suppose now that $u_1, u_2 \in \mathcal{D}'(\mathbb{R})$ are two primitives of v . Hence their difference $\tilde{u} = u_1 - u_2$ satisfies $\tilde{u}' = 0$, the prime standing for the derivative with respect to x . Using (38) in this case, we obtain that $\tilde{u} = C$, where C is a constant of the form $(\tilde{u}, \phi_0)$, $\phi_0 \in \mathcal{D}(\mathbb{R})$ normalized so that its integral over $\mathbb{R}$ is equal to 1. This concludes the proof. $\square$

The whole construction can be slavishly adapted to any open subset of $\mathbb{R}$.

7.2 Multiplication by a Smooth Function

In view of the discussion at the beginning of the section, we can define a second notable operation on distributions. Let $\Omega \subseteq \mathbb{R}^n$ and let $\phi \in C^\infty(\Omega)$. Observe that, for every $f \in \mathcal{D}(\Omega)$, $\phi f \in \mathcal{D}(\Omega)$ and, in addition there exists a sequence of constants $\{C_N\}_{N \in \mathbb{N}}$ such that

$$\sum_{|\alpha| \leq N} \sup |\partial^\alpha(\phi f)| \leq C_N \sum_{|\alpha| \leq N} \sup |\partial^\alpha f|.$$

In view of Definition 115 and of (34) in particular, the following definition is meaningful:

Definition 129. Let $u \in \mathcal{D}'(\Omega)$ and let $\phi \in C^\infty(\Omega)$. Then we call **product** between ϕ and u , the distribution $\phi u \in \mathcal{D}'(\Omega)$ defined via

$$(\phi u, f) \doteq (u, \phi f), \quad \forall f \in \mathcal{D}(\Omega).$$

Example 130. Consider $\Omega \subseteq \mathbb{R}^n$ such that $0 \in \Omega$ and let $u = \delta \in \mathcal{D}'(\Omega)$. For any $\phi \in C^\infty(\Omega)$, we observe that

$$(\phi \delta, f) = (\delta, \phi f) = \phi(0)f(0) = (\delta, \phi(0)f) = (\phi(0)\delta, f), \quad \forall f \in \mathcal{D}(\Omega)$$

where $\phi(0)$ plays the role of a constant besides that of the value of a function ϕ at $x = 0$. In other words,

$$\phi \delta = \phi(0)\delta. \quad \forall \phi \in C^\infty(\Omega)$$

Since, in Definition 129, ϕ represents a distribution of its own, it is natural to wonder whether this operation of multiplication and the one of derivation are compatible, namely if the Leibniz rule still holds true. This is the case:

Theorem 131. Let $\Omega \subseteq \mathbb{R}^n$, let $u \in \mathcal{D}'(\Omega)$, $\phi \in C^\infty(\Omega)$ and let α be a multi-index. Then

$$\partial^\alpha(\phi u) = \sum_{\substack{\beta, \gamma \\ \beta + \gamma = \alpha}} \frac{\alpha!}{\beta! \gamma!} \partial^\beta \phi \partial^\gamma u,$$

where the factorial of a multi-index $\alpha = (\alpha_1, \dots, \alpha_m)$ is defined as $\alpha! = \prod_{i=1}^m \alpha_i!$.

Proof. We proceed by induction. Since for $|\alpha| = 0$ there is nothing to prove, we start with $|\alpha| = 1$. In this case, for any $f \in \mathcal{D}(\Omega)$, we have

$$(\partial_i(\phi u), f) = -(\phi u, \partial_i f) = -(u, \phi(\partial_i f)).$$

Using the Leibnitz rule on the test function, we obtain

$$(u, \phi(\partial_i f)) = (u, \partial_i(\phi f) - f \partial_i \phi) = -(\partial_i u, \phi f) - (u, f \partial_i \phi).$$

Putting together the last two chains of identities we obtain the wanted result.

From this point onward one applies the induction step, namely assuming that the sought identity is valid up to an order $|\alpha| - 1$ and proving that it must be valid also for $|\alpha|$. This is a tedious albeit straightforward calculation which we omit. $\square$

This section is short, but it is dense of significance for at least three reasons which I would love (really I would!... just imagine how many more interesting questions I could ask at the exams!!!) to discuss in detail, but I do not have time.... as simple as that!

1. Since we have a notion of derivative on $\mathcal{D}'(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ and we have also a notion of multiplication by a smooth function, it is immediate to realize that one can make sense at the level of distributions of *partial differential operators*:

$$P(x, \partial) = \sum_{\alpha \in I} a_\alpha(x) \partial^\alpha : \mathcal{D}'(\Omega) \rightarrow \mathcal{D}'(\Omega),$$

where I is a finite collection of multi-indices and where, for each α , $a_\alpha \in C^\infty(\Omega)$. Finding solutions to $Pu = 0$, $u \in \mathcal{D}'(\Omega)$ is a very complicated, albeit ultimately very rewarding affair, which leads to advanced topics such as symbol calculus, pseudo-differential operators and Fourier integral operators. Unfortunately we do not have time to discuss them here, first of all because we miss one last key ingredient, the notion of Fourier transform. This we will learn!

2. We have learned that to make sense of the derivative of a distribution, we need to move the derivative operator to the level of test-functions, paying at most the price of a minus sign. In general this trick, which is also called a *duality (or transpose) relation*, allows to define at the level of distribution every operator mapping $\mathcal{D}(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ into itself, provided that it does not spoil sequential continuity.
3. We have learned how to multiply two distributions whenever one of them is a smooth function. This is the Pandora box. In general the product of two arbitrary distributions is ill-defined, the canonical example being δ^2 which is utterly meaningless. Yet, there are several instances, especially in quantum theories such as quantum field theory, where it is necessary to multiply two distributions, none of which is smooth. There exist sufficient criteria which allow us to discern whether such product is well-defined or not. This is a whole branch of functional analysis developed in the sixties. The brave one should learn and there is only one way to do it: Read and learn [2, Chapt. 8].

8 Distributions of Compact Support

As we have explained in the previous sections, the concept itself of a distribution in $\mathcal{D}'(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ is born from the interpretation of (31) as yielding for every $\phi \in C^\infty(\mathbb{R}^n)$ a linear functional $u_\phi : C_0^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$. It is not difficult to see that there is no reason why we could not exchange in this reasoning the role played by ϕ and by $f \in C_0^\infty(\mathbb{R}^n)$, namely (31) also yields for any such function a linear functional

$$u_f : C^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad f \in C_0^\infty(\mathbb{R}^n).$$

Generalizing this complementary viewpoint leads to a slightly different notion of distribution.

Definition 132. Let $\Omega \subseteq \mathbb{R}^n$. A linear functional $u : C^\infty(\Omega) \rightarrow \mathbb{C}$ is called continuous if there exist a compact set K , a constant $C_K \geq 0$ and $N \in \mathbb{N}$ such that

$$|(u, \phi)| \leq C_K \sum_{|\alpha| \leq N} \sup_{x \in K} |\partial^\alpha \phi|, \quad \forall \phi \in C^\infty(\Omega) \quad (39)$$

where α is a multi-index. These functionals are called **distributions of compact support** and they are indicated as $\mathcal{E}'(\Omega)$.

As for $\mathcal{D}'(\Omega)$, it is not always convenient to check directly (39). Hence an alternative equivalent characterization turns out to be useful:

Definition 133. Let $\Omega \subseteq \mathbb{R}^n$. A sequence $\{\phi_k\}_{k \in \mathbb{N}}$ of smooth functions on Ω is said to converge to 0 in $C^\infty(\Omega)$ if, for every multi-index α , the sequence $\{\partial^\alpha \phi_k\}_{k \in \mathbb{N}}$ converges to 0 uniformly on every compact subset of Ω .

Theorem 134. Let $\Omega \subseteq \mathbb{R}^n$. Then a linear functional u lies in $\mathcal{E}'(\Omega)$ if and only if $\lim_{k \rightarrow \infty} (u, \phi_k) = 0$ when $\{\phi_k\}_{k \in \mathbb{N}}$ is sequence of smooth functions in Ω such that $\lim_{k \rightarrow \infty} \phi_k = 0$ in the sense of Definition 133.

Proof. Mutatis mutandis the proof is identical to that of Theorem 116, hence we omit it. $\square$

The next proposition shows that there exist a relation between the spaces of distributions introduced so far:

Proposition 135. Let $\Omega \subseteq \mathbb{R}^n$. Then if $u \in \mathcal{D}'(\Omega)$ is such that $\text{supp}(u)$ is compact, there exists a unique member $\tilde{u} \in \mathcal{E}'(\Omega)$ for which

$$(\tilde{u}, f) = (u, f) \quad \forall f \in \mathcal{D}(\Omega).$$

Proof. Let us prove first of all existence. Pick any u as per hypothesis and choose $\rho \in C_0^\infty(\Omega)$ so that $\rho = 1$ on $\text{supp}(u)$. Define $\tilde{u} : C^\infty(\Omega) \rightarrow \mathbb{C}$ so that

$$\tilde{u}(\phi) \equiv (\tilde{u}, \phi) \doteq (u, \rho\phi). \quad (40)$$

This is a distribution since, in view of (34), for all compact sets $K \subset \Omega$, there exist $C \geq 0$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$|(\tilde{u}, \phi)| = |(u, \rho\phi)| \leq C_K \sum_{|\alpha| \leq N} \sup |\partial^\alpha(\rho\phi)|. \quad \forall \phi \in C^\infty(\Omega)$$

Applying the Leibniz rule to the right hand side, we obtain (39) where the role of the fixed compact set is played by $\text{supp}(\rho)$. In other words $\tilde{u} \in \mathcal{E}'(\Omega)$.

We prove uniqueness. Starting from $\tilde{u}$ as above, consider $\sigma \in C_0^\infty(\Omega)$ which is different from ρ but still 1 on a neighbourhood of $\text{supp}(u)$. Hence, for every $\phi \in C^\infty(\Omega)$,

$$(u, \rho\phi) - (u, \sigma\phi) = (u, (\rho - \sigma)\phi) = 0,$$

where we used Corollary 122, since $\text{supp}(u) \cap \text{supp}((\rho - \sigma)\phi) = \emptyset$. Since the only distribution which gives 0 when evaluated on all test-functions is the trivial one, we can conclude that $(\tilde{u}, f) = (u, f)$ for all $f \in \mathcal{D}(\Omega)$. In other words (40) extends u uniquely to $C^\infty(\Omega)$. $\square$

Exercise 27. Let $u \in \mathcal{E}'(\Omega)$, $\Omega \subseteq \mathbb{R}^n$. Prove that the restriction of u to $\mathcal{D}(\Omega)$ identifies an element of $\mathcal{D}'(\Omega)$.

In view of this last exercise as well as of the uniqueness statement of Proposition 135, it is customary to identify the elements of $\mathcal{E}'(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ as the **distributions of compact support**, that is those elements of $\mathcal{D}'(\Omega)$ whose support is compact. In other words

$$\mathcal{E}'(\Omega) \subset \mathcal{D}'(\Omega).$$

Actually the relation between these spaces is even stronger as this theorem shows.

Theorem 136. Let $\Omega \subseteq \mathbb{R}^n$. Then $C_0^\infty(\Omega)$ is dense in $C^\infty(\Omega)$ and $\mathcal{E}'(\Omega)$ is dense in $\mathcal{D}'(\Omega)$.

Proof. Choose a locally finite partition of unity¹² of Ω , $\{\psi_j\}_{j \in \mathbb{N}}$. For any $\phi \in C^\infty(\Omega)$, let

$$\phi_k = \sum_{j=1}^k \psi_j \phi.$$

Since each ψ_j is smooth and compactly supported then $\phi_k \in C_0^\infty(\Omega)$ and a direct calculation shows that

$$\lim_{k \rightarrow \infty} \phi_k = \phi,$$

¹²A partition of unity of an open subset $\Omega \subseteq \mathbb{R}^n$ is said to be *locally finite*, if, given an open cover $\{U_i\}_{i \in \mathbb{N}}$ of Ω , there exists $\{\psi_j\}_{j \in \mathbb{N}}$, where $\psi_j \in C_0^\infty(\Omega)$ and it is supported only in one U_i . In addition $\psi_j(x) \geq 0$, $\sum_{j=1}^\infty \psi_j = 1$ and any compact set intersects only a finite number of supports of the functions ψ_j .

which is tantamount to saying that $C_0^\infty(\Omega)$ is dense in $C^\infty(\Omega)$. With the very same argument, start from any $u \in \mathcal{D}'(\Omega)$ and construct

$$u_k = \sum_{j=1}^k \psi_j u \in \mathcal{E}'(\Omega).$$

Also in this case a direct computation shows that $\lim_{k \rightarrow \infty} u_k = u$, hence $\mathcal{E}'(\Omega)$ is dense in $\mathcal{D}'(\Omega)$. $\square$

A nice application of all the constructions that we have built so far is the following. Notice that it plays several important role in many physical models:

Theorem 137. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be such that $\text{supp}(u) = \{0\}$. Then, there exists $N \in \mathbb{N} \cup \{0\}$, such that*

$$u = \sum_{|\alpha| \leq N} c_\alpha \partial^\alpha \delta, \quad (41)$$

where α is a multi-index and $c_\alpha \in \mathbb{C}$ for every value of α .

Proof. We divide our analysis in two separate steps.

Part 1. We show that, if $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\text{supp}(u) = \{0\}$, then there exists $N \in \mathbb{N} \cup \{0\}$ such that $(u, f) = 0$ if $f \in \mathcal{D}(\mathbb{R}^n)$ and $\partial^\alpha f(0) = 0$ for all $|\alpha| \leq N$.

To start with, consider an auxiliary $h \in C_0^\infty(\mathbb{R}^n)$ so that $h = 1$ if $|x| < \frac{1}{2}$ and $h = 0$ if $|x| > 1$, where $|\cdot|$ is the Euclidean modulus on $\mathbb{R}^n$. For any $\epsilon \in (0, 1)$ it holds that, for all $f \in \mathcal{D}(\mathbb{R}^n)$ the function $f(x) - f(x)h(\frac{x}{\epsilon})$ vanishes for all $x \in \mathbb{R}^n$ such that $|x| < \frac{\epsilon}{2}$. In other words, since per hypothesis $\text{supp}(u) = \{0\}$,

$$(u, f) = (u, fh_\epsilon), \quad \forall f \in \mathcal{D}(\mathbb{R}^n)$$

where $h_\epsilon(x) \doteq h(\frac{x}{\epsilon})$. Equation 34 guarantees that, if we consider the closed unit ball $\overline{B_1(0)}$, there must exist $C \geq 0$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$|(u, f)| = |(u, fh_\epsilon)| \leq C \sum_{|\alpha| \leq N} \sup |\partial^\alpha (fh_\epsilon)|. \quad \forall f \in \mathcal{D}(\mathbb{R}^n)$$

Assume now, in addition, that $\partial^\alpha f(0) = 0$ whenever $|\alpha| \leq N$. Applying Taylor's expansion theorem¹³ to the derivatives of f , we obtain that, for any multi-index β such that $|\beta| \leq N$,

$$|\partial^\beta f(x)| \leq \epsilon^{N-|\beta|+1} \sum_{|\gamma|=N+1-|\beta|} \frac{1}{\gamma!} \sup \{ |\partial^\gamma f(x)| \mid |x| < \epsilon \},$$

¹³Recall that, if $\phi \in C^\infty(\mathbb{R})$, then $\phi(x) = \sum_{k=1}^{N-1} \frac{x^k}{k!} \frac{d^k \phi}{dx^k}(0) + R_N[\phi](x)$ where $R_N[\phi](x)$ is the remainder, which can be estimated as $R_N[\phi](x) \leq \frac{|x|^N}{N!} \sup \left\{ \left| \frac{d^N \phi(x')}{dx'^N} \right| \mid |x'| < |x| \right\}$. The generalization to $\mathbb{R}^n$ is straightforward.

where the factorial of a multi-index $\gamma = (\gamma_1, \dots, \gamma_k)$ is $\gamma! \doteq \prod_{i=1}^k \gamma_i!$. Using the Leibniz rule, see e.g. Theorem 131, we have

$$\partial^\alpha(fh_\epsilon) = \sum_{\substack{\beta, \gamma \\ \beta + \gamma = \alpha}} \frac{\alpha!}{\beta! \gamma!} \epsilon^{-|\gamma|} \partial^\beta f \partial^\gamma h.$$

Combining this data together and taking into account also the constraint $fh_\epsilon = 0$ if $|x| > \epsilon$, it holds that, for all multi-indices α with $|\alpha| \leq N$,

$$|\partial^\alpha(fh_\epsilon)| \leq C_\alpha \epsilon^{N-|\beta|+1-|\gamma|} = C_\alpha \epsilon^{N-|\alpha|+1},$$

where C_α is a constant independent from ϵ . As a by-product, it holds

$$|(u, f)| \leq C \sum_{|\alpha| \leq N} C_\alpha \epsilon^{N+1-|\alpha|} \leq C' \epsilon,$$

where C' is a constant. We can now take the limit as $\epsilon \rightarrow 0$ and we have proven the first part of the theorem.

Part 2. Choosing $N \in \mathbb{N} \cup \{0\}$ and h as in part 1., for any $f \in \mathcal{D}(\mathbb{R}^n)$, it holds

$$f(x) = h(x) \sum_{|\alpha| \leq N} \frac{x^\alpha}{\alpha!} \partial^\alpha f(0) + R_N[f](x),$$

where the remainder $R_N[f](x) \in \mathcal{D}(\mathbb{R}^n)$ as one can infer per subtraction of the left hand side and of the first term on the right hand side. In addition, per construction $\partial^\alpha R_N[f](0) = 0$ whenever $|\alpha| \leq N$. Hence, in view of part 1., $(u, R_N[f]) = 0$ and

$$(u, f) = \sum_{|\alpha| \leq N} \frac{\partial^\alpha f(0)}{\alpha!} (u, x^\alpha h) = \sum_{|\alpha| \leq N} c'_\alpha (\partial^\alpha f)(0),$$

where $c'_\alpha \doteq \frac{(u, x^\alpha h)}{\alpha!}$. By direct inspection of the right hand side we see that u can be chosen of the form (41) with $c_\alpha \doteq (-1)^{|\alpha|} c'_\alpha$. To conclude the theorem it suffices to show uniqueness. This is a rather easy step which we omit, leaving it as the next exercise. $\square$

Exercise 28. Let $\Omega \subseteq \mathbb{R}^n$. Prove that, if $u \in \mathcal{D}'(\Omega)$ is such that $(u, f) = 0$ for all $f \in \mathcal{D}(\Omega)$, then $u = 0$.

A similar, quite useful result can be also easily proven:

Exercise 29. Let $\Omega \subseteq \mathbb{R}^n$ and let $u \in \mathcal{D}'(\Omega)$. Let X be a dense subset of $\mathcal{D}(\Omega)$. Show that if $(u, f) = 0$ for all $f \in X$, then $u = 0$.

Before concluding the section, we discuss a notable property of distributions of compact support.

Definition 138. Let $\Omega \subseteq \mathbb{R}^n$ and let $u \in \mathcal{D}'(\Omega)$. We say that u is of **finite order** $N \in \mathbb{N} \cup \{0\}$, if the integer N of (34) is the same for all compact subsets $K \subset \Omega$.

Theorem 139. Let $\Omega \subseteq \mathbb{R}^n$ and let $u \in \mathcal{E}'(\Omega)$. Then it is a distribution of finite order N . Hence $(u, \phi) = 0$ for all $\phi \in C^\infty(\Omega)$ such that $\partial^\alpha \phi(x) = 0$ when $x \in \text{supp}(u)$ and α is a multi-index with $|\alpha| \leq N$.

Proof. Let $K = \text{supp}(u)$ and choose $\delta > 0$ such that

$$K_\delta \doteq \{x \in \mathbb{R}^n \mid d(x, K) \leq \delta\} \subset \Omega.$$

Choose $\rho \in C_0^\infty(\mathbb{R}^n)$ such that $\text{supp}(\rho) \subset B_1(0)$ and $\int_{\mathbb{R}^n} d^n x \rho(x) = 1$. Let χ_ϵ be the characteristic function of the set $K_{2\epsilon}$ where $0 < \epsilon < 4\delta$. Define

$$\sigma_\epsilon(x) \doteq \epsilon^{-n} \int_{\mathbb{R}^n} d^n y \chi_\epsilon(y) \rho\left(\frac{x-y}{\epsilon}\right).$$

Per construction $\sigma_\epsilon = 1$ on K_ϵ and, consequently,

$$|(u, \phi)| = |(u, \sigma_\epsilon \phi)| \leq C_{K_\delta} \sum_{|\alpha| \leq N} \sup |\partial^\alpha (\sigma_\epsilon \phi)|, \quad \forall \phi \in C^\infty(\mathbb{R}^n)$$

on account of (39). The same argument used in the proof of Theorem 137 yields the existence of a constant $B \geq 0$ such that $|(u, \phi)| \leq B\epsilon$. Notice that we used that $\partial^\gamma \sigma_\epsilon = O(\epsilon^{-|\gamma|})$. By taking the limit as $\epsilon \rightarrow 0$, the sought statements descend. $\square$

9 Miscellaneous Structures on Distributions

Goal of this section is to introduce additional structures which are associated to distributions and which play an important role in different contexts, both in physical models and in functional analysis. Contrary to the previous sections, there will not be a main topic, except partially in the next two subsections which are interconnected.

9.1 Tensor Product of Distributions

In this subsection, we study the structural properties of tensor products. Recall that both $\mathcal{D}'(\Omega)$ and $\mathcal{E}'(\Omega)$ are vector spaces and tensor products are naturally defined between elements of this class of spaces. A reader, who is not familiar with these concepts, should revise them on any sufficiently advanced book on linear algebra, *e.g.* [4, Ch. 13, §62].

As it is customary in this part of the course, let us start from a motivational example. Consider $\mathbb{R}^n$ and $\mathbb{R}^m$ or an open subset for each of them. Let $\phi \in C^\infty(\mathbb{R}^n)$ and $\chi \in C^\infty(\mathbb{R}^m)$. We define

$$(\phi \otimes \chi)(x, y) \doteq \phi(x)\chi(y), \quad \forall (x, y) \in \mathbb{R}^n \times \mathbb{R}^m$$

It is immediate to realize that $\phi \otimes \chi \in C^\infty(\mathbb{R}^n \times \mathbb{R}^m)$ and it identifies therefore a distribution $u_{\phi \otimes \chi} \in \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}^m)$ such that

$$(u_{\phi \otimes \chi}, g) = \int_{\mathbb{R}^n \times \mathbb{R}^m} d^n x d^m y \phi(x) \chi(y) g(x, y). \quad \forall g \in \mathcal{D}(\mathbb{R}^{n+m}).$$

As a direct consequence of this definition, if $g = h \otimes f$, with $h \in \mathcal{D}(\mathbb{R}^n)$ and $f \in \mathcal{D}(\mathbb{R}^m)$, it holds

$$(u_{\phi \otimes \chi}, hf) = (\phi, h)(\chi, f).$$

Our goal is to extend these structures to generic distributions consistently with the above examples. As a preliminary step, we discuss an operation which is from time to time known as *partial evaluation*.

Theorem 140. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $u \in \mathcal{D}'(\Omega)$ and let $\phi \in C^\infty(\mathbb{R}^n \times \mathbb{R}^m)$ be such that for every $y' \in \Omega'$, there exists an open subset of y' , say $\mathcal{O}_{y'} \subseteq \Omega'$, for which the support of the map $x \mapsto \phi(x, y)$ is contained in a compact set $K_{y'}$ whenever $y \in \mathcal{O}_{y'}$. Then*

$$(u, \phi(x, y)) \in C^\infty(\Omega'), \quad \text{and} \quad \partial^\alpha (u, \phi(x, y)) = (u, \partial_y^\alpha \phi(x, y)),$$

for all multi-indices α .

Proof. Let us choose ϕ as per hypothesis and let us call $\Upsilon(y) \doteq (u, \phi(x, y))$. For every $y' \in \Omega'$, $\exists \delta > 0$ such that, if $|y - y'| < \delta$, then $y \in \mathcal{O}_{y'}$. Let $a \in \mathbb{R}^m$ be such that $|a| < \delta$ and let

$$\phi_a(x, y') \doteq \phi(x, y' + a) - \phi(x, y').$$

Per linearity

$$\Upsilon(y' + a) - \Upsilon(y') = (u, \phi_a(x, y')).$$

Per hypothesis $\text{supp}(\phi_a(\cdot, y')) \subset K_{y'}$ where $K_{y'}$ is a compact set. Since ϕ is smooth, we know that, for every multi-index α , $\partial_x^\alpha \phi_a(x, y)$ tends to 0 uniformly as $a \rightarrow 0$ (compare with the definition of difference quotient). In other words $\phi_a(\cdot, y') \in C_0^\infty(\Omega)$ and

$$\lim_{a \rightarrow 0} (\Upsilon(y' + a) - \Upsilon(y')) = 0.$$

Hence Υ is continuous. Let us now prove that it is also continuously differentiable. Let e_j be the j -th standard Euclidean unit vector in $\mathbb{R}^m$. Let $0 < \epsilon < \delta$ and let

$$\chi_\epsilon(x, y') = \epsilon^{-1} (\phi(x, y' + \epsilon e_j) - \phi(x, y')) - \frac{\partial \phi(x, y')}{\partial y_j'}$$

In view of the definition of $\Upsilon(y)$, using χ_ϵ as test-function for u , we obtain

$$(u, \chi_\epsilon(x, y')) = \epsilon^{-1} (\Upsilon(x, y' + \epsilon e_j) - \Upsilon(x, y')) - \left(u, \frac{\partial \phi(x, y')}{\partial y_j} \right).$$

The choice of δ as above implies that $\text{supp}(\chi_\epsilon) \subset K_{y'}$ and, via Taylor's expansion theorem, for each multi-index α , $\partial_x^\alpha \chi_\epsilon(x, y')$ converges uniformly to 0 as $\epsilon \rightarrow 0$. This entails that the partial derivative of $\Upsilon(y)$ exists and (replacing y' with y)

$$\frac{\partial \Upsilon(y)}{\partial y_j} = \left(u, \frac{\partial \phi(x, y)}{\partial y_j} \right). \quad (42)$$

Notice that $\frac{\partial \phi(x, y)}{\partial y_j}$ satisfies the same hypotheses of ϕ . Hence the same argument which lead to conclude that Υ is continuous, ensures also that $\frac{\partial \Upsilon(y)}{\partial y_j}$ is continuous. Repeating the reasoning to all orders, yields that $\Upsilon \in C^\infty(\Omega')$. The second formula of the statement of the theorem is then a consequence of (42) reiterated to all orders. $\square$

There are two special sub-cases:

Corollary 141. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $u \in \mathcal{D}'(\Omega)$ and $g \in \mathcal{D}(\Omega \times \Omega')$. Then $(u, g) \in C_0^\infty(\Omega')$.*

Proof. Consider Theorem 140 and choose $\mathcal{O}_{y'} = \Omega'$, while $K_{y'} = \pi_\Omega(\text{supp}(g))$, where $\pi_\Omega : \Omega \times \Omega' \rightarrow \Omega$ is the projection on the first factor. Once more let $\Upsilon(y) \doteq (u, g(x, y))$. Its support is a subset of the projection of the support of ϕ on Ω' . Hence it is compact. $\square$

Corollary 142. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $u \in \mathcal{E}'(\Omega)$ and $\phi \in C^\infty(\Omega \times \Omega')$. Then $(u, \phi) \in C^\infty(\Omega')$.*

Proof. Let $\rho \in C_0^\infty(\Omega)$ be 1 on $\text{supp}(u)$. Following again Theorem 140, we call $\Upsilon(y) \doteq (u, \rho(x)\phi(x, y))$. The statement descends choosing $\mathcal{O}_{y'} = \Omega'$ and $K_{y'} = \text{supp}(\rho)$. $\square$

We are almost ready to define the tensor product of distributions, but we need a technical result whose proof is quite demanding and hence we shall omit it. The interested reader can refer to [1, Th. 4.3.1]:

Theorem 143. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Then the subspace of $C_0^\infty(\Omega \times \Omega')$ defined as the span of functions of the form $h \otimes f$, with $h \in C_0^\infty(\Omega)$ and $f \in C_0^\infty(\Omega')$ is dense in $C_0^\infty(\Omega \times \Omega')$.*

We can now state the main result of this section:

Theorem 144. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $u \in \mathcal{D}'(\Omega)$ and let $v \in \mathcal{D}'(\Omega')$. There exists a unique element $u \otimes v \in \mathcal{D}'(\Omega \times \Omega')$, called the **tensor product** of u and v such that*

$$(u \otimes v, f \otimes h) = (u, f)(v, h). \quad \forall f \in \mathcal{D}(\Omega) \text{ and } \forall h \in \mathcal{D}(\Omega') \quad (43)$$

Proof. In the statement of the theorem $u \otimes v$ is defined on the subspace $\mathcal{D}(\Omega) \otimes \mathcal{D}(\Omega')$. Let now $g \in \mathcal{D}(\Omega \times \Omega')$. Corollary 141 guarantees that $(u, g) \in C_0^\infty(\Omega')$ and thus it is natural to define

$$(u \otimes v, g) = (v, (u, g)) \quad \forall g \in \mathcal{D}(\Omega \times \Omega').$$

We have constructed a linear functional which satisfies the wanted constraints whenever tested against $g = f \otimes h$, $f \in \mathcal{D}(\Omega)$ and $g \in \mathcal{D}(\Omega')$. We need only to verify that $u \otimes v$ satisfies (34). Let $K \subset \Omega \times \Omega'$ be a compact set. Since a generic point of K is a pair (x, y) with $x \in \Omega$ and $y \in \Omega'$, we can introduce the projections on the two components, namely

$$\pi_1 : K \rightarrow K_1 \subset \Omega, \quad \text{and} \quad \pi_2 : K \rightarrow K_2 \subset \Omega'.$$

Let $g \in \mathcal{D}(\Omega \times \Omega')$ be such that $\text{supp}(g) \subseteq K$. Let $h(y) \doteq (u, g(x, y))$. This is a smooth and compactly supported function such that $\text{supp}(h) \subseteq K_2$. Since $v \in \mathcal{D}'(\Omega')$, (34) for v entails the existence of a constant $C_{K_2} \geq 0$ and of $N \in \mathbb{N} \cup \{0\}$ such that

$$|(v, h)| \leq C_{K_2} \sum_{|\beta| \leq N} \sup |\partial^\beta h|,$$

where β is an arbitrary multi-index. Theorem 140 entails also that

$$\partial^\beta h(y) = \partial^\beta (u, g(x, y)) = (u, \partial_y^\beta g).$$

If we recall that $u \in \mathcal{D}'(\Omega)$ and that the function $x \mapsto (\partial_y^\beta g)(x, y)$ is smooth, compactly supported, with support contained in K_1 , (34) entails the existence of a constant $C'_{K_1} \geq 0$ and of an integer $M \in \mathbb{N} \cup \{0\}$ such that

$$|(\partial^\beta h)(y)| \leq C'_{K_1} \sum_{|\alpha| \leq M} \sup \{ |(\partial_x^\alpha \partial_y^\beta g)(x, y)|, \mid x \in K_1 \} \leq C'_{K_1} \sum_{|\alpha| \leq M} \sup \left(|\partial_x^\alpha \partial_y^\beta g| \right),$$

which is nothing but (34) for $u \otimes v$. In other words $u \otimes v \in \mathcal{D}'(\Omega \times \Omega')$.

We need to establish uniqueness of this definition. Yet Theorem 143 entails that $\mathcal{D}(\Omega) \otimes \mathcal{D}(\Omega')$ is dense in $\mathcal{D}(\Omega \times \Omega')$. Uniqueness is then a direct consequence of the linearity of the space of distributions and of Exercise 29. $\square$

Exercise 30. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $u \in \mathcal{E}'(\Omega)$ and let $v \in \mathcal{E}'(\Omega')$. Prove that there exists a unique element $u \otimes v \in \mathcal{E}'(\Omega \times \Omega')$ such that

$$(u \otimes v, \phi \otimes \phi') = (u, \phi) (v, \phi'). \quad \forall \phi \in \mathcal{E}(\Omega) \text{ and } \forall \phi' \in \mathcal{E}(\Omega')$$

$u \otimes v$ is still called the tensor product of u and v .

The tensor product of distributions enjoys notable properties:

Proposition 145. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $u \in \mathcal{D}'(\Omega)$ and let $v \in \mathcal{D}'(\Omega')$. It holds that

$$(u \otimes v, g) = (u, (v, g)) = (v, (u, g)), \quad \forall g \in \mathcal{D}(\Omega \times \Omega').$$

In addition the following properties hold true:

1. $\text{supp}(u \otimes v) = \text{supp}(u) \times \text{supp}(v)$,

2. if α is a multi-index related to Ω and β one related to Ω' , then

$$\partial_x^\alpha \circ \partial_y^\beta (u \otimes v) = \partial^\alpha u \otimes \partial^\beta v,$$

3. The tensor product is a separately continuous bilinear form on $\mathcal{D}'(\Omega) \otimes \mathcal{D}'(\Omega')$.

Proof. Following the proof of Theorem 144, we have defined $u \otimes v$ so that $(u \otimes v, g) = (u, (v, g))$ for all $g \in \mathcal{D}(\Omega \times \Omega')$. The role of u and v can be exchanged obtaining a second definition of $u \otimes v$ such that $(u \otimes v, g) = (v, (u, g))$. Both definitions agree on (43). Hence the uniqueness of the extension proven in Theorem 144 entails that $(u, (v, g)) = (v, (u, g))$. Furthermore item 2. and item 3. are a direct consequence of this equality and of the definition of distributional derivative. We are left only with proving item 1.

We prove the double inclusion. Let us start with $\subseteq$. It holds per definition of $u \otimes v$ and per Corollary 122 that $(u \otimes v, g) = 0$ for all those g such that $\text{supp}(g) \cap (\text{supp}(u) \times \Omega') = \emptyset$ or such that $\text{supp}(g) \cap (\Omega \times \text{supp}(v)) = \emptyset$. In other words $\text{supp}(u \otimes v) \subseteq \text{supp}(u) \times \text{supp}(v)$.

We prove $\supseteq$. Suppose $x \in \text{supp}(u)$ and $y \in \text{supp}(v)$. There must exist $f \in \mathcal{D}(\Omega)$, with $x \in \text{supp}(f)$ and $h \in \mathcal{D}(\Omega')$ with $y \in \text{supp}(h)$ such that $(u, f) \neq 0$ and $(v, h) \neq 0$. At the same time $(u \otimes v, f \otimes h) \neq 0$, which entails that $(x, y) \in \text{supp}(u \otimes v)$. In other words $\text{supp}(u) \times \text{supp}(v) \subseteq \text{supp}(u \otimes v)$. $\square$

Exercise 31. Show that the result of Proposition 145 are still valid if $u \in \mathcal{E}'(\Omega)$ and $v \in \mathcal{E}'(\Omega')$, replacing $g \in \mathcal{D}(\Omega \times \Omega')$ with $\phi \in \mathcal{E}(\Omega \times \Omega')$.

Exercise 32. Let $\Theta(x_i)$, $i = 1, \dots, n$ be the Heaviside distribution. Prove that $\bigotimes_{i=1}^n \partial_{x_i} \Theta(x_i) = \delta(x_1, \dots, x_n)$ where $\delta = \delta(x_1, \dots, x_n) \doteq \bigotimes_{i=1}^n \delta(x_i)$. The dependence on the coordinates of the δ -distribution is meant to avoid confusion, not because we are Vikings.

9.2 Convolutions

In this subsection we introduce at a level of distributions an operation which is well-known in physics, especially in signal analysis. It goes under the name of convolution. Exactly as in all other cases, we will motivate our definitions looking at a very regular scenario. Let $\phi, \chi \in C^\infty(\mathbb{R}^n)$ and let $\text{supp}(\phi) \cap \text{supp}(\chi) \subseteq K$, where $K \subset \mathbb{R}^n$ is a compact set. We define

$$(\phi \star \chi)(x) \doteq \int_{\mathbb{R}^n} d^n y \phi(y) \chi(x - y),$$

where the integral on the right hand side converges thanks to the hypothesis on the supports of ϕ and χ . With a simple change of variable $y \mapsto z \doteq x - y$, one can realize that

$$(\phi \star \chi)(x) = \int_{\mathbb{R}^n} d^n y \phi(y) \chi(x - y) = \int_{\mathbb{R}^n} d^n z \phi(x - z) \chi(z) = (\chi \star \phi)(x).$$

If we want to interpret these identities at a level of distributions, we can even drop the hypothesis on the support of ϕ and of χ , defining $u_{\phi \star \chi} \in \mathcal{D}'(\mathbb{R}^n)$ via

$$(u_{\phi \star \chi}, f) \doteq \int_{\mathbb{R}^n} d^n x d^n y \phi(x) \chi(y) f(x+y). \quad (44)$$

In this case the integral is well defined thanks to the test-function f . In view of Theorem 144 it is tempting to interpret (44) as

$$(u_{\phi \star \chi}, f) = (\phi \otimes \chi, f(x+y)), \quad (45)$$

using the right hand side as the prototype to extend the definition to arbitrary elements of $\mathcal{D}'(\mathbb{R}^n)$. Unfortunately the problem is that, although $f \in C_0^\infty(\mathbb{R}^n)$, in order to account for the tensor product we need that the chain of maps $\mathbb{R}^n \times \mathbb{R}^n \ni (x, y) \mapsto x+y \mapsto f(x+y)$ is compactly supported. This is in general not the case. At the same time $(x, y) \mapsto f(x+y)$ is certainly smooth on $\mathbb{R}^n \times \mathbb{R}^n$. Hence (45) is well-defined as soon as we replace $\phi, \chi \in C^\infty(\mathbb{R}^n)$ with $u, v \in \mathcal{E}'(\mathbb{R}^n)$.

This succinct discussion suggests that defining a convolution between two distributions in $\mathcal{D}'(\mathbb{R}^n)$ might not be possible, but it is still not obvious that limiting ourselves to $\mathcal{E}'(\mathbb{R}^n)$ is the sharp solution. Indeed (please at this point, stop reading and watch an episode of Downton Abbey. I will not admit, you failing me, Mr Bates!), this is the case and we shall proceed with a constructive argument. Consider now $u \in \mathcal{E}'(\mathbb{R}^n)$ and $v \in \mathcal{D}'(\mathbb{R}^n)$. Let $\rho \in C_0^\infty(\mathbb{R}^n)$ such that $\rho = 1$ on $\text{supp}(u)$. In turn, for any $f \in \mathcal{D}(\mathbb{R}^n)$, $\text{supp}(\rho(x)f(x+y)) \subseteq \text{supp}(\rho) \times (\text{supp}(f) \setminus \text{supp}(\rho))$ is compact in $\mathbb{R}^n \times \mathbb{R}^n$. Here we are adopting the notation according to which, if $U \subset \mathbb{R}^n$, then $-U \doteq \{x \in \mathbb{R}^n, \mid -x \in U\}$ and, if $V \subset \mathbb{R}^n$, then $U + V = \{x \in \mathbb{R}^n, \mid \exists a \in U \text{ and } b \in V, \text{ such that } x = a + b\}$. Notice that a and b might not well be unique.

We can therefore define

$$(u \star v, f) \doteq (u \otimes v, \rho(x)f(x+y)). \quad \forall f \in \mathcal{D}(\mathbb{R}^n) \quad (46)$$

The right hand side is clearly well-defined and, moreover, it does not depend on the choice of the cut-off function. Let $\sigma \in C_0^\infty(\mathbb{R}^n)$ be equal to 1 on $\text{supp}(u)$. Then, since $(\rho(x) - \sigma(x))f(x+y) = 0$ on $\text{supp}(u) \times \text{supp}(v)$, we can use Proposition 145 and Corollary 122 to conclude that

$$(u \otimes v, \rho(x)f(x+y) - (u \otimes v, \sigma(x)f(x+y)) = (u \otimes v, (\rho(x) - \sigma(x))f(x+y)) = 0,$$

since $\text{supp}(u \otimes v) = \text{supp}(u) \times \text{supp}(v)$. It is also immediate to realize that $u \star v$ satisfies (34) on $\mathbb{R}^n$ since, if we fix a cut-off function ρ , there exists a compact subset $K \subset \mathbb{R}^n$ such that $\text{supp}(\rho(x)f(x+y)) \subseteq \text{supp}(\rho) \times (K \setminus \text{supp}(\rho))$. We have found a compact set in $\mathbb{R}^n \times \mathbb{R}^n$ and now we can use (34) for $u \otimes v$ to conclude that $u \star v \in \mathcal{D}'(\mathbb{R}^n)$. Summarizing and extending slightly this discussion we have

Theorem 146. *Let $u \in \mathcal{E}'(\mathbb{R}^n)$ and let $v \in \mathcal{D}'(\mathbb{R}^n)$. Then there exists the **convolution** between u and v defined as $u \star v \in \mathcal{D}'(\mathbb{R}^n)$ via (46):*

$$(u \star v, f) \doteq (u \otimes v, f(x+y)) = (v, (u, f(x+y))) = (u, (v, f(x+y))).$$

Exercise 33. Prove that the convolution is commutative, that is $u \star v = v \star u$ for all $u \in \mathcal{E}'(\mathbb{R}^n)$ and for all $v \in \mathcal{D}'(\mathbb{R}^n)$.

We study now a few properties of the convolution:

Proposition 147. *Let $u \in \mathcal{E}'(\mathbb{R}^n)$ and $v \in \mathcal{D}'(\mathbb{R}^n)$. Then*

$$\text{supp}(u \star v) \subset \text{supp}(u) + \text{supp}(v).$$

Proof. We start showing an ancillary result, namely that, if $K \subset \mathbb{R}^n$ is compact and $B \subseteq \mathbb{R}^n$ is closed, then $K + B$ is closed. In other words, we have to show that, for every converging sequence $\{x_j\}_{j \in \mathbb{N}}$ of elements in $K + B$, there exists $x \in K + B$ such that $\lim_{j \rightarrow \infty} x_j = x$. Since $\mathbb{R}^n$ is closed such x exists as an element of $\mathbb{R}^n$. To see that $x \in K + B$, we observe that each $x_j \in K + B$, hence $x_j = y_j + z_j$, with $y_j \in K$ and $z_j \in B$. Since K is compact there exists a subsequence of $\{y_j\}_{j \in \mathbb{N}}$ converging to an element $y \in K$. Without loss of generality and with a slight abuse of notation we call this sequence $\{y_j\}_{j \in \mathbb{N}}$. At the same time $z_j = x_j - y_j$ and, as $j \rightarrow \infty$, being B closed there exists $z = \lim_{j \rightarrow \infty} z_j$. Furthermore $z = x - y$, that is $x = y + z$ with $y \in K$ and $z \in B$. We have shown that $x \in K + B$.

We can apply this result to $\text{supp}(u)$ (compact) and to $\text{supp}(v)$ (closed) to infer that $\text{supp}(u) + \text{supp}(v)$ is closed. If we prove that $u \star v|_{\mathbb{R}^n \setminus (\text{supp}(u) \cup \text{supp}(v))} = 0$, the proposition is proven thanks to Definition 121. Yet, this is a straightforward consequence of the definition itself of convolution of distributions as given *e.g.* in Theorem 146. $\square$

Additional properties descend from the basic definitions of distributions and of the associated operations:

Exercise 34. Let $u \in \mathcal{E}'(\mathbb{R}^n)$ and $v \in \mathcal{D}'(\mathbb{R}^n)$. Then, if we consider the standard Euclidean coordinates over $\mathbb{R}^n$ and if we write ∂_j in place of $\frac{\partial}{\partial x_j}$, $j = 1, \dots, n$, it holds

$$\partial_j(u \star v) = \partial_j u \star v = u \star \partial_j v.$$

In addition let $a \in \mathbb{R}^n$ and let $\tau_a : \mathcal{D}'(\mathbb{R}^n) \rightarrow \mathcal{D}'(\mathbb{R}^n)$ be defined as

$$(\tau_a(u), f) = (u, \tau_{-a}(f)), \quad \forall f \in \mathcal{D}(\mathbb{R}^n)$$

where $\tau_{-a}f(x) \doteq f(x - a)$. It holds that

$$\tau_a(u \star v) = \tau_a u \star v = u \star \tau_a v.$$

Corollary 148. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$. Then $\delta \star u = u$.*

Proof. First of all observe that $\delta \in \mathcal{E}'(\mathbb{R}^n)$ since $\text{supp}(\delta) = \{0\}$ which is compact. For any $f \in \mathcal{D}(\mathbb{R}^n)$

$$(\delta \star u, f(x + y)) = (u, (\delta, f(x + y))) = (u, f(y)).$$

$\square$

Theorem 149. *The following statements hold true:*

(a) *Let $u \in \mathcal{E}'(\mathbb{R}^n)$ and let $\{v_j\}_{j \in \mathbb{N}}$ be a sequence of distributions converging to $v \in \mathcal{D}'(\mathbb{R}^n)$. Then, in $\mathcal{D}'(\mathbb{R}^n)$,*

$$\lim_{j \rightarrow \infty} u \star v_j = u \star v.$$

(b) *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and let $\{v_j\}_{j \in \mathbb{N}}$ be a sequence of distributions converging to $v \in \mathcal{D}'(\mathbb{R}^n)$. Let K be a compact subset of $\mathbb{R}^n$ and suppose $\text{supp}(v_j) \subset K$ for all j . Then, in $\mathcal{D}'(\mathbb{R}^n)$,*

$$\lim_{j \rightarrow \infty} u \star v_j = u \star v.$$

Proof. (a) Let $u \in \mathcal{E}'(\mathbb{R}^n)$ and let $f \in \mathcal{D}(\mathbb{R}^n)$. Then $(u, f(x+y))$ is compactly supported and it is smooth on account of Corollary 142 once we read f as the map $\mathbb{R}^n \times \mathbb{R}^n \ni (x, y) \mapsto f(x+y)$. From the definition of convolution as in Theorem 146, for all $f \in \mathcal{D}(\mathbb{R}^n)$,

$$\lim_{j \rightarrow \infty} (u \star v_j, f) = \lim_{j \rightarrow \infty} (v_j, (u, f(x+y))) = (v, (u, f(x+y))),$$

where we used the hypothesis of convergence of $\{v_j\}_{j \in \mathbb{N}}$.

(b) Let $\rho \in C_0^\infty(\mathbb{R}^n)$ be a cut-off function which is 1 in a neighbourhood of K . Still, on account of the definition of convolution as in Theorem 146, it holds that, for every $f \in \mathcal{D}(\mathbb{R}^n)$

$$\lim_{j \rightarrow \infty} (u \star v_j, f) = \lim_{j \rightarrow \infty} (v_j, \rho(y)(u, f(x+y))) = (u \star v, f),$$

where we used that $\rho(y)(u, f(x+y)) \in \mathcal{D}(\mathbb{R}^n)$. □

We conclude the section introducing the concept of regularization and proving a key result of the theory of distributions.

Theorem 150. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and let $\rho \in C_0^\infty(\mathbb{R}^n) \subset \mathcal{E}'(\mathbb{R}^n)$. Then $\rho \star u \in C^\infty(\mathbb{R}^n)$ and it is called **regularization** of u .*

Proof. Consider any $h \in C_0^\infty(\mathbb{R}^n)$. Since it is compactly supported, it identifies an element of $\mathcal{E}'(\mathbb{R}^n)$ which we indicate with the same symbol. Let $\sigma \in C_0^\infty(\mathbb{R}^n)$ be equal to 1 on $\text{supp}(h)$. The map $\mathbb{R}^n \times \mathbb{R}^n \ni (x, y) \mapsto \sigma(x)h(x-y)$ has compact support. It is thus meaningful the following:

$$(h \otimes u, \sigma(x)\rho(x-y)) = \int_{\mathbb{R}^n} d^n x h(x)(u, \rho(x-y)) = \left(u, \int_{\mathbb{R}^n} d^n x h(x)\rho(x-y) \right),$$

where, in the last equality, we used Proposition 145. Changing variable in the last integral as $x' \doteq x - y$, we obtain

$$\int_{\mathbb{R}^n} d^n x h(x)(u, \rho(x-y)) = \left(u, \int_{\mathbb{R}^n} d^n x' h(x'+y)\rho(x') \right) = (\rho \otimes u, h(x+y)) = (\rho \star u, h).$$

To conclude, we observe that Theorem 140 guarantees us that $(u, \rho(x - y)) \in C^\infty(\mathbb{R}^n)$. Since $h \in C_0^\infty(\mathbb{R}^n)$ is arbitrary, the statement descends. $\square$

Exercise 35. Let $h \in C_0^\infty(\mathbb{R}^n)$ be such that $\int_{\mathbb{R}^n} d^n x h(x) = 1$. Let $h_j(x) \doteq j^n h(jx)$, $j \in \mathbb{N}$. Prove that

$$\lim_{j \rightarrow \infty} h_j(x) = \delta_0.$$

From Theorem 150 it descends a key result:

Theorem 151. $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{D}'(\mathbb{R}^n)$.

Proof. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be arbitrary and let $h \in C_0^\infty(\mathbb{R}^n)$ be such that $\int_{\mathbb{R}^n} d^n x h(x) = 1$. Let $h_j(x) \doteq j^n h(jx)$, $j \in \mathbb{N}$. Observe that $\lim_{j \rightarrow \infty} h_j(x) = \delta$ in view of Exercise 35. In addition $\text{supp}(h_j) \subseteq \text{supp}(h)$ for all $j \in \mathbb{N}$. By item (b) of Theorem 149 and by Corollary 148, it holds that

$$\lim_{j \rightarrow \infty} h_j \star u = \delta \star u = u. \quad (47)$$

In addition, for all $j \in \mathbb{N}$, Theorem 150 guarantees that $h_j \star u \in C^\infty(\mathbb{R}^n)$.

Consider now another cut-off function, say $g \in C_0^\infty(\mathbb{R}^n)$ such that $g = 1$ if $|x| < 1$, $|\cdot|$ being the Euclidean length. Define

$$u_j \doteq g\left(\frac{x}{j}\right) h_j \star u. \quad \forall j \in \mathbb{N}$$

It holds that, for all $f \in \mathcal{D}(\mathbb{R}^n)$

$$(u_j, f) = \int_{\mathbb{R}^n} d^n x u_j f(x) = \int_{\mathbb{R}^n} d^n x (h_j \star u) f(x),$$

where the identity holds for j sufficiently large so to account for $\text{supp}(f)$. Notice that the use of the integral is legitimate since each $u_j \in C_0^\infty(\mathbb{R}^n)$ per construction. Using (47), the sought result descends. $\square$

Remark 10. Notice that, in this section, we are emphasizing the role of $\mathbb{R}^n$. It is nonetheless important to keep in mind that all the results that we have proven, especially Theorem 151 would adapt also to open subsets $\Omega \subseteq \mathbb{R}^n$. Yet, dealing with this more general case would require to control more technical details and we prefer omitting them.

9.2.1 Fundamental Solutions and The Structure Theorem

We have constructed the convolution of two distributions of which at least one is of compact support. As a result we have been able to prove that $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{D}'(\mathbb{R}^n)$. Yet, it would be desirable to go one step further and to understand whether it is possible to define the convolution between two non compactly supported distributions. As we will see, there are some

constraints and more work to do, but as a pay-back we will be able to introduce one of the main concept in the modern analysis of linear partial differential equations. For notational simplicity let us define the map ($m \geq 1$)

$$\mu : \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_m \rightarrow \mathbb{R}^n, \quad (x^{(1)}, \dots, x^{(m)}) \mapsto \mu(x^{(1)}, \dots, x^{(m)}) \doteq \sum_{i=1}^m x^{(i)}. \quad (48)$$

The role of this map is simply related to the idea that we have been able to define the convolution between distributions when one of them is compactly supported. A closer inspection to the construction unveils that the condition, that we have actually used, is that the intersection of the distributions involved in the convolution is compact.

Definition 152. Let $\bar{\Omega}_1, \dots, \bar{\Omega}_m$ be closed sets in $\mathbb{R}^m$. Then $\mu|_{\bar{\Omega}_1 \times \dots \times \bar{\Omega}_m}$ is **proper**¹⁴ if the pre-image of a compact subset is compact.

Exercise 36. Show that Definition 152 is equivalent to requiring that $\forall \delta > 0, \exists \delta' > 0$, such that $x^{(j)} \in \bar{\Omega}_j, j = 1, \dots, m$ and $\left| \sum_{i=1}^m x^{(i)} \right| \leq \delta$ implies $|x^{(j)}| \leq \delta'$ for all $j = 1, \dots, m$.

We show that the property of being a proper map is stable under suitable ϵ -deformations of the domain of definition:

Lemma 153. Let $\bar{\Omega}_1, \dots, \bar{\Omega}_m$ be closed sets in $\mathbb{R}^m$ and let $\mu|_{\bar{\Omega}_1 \times \dots \times \bar{\Omega}_m}$ be **proper**. For any $\epsilon > 0$ consider a closed ϵ -neighbourhood of $\bar{\Omega}_i, i = 1, \dots, m$ and call it $\bar{\Omega}_i^\epsilon$. Then $\mu|_{\bar{\Omega}_1^\epsilon \times \dots \times \bar{\Omega}_m^\epsilon}$ is proper.

Proof. The proof does not depend on the choice of m ; hence for simplicity we take $m = 2$. Per hypothesis of μ being proper on $\bar{\Omega}_1 \times \bar{\Omega}_2$, whenever $\delta > 0$ and $x \in \bar{\Omega}_1$ and $y \in \bar{\Omega}_2$ are such that $|x + y| \leq \delta$, then there exists $x' \in \bar{\Omega}_1$ and $y' \in \bar{\Omega}_2$ such that $|x - x'| \leq \epsilon$ and $|y - y'| \leq \epsilon$. As a consequence of the triangular inequality

$$|x' + y'| \leq |x' - x| + |y' - y| + |x + y| \leq 2\epsilon + \delta.$$

Still by hypothesis, there exists $\delta' > 0$ subordinated to $2\epsilon + \delta$ such that $|x'| \leq \delta'$ and $|y'| \leq \delta'$. In other words

$$|x| = |x - x' + x'| \leq |x - x'| + |x'| \leq \delta' + \epsilon,$$

and similarly for $|y|$. The arbitrariness of x and y entails the statement. $\square$

Let us now focus on distributions and let us proceed constructively. Let $u_i \in \mathcal{D}'(\mathbb{R}^n), i = 1, \dots, m$ and suppose that the map μ , as in (48), restricted to $\text{supp}(u_1) \times \dots \times \text{supp}(u_m)$ is proper. Choose $\epsilon > 0$ and $f \in \mathcal{D}(\mathbb{R}^n)$. Considering an ϵ -neighbourhood of the support of each distribution, Lemma 153 guarantees us that

$$K_\epsilon(f) = (\text{supp}(u_1)^\epsilon \times \dots \times \text{supp}(u_m)^\epsilon) \cap \mu^{-1}(\text{supp}(f)), \quad (49)$$

¹⁴Observe that, by standard arguments of topology, the property of a map being proper is equivalent to saying that the image of every open set is open.

is compact since the pre-image of a compact set is compact, being μ proper. Choose now $\rho_i \in C_0^\infty(\mathbb{R}^n)$, $i = 1, \dots, m$ such that $\text{supp}(\rho_1 \otimes \dots \otimes \rho_m) \subset K_\epsilon(f)$ and $\rho \doteq \rho_1 \otimes \dots \otimes \rho_m$ is equal to 1 on $K_0(f)$. Notice that the map

$$\mathcal{D}(\mathbb{R}^n) \ni f \mapsto (u_1 \otimes \dots \otimes u_m, \rho(x_1, \dots, x_m)f(\mu(x_1, \dots, x_m))) \in \mathbb{C}, \quad (50)$$

is well-defined since it is nothing but the application of Theorem 146 to $\rho_1 u_1, \dots, \rho_m u_m$ which are all elements of $\mathcal{E}'(\mathbb{R}^n)$.

Exercise 37. Prove that (50) does not depend on the choice of the cut-off functions $\rho(x_i)$, $i = 1, \dots, m$.

Using by now standard arguments, one can show that (50) defines a sequentially continuous linear functional, hence a distribution. In other words we have proven the following:

Theorem 154. *Let $u_1, \dots, u_m \in \mathcal{D}'(\mathbb{R}^n)$ and let the restriction of the map (48) to $\text{supp}(u_1) \times \dots \times \text{supp}(u_m)$ be proper. Then the convolution $u_1 \star \dots \star u_m$, defined via (50) is a well-defined element of $\mathcal{D}'(\mathbb{R}^n)$. In addition it reduces to Theorem 146 if $m = 2$ and $u_1, u_2 \in \mathcal{E}'(\mathbb{R}^n)$.*

The extension of the convolution to $\mathcal{D}'(\mathbb{R}^n)$ enjoys several notable, structural properties:

Proposition 155. *Under the same assumptions of Theorem 154, it holds*

1. $u_1 \star \dots \star u_m$ is defined recursively: For all $f \in \mathcal{D}(\mathbb{R}^n)$

$$(u_1 \star \dots \star u_m, f) = (u_1, f_{m-1}),$$

where $f_0 = f$ and $f_{j+1}(x) \doteq (u_{m-j}, f_j(x + y))$, where $j = 0, 1, \dots, m-2$,

2. $u_1 \star \dots \star u_m$ is associative: if I and J are disjoint subsets of $\{1, \dots, m\}$, then

$$u_1 \star \dots \star u_m = (\star u_i)_{i \in I} \star (\star u_j)_{j \in J},$$

3. $\text{supp}(u_1 \star \dots \star u_m) \subseteq \text{supp}(u_1) + \dots + \text{supp}(u_m)$, where the $+$ -sign stands for the set sum in a vector space,

4. Suppose that $\{v_j\}_{j \in \mathbb{N}}$ is a sequence of distributions of $\mathcal{D}'(\mathbb{R}^n)$, all supported in $\bar{\Omega}_1$ such that $\lim_{j \rightarrow \infty} v_j = u_1$ in the sense of $\mathcal{D}'(\mathbb{R}^n)$. Then

$$\lim_{j \rightarrow \infty} v_j \star u_2 \star \dots \star u_m = u_1 \star \dots \star u_m,$$

where the limit is taken again in the sense of $\mathcal{D}'(\mathbb{R}^n)$.

Proof. All the statements descend from the standard properties of the convolution and from the defining formula (50). Hence we can safely omit them. $\square$

In the following we will use all these last results extensively. We will not verify explicitly that the hypotheses are fulfilled, especially that the map μ restricted to the relevant domains is proper. We leave this endeavour (Freedom and Independence, thirty seconds to go! Who got the quote?) to the reader.

Consider for simplicity the one dimensional scenario. Choose therefore $u \in \mathcal{D}'(\mathbb{R})$ and $\rho \in C^\infty(\mathbb{R})$ such that $\rho = 0$ for all $x < a$ and $\rho = 1$ for all $x > b$. We call

$$u^+ \doteq \rho u, \quad u^- \doteq (1 - \rho)u.$$

The above construction entails the following notable result:

Exercise 38. Prove that

$$v = \Theta(x) \star u^+ - (1 - \Theta(x)) \star u^-$$

is a well-defined element of $\mathcal{D}'(\mathbb{R})$ such that $\partial v = u$.

In order to generalize the whole construction to n -dimensions, let $x \in \mathbb{R}^n$ and let $x = (x', x_n)$, *i.e.*, x_n is the last coordinate. Define the elements of $\mathcal{D}'(\mathbb{R}^n)$

$$E^+ = \delta_{n-1} \otimes \Theta(x_n), \quad E^- = \delta_{n-1} \otimes (\Theta(x_n) - 1),$$

where δ_{n-1} is the δ -distribution on the remaining subspace $\mathbb{R}^{n-1}$. Using Example 126, it holds that

$$\partial_{x_n} E^+ = \partial_{x_n} E^- = \delta_n.$$

Let now $u \in \mathcal{D}'(\mathbb{R}^n)$ and choose $\rho \equiv \rho(x_n)$ as above. In full analogy to the one dimensional scenario, let

$$u^+ = u\rho(x_n), \quad u^- = u(1 - \rho(x_n)).$$

As a consequence of Exercise 38, it holds that

$$v = E^+ \star u^+ + E^- \star u^-,$$

is a well defined element of $\mathcal{D}'(\mathbb{R}^n)$ such that $\partial_{x_n} v = u$.

It is a legitimate question what is the purpose of the above digression. To start answering, suppose that $k \in \mathcal{D}'(\mathbb{R}^n)$ has been assigned. It defines a map

$$K : D(K) \subseteq \mathcal{D}'(\mathbb{R}^n) \rightarrow \mathcal{D}'(\mathbb{R}^n), \quad u \mapsto K(u) = k \star u \tag{51}$$

where the domain of K is defined according to Theorem 154. Suppose now that $v \in \mathcal{D}'(\mathbb{R}^n)$ has been assigned and that we are looking for those $u \in \mathcal{D}'(\mathbb{R}^n)$ solving

$$K(u) = v.$$

The answer to this question is quite surprising since it is not dependent on v .

Definition 156. We call **fundamental solution** of the operator K as in (51), $E \in \mathcal{D}'(\mathbb{R}^n)$ such that

$$K(E) = k \star E = \delta.$$

It is immediate to realize that E must lie in $D(K)$ to be well-defined and that it allows for the construction of a solution of $K(u) = v$ as

$$u = E \star v,$$

where the left hand side exists whenever the right hand side does, for example whenever $v \in \mathcal{E}'(\mathbb{R}^n)$. Observe that it is of paramount relevance the associativity of the convolution since

$$K(u) = k \star (E \star v) = (k \star E) \star v = \delta \star v = v.$$

The most relevant application of fundamental solutions is at the level of differential operators. For simplicity let us consider

$$P = \sum_{|\alpha| \leq m} a_\alpha \partial^\alpha, \quad (52)$$

where α is any multi-index and $a_\alpha \in \mathbb{C}$ for all α . Bear in mind that the requirement of having constant coefficients can be somehow relaxed, but the analysis would become much more complicated and, thus, we avoid it. The following proposition might be at first glance surprising

Proposition 157. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and let P as in (52). Then P is equivalent to a convolution, namely, if $k \doteq P\delta \in \mathcal{D}'(\mathbb{R}^n)$

$$P(u) = k \star u.$$

Proof. Since $\delta \in \mathcal{E}'(\mathbb{R}^n)$ the convolution is well-defined as per Theorem 146. Applying Exercise 34 and Corollary 148, it holds that

$$k \star u = P\delta \star u = \delta \star Pu = Pu,$$

which is the sought identity. □

In view of this proposition, it is meaningful to ask which is the fundamental solution of a given partial differential operator with constant coefficients (52). In other words, for all $v \in \mathcal{E}'(\mathbb{R}^n)$,

$$Pu = v \implies u = E \star v,$$

that is, we have constructed a particular solution to the PDE $Pu = v$. This entails that solving such partial differential equation restricts to constructing the fundamental solution E of P and to solving the homogeneous equation $Pu = 0$.

Remark 11. It is noteworthy that we have never stated uniqueness of the fundamental solution. Suppose then that there exist two linearly independent fundamental solutions, say $E_1, E_2 \in \mathcal{D}'(\mathbb{R}^n)$. Since $PE_i = \delta$, $i = 1, 2$, per linearity $P(E_1 - E_2) = 0$. In addition, using the associativity of the convolution, it holds that, not only $E_1 - E_2$ is a solution of the homogeneous equation, but also all $(E_1 - E_2) \star v$, with $v \in \mathcal{E}'(\mathbb{R}^n)$. An interesting question is whether all solutions to the homogeneous equation can be characterized in this way. The answer is sometimes negative, e.g. the Helmholtz equation, and sometimes positive, e.g., the wave equation.

As a last step we show a structure theorem for $\mathcal{D}'(\mathbb{R}^n)$ and we give a notable example of fundamental solution, which has important applications to physical models, the Poisson equation in particular.

Theorem 158 (Structure Theorem for $\mathcal{D}'(\mathbb{R}^n)$). *The restriction of any $u \in \mathcal{D}'(\mathbb{R}^n)$ to a bounded open set $X \subset \mathbb{R}^n$ is a derivative of finite order of a continuous function.*

Proof. Since X is bounded, there exists $\sigma \in C_0^\infty(\mathbb{R}^n)$ such that $\sigma|_X = 1$. Hence $u|_X = \sigma u$. Let $x_+ \doteq x\Theta(x)$ and let $E_k \doteq \frac{(x_1)_+^k \dots (x_n)_+^k}{((k-1)!)^n}$ where $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ and $k \in \mathbb{N}$. A direct computation shows that $(\partial_{x_1} \dots \partial_{x_n})^k E_k = \delta$.

Since $\sigma u \in \mathcal{E}'(\mathbb{R}^n)$, we can apply Theorem 139. Hence there exists $N \in \mathbb{N}$ such that

$$\sigma u = (\partial_{x_1} \dots \partial_{x_n})^{N+2} (E_{N+2} \star (\sigma u)),$$

as can verify readily using Exercise 34. To conclude the proof we need to show that $E_{N+2} \star (\sigma u)$ is a continuous function. Let $\rho \in C_0^\infty(\mathbb{R}^n)$ be chosen as in Proposition 113 and consider the associated regularization – see Theorem 150,

$$\phi_\epsilon \doteq (E_{N+2} \star (\sigma u)) \star \rho_\epsilon,$$

where $\rho_\epsilon = \epsilon^{-n} \rho(\frac{x}{\epsilon})$. Using the associativity and the symmetry under exchange of the factors of the convolution

$$\phi_\epsilon = (E_{N+2} \star (\sigma u)) \star \rho_\epsilon = \sigma u \star (E_{N+2} \star \rho_\epsilon) = (\sigma u, (E_{N+2} \star \rho_\epsilon)(x - y)).$$

Still Theorem 150 (rather its proof) entails that $E_{N+2} \star \rho_\epsilon(z)$ converges to $E_{N+2}(z)$ in $C^N(\mathbb{R}^n)$ as $\epsilon \rightarrow 0$. Since σu is a distribution of order N , (34) applied to ϕ_ϵ shows that, uniformly on a compact set

$$\lim_{\epsilon \rightarrow 0} \phi_\epsilon = (u, E_{N+2}(x - y)),$$

which as a partial evaluation yielding a continuous function in x . At the same time, per construction ϕ_ϵ converges to $(\sigma u) \star E_{N+2}$ as $\epsilon \rightarrow 0$. Since the two limits must coincide, the theorem is proven. $\square$

Example 159. We construct the fundamental solution of the *Laplace operator* over $\mathbb{R}^n$

$$\Delta = \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2},$$

where $\{x_i\}_{i=1, \dots, n}$ are the Euclidean coordinates over $\mathbb{R}^n$. Switching from the Euclidean to the spherical coordinates $(r, \theta_1, \dots, \theta_{n-1})$, we can write the Laplace operator as

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{n-1}{r} \frac{\partial}{\partial r} + \frac{\Delta_{\mathbb{S}^{n-1}}}{r^2},$$

where $\Delta_{\mathbb{S}^{n-1}}$ is the Laplace-Beltrami operator on the unit $(n-1)$ -sphere $\mathbb{S}^{n-1}$. Smooth solutions on $(0, \infty) \times \mathbb{S}^{n-1}$ of $\Delta u = 0$ can be easily constructed yielding

$$u = A + Br^{2-n}, \quad n > 2 \quad \text{or} \quad u = A + B \ln r, \quad n = 2,$$

where A, B are arbitrary constants. Yet, these solutions are not defined at $0 \in \mathbb{R}^n$. Let us thus focus on $n > 2$ and let us consider the map $x \mapsto |x|^{2-n}$. This lies in $L^1_{loc}(\mathbb{R}^n)$, hence it identifies an element of $\mathcal{D}'(\mathbb{R}^n)$. Consider now $u \in \mathcal{D}'(\mathbb{R}^n)$ such that

$$u = \Delta |x|^{2-n}.$$

Since $|x| = r$, except at $0 \in \mathbb{R}^n$, $u = 0$. Hence $\text{supp}(u) = \{0\}$. Using Theorem 137, there exists $N \in \mathbb{N} \cup \{0\}$ such that

$$u = \sum_{|\alpha| \leq N} c_\alpha \partial^\alpha \delta,$$

where α is a multi-index and $c_\alpha \in \mathbb{C}$ for every value of α . In order to fix the constants, we observe that $|x|^{2-n}$ is homogeneous of degree $2-n$ and thus u is homogeneous of degree $-n$. At a distributional level, the standard homogeneity condition $\phi(tx) = t^\lambda \phi(x)$, $t > 0$ and $\phi \in C^\infty(\mathbb{R}^n)$ homogeneous of degree λ , translates to requiring that $u \in \mathcal{D}'(\mathbb{R}^n)$ is such that

$$(u, f(x)) = t^{n+\lambda}(u, f(tx)), \quad t > 0, \quad \text{and} \quad \forall f \in \mathcal{D}(\mathbb{R}^n).$$

If we require that the distribution u is homogeneous of degree $-n$, this last formula imposes a constraint to the coefficients c_α :

$$\sum_{|\alpha| \leq N} c_\alpha \partial^\alpha \delta - \sum_{|\alpha| \leq N} t^{-|\alpha|} c_\alpha \partial^\alpha \delta = 0, \quad t > 0$$

which entails that $c_\alpha = 0$ for all α except $\alpha = 0$. This can be proven for example testing the above equality against $\frac{x^\alpha}{\alpha!} f$, with $f \in \mathcal{D}(\mathbb{R}^n)$. In other words we have obtained that

$$\Delta |x|^{2-n} = c\delta,$$

where c is a constant. To compute it, consider $h \in C_0^\infty(\mathbb{R})$ such that h is 1 in a neighbourhood of the origin. Hence

$$c = (\Delta |x|^{2-n}, h) = (|x|^{2-n}, \Delta h) = \text{Vol}(\mathbb{S}^{n-1}) \int_0^\infty dr \, r \left(\frac{d^2 h}{dr^2} + \frac{n-1}{r} \frac{dh}{dr} \right) = (2-n) \text{Vol}(\mathbb{S}^{n-1}).$$

A standard computation shows that $\text{Vol}(\mathbb{S}^{n-1}) = \frac{(2\pi)^{\frac{n}{2}}}{\Gamma(\frac{n}{2})}$. This entails that a fundamental solution of the Laplace operator is for $n > 2$

$$E_\Delta = \frac{\Gamma(\frac{n}{2}) |x|^{2-n}}{(2-n)(2\pi)^{\frac{n}{2}}}.$$

Exercise 39. Prove that the fundamental solution for the Laplace operator in $2D$ is $E_\Delta = -\frac{1}{2\pi} \ln |x| \in \mathcal{D}'(\mathbb{R}^2)$.

Exercise 40. On $\mathbb{R}^n \equiv \mathbb{R} \times \mathbb{R}^{n-1} \ni (t, x)$, let the **heat equation** be $H = \frac{\partial}{\partial t} + \nabla$ where ∇ is the Laplace operator on $\mathbb{R}^{n-1}$. Verify that the fundamental solution for H is

$$E_H = \Theta(t) \frac{e^{-\frac{|x|^2}{4t}}}{(2\sqrt{\pi t})^n},$$

where $\Theta(t)$ is the Heaviside function.

9.3 Distributional Kernels

In this subsection we discuss a topic which is often briefly mentioned, but it is actually of paramount relevance, especially in theoretical physics. To wit we outline the theory of distributional kernels which are at the heart of important objects such as Green operators. Furthermore these kernels are also one, if not the main reason, why in a less mathematically inclined literature, distributions are always written and implicitly also treated as ordinary functions (with singularities somewhere).

In order to understand the concept underneath distributional kernels, we start once more from a rather simple example where all objects involved are as much regular as possible. In particular we are interested in the so-called *integral transforms*. A simple (and not at all general) example can be constructed as follows. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. For every $k \in C^\infty(\Omega \times \Omega')$, we define a map

$$\mathcal{K} : C_0^\infty(\Omega') \rightarrow C^\infty(\Omega), \quad f \mapsto \mathcal{K}(f)(x) \doteq \int_{\Omega'} d^n y \, k(x, y) f(y). \quad (53)$$

For the applications in physics it is important to be able to extend the target space from the smooth functions to the distributions over Ω . Our goal is to understand when it is possible and we will realize that the most general starting point is actually a distribution $k \in \mathcal{D}'(\Omega \times \Omega')$.

9.3.1 Schwartz Kernels

Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let now $k \in \mathcal{D}'(\Omega \times \Omega')$. Per restriction k defines a linear map on $\mathcal{D}(\Omega) \times \mathcal{D}(\Omega')$ via the defining relation

$$\mathcal{D}(\Omega) \times \mathcal{D}(\Omega') \ni (f, h) \mapsto (k, f \otimes h) \in \mathbb{C}.$$

With a slight abuse of notation we will keep the symbol k in place of the less appealing $k|_{\mathcal{D}(\Omega) \times \mathcal{D}(\Omega')}$. Notice that, being $k \in \mathcal{D}'(\Omega \times \Omega')$, even its restriction to $\mathcal{D}(\Omega) \times \mathcal{D}(\Omega')$ satisfies (34): There exist two compact sets $K \subset \Omega$ and $K' \subset \Omega'$, a constant $C_{K, K'} \geq 0$ and an integer $N \in \mathbb{N} \cup \{0\}$ such that, for all $f \in \mathcal{D}(\Omega)$ and for all $h \in \mathcal{D}(\Omega')$

$$|(k, f \otimes h)| \leq C_{K, K'} \sum_{|\alpha| + |\beta| \leq N} \sup |\partial^\alpha f \partial^\beta h| \leq C_{K, K'} \left(\sum_{|\alpha| \leq N} \sup |\partial^\alpha f| + \sum_{|\beta| \leq N} \sup |\partial^\beta h| \right). \quad (54)$$

As a consequence, if we fix one of the two test functions, we can construct two additional distributions, namely

$$k_f \in \mathcal{D}'(\Omega'), \quad \mathcal{D}(\Omega') \ni h \mapsto k_f(h) \doteq (k, f \otimes h), \quad (55)$$

and equivalently

$$k_h \in \mathcal{D}'(\Omega), \quad \mathcal{D}(\Omega) \ni f \mapsto k_h(f) \doteq (k, f \otimes h).$$

In addition (54) guarantees us also that the assignment $C_0^\infty(\Omega) \ni f \mapsto k_f$ (and equivalently $C_0^\infty(\Omega') \ni h \mapsto k_h$) is sequentially continuous – see Definition 116. In other words, if $\{f_j\}_{j \in \mathbb{N}}$ is a sequence of elements in $C_0^\infty(\Omega)$ such that $\lim_{j \rightarrow \infty} f_j = 0$, then $\lim_{j \rightarrow \infty} k_{f_j} = 0$ in $\mathcal{D}'(\Omega')$. In view of all these properties, the following definition is justified:

Definition 160. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. We call **Schwartz (or distribution) kernel** any $k \in \mathcal{D}'(\Omega \times \Omega')$ which generates a distribution $k_f \in \mathcal{D}'(\Omega')$ defined as in (55).*

Any Schwartz kernel comes accompanied with a companion kernel which can be defined as follows. Observe that the swap $\Omega \times \Omega' \ni (x, y) \mapsto (y, x) \in \Omega' \times \Omega$ induces a counterpart at the level of test functions (a pull-back map – wait, we shall define it later... later):

$${}^t: C_0^\infty(\Omega' \times \Omega) \rightarrow C_0^\infty(\Omega \times \Omega'), \quad g(y, x) \mapsto {}^t(g)(x, y) \equiv g^t(x, y) \doteq g(y, x). \quad (56)$$

Notice that t can be inverted and, without much risk of causing confusion, we will write the inverse still with the same symbol omitting the superscript -1 . Using a duality argument exactly as in the previous sections,

Definition 161. *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let $k \in \mathcal{D}'(\Omega \times \Omega')$ be a Schwartz kernel. We call **transpose kernel** ${}^t k \in \mathcal{D}'(\Omega' \times \Omega)$ the distribution defined as*

$$({}^t k, g) \doteq (k, {}^t g), \quad \forall g \in \mathcal{D}(\Omega' \times \Omega).$$

Remark 12. *Notice that any transpose kernel ${}^t k$ generates a map ${}^t \mathcal{K} : C_0^\infty(\Omega) \rightarrow \mathcal{D}'(\Omega')$ such that*

$$({}^t \mathcal{K}(f), h) = (k, f \otimes h) = (\mathcal{K}(h), f). \quad \forall f \in \mathcal{D}(\Omega) \text{ and } \forall h \in \mathcal{E}(\Omega') \quad (57)$$

Notice that ${}^t k$ satisfies (34) since it is inherited from the one of k which is per hypothesis a distribution. It is also straightforward to see that the definition of distribution kernel coincides with (53) whenever $k \in C^\infty(\Omega \times \Omega') \subset \mathcal{D}'(\Omega')$. The overall idea of Schwartz kernels is to identify a sequentially continuous map from $C_0^\infty(\Omega) \rightarrow \mathcal{D}'(\Omega \times \Omega')$ whenever we start from $k \in \mathcal{D}'(\Omega \times \Omega')$. The content of the next, very famous (well, at least, in functional analysis... we are not talking about singers. By the way, this morning I read that one of the members of the Iron Maiden band is also a pilot for Meridiana. Did you know that? Cool, isn't it?) theorem proves that also the converse is true:

Theorem 162 (Schwartz Kernel Theorem). *Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let $\mathcal{K} : C_0^\infty(\Omega') \rightarrow \mathcal{D}'(\Omega)$. It is a sequentially continuous, linear map if and only if it is generated by a Schwartz kernel $k \in \mathcal{D}'(\Omega \times \Omega')$ via the defining identity:*

$$(\mathcal{K}(f), h) = (k, f \otimes h). \quad \forall f \in \mathcal{D}(\Omega') \text{ and } \forall h \in \mathcal{D}(\Omega)$$

In addition k is completely determined by $\mathcal{K}$.

Proof. We need only to prove $\implies$ since the reverse implication has been already proven in the first part of the section.

WARNING: THE MOST DIFFICULT PROOF OF THE COURSE.

Let $\mathcal{K}$ be as per hypothesis and suppose that we have identified a distribution $k \in \mathcal{D}'(\Omega \times \Omega')$ as required. Uniqueness is then guaranteed by Theorem 143 since $\mathcal{D}(\Omega) \times \mathcal{D}(\Omega')$ is dense in $\mathcal{D}(\Omega \times \Omega')$. Hence we need only to establish existence and, for simplicity we assume that $\Omega = \mathbb{R}^n$ and $\Omega' = \mathbb{R}^m$. The general case can be obtained using a partition of unity, but it would complicate an already heavy proof, hence we avoid it.

Let us assume thus that $\mathcal{K} : C_0^\infty(\mathbb{R}^m) \rightarrow \mathcal{D}'(\mathbb{R}^n)$. We define a bilinear form $B_{\mathcal{K}} : C_0^\infty(\mathbb{R}^m) \times C^\infty(\mathbb{R}^n) \rightarrow \mathbb{C}$ as

$$B_{\mathcal{K}}(h, f) \doteq (\mathcal{K}(h), f). \quad \forall f \in C_0^\infty(\mathbb{R}^n) \text{ and } \forall h \in C_0^\infty(\mathbb{R}^m).$$

As a first step, we look for an estimate similar to (34). Consider the following two compact sets: For any but fixed $b > 0$, let

$$K_b \doteq \{x \in \mathbb{R}^n \mid |x_i| \leq b, i = 1, \dots, n\},$$

where $\{x_i\}_{i=1, \dots, n}$ stand for the standard Euclidean coordinates on $\mathbb{R}^n$ and let

$$K'_b \doteq \{y \in \mathbb{R}^m \mid |y_j| \leq b, j = 1, \dots, m\},$$

where $\{y_j\}_{j=1, \dots, m}$ are the standard Euclidean coordinates on $\mathbb{R}^m$. Since $\mathcal{K}(h) \in \mathcal{D}'(\mathbb{R}^n)$, (34) for this distribution entails the existence of a constant $C_{K_b}(h)$, depending also on the choice of h , and of $N(h) \in \mathbb{N} \cup \{0\}$ such that

$$|B_{\mathcal{K}}(h, f)| \leq C_{K_b}(h) \sum_{|\alpha| \leq N(h)} \sup |\partial^\alpha f|, \quad \forall f \in C^\infty(K_b)$$

where α is a suitable multi-index. In addition we know that $\mathcal{K}$ is per hypothesis sequentially continuous. Hence the map $C_0^\infty(\mathbb{R}^m) \ni h \mapsto B_{\mathcal{K}}(f, h)$ identifies for all $f \in C_0^\infty(\mathbb{R}^n)$ an element of $\mathcal{D}'(\mathbb{R}^m)$. In other words, still (34) entails the existence of a constant $C'_{K'_b}(f)$ and of $M(f) \in \mathbb{N} \cup \{0\}$ such that

$$|B_{\mathcal{K}}(h, f)| \leq C'_{K'_b}(f) \sum_{|\beta| \leq M(f)} \sup |\partial^\beta h|, \quad \forall h \in C^\infty(K'_b)$$

where β is a suitable multi-index. We have shown that $B_{\mathcal{K}}$ is separately continuous on $C^\infty(K_b) \times C^\infty(K'_b)$. This entails that $B_{\mathcal{K}}$ is jointly continuous¹⁵, namely there exists a constant $C_{K, K'}$ and

¹⁵We are using a rather technical result on the theory of locally convex vector spaces, c.f. [6, §34.2].

an integer $N' \in \mathbb{N} \cup \{0\}$ such that

$$|B_{\mathcal{K}}(h, f)| \leq C_{K, K'} \sum_{|\alpha| \leq N'} \sup |\partial^\alpha f| \sum_{|\beta| \leq N'} \sup |\partial^\beta h|. \quad \forall f \in C^\infty(K_b) \text{ and } \forall h \in C^\infty(K'_b)$$

The next step consists of extending $B_{\mathcal{K}}$ to a distribution on $\mathbb{R}^n \times \mathbb{R}^m$. Consider now $K_a \times K'_a$ with $0 < a < b$. It is still a compact set. Let $\chi \in C^\infty(K_a \times K'_a)$ and choose two cut-off functions, say $\rho \in C^\infty(K_b)$ and $\sigma \in C^\infty(K'_a)$ which are respectively 1 on K_a and K'_a . Let us extend $\chi \equiv \chi(x, y)$ to a function on the whole $\mathbb{R}^n \times \mathbb{R}^m$ which is periodic of period b separately on each argument. With a slight abuse of notation, we refer to this function still as χ . The Fourier expansion of χ is

$$\chi(x, y) = \sum_{\mathbb{Z}^n \times \mathbb{Z}^m} \hat{\chi}_{s, t} \rho(x) \sigma(y) E(s \cdot x) E(t \cdot y),$$

where

$$E(\lambda) \doteq \exp \left(2\pi i \frac{\lambda}{b} \right),$$

and where $s \in \mathbb{Z}^n$, while $t \in \mathbb{Z}^m$. The symbol $\cdot$ indicates the standard Euclidean inner product one the integers are embedded in $\mathbb{R}$. In addition the Fourier theory tells us that

$$\hat{\chi}_{s, t} = b^{-(n+m)} \int_{K_b \times K'_b} d^n x d^m y \chi(x, y) E(-s \cdot x) E(-t \cdot y).$$

Integrating this expression by parts, we obtain that, for any $M \in \mathbb{N} \cup \{0\}$ there exists a constant $C_1(M) \geq 0$ such that

$$(1 + |s|)^M (1 + |t|)^M |\hat{\chi}| \leq C_1(M) \sum_{\substack{|\alpha| \leq M \\ |\beta| \leq M}} \sup |\partial_x^\alpha \partial_y^\beta \chi|. \quad (s, t) \in \mathbb{Z}^n \times \mathbb{Z}^m$$

In addition, since $B_{\mathcal{K}}$ is jointly continuous we know that there must exist a constant $C_2 \geq 0$ and $N' \in \mathbb{N} \cup \{0\}$ (the same as above) such that

$$|B_{\mathcal{K}}(\rho(x) E(s \cdot x), \sigma(y) E(t \cdot y))| \leq C_2 (1 + |s|)^{N'} (1 + |t|)^{N'}. \quad (s, t) \in \mathbb{Z}^n \times \mathbb{Z}^m$$

These two last estimates can be combined to obtain a third constant $C_3 \geq 0$ such that

$$\sum_{\mathbb{Z}^n \times \mathbb{Z}^m} |\hat{\chi}_{s, t} B_{\mathcal{K}}(\rho(x) E(s \cdot x), \sigma(y) E(t \cdot y))| \leq C_3 \sum_{\substack{|\alpha| \leq M \\ |\beta| \leq M}} \sup |\partial_x^\alpha \partial_y^\beta \chi|, \quad (58)$$

where now M is not arbitrary, but sufficiently large. Equation (58) suggests to define a continuous linear functional $k : C^\infty(K_a \times K'_a) \rightarrow \mathbb{C}$ as

$$(k, g) \doteq \sum_{\mathbb{Z}^n \times \mathbb{Z}^m} \hat{g}_{s, t} B_{\mathcal{K}}(\rho(x) E(s \cdot x), \sigma(y) E(t \cdot y)), \quad (59)$$

where g plays the role of χ . This definition is meaningful since the series converges absolutely and the sum on the right hand side is bounded by (58).

We are almost done. First of all we notice that (59) satisfies the identity of the statement of the theorem, namely for every $f \in C^\infty(K_a)$ and for every $h \in C^\infty(K'_a)$, we have that

$$(k, f \otimes h) = \sum_{\mathbb{Z}^n \times \mathbb{Z}^m} (\mathcal{K}(\psi_h), \phi(f)), \quad (60)$$

where

$$\phi_f \doteq \widehat{f}\rho(x)E(s \cdot x), \quad \psi_h \doteq \widehat{h}\sigma(y)E(t \cdot y),$$

while

$$\widehat{f} = b^{-n} \int_{K_b} d^n x f(x) E(-s \cdot x) \quad \text{and} \quad \widehat{h} = b^{-n} \int_{K'_b} d^n y h(y) E(-t \cdot y).$$

The construction can be carried over to obtaining a sequence of distributions k_j built over $K_j \times K'_j$. We can now use Theorem 143 to conclude that the restriction of k_j to $K_l \times K'_l$ with $l \leq j$ agrees with k_l . Taking the limit for $j \rightarrow \infty$, we conclude the existence of the sought distribution $k \in \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}^m)$. $\square$

9.3.2 Regular Kernels

In the plethora of distribution kernels, we can identify a notable subclass, which is especially relevant for applications to analysing the structural properties of certain partial differential operators, such as for example those ruling the wave equation.

Definition 163. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. A map $k : C_0^\infty(\Omega') \rightarrow C^\infty(\Omega)$ is called **continuous** if, for any compact subsets $K \subset \Omega$ and $K' \subset \Omega'$, for any multi-index α , there exist a constant $C_{K,K'} \geq 0$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$\sup_{x \in K} \{|\partial^\alpha k[h](x)|\} \leq C_{K,K'} \sum_{|\beta| \leq N} \sup |\partial^\beta h|. \quad \forall h \in C_0^\infty(K')$$

As a consequence of this definition we can prove the following result:

Theorem 164. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let $k \in \mathcal{D}'(\Omega \times \Omega')$ be a Schwartz kernel. Suppose that ${}^t\mathcal{K} : C_0^\infty(\Omega) \rightarrow C^\infty(\Omega')$ is a continuous map in the sense of Definition 163. Then $\mathcal{K}$ can be extended to $\mathcal{K} : \mathcal{E}'(\Omega') \rightarrow \mathcal{D}'(\Omega)$. In addition $\mathcal{K}$ is sequentially continuous as follows: If a sequence $\{u_j\}_{j \in \mathbb{N}}$ converges to $u \in \mathcal{E}'(\Omega)$ as $j \rightarrow \infty$ in $\mathcal{D}'(\Omega)$ and if all $\text{supp}(u_j)$, $j \in \mathbb{N}$ are contained in the same compact set, then

$$\lim_{j \rightarrow \infty} \widetilde{\mathcal{K}}(u_j) = \widetilde{\mathcal{K}}(u),$$

where the limit is taken in $\mathcal{D}'(\Omega)$.

Proof. Let $u \in \mathcal{E}'(\Omega')$ and define $\tilde{\mathcal{K}}(u) \in \mathcal{D}'(\Omega)$ as follows:

$$(\tilde{\mathcal{K}}(u), f) = (u, {}^t\mathcal{K}(f)). \quad \forall f \in \mathcal{D}(\Omega)$$

Since ${}^t\mathcal{K}$ is continuous per hypothesis, we can infer that $\tilde{\mathcal{K}}(u)$ is a well-defined distribution in $\mathcal{D}'(\Omega)$. In addition, being defined $\tilde{\mathcal{K}}(u)$ per duality, it is sequentially continuous since the right hand side enjoys this property.

We need only to prove that $\tilde{\mathcal{K}}$ extends $\mathcal{K}$. Let $u \in C_0^\infty(\Omega') \subset \mathcal{E}'(\Omega')$. Per definition

$$(\tilde{\mathcal{K}}(u), f) = (u, {}^t\mathcal{K}(f)) = \int_{\Omega'} d^m y u(y) {}^t\mathcal{K}[f](y).$$

Since ${}^t\mathcal{K}[f] \in C^\infty(\Omega') \subset \mathcal{D}'(\Omega')$, we can reinterpret the last chain of equalities as yielding

$$(\tilde{\mathcal{K}}(u), f) = \int_{\Omega'} d^m y u(y) {}^t\mathcal{K}[f](y) = ({}^t\mathcal{K}(f), u) = (k, u \otimes f) = (\mathcal{K}(u), f),$$

where we used (57). □

Notice that, with a slight abuse of notation, the tilde symbol on top of k is often dropped and we will adopt this convention.

Definition 165. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let $k \in \mathcal{D}'(\Omega \times \Omega')$ be a Schwartz kernel. If $\mathcal{K} : C_0^\infty(\Omega') \rightarrow C^\infty(\Omega)$ and ${}^t\mathcal{K} : C_0^\infty(\Omega) \rightarrow C^\infty(\Omega')$, generated from ${}^t k$, are both continuous in the sense of Definition 163, then k is called a **regular kernel**.

Corollary 166. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let $k \in \mathcal{D}'(\Omega \times \Omega')$ be a regular kernel. Both k and ${}^t k$ extend to sequentially continuous maps respectively from $\mathcal{E}'(\Omega') \rightarrow \mathcal{D}'(\Omega)$ and from $\mathcal{E}'(\Omega) \rightarrow \mathcal{D}'(\Omega')$.

Proof. This is a straightforward consequence of Theorem 164. □

Theorem 167. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$ and let $\mathcal{K} : C_0^\infty(\Omega') \rightarrow C^\infty(\Omega)$ be a continuous map. Then its Schwartz kernel is

$$(k, g) \doteq \int_{\Omega} d^n x \mathcal{K}[g](x), \quad \forall g \in \mathcal{D}(\Omega \times \Omega')$$

where $\mathcal{K}[g](x)$ is the image of $\Omega' \ni y \mapsto g(x, y)$ under the action of $\mathcal{K}$.

Proof. Following the same notation as in the proof of the Schwartz kernel theorem 160, we focus on $g \in C^\infty(K_a \times K'_a)$ where $a > 0$ and $K_a \times K'_a$ is compact. The general case can be obtained from this one using a partition of unity. Using (60)

$$(k, g) = \sum_{\mathbb{Z}^n \times \mathbb{Z}^m} \int_{K_a} d^n x \hat{g}_{s,t} \rho(x) E(s \cdot x) \mathcal{K}(\sigma(y) E(t \cdot y)).$$

Definition 163 entails that there exists a constant $C \geq 0$ depending on $\text{supp}(\rho)$ and on $\text{supp}(\sigma)$ such that

$$|\rho(x)E(s \cdot x)\mathcal{K}(\sigma(y)E(t \cdot y))| \leq C \sum_{|\alpha| \leq N} \sup |\partial^\alpha(\sigma(y)E(t \cdot y))|,$$

where α is a suitable multi-index and $N \in \mathbb{N} \cup \{0\}$. We can find a second constant $C_1 \geq 0$ such that

$$|\rho(x)E(s \cdot x)\mathcal{K}(\sigma(y)E(t \cdot y))| \leq C_1(1 + |s|)^N(1 + |t|)^N.$$

This estimate entails that

$$\sum_{\mathbb{Z}^n \times \mathbb{Z}^m} \widehat{g}_{s,t} \rho(x)E(s \cdot x)\mathcal{K}(\sigma(y)E(t \cdot y))$$

converges uniformly to $\mathcal{K}[g](x)$. We are thus allowed to exchange the sum with the integral in (k, g) obtaining the sought result. $\square$

9.4 Coordinate Transformations on Distributions

There is only one last operation that we need to define at the level of distributions. Let us explain it when we are working with regular functions. For simplicity we consider $\phi \in C^\infty(\mathbb{R})$ and the associated distribution $u_\phi \in \mathcal{D}'(\mathbb{R})$ such that

$$(u_\phi, f) = \int_{\mathbb{R}} dx \phi(x) f(x). \quad \forall f \in \mathcal{D}(\mathbb{R})$$

Suppose that we want to make a change of coordinates, say $y = y(x)$ where y is assumed to have range over $\mathbb{R}$. For any $f \in \mathcal{D}(\mathbb{R})$, the above expression becomes

$$(u_\phi, f) = \int_{\mathbb{R}} dx \phi(x) f(x) = \int_{\mathbb{R}} dy \frac{dx}{dy} \phi(x(y)) f(x(y)).$$

Since $y = y(x)$ and its inverse $x = x(y)$ are smooth $f(x(y))$ is still smooth and compactly supported and $\frac{dx}{dy} \phi(x(y)) \in C^\infty(\mathbb{R})$. The rightmost hand side defines thus again a distribution obtained from the change of coordinates.

Our goal is to understand how distributions behave under the action of a coordinate transformation or of a more general class of maps that we will be defining.

Definition 168. Let $\Omega \subseteq \mathbb{R}^n$ and $\Omega' \subseteq \mathbb{R}^m$. A map $\iota \in C^\infty(\Omega, \Omega')$ is called a **diffeomorphism** if it is a bijection with inverse map ι^{-1} smooth.

It is immediate from this definition that the requirement of being a bijection forces $m = n$ and the determinant of the Jacobian of the map¹⁶ $J[\iota](x) = \det[\mathcal{J}[\iota]](x)$ does not vanish on Ω . The following theorem characterizes the action of a diffeomorphism on a distribution. Prior to it we need an ancillary definition and result:

¹⁶Recall that the Jacobian of ι is nothing but the matrix $\frac{\partial \iota_j}{\partial x_i}$.

Definition 169. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. A homomorphism $\mu : C_0^\infty(\Omega) \rightarrow C_0^\infty(\Omega')$ is said to be **continuous** if

1. for every compact set $K' \subset \Omega'$, there exists a compact set $K \subset \Omega$ such that $\text{supp}(\mu(f)) \subseteq K'$ for all $f \in C_0^\infty(\Omega)$,
2. for every compact set $K' \subset \Omega'$ and for any $N \in \mathbb{N} \cup \{0\}$, there exists a constant $C_{K',N}$ and an integer $M \equiv M(K', N) \in \mathbb{N} \cup \{0\}$ such that

$$\sum_{|\alpha| \leq N} \sup |\partial^\alpha(\mu(f))| \leq C_{K',N} \sum_{|\beta| \leq M} \sup |\partial^\beta f|, \quad \forall f \in C_0^\infty(\Omega) \text{ with } \text{supp}(f) \subseteq K$$

where α and β are suitable multi-indices.

Corollary 170. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. Let $\mu : C_0^\infty(\Omega) \rightarrow C_0^\infty(\Omega')$ be a continuous homomorphism. Let ${}^t\mu : \mathcal{D}'(\Omega') \rightarrow \mathcal{D}'(\Omega)$ be defined as

$$({}^t\mu(v), f) = (v, \mu(f)). \quad \forall v \in \mathcal{D}'(\Omega') \text{ and } \forall f \in \mathcal{D}(\Omega)$$

Then ${}^t\mu$ is sequentially continuous.

Proof. Let $K \subseteq \Omega$ be a compact set and let $f \in \mathcal{D}(\Omega)$ with $\text{supp}(f) \subseteq K$. On account of Definition 169, there exists $K' \subseteq \Omega'$ such that $\mu(f)$ and $\text{supp}(\mu(f)) \subseteq K'$. Since $v \in \mathcal{D}'(\Omega')$ per hypothesis, (34) entails that there must exist a constant $C_{K'} \geq 0$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$|(v, \mu(f))| \leq C_{K'} \sum_{|\alpha| \leq N} \sup |\partial^\alpha(\mu(f))|,$$

where α is a suitable multi-index. If we combine this fact with the estimate in Definition 169, we obtain (34) for ${}^t\mu(v)$. hence ${}^t\mu(v)$ defines an element of $\mathcal{D}'(\Omega)$. It is now straightforward from the construction that, if $\{v_k\}_{k \in \mathbb{N}}$ is a sequence of distributions in $\mathcal{D}'(\Omega')$ such that $\lim_{k \rightarrow \infty} v_k = v$, then $\lim_{k \rightarrow \infty} {}^t\mu(v_k) = {}^t\mu(v)$ in $\mathcal{D}'(\Omega)$. This is sequential continuity, as sought. $\square$

Theorem 171. Let $\Omega, \Omega' \subseteq \mathbb{R}^n$ and let $\iota : \Omega \rightarrow \Omega'$ be a diffeomorphism. For any $v \in \mathcal{D}'(\Omega')$ the **pull-back** of v is $\iota^*v \in \mathcal{D}'(\Omega)$ defined as

$$(\iota^*v, f) \doteq (v, |J[\iota^{-1}]| \iota_*f), \quad \forall f \in \mathcal{D}(\Omega)$$

where ι_*f is the **push-forward** of f , namely the pull-back¹⁷ via ι^{-1} , defined as

$$(\iota_*f)(y) \doteq (f \circ \iota^{-1})(x).$$

¹⁷We recall that, if $\gamma \in C^\infty(\mathbb{R}^n; \mathbb{R}^m)$, the pull-back via γ is a map $\gamma^* : C^\infty(\mathbb{R}^m) \rightarrow C^\infty(\mathbb{R}^n)$ such that $\gamma^*\phi \doteq \phi \circ \gamma$ for all $\phi \in C^\infty(\mathbb{R}^m)$.

Proof. Since ι^{-1} is smooth, hence in particular continuous, as per Definition 168, it maps compact sets in Ω' into compact sets in Ω . A standard result from basic calculus tells us that $J[\iota^{-1}] = (\iota^{-1})^* (J[\iota]^{-1})$ on Ω' . In addition $C_0^\infty(\Omega) \ni f \mapsto (\iota^{-1})^* f |J[\iota^{-1}]|$ is a continuous map from $C_0^\infty(\Omega) \rightarrow C_0^\infty(\Omega')$ in view of Definition 169. Thanks to Corollary 170, ι^* is a sequentially continuous map from $\mathcal{D}'(\Omega') \rightarrow \mathcal{D}'(\Omega)$. $\square$

Remark 13. Under the hypotheses of the previous theorem, notice that if $v \in C^\infty(\Omega)$, then, for every $f \in \mathcal{D}(\Omega)$, it holds

$$(\iota^* v, f) = \int_{\Omega'} d^n y v(y) J[\iota^{-1}](y) [\iota_*(f)](y) = \int_{\Omega} d^n x (v \circ \iota)(x) f(x),$$

where we changed coordinates as $y = \iota(x)$.

Exercise 41. Under the assumptions of Theorem 171, show that the derivatives of $\iota^* u$ satisfy the chain rule.

Example 172. An interesting consequence of Theorem 171 with manifold applications in several concrete calculations is the following. Under the assumption of the theorem just mentioned, assume that $u = \delta$ and, consequently that $0 \in \Omega'$.

In this case, it holds that, for any $f \in \mathcal{D}(\Omega)$,

$$(\iota^* \delta, f) = (\delta, |J[\iota^{-1}]| \iota_*^{-1} f) = J[\iota^{-1}](0) f(\iota^{-1}(0)) = (|J[\iota^{-1}]|(0) \delta_{\iota^{-1}(0)}, f),$$

where we used that $|J[\iota]|$ is continuous and thus the results of Example 130 are applicable, up to the straightforward extension of the results from smooth to continuous functions. In particular we have shown that

$$\iota^* \delta = |J[\iota^{-1}]|(0) \delta_{\iota^{-1}(0)}.$$

This formula is often found in the literature in a slightly, albeit equivalent, form. Let $\iota^{-1}(x)$ be indicated for simplicity as $\phi(x)$. Let $\{x_i\}_{i=1, \dots, n}$ be the points for which $\phi(x_i) = 0$. Then

$$\phi^* \delta \equiv \delta(\phi(x)) = \sum_{i=1}^n \frac{\delta(x - x_i)}{|J[\phi]|(x_i)},$$

where we used the Viking notation $\delta(x - x_i)$ in place of δ_{x_i} since it is more common in the literature. Notice that, in the special case of $\Omega = \mathbb{R}$, the formula becomes even nicer namely

$$\delta(\phi(x)) = \sum_{i=1}^n \frac{\delta(x - x_i)}{|\phi'(x_i)|},$$

where the prime symbol stands for the derivative with respect to the Euclidean coordinate x along $\mathbb{R}$.

A slight disadvantage of Theorem 171 lies in the constraint of taking the source and the target manifold of the same dimension. It is possible to relax slightly this requirement considering a more general class of functions.

Definition 173. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. A map $\Phi \in C^\infty(\Omega, \Omega')$ is called a **submersion** if $m \leq n$ and the Jacobian $\mathcal{J}[\Phi](x)$ has maximal rank for all $x \in \Omega$.

Theorem 174. Let $\Omega \subseteq \mathbb{R}^n$ and let $\Omega' \subseteq \mathbb{R}^m$. If $\Phi : \Omega \rightarrow \Omega'$ is a submersion, then there exists a unique, sequentially continuous pull-back map $\Phi^* : \mathcal{D}'(\Omega') \rightarrow \mathcal{D}'(\Omega)$.

Proof. Suppose that $m = n$. In this case, we can use the inverse function theorem to infer that the submersion is locally a diffeomorphism. Up to the use of a partition of unity and of the Heine-Borel theorem, the proof is that of Theorem 171.

Let $m < n$ and let us write

$$\{\Phi(x) < t\} \doteq \{x \in \Omega \mid \Phi_j(x) < t_j, \quad j = 1, \dots, m\}.$$

For any $f \in C_0^\infty(\Omega)$ and for any $t \in \Omega'$,

$$f_\Phi(t) = \frac{\partial^m}{\partial t_1 \dots \partial t_m} \int_{f(x) < t} d^n x f(x).$$

Take a point $x_0 \in \Omega$. Since f is a submersion, we can use the implicit function theorem to infer the existence of a neighbourhood $U_0 \ni x_0$ in Ω and of a diffeomorphism $\lambda : U_0 \rightarrow \lambda(U_0)$ such that, for any $x \in U$, $\lambda(x) = (\zeta', \zeta'')$ where $\zeta' \in \mathbb{R}^{n-m}$ and $\zeta'' = \Phi(x) \in \mathbb{R}^m$. One has

$$f_\Phi(\zeta'') = \int_{\Omega' \cap \lambda(U)} d^m \zeta' (\lambda^{-1})^* f(\zeta', \zeta'') |J[\lambda^{-1}](\zeta', \zeta'')|,$$

where $\zeta'' = f(x)$. From this formula it descends that on $\lambda(U)$,

$$\Phi^*(u) = |J[\lambda^{-1}](\zeta)| 1(\zeta') \otimes u(\zeta''),$$

where $1(\zeta')$ stands for the constant distribution on $\lambda(U) \cap \mathbb{R}^{n-m}$. This is the restriction of the pull-back map $\Phi^* : \mathcal{D}'(\Omega') \rightarrow \mathcal{D}'(U)$. To obtain a global result, we use Theorem 123 to patch together distributions defined locally obtaining also that

$$(\Phi^* u, f) = (u, f_\Phi). \quad \forall u \in \mathcal{D}'(\Omega') \text{ and } \forall f \in \mathcal{D}(\Omega)$$

As a consequence $\Phi^* : \mathcal{D}'(\Omega') \rightarrow \mathcal{D}'(\Omega)$ is sequentially continuous. □

10 Tempered Distributions and the Fourier Transform

Goal of this section is twofold: The first is to introduce a third space of test functions and of distributions which sits in between the rather restrictive $\mathcal{E}'(\mathbb{R}^n)$ and the huge $\mathcal{D}'(\mathbb{R}^n)$. The reason for doing it is mainly related to the second goal, outlining the theory of Fourier transforms. Let us start from giving an heuristic reason for this new operation: generalizing the standard Fourier analysis. As one learns usually in basic calculus, if we consider any complex valued $\phi \in C^\infty(\mathbb{R})$ which is periodic of period 1 (both the period and the domain $\mathbb{R}$ are chosen for simplicity and not for the sake of generality), we can decompose it as follows:

$$\phi(x) = \sum_{n=-\infty}^{\infty} \phi_n e^{inx},$$

where the coefficients $\phi_n \in \mathbb{C}$ are defined as

$$\phi_n = \int_{\mathbb{R}} dx \phi(x) e^{-inx}.$$

The convergence of the Fourier series requires a detailed analysis which is beyond the scopes of these notes. Instead we mentioned it because it leads to the natural question whether a similar expansion exists for functions which are not periodic. The answer is positive and it is at the hearth of the theory of Fourier transforms which appears in physics in so many contexts both at a theoretical and at an applied level.

In order to outline the main concepts, let us start from the simplest case:

Definition 175. Let $\phi \in L^1(\mathbb{R}^n)$. We call **Fourier transform** of ϕ the function

$$\widehat{\phi}(k) \doteq \int_{\mathbb{R}^n} d^n x e^{ik \cdot x} \phi,$$

where $\cdot$ stands for the Euclidean inner product.

Remark 14. The Fourier transform is defined up to a convention (namely choosing between i and $-i$ at the exponential) and to a normalization factor. The latter plays no role in its structural properties, except for one. We will see that the Fourier transform can be extended to a unitary operator on $L^2(\mathbb{R}^n)$ provided that we replace $d^n x$ with $\frac{d^n x}{(2\pi)^{\frac{n}{2}}}$. Starting from the beginning with this factor would force us to bring along several normalization factors, which are not playing a role, but they are easily mishandled.

It is clear from the definition that several questions are open, namely

- Does Definition 175 make sense?
- If it does, in which space lies $\widehat{\phi}$?

- Can we extend the range of applicability from just $L^1(\mathbb{R}^n)$?

The goal of the rest of this section is to give an answer to these questions and to outline some further applications of these answers. Let us start from the case at hand proving that the above definition is well-founded. The result is an application of Fubini theorem whose statement and proof for integrable functions can be found in any textbook of calculus.

We start with a preliminary result:

Theorem 176. *Let $\Omega \subseteq \mathbb{R}^n$ and let $I \subseteq \mathbb{R}$ with coordinate t . Suppose that $\phi : \Omega \times I \rightarrow \mathbb{C}$ is such that*

- (a) $\partial_t \phi(x, t)$ exists as a continuous function for almost all $x \in X$,
- (b) $\exists g \in L^1(X)$ with $g \geq 0$ such that $|\partial_t \phi| \leq g$ for all $t \in I$,
- (c) There exists $t_0 \in I$ such that $\int_{\Omega} d^n x \phi(x, t_0)$ exists.

Then it holds that

$$\Phi(t) \doteq \int_{\Omega} d^n x \phi(x, t),$$

exists for all $t \in I$ and it is continuously differentiable with

$$\frac{d\Phi}{dt} = \int_{\Omega} d^n x \partial_t \phi(x, t).$$

Proof. Condition (b) guarantees that $\int_{\Omega} d^n x \partial_t \phi(x, t)$ is well defined for all $t \in I$. If we call such function $F_1(t)$, we can use condition (a) and Lebesgue dominated convergence (actually, if you know it, Lebesgue bounded convergence suffices) to conclude that F_1 is a continuous function on I . In addition thereon

$$\int_{t_0}^t ds F_1(s) = \int_{t_0}^t ds \int_{\Omega} d^n x \partial_s \phi(x, s) = \int_{\Omega} d^n x \int_{t_0}^t ds \partial_s \phi(x, s) = \int_{\Omega} d^n x (\phi(x, t) - \phi(x, t_0)),$$

for almost all $x \in \Omega$. Observe that we used Fubini theorem in the second equality and the fundamental theorem of calculus in the third. By linearity of the integral, using condition (c) to pick a distinguished $t_0 \in I$, we obtain

$$\Phi(t) = \int_{\Omega} d^n x \phi(x, t_0) + \int_{t_0}^t ds F_1(s).$$

Since both terms on the right hand side exist, so it does $\Phi(t)$. The remaining statements are a direct consequence of F_1 being continuous. $\square$

Exercise 42. Let $\phi \in L^1(\mathbb{R}^n)$ and $\phi' \in L^\infty(\mathbb{R}^n)$. Prove that $\phi\phi' \in L^1(\mathbb{R}^n)$. (Hint: Do not use Hölder inequality, go with brute force!)

As a consequence

Theorem 177. Let $\phi \in L^1(\mathbb{R}^n)$. Then the Fourier transform as in Definition 175 exists, it is continuous and bounded ($\widehat{\phi} \in L^\infty(\mathbb{R}^n)$) as

$$|\widehat{\phi}| \leq \|\phi\|_{L^1}.$$

Furthermore

1. If $\phi, \phi' \in L^1(\mathbb{R}^n)$, then

$$\int_{\mathbb{R}^n} d^n x \phi(x) \widehat{\phi'}(x) = \int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) \phi'(k),$$

2. If $\phi, \phi' \in L^1(\mathbb{R}^n)$, then the convolution $\phi \star \phi'$ exists for almost all $x \in \mathbb{R}^n$ and $\phi \star \phi' \in L^1(\mathbb{R}^n)$.

3. If $\phi, \phi' \in L^1(\mathbb{R}^n)$, then

$$\widehat{\phi \star \phi'} = \widehat{\phi} \widehat{\phi'}.$$

Proof. Let $\phi \in L^1(\mathbb{R}^n)$. Then its Fourier transform is continuous due to Lebesgue dominated convergence theorem. At the same time it is bounded since

$$|\widehat{\phi}(k)| = \left| \int_{\mathbb{R}^n} d^n x \phi(x) e^{ik \cdot x} \right| \leq \int_{\mathbb{R}^n} d^n x |\phi(x) e^{ik \cdot x}| = \int_{\mathbb{R}^n} d^n x |\phi(x)| = \|\phi\|_{L^1}.$$

Furthermore continuity and boundedness entail existence of $\widehat{\phi}$. Let us focus on the remaining points:

1. Let $\phi, \phi' \in L^1(\mathbb{R}^n)$. Notice that, since $\widehat{\phi'} \in L^\infty(\mathbb{R}^n)$, then, by Exercise 42, $\phi(x) \widehat{\phi'}(x) \in L^1(\mathbb{R}^n)$. In addition, per Definition 175, it holds

$$\begin{aligned} \int_{\mathbb{R}^n} d^n x \phi(x) \widehat{\phi'}(x) &= \int_{\mathbb{R}^n} d^n x \phi(x) \int_{\mathbb{R}^n} d^n k \phi'(k) e^{ik \cdot x} = \\ &= \int_{\mathbb{R}^n} d^n k \phi'(k) \int_{\mathbb{R}^n} d^n x \phi(x) e^{ik \cdot x} = \int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) \phi'(k), \end{aligned}$$

where we used once more Fubini theorem.

2. Observe that the following integral exists,

$$\int_{\mathbb{R}^n \times \mathbb{R}^n} d^n x d^n y |\phi(x)\phi'(y-x)| \leq \|\phi\|_{L^1} \|\phi'\|_{L^1},$$

where the right hand side is finite per hypothesis. This inequality entails that $\phi(x)\phi'(y-x)$ is measurable, hence the convolution exists for almost all y on account of Fubini theorem and it identifies an element of $L^1(\mathbb{R}^n)$.

3. Per definition of Fourier transform

$$\widehat{\phi \star \phi'}(k) = \int_{\mathbb{R}^n} d^n y e^{ik \cdot y} \int_{\mathbb{R}^n} d^n x \phi(x)\phi'(y-x) = \int_{\mathbb{R}^n} d^n x \phi(x) \int_{\mathbb{R}^n} d^n y e^{ik \cdot y} \phi'(y-x).$$

Changing coordinate as $y \mapsto y' \doteq y - x$ and observing that $e^{ik \cdot y} = e^{ik \cdot y'} e^{ik \cdot x}$, we obtain

$$\widehat{\phi \star \phi'}(k) = \int_{\mathbb{R}^n} d^n x \phi(x) \int_{\mathbb{R}^n} d^n y e^{ik \cdot y'} e^{ik \cdot x} \phi'(y') = \int_{\mathbb{R}^n} d^n x \phi(x) e^{ik \cdot x} \int_{\mathbb{R}^n} d^n y e^{ik \cdot y'} \phi'(y') = \widehat{\phi}(k) \widehat{\phi'}(k),$$

which is the sought result. $\square$

Exercise 43. Prove the **Riemann-Lebesgue lemma**, namely that, if $\phi \in L^1(\mathbb{R}^n)$, then $\lim_{k \rightarrow \infty} \widehat{\phi}(k) = 0$. (Hint: Start from $n = 1$ and consider the sequence of characteristic functions, defined on the intervals $[-n, n]$, $n \in \mathbb{N}$.)

Notice that, since $L^1(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ we have already defined the Fourier transform on a large class of distributions. At the same time since $\mathcal{D}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$, one could play the trick of extending the Fourier transform on the whole $\mathcal{D}'(\mathbb{R}^n)$ by duality defining it first on test functions. This cannot work because the Fourier transform of a smooth, compactly supported function, is no longer compactly supported. To convince yourself that the problem lies in the support, solve the following exercise, in which, for making the computation easy, we give up on smoothness (the problem is not there anyway):

Exercise 44. Compute the Fourier transform of $\chi_{a,b} \in C_0^0(\mathbb{R})$ which is the characteristic function of the interval $[a,b]$, i.e.:

$$\chi(x) \doteq \begin{cases} 1 & x \in [a, b] \\ 0 & \text{elsewhere} \end{cases}$$

In other words we have to grin and bear. The only partial way out is to define another space of test-functions which is explicitly built to be invariant under Fourier transform. Starting from it, we will construct the associated continuous linear functionals, a priori knowing that we will obtain a set smaller than $\mathcal{D}'(\mathbb{R}^n)$, but, hopefully, larger than $L^1(\mathbb{R}^n)$.

10.1 The Rapidly Decreasing Test Functions

Let's hurry and get it started. We introduce the new space of test functions:

Definition 178. A function $\phi \in C^\infty(\mathbb{R}^n)$ is called **rapidly decreasing** if, for all pairs of multi-indices α, β

$$\|\phi\|_{\alpha, \beta} \doteq \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta \phi| < \infty. \quad (61)$$

The space of all these functions is denoted as $\mathcal{S}(\mathbb{R}^n)$.

In addition we can give a natural notion of limit of sequences:

Definition 179. Let $\{\phi_j\}_{j \in \mathbb{N}}$ be a sequence of elements in $\mathcal{S}(\mathbb{R}^n)$. We say that it converges to $\phi \in \mathcal{S}(\mathbb{R}^n)$ and we write $\lim_{j \rightarrow \infty} \phi_j = \phi$ if, for all multi-indices α, β ,

$$\lim_{j \rightarrow \infty} \|\phi_j - \phi\|_{\alpha, \beta} = 0.$$

Two important remarks are in due course:

- The terminology *rapidly decreasing* is not used by chance (obviously... they were invented almost 60 years ago. If there were a better name, someone would have already come up with it!). As a matter of fact, for any $\phi \in \mathcal{S}(\mathbb{R}^n)$ the condition of rapid decrease entails that, for any $N \in \mathbb{N} \cup \{0\}$,

$$\lim_{|x| \rightarrow \infty} |x|^N \partial^\beta \phi = 0,$$

for all multi-indices β . In other words ϕ and its derivatives tend to 0 as $x \rightarrow \pm\infty$ faster than any power of x . It is thus very easy to construct example of functions lying in $\mathcal{S}(\mathbb{R}^n)$ as the exponential is known to satisfy to these conditions at least at one end. Hence $\phi(x) = e^{-c|x|^2} \in \mathcal{S}(\mathbb{R}^n)$ for all $c \in \mathbb{C}$ such that $\Re(c) > 0$.

- For the mathematically inclined reader, we observe that Definition 179 endows $\mathcal{S}(\mathbb{R}^n)$ with a natural family of seminorms. Hence $\mathcal{S}(\mathbb{R}^n)$ is a locally convex topological vector space, complete with respect to such topology. One can prove that, in addition, $\mathcal{S}(\mathbb{R}^n)$ is metrizable. In other words $\mathcal{S}(\mathbb{R}^n)$ is a Fréchet space – see for example [6] or the appendix of [1].

Let us review a few basic properties of this new space of functions.

Proposition 180. The following properties hold true:

1. $\mathcal{S}(\mathbb{R}^n)$ is **stable**¹⁸ under differentiation and under multiplication by polynomials. In addition if p and q are two polynomials the map $\mathcal{S}(\mathbb{R}^n) \ni \phi \mapsto p(x)q(D)\phi \in \mathcal{S}(\mathbb{R}^n)$ is continuous. Here $q(D)$ means the following: Let $q(x) = \sum_{|\alpha| \leq N} c_\alpha x^\alpha$, where α is a multi-index, $c_\alpha \in \mathbb{C}$ for each α and $N \in \mathbb{N} \cup \{0\}$. Then $q(D) = \sum_{|\alpha| \leq N} c_\alpha \partial^\alpha$.

¹⁸A vector space $V \subset W$ is called stable under the action of a linear operator $A : W \rightarrow W$ if $A(V) \subseteq V$.

2. The injection $C_0^\infty(\mathbb{R}^n) \hookrightarrow \mathcal{S}(\mathbb{R}^n)$ is continuous,

3. $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$,

4. The injection $\mathcal{S}(\mathbb{R}^n) \hookrightarrow L^1(\mathbb{R}^n)$ is continuous.

Proof. 1. It is a direct consequence of Definition 178 and in particular of the arbitrariness of the choice of multi-indices in $\|\phi_{\alpha,\beta}\|$, $\phi \in \mathcal{S}(\mathbb{R}^n)$.

2. Notice that for, per definition, every $f \in C_0^\infty(\mathbb{R}^n)$ must vanish together with all its derivatives outside a compact subset and that taking derivatives of f or multiplying it by any polynomial yields another smooth and compactly supported function. Hence, for any but fixed α and β $\|f\|_{\alpha,\beta}$ is nothing but the supremum of a smooth and compactly supported function on its support. This is finite. To prove that the injection $C_0^\infty(\mathbb{R}^n) \hookrightarrow \mathcal{S}(\mathbb{R}^n)$ is continuous, it suffices to recall Definition 114 to realize that, for every $\{f_j\}_{j \in \mathbb{N}}$ such that $f_j \in C_0^\infty(\mathbb{R}^n)$ and $\lim_{j \rightarrow \infty} f_j = f$ in $C_0^\infty(\mathbb{R}^n)$ entails per comparison with (61) that $\lim_{j \rightarrow \infty} \|f_j - f\|_{\alpha,\beta} = 0$ for all multi-indices β . Since, for any multi-index α , $x^\alpha f_j \in C_0^\infty(\mathbb{R}^n)$ for all $j \in \mathbb{N}$ and $\lim_{j \rightarrow \infty} x^\alpha f_j = x^\alpha f$ in $C_0^\infty(\mathbb{R}^n)$, the same reasoning leads to the convergence to 0 in $\mathcal{S}(\mathbb{R}^n)$ of $\|f_j - f\|_{\alpha,\beta}$.

3. Let $\phi \in \mathcal{S}(\mathbb{R}^n)$ and let $\rho \in C_0^\infty(\mathbb{R}^n)$ be equal to 1 for all $x \in \mathbb{R}^n$ such that $|x| \leq 1$. Let $\phi_j(x) \doteq \rho\left(\frac{x}{j}\right) \phi(x)$, $j \in \mathbb{N}$. Per construction $\phi_j \in C_0^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ for all $j \in \mathbb{N}$. Since $\mathcal{S}(\mathbb{R}^n)$ is a vector space, for all j , $\phi - \phi_j \in \mathcal{S}(\mathbb{R}^n)$. Hence there must exist $C_N \geq 0$ for all $N \in \mathbb{N} \cup \{0\}$

$$\sum_{\substack{|\alpha| \leq N \\ |\beta| \leq N}} \|\phi - \phi_j\|_{\alpha,\beta} \leq C_N \sum_{\substack{|\alpha| \leq N \\ |\beta| \leq N}} \sup\{|x^\alpha \partial^\beta \phi(x)| \mid |x| \geq j\},$$

where the inequality $|x| \geq j$ accounts for the support properties of ρ . As $j \rightarrow \infty$ the right hand side tends to 0 since the argument of the supremum obeys the following inequality due to (61)

$$|x^\alpha \partial^\beta \phi| \leq |x|^{-1} \sum_{|\gamma|=|\alpha|+1} \|\phi\|_{\gamma,\beta}.$$

Since when $j \rightarrow \infty$ also $|x| \rightarrow \infty$, while the sum is finite, we have justified the previous statement and this part of the proposition is proven.

4. Let $N > \frac{n}{2}$ be an integer. It holds that $\langle x \rangle^{-N} \doteq (1 + |x|^2)^{-N} \in L^1(\mathbb{R}^n)$. If $\phi \in \mathcal{S}(\mathbb{R}^n)$, then there exists $C \geq 0$ such that

$$\langle x \rangle^N |\phi(x)| \leq C \sum_{|\alpha| \leq 2N} \|\phi\|_{\alpha,0} \implies |\phi(x)| \leq \frac{C \sum_{|\alpha| \leq 2N} \|\phi\|_{\alpha,0}}{\langle x \rangle^N}.$$

Since the right hand side of the inequality lies in L^1 so it does the left hand side. In addition we obtain that there must exist a constant $C' \geq 0$ such that

$$\|\phi\|_{L^1} \leq C' \sum_{|\alpha| \leq 2N} \|\phi\|_{\alpha,0}.$$

This entails that a sequence of elements $\{\phi_j\}_{j \in \mathbb{N}}$ converging to ϕ in $\mathcal{S}(\mathbb{R}^n)$ must converge to the same element in $L^1(\mathbb{R}^n)$. $\square$

We are still falling short of one key goal: proving that $\mathcal{S}(\mathbb{R}^n)$ is stable under Fourier transform, the key motivation for introducing this space. We start from an auxiliary lemma, which tells us that the derivation and multiplication by polynomials are dual to each other with respect to Fourier transform.

Lemma 181. *Let $\phi \in \mathcal{S}(\mathbb{R}^n)$ and let $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow L^\infty(\mathbb{R}^n)$ indicate the Fourier transform, namely $\mathcal{F}(\phi) = \widehat{\phi}$. Then, for every multi-index α*

$$\mathcal{F} \circ \partial^\alpha = (-i)^{|\alpha|} k^\alpha \circ \mathcal{F}, \quad \text{and} \quad \mathcal{F} \circ x^\alpha = (-i)^{|\alpha|} \partial^\alpha \circ \mathcal{F}.$$

Proof. Since $\phi \in \mathcal{S}(\mathbb{R}^n)$ and since $\mathcal{S}(\mathbb{R}^n)$ is stable under derivation and multiplication by any polynomial – see Proposition 180, the Fourier transform of ϕ , $x^\alpha \phi$ and $\partial^\alpha \phi$ always exists, no matter the choice of the multi-index α . The first of the two identities descends from a reiterated integration by parts, the boundary terms being always vanishing. The second one includes a differentiation under the sign of integral, which is justified by Theorem 176. $\square$

Notice that the target space of $\mathcal{F}$ is $L^\infty(\mathbb{R}^n)$ since we could only draw a conclusion by using item 4. in Proposition 180 and Theorem 177. The next proposition shows that we can replace $L^\infty(\mathbb{R}^n)$ with $\mathcal{S}(\mathbb{R}^n)$ in Lemma 181.

Proposition 182. *The Fourier transform is a continuous map from $\mathcal{S}(\mathbb{R}^n)$ to $\mathcal{S}(\mathbb{R}^n)$.*

Proof. Observe that Lemma 181 and the last item of Proposition 180 entail that, for every multi-index α , $\partial^\alpha \phi$ and $x^\alpha \phi$ are the Fourier transform of a function in $L^1(\mathbb{R}^n)$ provided that $\phi \in \mathcal{S}(\mathbb{R}^n)$. Theorem 177 entails also that the Fourier transform is continuous. In particular this entails that $\partial^\beta \widehat{\phi}$ is continuous for all multi-indices β . Hence $\widehat{\phi} \in C^\infty(\mathbb{R}^n)$. The estimate $|\widehat{\phi}| \leq \|\phi\|_{L^1}$ together with Lemma 181 yields that

$$\|\widehat{\phi}\|_{\alpha, \beta} = \|(-1)^{|\alpha|+|\beta|} \widehat{\partial^\alpha (x^\beta \phi)}\| \leq \|\partial^\alpha (x^\beta \phi)\|_{L^1},$$

where $\|\cdot\|_{\alpha, \beta}$ is given in (61). In other words $\widehat{\phi} \in \mathcal{S}(\mathbb{R}^n)$. Continuity is now a direct consequence of Proposition 180, of item 1. and 4. in particular. $\square$

Exercise 45. Show that $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ admits $(2\pi)^n$ as eigenvalue and that the associated eigenvector is the Gaussian function¹⁹ $e^{-\frac{|x|^2}{2}}$.

A further, at this stage not surprising, notable property of the Fourier transform is the following: It is invertible!

¹⁹In a physics language, we are saying that the Fourier transform of a Gaussian function is still a Gaussian function.

Proposition 183. *Let $\phi \in \mathcal{S}(\mathbb{R}^n)$. It holds*

$$\phi(x) = \int_{\mathbb{R}^n} \frac{d^n k}{(2\pi)^n} \widehat{\phi}(k) e^{-ix \cdot k}. \quad (62)$$

In other words there exists the inverse of $\mathcal{F}$, namely $\mathcal{F}^{-1} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ and it is a continuous isomorphism of topological vector spaces. Conventionally we write $\mathcal{F}^{-1}(\phi) = \check{\phi}$ and

$$\widehat{\widehat{\phi}} = (2\pi)^n \iota_- \circ \phi \quad \text{and} \quad \hat{\check{\phi}} = \check{\hat{\phi}} = \phi,$$

where $\iota_- : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the map $\iota_-(x) = -x$.

Proof. We divide the proof in three different parts.

Part 1: We use Theorem 177 to infer that, since $\mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$ then, for every $\phi, \phi' \in \mathcal{S}(\mathbb{R}^n)$, it holds

$$\int_{\mathbb{R}^n} d^n x \phi(x) \widehat{\phi'}(x) = \int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) \phi'(k).$$

Choose $\phi'(k) \doteq e^{-\frac{\epsilon^2}{2}|k|^2}$ and, in view of Exercise 45, the above equality becomes

$$\int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) e^{-\frac{\epsilon^2}{2}|k|^2} = \left(\frac{\sqrt{2\pi}}{\epsilon} \right)^n \int_{\mathbb{R}^n} d^n x \phi(x) e^{-\frac{|x|^2}{2\epsilon^2}} = (2\pi)^{\frac{n}{2}} \int_{\mathbb{R}^n} d^n z \phi(\epsilon z) e^{-\frac{|z|^2}{2}},$$

where we introduced $z \doteq \epsilon x$ in the last equality. Since $\widehat{\phi} \in \mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$ and, for all $\epsilon \geq 0$, $e^{-\frac{\epsilon^2}{2}|k|^2}$ takes values in $(0, 1]$ we can apply Lebesgue dominated convergence theorem taking the limit as $\epsilon \rightarrow 0$. For the same reason we can take the same limit in the rightmost integral (observe that $\phi(\epsilon z)$ is bounded for all $\epsilon \geq 0$), hence obtaining

$$\int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) = (2\pi)^{\frac{n}{2}} \phi(0) \int_{\mathbb{R}^n} d^n x e^{-\frac{|x|^2}{2}} = (2\pi)^n \phi(0).$$

To obtain the sought result it is sufficient to change variable of integration from x to $x' = x + x_0$, where $x_0 \in \mathbb{R}$ is arbitrary but fixed. Repeating the above calculation we obtain exactly that

$$\int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) e^{-ik \cdot x_0} = (2\pi)^n \phi(x_0),$$

which corresponds to the first statement of the proposition since x_0 is arbitrary.

Part 2: To show that $\mathcal{F}$ is an isomorphism it suffices to show that it is injective since we have already exhibited an inverse, namely (62). Suppose that $\widehat{\phi} = 0$. Then we can use (62) to infer that $\phi = 0$. Furthermore, the very same formula guarantees su that if $\{\phi_j\}_{j \in \mathbb{N}}$ is a sequence of

functions in $\mathcal{S}(\mathbb{R}^n)$ converging to $\phi \in \mathcal{S}(\mathbb{R}^n)$ as $j \rightarrow \infty$, then also $\widehat{\phi}_j \doteq \mathcal{F}(\phi_j)$ converges to $\widehat{\phi}$ as $j \rightarrow \infty$. Hence we have individuated an isomorphism of topological vector spaces.

Part 3: We need to show that $\mathcal{F}^{-1}$ is continuous. Yet (62) tells us that $\mathcal{F}^{-1} = \mathcal{F} \circ \iota_-$ where $\iota_- : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is such that $\iota_-(k) = -k$. Since ι_- is continuous per construction and $\mathcal{F}$ is continuous per Proposition 182, so must be $\mathcal{F}^{-1}$ being the composition of continuous maps. $\square$

Notice that contrary to the Fourier transform, there are no conventions in the choice of its inverse, once $\mathcal{F}$ has been fixed.

10.2 Tempered Distributions

We are now ready to introduce linear functionals on the new space of test functions that we have introduced. There are two fully equivalent ways to proceed. The first mimics what we did in the preceding sections.

Definition 184. We call **tempered distributions** the linear functionals $u : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$ such that, there exists a constant $C \geq 0$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$|(u, \phi)| \leq C \sum_{|\alpha|, |\beta| \leq N} \sup |x^\alpha \partial^\beta \phi|,$$

for all multi-indices α, β . The collection of all tempered distributions is the vector space $\mathcal{S}'(\mathbb{R}^n)$.

$\mathcal{S}'(\mathbb{R}^n)$ shares many properties with the other spaces of distributions such as $\mathcal{D}'(\mathbb{R}^n)$ and $\mathcal{E}'(\mathbb{R}^n)$. Since the analysis and the proofs are similar we omit them and we offer them as exercises.

Exercise 46. Prove that a linear functional $u : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$ is continuous if and only if it is sequentially continuous. (Hint: Follow the proof of Theorem 116).

The second way to introduce tempered distributions stems from Proposition 180, in particular from the continuous embedding $C_0^\infty(\mathbb{R}^n) \hookrightarrow \mathcal{S}(\mathbb{R}^n)$. Every linear functional on $\mathcal{S}(\mathbb{R}^n)$ can thus be restricted to $C_0^\infty(\mathbb{R}^n)$ raising the natural question whether one obtains or not an element of $\mathcal{D}'(\mathbb{R}^n)$. This justifies the following alternative definition

Definition 185. We call *space of tempered distributions* $\mathcal{S}'(\mathbb{R}^n)$ the subspace of $\mathcal{D}'(\mathbb{R}^n)$ consisting of distributions extending to continuous linear forms on $\mathcal{S}(\mathbb{R}^n)$.

The two definitions are clearly equivalent since from Definition 185 we can move via a restriction to Definition 184 on account of item 2. in Proposition 180. At the same time item 3. of the same proposition tells us that $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$. Hence every continuous linear functional on the first space can be uniquely extended to one on the second.

Another important aspect of tempered distributions consists of the realization that they are nothing but continuous functions of polynomial growth at infinity. This is the hearth of the

so-called **structure theorem**²⁰ for $\mathcal{S}'(\mathbb{R}^n)$. We will omit its proof and an interested reader can refer to [1, Th. 8.3.1]:

Theorem 186. *Every tempered distribution is a derivative of finite order of a continuous function of polynomial growth.*

Exercise 47. Show that $L^1(\mathbb{R}^n)$ and $\mathcal{E}'(\mathbb{R}^n)$ are continuously embedded in $\mathcal{S}'(\mathbb{R}^n)$

The next step consists of studying the extension of the notion of Fourier transform to tempered distributions. The procedure relies on Proposition 182 and on a duality argument.

Definition 187. *The Fourier transform of a tempered distribution $u \in \mathcal{S}'(\mathbb{R}^n)$ is $\widehat{u} \in \mathcal{S}'(\mathbb{R}^n)$ such that*

$$(\widehat{u}, \phi) = (u, \widehat{\phi}). \quad \forall \phi \in \mathcal{S}(\mathbb{R}^n) \quad (63)$$

Let us characterize the properties of this transform.

Theorem 188. *Let $\mathcal{F} : \mathcal{S}'(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n)$ be the map defined via (63). It is a sequentially continuous map with sequentially continuous inverse.*

Proof. The Fourier transform is linear per construction. Definition 187 in combination with Proposition 183 entails that, for every $\phi \in \mathcal{S}(\mathbb{R}^n)$ and for every $u \in \mathcal{S}'(\mathbb{R}^n)$,

$$(\widehat{\widehat{u}}, \phi) = (\widehat{u}, \widehat{\phi}) = (u, \widehat{\widehat{\phi}}) = (2\pi)^n (u, \iota_- \phi), \quad (64)$$

from which it descends the inversion formula $(u, \iota_- \phi) = (2\pi)^{-n} (\widehat{\widehat{u}}, \widehat{\phi})$. In addition this formula guarantees that $\mathcal{F}$ is injective as $\widehat{u} = \mathcal{F}(u) = 0$ entails that $u = 0$ in view of the arbitrariness of ϕ and of Proposition 183. Since we have an explicit inversion formula, namely $\mathcal{S}'(\mathbb{R}^n) \ni u = \widehat{\widehat{u}}$, we have proven that $\mathcal{F}$ is an isomorphism. Sequential continuity of $\mathcal{F}$ is a consequence of (63) and that of $\mathcal{F}^{-1}$ of (64). $\square$

One might complain that we have used the same symbol $\mathcal{F}$ (did you complain, isn't it? Beatrice, at least you did, no?) for the Fourier transform on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$ without giving any justification. We remedy to this misbehaviour. Observe first that any $\phi' \in \mathcal{S}(\mathbb{R}^n)$ induces an element of $\mathcal{S}'(\mathbb{R}^n)$, $u_{\phi'}$ such that $u_{\phi'}(\phi) = (u_{\phi'}, \phi) \equiv (\phi', \phi)$.

Corollary 189. *The Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$ is the extension of the map $\mathcal{S}(\mathbb{R}^n) \ni \phi \mapsto (\widehat{\phi'}, \phi) \in \mathbb{C}$, where $\widehat{\phi'}$ is the standard Fourier transform of $\phi' \in \mathcal{S}(\mathbb{R}^n)$*

Proof. This is a straightforward consequence of sequential continuity. $\square$

Exercise 48. Prove that the Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$ satisfies the following properties: For all $u \in \mathcal{S}'(\mathbb{R}^n)$,

²⁰Exactly as per $\mathcal{D}'(\mathbb{R}^n)$, see Theorem 158, a structure theorem is nothing but a theorem which characterizes the elements of a rather abstract space in terms of more concrete objects.

1. $\widehat{\partial^\alpha u} = (-i)^{|\alpha|} k^\alpha \widehat{u}$,
2. $\widehat{x^\alpha u} = (-i)^{|\alpha|} \partial^\alpha \widehat{u}$
3. $\widehat{\tau_h u} = \widehat{u} e^{ik \cdot h}$
4. $\widehat{u e^{-ik \cdot h}} = \tau_h \widehat{u}$,

where $h \in \mathbb{R}^n$ and τ_h is the translation operator defined on any $\phi \in C^\infty(\mathbb{R}^n)$ as $(\tau_h \phi)(x) \doteq \phi(x + h)$ and on distributions per duality.

Exercise 49. Prove that the following Fourier transforms are meaningful and compute them:

1. $\widehat{\delta} = 1$ (regardless of the dimension of the underlying space),
2. $\widehat{\Theta}(k) = -\frac{i}{k} + \pi \delta$ where $\Theta(x)$ is the Heaviside function.
3. Let $\text{sign}(x) \doteq \Theta(x) - \Theta(-x)$ be the sign function. Then $\widehat{\text{sign}}(k) = -\frac{2i}{k}$.

The inversion formula of the Fourier transform has been proven using a 1-parameter families of Gaussian functions and the convolution. Using the Fourier transform of the δ -distribution, one can simplify considerably the proof.

Exercise 50. Prove the inversion formula for the Fourier transform on $\mathcal{S}(\mathbb{R}^n)$ via a brute force method. (Hint: As said, you need the Fourier transform of the δ -distribution computed in Exercise 49.)

We are falling short of one last topic, namely extending the Fourier transform to the convolution of distributions, one of them being in $\mathcal{S}'(\mathbb{R}^n)$. We start from a preliminary result which characterizes the Fourier transform of a distribution of compact support

Theorem 190. Let $u \in \mathcal{E}'(\mathbb{R}^n)$. Then $\widehat{u}(k) = (u, e^{ik \cdot x})$ and $\widehat{u} \in C^\infty(\mathbb{R}^n)$. Furthermore $\widehat{u}$ is a so called **multiplier**²¹ on $\mathcal{S}'(\mathbb{R}^n)$, that is $\partial^\alpha \widehat{u}$ is a smooth function of polynomial growth for all multi-indices α .

Proof. The map $\mathbb{R}^n \ni k \mapsto (u, e^{ik \cdot x})$ is smooth on account of Corollary 142. In addition, in view of Exercise 47, we know that $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$. Hence u admits Fourier transform $\widehat{u} \in \mathcal{S}'(\mathbb{R}^n)$ in view of Definition 187. Since $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$ as per Proposition 180, $\widehat{u}$ is uniquely determined by its restriction to $C_0^\infty(\mathbb{R}^n)$. Exercise 30 guarantees us that, for all $f \in C_0^\infty(\mathbb{R}^n) \subset \mathcal{E}'(\mathbb{R}^n)$, $u \otimes f$ exists since $u \in \mathcal{E}'(\mathbb{R}^n)$. Applying the results of Exercise 31, we obtain that

$$(u \otimes f, e^{ik \cdot x}) = (u, (f, e^{ik \cdot x})) = \left(u, \int_{\mathbb{R}^n} d^n k e^{ik \cdot x} f(k) \right) = (u, \widehat{f}) = (\widehat{u}, f),$$

²¹The terminology is justified by the fact that the product of a tempered distribution by any smooth function of polynomial growth with all its derivatives is still a tempered distribution.

where we used the definition of Fourier transform on $\mathcal{S}'(\mathbb{R}^n)$. At the same time still Exercise 31 guarantees that

$$(u \otimes f, e^{ik \cdot x}) = (u, (f, e^{ik \cdot x})) = (f, (u, e^{ik \cdot x})) = \int_{\mathbb{R}^n} d^n k f(k) (u, e^{ik \cdot x}) = ((u, e^{ik \cdot x}), f),$$

where we used that $(u, e^{ik \cdot x})$ as a function of k is smooth. Comparing the last two chains of equalities we obtain $\widehat{u}(k) = (u, e^{ik \cdot x})$, where the pointwise evaluation of the left hand side is meaningful having proven that the left hand side is smooth.

To prove the last statement of theorem, we recall that, thanks to Theorem 140, for every multi-index α ,

$$\partial^\alpha \widehat{u}(k) = \partial^\alpha (u, e^{ik \cdot x}) = (u, \partial^\alpha (e^{ik \cdot x})) = (u, i^{|\alpha|} x^\alpha e^{ik \cdot x}) = i^{|\alpha|} (x^\alpha u, e^{ik \cdot x}),$$

where we used that, being x^α smooth, we can use Definition 129. In addition, since $x^\alpha u \in \mathcal{E}'(\mathbb{R}^n)$, (39) entails the existence for each multi-index α of a compact set K_α and of a constant C_{K_α} , as well as of $N_\alpha \in \mathbb{N} \cup \{0\}$ such that

$$|\partial^\alpha \widehat{u}(k)| \leq C_{K_\alpha} \sum_{|\beta| \leq N_\alpha} \sup_{x \in K_\alpha} \{|\partial^\beta (x^\alpha e^{ik \cdot x})|\},$$

where β is a suitable multi-index. A direct application of the Leibniz rule entails the existence of a second constant C'_{K_α} such that

$$|\partial^\alpha \widehat{u}(k)| \leq C'_{K_\alpha} (1 + |k|)^{N_\alpha}.$$

This entails that $\partial^\alpha \widehat{u}$ has at most polynomial growth. □

Following [2, Th. 7.1.14]

Corollary 191. *Let $u \in \mathcal{E}'(\mathbb{R}^n)$. Then $\widehat{u}(k) = (u, e^{ik \cdot x})$ can be extended to any $k \in \mathbb{C}$ and $\widehat{u}(k)$ is entire analytic²².*

Proof. One can repeat slavishly the proof of Theorem 190 obtaining that $\widehat{u}(k) \in C^\infty(\mathbb{C}^n)$. A direct computation shows also that $\widehat{u}(k)$ satisfies the Cauchy-Riemann equations, since we can differentiate directly the exponent. Hence $\widehat{u}$ is entire, analytic. □

Corollary 192. *Let $u, v \in \mathcal{E}'(\mathbb{R}^n)$. Then $\widehat{u \star v} = \widehat{u} \widehat{v}$.*

Proof. Combining Theorem 146 with Proposition 147, we conclude that $u \star v \in \mathcal{E}'(\mathbb{R}^n)$. Hence it admits Fourier transform since $\mathcal{E}'(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$. Furthermore Theorem 190 tells us that

$$\widehat{u \star v} = (u \star v, e^{ik \cdot x}) = (u \otimes v, e^{ik \cdot (x+y)}) = (u, (v, e^{ik \cdot x} e^{ik \cdot y})) = (u, e^{ik \cdot x}) (v, e^{ik \cdot y}) = \widehat{u} \widehat{v},$$

where we used Theorem 146 and the $(\cdot, \cdot) : \mathcal{E}'(\mathbb{R}^n) \times \mathcal{E}(\mathbb{R}^n) \rightarrow \mathbb{C}$. □

²²The extension to the whole complex plane of k goes also under the name of **Fourier-Laplace** transform.

Observe that, in the last corollary, it played a key role the fact that both $\widehat{u}$ and $\widehat{v}$ are smooth, since their product is well-defined. As we mentioned several times, the product of two-distributions, even if in $\mathcal{S}'(\mathbb{R}^n)$ or in $\mathcal{E}'(\mathbb{R}^n)$, is in general ill-defined. Hence we cannot expect that Corollary 192 is true for all possible distributions. Yet it is clear, at least at an heuristic level, that it is not necessary that both members at the right hand side are smooth, but just that one of them is, so that we can appeal to Definition 129. To make precise this idea, we conclude with one last extension:

Theorem 193. *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ and let $v \in \mathcal{E}'(\mathbb{R}^n)$. Then $u \star v \in \mathcal{S}'(\mathbb{R}^n)$ and $\widehat{u \star v} = \widehat{u} \widehat{v}$.*

Proof. Observe that $\widehat{u} \in \mathcal{S}'(\mathbb{R}^n)$ and, on account of Theorem 190, $\widehat{v} \in C^\infty(\mathbb{R}^n)$. The product $\widehat{u} \widehat{v}$ is well-defined per Definition 129. In addition Theorem 190 also guarantees us that $\widehat{v}$ is at most of polynomial growth, hence $\widehat{u} \widehat{v} \in \mathcal{S}'(\mathbb{R}^n)$. Since the Fourier transform is an invertible map, there must exist $w \in \mathcal{S}'(\mathbb{R}^n)$ such that $\widehat{w} = \widehat{u} \widehat{v}$. Using (64), we obtain that, for all $\phi \in \mathcal{S}(\mathbb{R}^n)$,

$$(w, \check{\phi}) = (2\pi)^{-n}(\widehat{w}, \widehat{\phi}) = (2\pi)^{-n}(\widehat{u} \widehat{v}, \widehat{\phi}) = (2\pi)^{-n}(\widehat{u}, \widehat{v \star \phi}),$$

where we used Definition 129 since $\widehat{v}$ is smooth. Restricting ϕ to $C_0^\infty(\mathbb{R}^n)$, we have $\phi \in \mathcal{E}'(\mathbb{R}^n)$. Hence Corollary 192 tells us that $\widehat{v \star \phi} = \widehat{v} \widehat{\phi}$. It follows

$$(w, \check{\phi}) = (2\pi)^{-n}(\widehat{u}, \widehat{v \star \phi}) = (u \star v, \check{\phi}), \quad \forall \phi \in C_0^\infty(\mathbb{R}^n).$$

Since $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$ per Proposition 180, the above chain of equalities identifies uniquely w as $u \star v \in \mathcal{S}'(\mathbb{R}^n)$, from which the statement descends since $\widehat{w} = \widehat{u} \widehat{v}$. $\square$

10.3 The Fourier-Plancherel Transform on L^2

The last topic (not really the last, but, you know... at some point the course has to finish!) that we will outline quite in detail concerns the structural properties of the Fourier transform on $L^2(\mathbb{R}^n)$. First of all we make the following important observation. Recall that the spaces $L^p(\mathbb{R}^n)$, $p \in \mathbb{N}$ are all Banach spaces including $p = \infty$ and only if $p = 2$ we have a Hilbert space.

Proposition 194. *The following statements hold true:*

1. $\mathcal{S}(\mathbb{R}^n)$ is continuously and injectively embedded in $L^p(\mathbb{R}^n)$ for all $p \in \mathbb{N}$,
2. $L^q(\mathbb{R}^n)$ is continuously and injectively embedded in $\mathcal{S}'(\mathbb{R}^n)$.

Proof. 1. Let $\phi \in \mathcal{S}(\mathbb{R}^n)$. Per definition, it is smooth and it tends to 0 as $|x| \rightarrow \infty$, faster than any power. Hence, switching to polar coordinates $(|x|, \theta_1, \dots, \theta_{n-1})$, for all $p \in \mathbb{N}$,

$$\int_{\mathbb{R}^n} d^n x |\phi(x)|^p = \int_{\mathbb{S}^{n-1}} d\mathbb{S}^{n-1}(\theta_1, \dots, \theta_{n-1}) \int_0^\infty d(|x|) |x|^{n-1} |\phi(|x|, \theta_1, \dots, \theta_n)|^p,$$

where $d\mathbb{S}^{n-1}(\theta_1, \dots, \theta_{n-1})$ is the standard volume element over the unit $(n-1)$ -sphere. Dividing the integral over $|x|$ between $(0, a)$ and (a, ∞) with $a > 0$ sufficiently large, the first integral is

finite, while, for the second one, we can use that $|\phi(|x|, \theta_1, \dots, \theta_n)| \leq \frac{C(\theta_1, \dots, \theta_n)}{|x|^q}$. Observe that $C(\theta_1, \dots, \theta_n)$ is a strictly positive, continuous function over $\mathbb{S}^{n-1}$. Hence its integral to the p -th power over $\mathbb{S}^{n-1}$ is finite being the $(n-1)$ -sphere compact. By choosing a accordingly also to the power p and to the dimension n , we can make q big enough so that the integral over $|x|$ of $|x|^{n-1}|\phi(|x|, \theta_1, \dots, \theta_n)|^p$ becomes finite. Hence $\phi \in L^p(\mathbb{R}^n)$ and the embedding is per construction injective. The same construction tells us that, if we take a sequence $\{\phi_j\}_{j \in \mathbb{N}}$ of elements in $\mathcal{S}(\mathbb{R}^n)$ converging to ϕ as per Definition 179, then $\|\phi_j - \phi\|_{L^p}$ tends to 0 as $j \rightarrow \infty$. Hence the embedding is continuous.

2. Consider an equivalence class of functions $[\psi] \in L^q(\mathbb{R}^n)$, $q \in \mathbb{N}$. It identifies a functional $u_\psi : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}$ as follows

$$u_\psi(\phi) = \int_{\mathbb{R}^n} d^n x \psi(x) \phi(x), \quad \forall \phi \in \mathcal{S}(\mathbb{R}^n). \quad (65)$$

First of all we observe that the integral is finite. In point 1. of this proposition we have shown that $\phi \in L^p(\mathbb{R}^n)$ for all $p \in \mathbb{N}$. Hence we can choose p so that $\frac{1}{p} + \frac{1}{q} = 1$ and Hölder inequality guarantees us that $\psi\phi \in L^1(\mathbb{R}^n)$. Hence $\psi\phi$ is integrable. In addition the definition does not depend on the choice of representative of $[\psi]$ since, different elements coincide up to a set of zero measure. Since the right-hand side of (65) vanishes if ψ vanishes almost everywhere, the statement is proven. By using that $|u_\psi(\phi)|$ is finite, one can easily find a constant C so that u_ψ satisfies the estimate required by Definition 184. In other words $u_\psi \in \mathcal{S}'(\mathbb{R}^n)$. At the same time (65) tell us that if $u_\psi = 0$, then $[\psi]$ must also vanish. Hence the embedding $L^p(\mathbb{R}^n)$ is injective. Continuity follows by the by now usual, standard arguments and we omit it. $\square$

In view of this proposition, we will no longer represent elements of $L^p(\mathbb{R}^n)$ as equivalence classes and we will indicate the associated distribution simply as $(\psi, \cdot)$ rather than with u_ψ . As next step, we prove a technical, useful (well... if it were not useful... would we be proving it? Yes, if I were mean... am I mean? Maybe!)

Lemma 195. *Let $\rho \in C_0^\infty(\mathbb{R}^n)$ be a positive function such that $\int_{\mathbb{R}^n} d^n x \rho(x) = 1$. Let $\epsilon > 0$ and let*

$$\rho_\epsilon(x) \doteq \epsilon^{-n} \rho\left(\frac{x}{\epsilon}\right).$$

For any $\psi \in L^2(\mathbb{R}^n)$, let

$$\psi_\epsilon(x) \doteq \psi \star \rho = \int_{\mathbb{R}^n} d^n y \psi(y) \rho_\epsilon(x - y).$$

Then $\psi_\epsilon \in C^\infty(\mathbb{R}^n)$ and $\lim_{\epsilon \rightarrow 0} \psi_\epsilon = \psi$ in $L^2(\mathbb{R}^n)$. Furthermore, for all $\epsilon > 0$, $\|\psi_\epsilon\|_{L^2} \leq \|\psi\|_{L^2}$.

Proof. Since $\psi \in L^2(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ and $\rho_\epsilon \in C_0^\infty(\mathbb{R}^n) \subset \mathcal{E}'(\mathbb{R}^n)$, $\epsilon > 0$, we know that $\psi \star \rho_\epsilon$ exists and it is a smooth function thanks to Theorem 150. In addition the integral exists since

$\|\psi(y)\rho_\epsilon(x-y)\|_{L^1} \leq \|\psi(y)\|_{L^2} \|\rho_\epsilon(x-y)\|_{L^2}$ by Hölder inequality. A clever use of the Cauchy-Schwartz inequality yields

$$|\psi_\epsilon(x)|^2 \leq \int_{\mathbb{R}^n} d^n y |\psi(x-y)|^2 \rho_\epsilon(y) \int_{\mathbb{R}^n} d^n z \rho_\epsilon(z) = \int_{\mathbb{R}^n} d^n y |\psi(x-y)|^2 \rho_\epsilon(y).$$

Integrating over the variable x , we obtain

$$\|\psi_\epsilon\|_{L^2}^2 \leq \int_{\mathbb{R}^n} d^n x \int_{\mathbb{R}^n} d^n y |\psi(x-y)|^2 \rho_\epsilon(y) = \int_{\mathbb{R}^n} d^n y \rho_\epsilon(y) \int_{\mathbb{R}^n} d^n z |\psi(z)|^2 = \|\psi\|_{L^2}^2,$$

where, in the last equality, we have implemented the change of variable $x \mapsto z \doteq x - y$. As last point, we control the limit as $\epsilon \rightarrow 0$:

$$|\psi(x) - \psi_\epsilon(x)|^2 = \left| \int_{\mathbb{R}^n} d^n y (\psi(x) - \psi(x-y)) \rho_\epsilon(y) \right|^2 \leq \int_{\mathbb{R}^n} d^n y |\psi(x) - \psi(x-y)|^2 \rho_\epsilon(y).$$

Applying once again Fubini theorem and changing coordinate $y \mapsto y' \doteq \epsilon y$, we obtain

$$\|\psi - \psi_\epsilon\|_{L^2}^2 \leq \int_{\mathbb{R}^n} dy' \rho(y') \|\tau_{\epsilon y} \psi - \psi\|_{L^2}^2,$$

where $\tau_{\epsilon y} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ is the translation operator $\tau_{\epsilon y} \psi(x) = \psi(x + \epsilon y)$. To conclude the proof it is sufficient to show that the norm under the sign of integral tends to 0 as $\epsilon \rightarrow 0$, applying subsequently the dominated convergence theorem. This is tantamount to showing that the translation operator is continuous and we leave this part to the reader as an exercise. $\square$

Exercise 51. Prove that the translation operator $\tau_a : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$, $a \in \mathbb{R}$, such that $[\tau_a \psi](x) = \psi(x + a)$ is continuous.

We can now prove in sequence the two main theorems of this section.

Theorem 196. $C_0^\infty(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$. As a consequence we have the following chain of continuous, injective embeddings:

$$C_0^\infty(\mathbb{R}^n) \hookrightarrow \mathcal{S}(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n) \hookrightarrow \mathcal{S}'(\mathbb{R}^n).$$

Proof. We need only to prove that $C_0^\infty(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ since the other statements are a consequence of Proposition 194.

Let $\psi \in L^2(\mathbb{R}^n)$ and let $\rho \in C_0^\infty(\mathbb{R}^n)$ be a positive function such that $\int_{\mathbb{R}^n} d^n x \rho(x) = 1$. Let $\epsilon > 0$ and $\rho_\epsilon(x) \doteq \epsilon^{-n} \rho\left(\frac{x}{\epsilon}\right)$. Define

$$\psi_\epsilon = \psi \star \rho_\epsilon.$$

Choose now $\sigma \in C_0^\infty(\mathbb{R}^n)$ such that $0 \leq \sigma \leq 1$ and $\sigma = 1$ for all $x \in \mathbb{R}^n$ such that $|x| \leq 1$. Let $\phi_\epsilon(x) \doteq \sigma(\epsilon x)\psi_\epsilon(x)$. Since $\psi_\epsilon \in C^\infty(\mathbb{R}^n)$ on account of Lemma 195, $\phi_\epsilon \in C_0^\infty(\mathbb{R}^n)$. In addition, by the triangular inequality,

$$\|\psi - \phi_\epsilon\|_{L^2} \leq \|(\psi - \psi_\epsilon)\sigma(\epsilon x)\|_{L^2} + \|(1 - \sigma(\epsilon x))\psi\|_{L^2} \leq \|\psi - \psi_\epsilon\|_{L^2} + \left(\int_{|x| > \frac{1}{\epsilon}} d^n x |\psi(x)|^2 \right)^{\frac{1}{2}}.$$

When $\epsilon \rightarrow 0$, the first term vanishes thanks to Lemma 195, while the second vanishes since $\psi \in L^2(\mathbb{R}^n)$. $\square$

One might argue that Theorem 196 is redundant having already established in Theorem 151 that $C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{D}'(\mathbb{R}^n)$. This information, combined with the already established continuous, injective embeddings $L^2(\mathbb{R}^n) \hookrightarrow \mathcal{S}'(\mathbb{R}^n) \hookrightarrow \mathcal{D}'(\mathbb{R}^n)$ would have allowed us to draw the same conclusion. This is indeed true, but, here, we offered a new, direct and, by all means shorter proof.

Theorem 197 (Plancherel Theorem). *Let $\psi \in L^2(\mathbb{R}^n)$. Then its Fourier transform in the sense of Definition 187 exists and*

$$\|\widehat{\psi}\|_{L^2}^2 = (2\pi)^n \|\psi\|_{L^2}^2. \quad (66)$$

Proof. We start establishing (66) for $\phi \in \mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$. Let also $\phi' \in \mathcal{S}(\mathbb{R}^n)$. In particular, since Proposition 194 entails that $\mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$, we can use Theorem 177 to infer that

$$\int_{\mathbb{R}^n} d^n x \phi(x) \widehat{\phi}'(x) = \int_{\mathbb{R}^n} d^n k \widehat{\phi}(k) \phi'(k).$$

If we choose $\phi'(k) = \overline{\widehat{\phi}(k)}$, by (62), this equality becomes

$$(2\pi)^n \|\phi\|_{L^2}^2 = \|\widehat{\phi}\|_{L^2}^2. \quad (67)$$

Consider now $\psi \in L^2(\mathbb{R}^n)$. By the definition of Fourier transform 187 and by the Cauchy-Schwartz inequality, it holds

$$|(\widehat{\psi}, \phi)| = |(\psi, \widehat{\phi})| \leq \|\psi\|_{L^2} \|\widehat{\phi}\|_{L^2} = (2\pi)^n \|\psi\|_{L^2} \|\phi\|_{L^2}. \quad (68)$$

Since we have established in Theorem 196 that $C_0^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$, we can extend this chain of inequalities to all $\phi \in L^2(\mathbb{R}^n)$. The map $L^2(\mathbb{R}^n) \ni \phi \mapsto (\widehat{\psi}, \phi)$ identifies a continuous linear functional on $L^2(\mathbb{R}^n)$ since (68) entails that the functional is bounded, hence continuous per Theorem 24. By Riesz representation theorem 17, there must exist $\psi' \in L^2(\mathbb{R}^n)$ such that

$$(\widehat{\psi}, \phi) = \int_{\mathbb{R}^n} d^n x \overline{\psi'} \phi, \quad \forall \phi \in L^2(\mathbb{R}^n)$$

where $\|\psi'\|_{L^2} \leq (2\pi)^{\frac{n}{2}} \|\psi\|_{L^2}$. Hence $\widehat{\psi} = \overline{\psi'} \in L^2(\mathbb{R}^n)$ and $\|\widehat{\psi}\|_{L^2} \leq (2\pi)^{\frac{n}{2}} \|\psi\|_{L^2}$. Repeating the same line of reasoning replacing ψ with $\widehat{\psi}$, we obtain $\|\widehat{\widehat{\psi}}\|_{L^2} \leq (2\pi)^{\frac{n}{2}} \|\widehat{\psi}\|_{L^2} \leq (2\pi)^n \|\psi\|_{L^2}$. At the same time (64) guarantees that $\widehat{\widehat{\psi}} = (2\pi)^n \iota_-(\psi)$. Yet since, up to a multiplicative constant, the inverse Fourier transform $\mathcal{F}^{-1}$ is nothing but $\mathcal{F}$ composed with the inversion map $k \rightarrow -k$ which is a unitary operator on $L^2(\mathbb{R}^n)$, it holds $\|\psi\|_{L^2} = \|\widehat{\psi}\|_{L^2}$. The last chain of inequalities becomes

$$(2\pi)^n \|\psi\|_{L^2} \leq (2\pi)^{\frac{n}{2}} \|\widehat{\psi}\|_{L^2} \leq (2\pi)^n \|\psi\|_{L^2},$$

which entails the sought result. $\square$

We prove a series of (almost) straightforward consequences of Plancherel theorem:

Corollary 198. *The (modified) Fourier transform $\mathcal{F}' \doteq (2\pi)^{-\frac{n}{2}} \mathcal{F}$ is a unitary operator on $L^2(\mathbb{R}^n)$.*

Proof. It is an immediate consequence of (66), which guarantees that $\mathcal{F}'$ is an isometry and a bijective map. $\square$

Corollary 199. *Let $\psi, \psi' \in L^2(\mathbb{R}^n)$. Then*

$$(\psi, \psi')_{L^2} = (\widehat{\psi}, \widehat{\psi'})_{L^2}.$$

Proof. The statement descends from Plancherel theorem 197, which guarantees that $\widehat{\psi}, \widehat{\psi'} \in L^2(\mathbb{R}^n)$, in combination with Corollary 198 and with the polarization identity (5). $\square$

Lemma 200. *Let $\psi, \psi' \in L^2(\mathbb{R}^n)$. Then*

$$(\psi \star \psi')(x) = \int_{\mathbb{R}^n} d^n y \, \psi(y) \psi'(x - y),$$

is a bounded continuous function

Proof. Since both ψ and ψ' are square-integrable, the integral is well-defined and, by the Cauchy-Schwartz inequality

$$|(\psi \star \psi')(x)| \leq \|\psi\|_{L^2} \|\psi'\|_{L^2}. \quad \forall x \in \mathbb{R}^n$$

Hence the convolution is bounded. To prove that it is a continuous function, we can compute directly the difference quotient. Let $h \in \mathbb{R}^n$, then, by Hölder inequality

$$|(\psi \star \psi')(x + h) - (\psi \star \psi')(x)| \leq \|\psi\|_{L^2} \|\tau_h(\psi') - \psi'\|_{L^2},$$

where $\tau_h : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ is the translation operator, proven to be continuous in Exercise 51. Continuity in particular shows that the left hand side of the above inequality tends to 0 as $h \rightarrow 0$. This is the sought result. $\square$

Theorem 201. *Let $\psi, \psi' \in L^2(\mathbb{R}^n)$. Then*

$$\widehat{\psi \star \psi'} = \widehat{\psi} \widehat{\psi'}.$$

Proof. On account of Plancherel theorem 197, $\widehat{\psi}, \widehat{\psi'} \in L^2(\mathbb{R}^n)$. Hence, by Hölder inequality $\widehat{\psi} \widehat{\psi'} \in L^1(\mathbb{R}^n)$. We can take its inverse Fourier transform (62):

$$\mathcal{F}^{-1}(\widehat{\psi} \widehat{\psi'})(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} d^n k \widehat{\psi}(k) \widehat{\psi'}(k) e^{-ik \cdot x}.$$

Let $h \in \mathbb{R}^n$. Since $\psi' e^{-ik \cdot z} \in L^2(\mathbb{R}^n)$, we can define as an element of $L^2(\mathbb{R}^n)$

$$\gamma_z \doteq \mathcal{F}^{-1}(\widehat{\psi'}(k) e^{ik \cdot z}).$$

Using Corollary 199, we obtain

$$\mathcal{F}^{-1}(\widehat{\psi} \widehat{\psi'})(x) = (\gamma_z, \psi)_{L^2}. \quad (69)$$

Let us compute explicitly γ_z : Let $\phi \in \mathcal{S}(\mathbb{R}^n)$, then

$$(\gamma_z, \check{\phi}) = (2\pi)^{-n} (\widehat{\gamma_z}, \widehat{\phi}) = (2\pi)^{-n} \int_{\mathbb{R}^n} d^n k \overline{\widehat{\psi'}(k)} \widehat{\phi}(k) e^{iz \cdot k}.$$

Using once again Corollary 199, we obtain

$$(\gamma_z, \check{\phi}) = \int_{\mathbb{R}^n} d^n x \overline{\psi'(x)} \widehat{\phi}(x - z) = \int_{\mathbb{R}^n} d^n x \overline{\psi'(z - x)} \check{\phi}(x),$$

from which it descends $\gamma_z(x) = \overline{\psi'(z - x)}$. Inserting this result in (69), one obtains the sought $\mathcal{F}^{-1}(\widehat{\psi} \widehat{\psi'}) = \psi \star \psi'$. $\square$

10.3.1 The Position and the Momentum Operator (Again)

In this subsection we solve a problem left open from the first part of the course, namely proving Propositions 91 and 93. Let us start from the former. Bearing in mind (18) and (19), we can slightly improve Proposition 91 as follows:

Proposition 202. *Each position operator X_i , $i = 1, \dots, n$ on $D(X_i) \subset L^2(\mathbb{R}^n)$ is self-adjoint and both $C_0^\infty(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ are cores for X_i .*

Proof. Since $C_0^\infty(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ per Theorem 196 and $C_0^\infty(\mathbb{R}^n) \subset D(X_i)$ for all $i = 1, \dots, n$, $D(X_i)$ is dense in $L^2(\mathbb{R}^n)$. Hence, per Definition 81, X_i admits an adjoint. In addition for all $\psi, \psi' \in D(X_i)$ it holds

$$(\psi, X_i \psi') = \int_{\mathbb{R}^n} d^n x \overline{\psi(x)} x_i \psi'(x) = (X_i \psi, \psi').$$

Hence X_i is a symmetric operator. This entails that $X_i \subseteq X_i^*$. We need to prove the converse inclusion. We observe that $\psi \in D(X_i^*)$ if and only if there exists $\tilde{\psi} \in L^2(\mathbb{R}^n)$ such that

$$\int_{\mathbb{R}^n} d^n x \bar{\psi}(x) x_i \psi'(x) = \int_{\mathbb{R}^n} d^n x \tilde{\psi}(x) \psi'(x). \quad \forall \psi' \in D(X_i) \quad (70)$$

This entails that $\tilde{\psi}(x)$ coincides almost everywhere with $x_i \psi(x)$, but this identity is meaningful if and only if $x_i \psi(x) \in L^2(\mathbb{R}^n)$. Hence

$$D(X_i^*) = \{\psi \in L^2(\mathbb{R}^n) \mid x_i \psi \in L^2(\mathbb{R}^n)\}.$$

Comparing with (19), $D(X_i^*) = D(X_i)$, which, in combination with the information that $D(X_i)$ is dense and that $X_i = X_i^*$ thereon, entails that X_i is self-adjoint.

Since both $C_0^\infty(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ are dense in $L^2(\mathbb{R}^n)$ and thus on $D(X_i)$, they are cores for the restriction of X_i thereon is essentially self-adjoint. We show it for $C_0^\infty(\mathbb{R}^n)$, the proof for $\mathcal{S}(\mathbb{R}^n)$ following suit. Observe that, for all $i = 1, \dots, n$, $X_{i,0} \doteq X_i|_{C_0^\infty(\mathbb{R}^n)}$ admits adjoint $X_{i,0}^*$ and $X_{i,0} = X_{i,0}^*$ on account of (70). Since we know that X_i is self-adjoint, item 2. of Theorem 87 guarantees that $\ker(X_i^* \pm i\mathbb{I}) = \{0\}$. The triviality of the kernel is true also if we replace X_i^* with $X_{i,0}^*$ as the two operators coincide on $C_0^\infty(\mathbb{R}^n)$. Theorem 88 guarantees that $X_{i,0}$ is essentially self-adjoint. $\square$

The last step is to focus on the momentum operator and, in view both of (20), of (21) and of what we learned, we can improve slightly also Proposition 93. Yet, the first step is to start from a different definition of momentum operator:

Definition 203. Let $K_j : D(K_j) \rightarrow L^2(\mathbb{R}^n)$ be the multiplication operator by k_j defined as in (18), replacing x with k . We call **momentum operator**

$$P_j \doteq \mathcal{F} \circ K_j \circ \mathcal{F}^{-1},$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ are respectively the Fourier and inverse Fourier transform on $L^2(\mathbb{R}^n)$.

First of all we observe that since $\mathcal{F}$ and its inverse are unitary as a consequence of Plancherel theorem 197, it holds that

$$D(P_j) = \{\psi \in L^2(\mathbb{R}^n) \mid \mathcal{F}^{-1}\psi \in D(K_j)\}. \quad (71)$$

At this stage one can ask why we are using specifically the symbol P_j . The answer lies in the following proposition whose proof is rather complicated, involving essentially the whole [5, §5.3.2]:

Proposition 204. The operators P_j of Definition 20 and of Definition 203 coincide.

We are now ready to state the new and improved version of Proposition 93

Proposition 205. *Each momentum operator P_j , $j = 1, \dots, n$ with domain (71) is self-adjoint and both $C_0^\infty(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ are cores for P_j .*

Proof. Let $\psi \in D(P_j)$. On account of Proposition 202 we know that K_j is self-adjoint and, by Plancherel theorem 197 that $\mathcal{F}$ is unitary. In addition, per direct inspection of (71), $D(P_j) = D(\mathcal{F} \circ K_j)$. Suppose that $\psi \in D(\mathcal{F} \circ K_j)$. As a consequence we are entitled to use Exercise 21 to conclude that P_j is self-adjoint.

To conclude, we recall that both $C_0^\infty(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ are cores for the multiplication operator as per Theorem 202. Hence they are cores also for K_j . We can use once more Exercise 21 to conclude that P_j restricted to either $C_0^\infty(\mathbb{R}^n)$ or $\mathcal{S}(\mathbb{R}^n)$ is essentially self-adjoint. Hence they are both cores for P_j . $\square$

10.3.2 The Creation, the Annihilation and the Number Operators

In this short last section, we introduce three operators, which play a key role in quantum mechanics and in quantum field theory. For the sake of simplicity we consider the one dimensional scenario, namely $\mathcal{H} = L^2(\mathbb{R})$. We call X and P the position (18) and the momentum operator (203) respectively.

Definition 206. *Let $\mathcal{H} = L^2(\mathbb{R})$ and let X, P be the position and the momentum operator respectively. We call²³*

- **annihilation operator** $A \doteq X + iP$, with natural domain $D(A) = D(X) \cap D(P)$,
- **creation operator** $A^* \doteq X - iP$, with natural domain $D(A^*) = D(X) \cap D(P)$
- **number operator** $N = A^*A$, with natural domain $D(N) = \{\psi \in L^2(\mathbb{R}) \mid A\psi \in D(A^*)\}$.

We start investigating the properties of these operators:

Proposition 207. *The creation and annihilation operators are closed and $A^* = A^*$.*

Proof. In view of (19) and of (71), we know that both A and A^* are densely defined since $\mathcal{S}(\mathbb{R}^n) \subset D(X), D(P)$. Theorem 82 entails that A^* and $(A^*)^*$ are closed. Yet, since both X and P are self-adjoint and $D(A) = D(A^*)$, we can conclude that $A^* = A^*$ and $A = (A^*)^* = (A^*)^*$. $\square$

Corollary 208. *The number operator N is self-adjoint and positive.*

Proof. Observe that $\mathcal{S}(\mathbb{R}^n) \subset D(N)$. Hence N is densely defined. The adjoint is $N^* = (AA^*)^* = (A^*)^*A^* = AA^*$ where we used Proposition 207, according to which $A^* = A^*$ and also $(A^*)^* = A$ since $D(A) = D(A^*)$. To conclude, observe that, for all $\psi \in D(N)$,

$$(\psi, N\psi)_{L^2} = (\psi, AA^*\psi)_{L^2} = (A^*\psi, A^*\psi)_{L^2} = (A^*\psi, A^*\psi)_{L^2} = \|A^*\psi\|_{L^2}^2 \geq 0.$$

In other words N is positive. $\square$

²³In comparison to the physics literature we are putting equal to 1 all dimensional constant, the mass, $\hbar$ and the frequency in particular.

The name *number operator* is given for a reason. Let us look for eigenvectors of N in $L^2(\mathbb{R})$. We observe first of all, that, for every $\psi \in \mathcal{S}(\mathbb{R}^n) \subseteq L^2(\mathbb{R}^n)$,

$$[A, A^*]\psi = (AA^* - A^*A)\psi = \psi. \implies [A, A^*] = \mathbb{I},$$

where the last equality holds also on the whole $D([A, A^*])$.

Exercise 52. Prove that the commutator relation $[A, A^*] = \mathbb{I}$ on $D([A, A^*])$ is equivalent to the *canonical commutation relations* $[X, P] = i\mathbb{I}$ on $D([X, P])$.

Secondly if we consider the Gaussian function

$$\psi_0(x) = (\pi)^{-\frac{1}{4}} e^{-\frac{x^2}{2}} \implies N\psi_0 = 0.$$

In other words $\psi_0(x) \in \mathcal{S}(\mathbb{R}^n)$ is an eigenvector of norm 1 and eigenvalue 0. Consider now

$$\psi_k(x) = \frac{(A^*)^k}{\sqrt{k!}} \psi_0(x) \quad \forall k \in \mathbb{N} \cup \{0\}.$$

This is still an element of $\mathcal{S}(\mathbb{R}^n)$ in view of the definition of the creation operator and it coincides with the so called *Hermite functions of order k* . By using the commutator relation between A and A^* , a straightforward calculation unveils that

$$N\psi_k(x) = k\psi_k(x),$$

that is $\psi_k(x)$ is a norm 1 eigenvector of the number operator with eigenvalue $k \in \mathbb{N} \cup \{0\}$. Yet Hermite polynomials are known to be an orthogonal basis of $L^2(\mathbb{R})$ – see [5, Example 3.32(4)]. Hence we have constructed an orthonormal basis of $L^2(\mathbb{R})$ made of eigenvectors of the number operator, all with positive, integer eigenvalues. This is the reason for the name *number operator*.

A Topology: Basic Definitions and Structures

Topology is a whole branch of pure mathematics which focuses on making precise the heuristic notions of open and closed sets. Understanding it is of supreme relevance as it is at the hearth of key structures and concepts, the prime example being the continuity of maps between sets. Goal of this appendix is not to make a thorough exposition of the state of the art in the topology, but to recall a few elementary, basic facts and results which are used in different points of these notes. A recollection of what follows can be found in any book on topology since we will not dwell into any particularly advanced topic. We start with the main definition:

Definition 209. Let $(X, \mathcal{T})$ be a pair consisting of a set X and a collection of subsets $\mathcal{T}$ lying in X . We call it a **topological space** if the following three conditions are satisfied:

1. $\emptyset, X \in \mathcal{T}$,

2. the intersection of an arbitrary, finite number of elements of $\mathcal{T}$ lies in $\mathcal{T}$,
3. the union of an arbitrary, countable number of elements of $\mathcal{T}$ lies in $\mathcal{T}$.

$\mathcal{T}$ is called a **topology** on X and its elements are denoted as **open sets**.

Having made precise the notion of topology, we can refine further the nomenclature as follows:

Definition 210. Let $(X, \mathcal{T})$ be a topological space. We call

- **open neighbourhood** of a point $p \in X$ any $A \in \mathcal{T}$ such that $p \in A$,
- **interior point** of a subset $W \subset X$ any $p \in W$ such that there exists an open neighbourhood A of p with $A \subset W$,
- the **interior** of a set $W \subset X$

$$\text{Int}(W) \doteq \{x \in X \mid x \text{ is an interior point of } W\},$$

- the **exterior** of a set $W \subset X$

$$\text{Ext}(W) \doteq \{x \in X \mid x \text{ is an interior point of } X \setminus W\},$$

where $\setminus$ stands for the complement.

- the **frontier** of a set $W \subset X$

$$\partial W \doteq X \setminus (\text{Int}(W) \cup \text{Ext}(W)).$$

- a **closed set**, any subset $W \subseteq X$ such that $X \setminus W \in \mathcal{T}$.

It is clear from the definition that a topology $\mathcal{T}$ can contain a huge, if not infinite, number of elements. Much in the same spirit of vector spaces, it is worth considering whether there exists a limited number of elements of $\mathcal{T}$ out of which one can reconstruct all other. This leads to

Definition 211. Let $(X, \mathcal{T})$ be a topological space. We call **basis** of the topology a collection $\mathcal{B}$ of elements of $\mathcal{T}$ such that each element of $\mathcal{T}$ can be written as the union of countable elements of $\mathcal{B}$.

The notion of a topological space is rather generic, but, at the same time, it is devised to behave well under all operations that one would like to perform. The most notable example consists of considering any subset W of a set X endowed with a topology $\mathcal{T}$. The latter induces a counterpart on W by considering $\mathcal{T}_W \doteq \{A \cap W, \quad \forall A \in \mathcal{T}\}$. This is called *induced topology*. In addition topologies are compatible with simple operations such as the Cartesian product.

Definition 212. Let $(X, \mathcal{T})$ and $(X', \mathcal{T}')$ be two topological spaces and let $X \times X'$ be their Cartesian product. Then we call **product topology** $\tilde{\mathcal{T}}$ the collection of all pairs (A, A') with $A \in \mathcal{T}$ and $A' \in \mathcal{T}'$.

At the same time Definition 209 is so generic that it is to devise example which are mathematically well-defined, but behave pathologically.

Example 213. Let X be the so-called **line with two origins**. It is constructed as follows: Take two copies of $\mathbb{R}$, say $\mathbb{R}_1$ and $\mathbb{R}_2$ and identify each point $x_1 \in \mathbb{R}_1$ with the point $x_2 \in \mathbb{R}_2$ such that $x_1 = x_2$ except when $x_1 = 0$. The set X can be made easily a topological space. The standard topology of $\mathbb{R}$ is built out of the union of all possible open intervals $(a, b) \subset \mathbb{R}$. We can induce a topology on X by considering all these intervals whenever the origin of either $\mathbb{R}_1$ or $\mathbb{R}_2$ does not line therein. In this case we have to consider instead two copies, one containing the first origin and one the second.

It is easy to realize that this space is a mess, since it fails to enjoy a simple property, namely that for every pair of points p, q there must exist two disjoint open sets, the first containing p , the second q . This is not possible if p and q are the two origins.

To avoid the problem outlined in this example we require an additional property:

Definition 214. Let $(X, \mathcal{T})$ be a topological space. We say that it is a **Hausdorff space**, if for every $x, x' \in X$ with $x \neq x'$, there exists $A, A' \in \mathcal{T}$ such that $x \in A$, $x' \in A'$ and $A \cap A' = \emptyset$.

From now on all topological spaces will be assumed to be Hausdorff even if the condition is not spelt out explicitly. An additional structure, which plays a key role in these notes is the following:

Definition 215. Let $(X, \mathcal{T})$ be a topological space and let $W \subseteq X$. We call **closure** of W the set

$$\overline{W} \doteq \cap \{C \mid W \subseteq C \subseteq X \text{ and } C \text{ is closed}\}.$$

In particular we say that W is **dense** in X if $\overline{W} = X$.

If you believe to have understood the above definitions, it is recommended to solve the following

Exercise 53. Let $(X, \mathcal{T})$ be a topological space and let $W \subseteq X$. Prove that

1. $\overline{W}$ is closed,
2. W is closed if and only if $\overline{W} = W$,
3. $\overline{\overline{W}} = \overline{W}$,
4. If W' is such that $W \subset W'$, then $\overline{W} \subset \overline{W'}$.

For the sake of completeness (just morally speaking of course), it is interesting to emphasize that the standard notions of convergence and continuity of functions are a by-product of that of topological space. This might be totally antithetic to the standard ϵ, δ prescriptions learned in the course of calculus, but it is not the case:

Definition 216. Let $(X, \mathcal{T})$ be a topological space. We say that a sequence $x_n \in X$, $n \in \mathbb{N}$ **converges** to $x \in X$, if for every open neighbourhood A of x , there exists $n_A \in \mathbb{N}$ such that, for all $n > n_A$, $x_n \in A$. In this case we write

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{or also} \quad x_n \rightarrow x \text{ as } n \rightarrow \infty.$$

In addition we call $x \in X$ a **limit point** of a subset $W \subseteq X$ if any neighbourhood of x contains a point of $W \setminus \{x\}$.

A direct consequence of this definition, whose proof can be found in all textbooks of topology, is the following:

Proposition 217. Let $(X, \mathcal{T})$ be a topological space and let $W \subseteq X$. Then $\overline{W}$ coincides with the union of S and of all its limit points.

Definition 218. Let $(X, \mathcal{T})$ and $(X', \mathcal{T}')$ be two topological spaces and let $f : X \rightarrow X'$ be any function. We say that

- f is **continuous** if $f^{-1}(A') \in \mathcal{T}$ for all $A' \in \mathcal{T}'$,
- f is an **homeomorphism** if f is continuous, bijective and f^{-1} is continuous.

Exercise 54. Prove that the standard notion of continuous function over $\mathbb{R}$ is equivalent to the one of Definition 218.

The last structure, which stems from the definition of a topology and which plays a key role in these notes, is that of a compact set.

Definition 219. Let $(X, \mathcal{T})$ be a topological space and let $W \subseteq X$. We say that

- W is **compact** if any open covering of K admits a finite sub-covering, i.e., let $\{A_n\}_{n \in I} \in \mathcal{T}$, I being an index set, be such that $K \subseteq \bigcup_{n \in I} A_n$ (covering), then there exists a finite collection of indices $J \subseteq I$ such that $K \subseteq \bigcup_{n \in J} A_n$.
- W is **relatively compact** if $\overline{W}$ is compact,
- X is **locally compact**, if for any $x \in X$, there exist a relatively compact subset W such that $x \in W$.

To conclude we stress that, although a given set X can be endowed with different topologies, barring in this appendix, we will leave the specification of $\mathcal{T}$ implicit since it will be always clear from the context. Hence we will write simply X in place of $(X, \mathcal{T})$.

B Extreme Topic: The Wavefront Set

We are left with one rather unsatisfactory feature of the theory of distributions, namely their multiplication is in general ill-defined. We have learned the the product uv , $u, v \in \mathcal{D}'(\mathbb{R})$ is meaningful as soon at least one of the two distributions is actually a smooth function. This is clearly non exhaustive as one can infer from the Hölder inequality or from Exercise 42 recalling that $L^p(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ for all integers p and for $p = \infty$ on account of Proposition 194 combined with Definition 185. At the same time it would be desirable to identify a criterion whether two distributions can be multiplied even when they are not part of any L^p -space. If this were not possible, we would have to resort to a case-by-case analysis, a dreadful prospective, especially for the applications to the theory of PDEs and of quantum fields (Feynmann integrals in particular).

Goal of this Appendix is to outline the theory which identifies a *sufficient* criterion for the multiplication of distributions by giving a more detailed characterization of their singularities. A reader interested in this topic should be brave, daring and recalling all trainings in all animes, be ready for suffering before being rewarded.

Although for conceptual and notational consistency, we shall follow mainly [1], a reader is strongly encouraged to refer to [2, Chapt. 8]. Notice that we will be omitting most of the proofs since they can be complicated. We refer to these references for all missing proofs. We start from two preliminary definitions:

Definition 220. A subset $\Gamma \subset \mathbb{R}^n \setminus \{0\}$ is called **conic** if for all $k \in \Gamma$, $\lambda k \in \Gamma$ with $\lambda > 0$. A conic open neighbourhood of a point $x \in \mathbb{R}^n$ is a conic open subset Γ such that $x \in \Gamma$.

If we recall that Corollary 191 guarantees us that the Fourier transform of a distribution in $\mathcal{E}'(\mathbb{R}^n)$ is an entire analytic function, we can introduce a net division in the Fourier space $\mathbb{R}^n \ni k$. Heuristically we look for those direction along which the Fourier transform behaves as if it were a rapidly decreasing function. The idea is motivated from the embedding $C_0^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ which guarantees us that the Fourier transform of a smooth and compactly supported function is both rapidly decreasing (Proposition 182) and analytic (Corollary 191 since $C_0^\infty(\mathbb{R}^n) \subset \mathcal{E}'(\mathbb{R}^n)$).

Definition 221. Let $u \in \mathcal{E}'(\mathbb{R}^n)$. We say that k_0 is a **regular direction** for u if there exists a fixed conic neighbourhood $\Gamma \ni k_0$ such that, for all $N \in \mathbb{N}$, there exists $C_N \geq 0$ such that

$$|\widehat{u}(k)| \leq C_N \langle k \rangle^{-N}, \quad \forall k \in \Gamma \quad (72)$$

where $\langle k \rangle \doteq (1 + |k|^2)^{\frac{1}{2}}$. We call $\Sigma(u)$ the complement in $\mathbb{R}^n$ of all regular directions.

The following lemma shows that Definition 221 is well-founded

Lemma 222. Let $u \in \mathcal{E}'(\mathbb{R}^n)$. Then $u \in C_0^\infty(\mathbb{R}^n)$ if and only if all directions of $\widehat{u}(k)$ are regular.

Proof. Suppose $u \in C_0^\infty(\mathbb{R}^n)$. Then, on account of Proposition 182 $\widehat{u} \in \mathcal{S}(\mathbb{R}^n)$. Hence, for all $k \in \mathbb{R}^n$ (72) is satisfied.

Suppose now that all directions of $\widehat{u}(k)$ are regular. Then $u = \mathcal{F}^{-1}(u) = \int_{\mathbb{R}^n} \frac{d^n k}{(2\pi)^n} \widehat{u}(k) e^{-ik \cdot x}$. Since the integral is uniformly convergent, one can compute explicitly the difference quotient inferring that u is continuous. Taking any derivative is tantamount to multiplying $\widehat{u}$ by a suitable polynomial. Yet (72) remain valid also for $k^\alpha \widehat{u}(k)$ for all multi-indices α . The same argument used to prove continuity shows that u is smooth. Since $u \in \mathcal{E}'(\mathbb{R}^n)$ per assumption, then u is also compactly supported. $\square$

Definition 223. Let $u \in \mathcal{E}'(\mathbb{R}^n)$. We call **singular support** of u , the set $\text{singsupp}(u)$ which is the collection of all $x \in \mathbb{R}^n$, such that there does not exist an open neighbourhood $\Omega \subseteq \mathbb{R}^n$ with $x \in \Omega$ and a smooth function $u|_\Omega \in C^\infty(\Omega)$.

Before introducing the main object of this section, we observe that we are working only with distributions of compact support, whereas we would like to consider $\mathcal{D}'(\mathbb{R}^n)$. Yet, if we multiply $u \in \mathcal{D}'(\mathbb{R}^n)$ by any $\phi \in C_0^\infty(\mathbb{R}^n)$, $\phi u \in \mathcal{E}'(\mathbb{R}^n)$. The problem is that there are infinite possible choices for ϕ . This motivates what follows:

Definition 224. Let $u \in \mathcal{D}'(\mathbb{R}^n)$. We call **wavefront set** of u , the set $WF(u)$ of all points $(x_0, k_0) \in \mathbb{R}^n \times \mathbb{R}^n \setminus \{0\}$ such that there does not exist $\phi \in C_0^\infty(\mathbb{R}^n)$ with $\phi(x_0) = 0$ such that $k_0 \in \Sigma(\phi u)$.

There are at least two doubts to dispel. The first concerns the question on the effect of the multiplication by a compactly supported smooth function on the regular directions. It is (not so) easy to show that their number does not increase:

Lemma 225. Let $u \in \mathcal{E}'(\mathbb{R}^n)$. Then $\Sigma(\phi u) \subseteq \Sigma(u)$ for all $\phi \in C_0^\infty(\mathbb{R}^n)$.

The second doubt concerns instead the role of the point x_0 . If we observe that the wavefront set is a subset of a Cartesian product, we can introduce the natural projections $\pi_1, \pi_2 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ on the first and on the second component.

Proposition 226. Let $u \in \mathcal{D}'(\mathbb{R}^n)$. Then $\pi_1(WF(u)) = \text{suppsing}(u)$.

As a consequence it holds the following:

Lemma 227. Let $u \in \mathcal{D}'(\mathbb{R}^n)$. Then $u \in C^\infty(\mathbb{R}^n)$ if and only if $WF(u) = \emptyset$.

The wavefront set enjoys notable properties under the action of a multiplication and of a derivative operator:

Proposition 228. Let $u \in \mathcal{D}'(\mathbb{R}^n)$. Then

1. for all $f \in C^\infty(\mathbb{R}^n)$, $WF(fu) \subseteq WF(u)$,
2. Let $P = P(x, D)$ be a differential operator with smooth coefficients. $WF(Pu) \subseteq WF(u)$.

The third operation, that we need, is instead related to the tensor product of two-distributions $u, v \in \mathcal{D}'(\mathbb{R}^n)$. The wavefront set shows us that the resulting distribution has singular directions ruled by those of u and v :

Theorem 229. *Let $u, v \in \mathcal{D}'(\mathbb{R}^n)$. Then*

$$WF(u \otimes v) \subseteq (WF(u) \times WF(v)) \cup (WF(u) \times (\text{supp}(v) \times \{0\})) \cup ((\text{supp}(u) \times \{0\}) \times WF(v)).$$

We can now state the first of two key theorems of the theory of distributions, proven with microlocal techniques, that is by looking at the underlying wavefront sets. The first refers to [2, Th. 8.2.4]

Theorem 230. *Let $u \in \mathcal{D}'(\Omega')$ where $\Omega' \subset \mathbb{R}^m$. Let $\Omega \subseteq \mathbb{R}^n$ and let $\lambda \in C^\infty(\Omega, \Omega')$. Denote the set of normals of λ as*

$$N_\lambda = \{(\lambda(x), k) \in \Omega' \times \mathbb{R}^n \mid \nabla \lambda \cdot k = 0\}.$$

Then, if $N_\lambda \cap WF(u) = \emptyset$, $\exists! \lambda^ u \in \mathcal{D}'(\Omega)$ such that*

1. $\lambda^*(u) = u \circ \lambda$ if $u \in C^\infty(\Omega')$,
2. *for any closed conic subset $\Gamma \subset \Omega' \times (\mathbb{R}^n \setminus \{0\})$ such that $\Gamma \cap N_\lambda = \emptyset$, there is a continuous map indicated as $\lambda^* : \mathcal{D}'_\Gamma(\Omega') \rightarrow \mathcal{D}'_{\lambda^*\Gamma}(\Omega)$. Here $\mathcal{D}'_\Gamma(\Omega) = \{u \in \mathcal{D}'(\Omega) \mid WF(u) \subset \Gamma\}$ while²⁴*

$$\lambda^*\Gamma = \{(x, \nabla \lambda(x) \cdot k) \mid (\lambda(x), k) \in \Gamma\}.$$

In addition, when defined, $\lambda^ u$ is such that $WF(\lambda^*(u)) \subseteq WF(u)$.*

At last we can use Theorem 230 together with Theorem 229 to establish a *sufficient* criterion for the multiplication of two distributions, see [2, Th. 8.2.10].

Theorem 231 (multiplication of distributions). *If $u, v \in \mathcal{D}'(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ and if $\delta : \Omega \rightarrow \Omega \times \Omega$ is the diagonal map $\delta(x) = (x, x)$, $x \in \Omega$, then $uv = \delta^*(u \otimes v)$ is well-defined provided that if $(x, k) \in WF(u)$, then $(x, -k) \notin WF(v)$. In addition*

$$WF(uv) \subseteq \{(x, k + k') \in \Omega \times \mathbb{R}^n \setminus \{0\}\},$$

such that $(x, k) \in WF(u)$ or $x = 0$ and $(x, k') \in WF(v)$ or $k' = 0$.

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²⁴Notice that here $\nabla \lambda(x)$ means the gradient of λ evaluated at $x \in \mathbb{R}^n$.

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